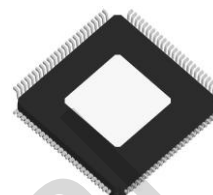


## Features

- AEC-Q100 qualified
- Engineered for ISO26262 compliant system
- 1 pre boost regulator with 2.4 A output current
- 1 pre buck regulator with 2.3 A output current
- 1 linear 5 V regulator with 1.6 A output current
- 3 independent self-protection 5 V tracking regulator with 150 mA output.
- 1 input voltage pin for monitor external tracking.
- Coordinated soft start-up of all regulators
- 4 channels LS injector LS drivers
- 2 channels LS drivers for O2H load with current sense
- 2 channels LS camshaft or solenoid drivers
- 5 channels LS relay drivers
- 2 channels LS LED drivers
- 3 channels LS/HS drivers with low battery function for smart start
- 1 channel LS main relay driver (MRD) with internal diode for reverse battery protection
- 5 channels pre-drivers for external FET drivers. Pre-driver 1&3 configurable for O2H load with external Rshunt-on the source of Ext.N-channel MOS
- 6 channels pre-drivers for internal or external igniter drivers
- 1 K-Line ISO9141/LIN 2.1 compliant
- Integrated charge-pump
- VRS-interface
- Watchdog
- Wake-up pin
- Temperature sensor and monitoring
- Stop-counter with wake up
- Dual bandgap reference & oscillator
- Supports SPI communication mode
- Micro-second-channel MSC for differential single ended mode

- SEO function
- CAN-FD with wake up by CAN function
- Package ETQFP exposed pad



## Description

The HE9788 is an integrated circuit designed for automotive engine management system.

HE9788 is a device realized in BCD proprietary technology, able to provide the full set of power supplies and signal preprocessing peripherals needed to control a 4 cylinders internal combustion engine.



| Part Number | Package  | Marking | Packing        |
|-------------|----------|---------|----------------|
| HE9788RIAJ  | ETQFP100 | HE9788  | 1000/Tape&Reel |

\*Not sure if you got the latest datasheet or not. Please kindly contact the sales of KKChips to confirm. Thanks!

\*If you would like to know about the SPI communication mode, please contact us.

## Contents

|                                                                                                            |           |
|------------------------------------------------------------------------------------------------------------|-----------|
| <b>Features.....</b>                                                                                       | <b>1</b>  |
| <b>Description.....</b>                                                                                    | <b>1</b>  |
| <b>Contents.....</b>                                                                                       | <b>2</b>  |
| <b>List of tables.....</b>                                                                                 | <b>7</b>  |
| <b>List of figures.....</b>                                                                                | <b>10</b> |
| <b>1 Block diagram and pin description.....</b>                                                            | <b>13</b> |
| 1.1 Block diagram.....                                                                                     | 13        |
| 1.2 Pin description.....                                                                                   | 14        |
| <b>2 Absolute maximum ratings.....</b>                                                                     | <b>17</b> |
| 2.1 Latch-up trials.....                                                                                   | 20        |
| 2.2 ESD.....                                                                                               | 20        |
| 2.3 Temperature ranges and thermal data.....                                                               | 21        |
| 2.4 Functional range.....                                                                                  | 23        |
| <b>3 Operation behavior.....</b>                                                                           | <b>24</b> |
| 3.1 Power up/down.....                                                                                     | 24        |
| 3.1.1 Power up/down state diagram chart.....                                                               | 24        |
| 3.1.2 Power up sources and actions summary.....                                                            | 24        |
| 3.1.3 Power up from KEY_IN.....                                                                            | 25        |
| 3.1.4 Power down from KEY_IN when PHOLD = 0.....                                                           | 25        |
| 3.1.5 Power up from WK_IN.....                                                                             | 25        |
| 3.1.6 Power down from WK_IN when PHOLD=0.....                                                              | 26        |
| 3.1.7 Power up from WAKE_UP_TIMER.....                                                                     | 27        |
| 3.1.8 Power on with PHOLD_EN (EOT function 3).....                                                         | 31        |
| 3.1.9 Power down from WAKE_UP_TIMER when PHOLD_EN=0.....                                                   | 32        |
| 3.1.10 Power down from PHOLD.....                                                                          | 32        |
| 3.1.11 Power up from CAN.....                                                                              | 32        |
| 3.1.12 Power down from WAKE_UP_CAN when PHOLD_EN=0.....                                                    | 33        |
| 3.1.13 Finish wake function.....                                                                           | 33        |
| 3.2 Secure Engine OFF(SEO) function.....                                                                   | 35        |
| 3.3 Reset strategies.....                                                                                  | 37        |
| 3.4 Smart reset circuit.....                                                                               | 38        |
| 3.4.1 Smart reset RSTN pin.....                                                                            | 38        |
| 3.4.2 Smart Reset RSTC pin.....                                                                            | 39        |
| 3.4.3 After run reset.....                                                                                 | 39        |
| 3.5 Thermal shutdown.....                                                                                  | 41        |
| <b>4 Watchdog.....</b>                                                                                     | <b>43</b> |
| 4.1 Basic feature.....                                                                                     | 43        |
| 4.1.1 Monitoring module - WDA Functionality.....                                                           | 44        |
| 4.1.2 ERR CNT (EC) and reactions, PWR COUNTER (PWR_CNT) and generation of the monitoring module reset..... | 46        |
| 4.1.3 Generation of a monitoring module reset.....                                                         | 47        |
| 4.1.4 Question generation.....                                                                             | 48        |
| 4.1.5 Response comparison.....                                                                             | 48        |
| 4.1.6 Reset behavior.....                                                                                  | 49        |
| 4.1.7 Access during a sequencer-run.....                                                                   | 50        |
| 4.1.8 Clock and time references.....                                                                       | 50        |
| <b>5 Main relay driver (low-side driver).....</b>                                                          | <b>52</b> |
| 5.1 ON state diagnostic.....                                                                               | 53        |
| 5.1.1 Overcurrent protection.....                                                                          | 53        |

|           |                                                                  |            |
|-----------|------------------------------------------------------------------|------------|
| 5.1.2     | Thermal protection .....                                         | 54         |
| 5.2       | OFF state diagnostic .....                                       | 54         |
| 5.3       | Error handling .....                                             | 56         |
| <b>6</b>  | <b>Multi-voltage regulator supplies .....</b>                    | <b>57</b>  |
| 6.1       | Pre-boost regulator .....                                        | 59         |
| 6.2       | Pre-buck regulator with internal MOS .....                       | 62         |
| 6.3       | Tracking regulator for sensors supply .....                      | 65         |
| 6.4       | External tracking regulator monitor .....                        | 68         |
| 6.5       | VDD5 linear regulator with external MOSFET .....                 | 69         |
| 6.6       | VDD_IO .....                                                     | 71         |
| 6.7       | VB_IN and VB_IN_SW SUPPLY .....                                  | 72         |
| 6.8       | Charge pump .....                                                | 74         |
| 6.9       | Main oscillator .....                                            | 76         |
| <b>7</b>  | <b>Low-side drivers .....</b>                                    | <b>78</b>  |
| 7.1       | Diagnosis .....                                                  | 78         |
| 7.1.1     | ON state - overcurrent protection .....                          | 78         |
| 7.1.2     | ON state - thermal protection .....                              | 78         |
| 7.1.3     | ON/OFF state - Error in on status diagnosis .....                | 79         |
| 7.1.4     | OFF state - short load and open load .....                       | 80         |
| 7.1.5     | Error handling .....                                             | 83         |
| 7.2       | Low-side driver - INJECTOR INJ[1:4] .....                        | 84         |
| 7.2.1     | Enable pin INJ_ENA .....                                         | 87         |
| 7.3       | Low-side driver - O2 HEATER O2H[1:2] .....                       | 87         |
| 7.4       | Low-side driver - SOLENOID SOL [1:2] .....                       | 92         |
| 7.5       | Low-side driver - RELAY RLY [1:5] .....                          | 95         |
| 7.6       | Low-side driver - LED[1:2] .....                                 | 99         |
| <b>8</b>  | <b>High/low-side drivers - STARTER STR[1:3] .....</b>            | <b>102</b> |
| 8.1       | ON state diagnosis .....                                         | 103        |
| 8.2       | ON/OFF state - Error in on status diagnosis .....                | 103        |
| 8.3       | OFF state diagnosis LS mode .....                                | 103        |
| 8.4       | OFF state diagnosis HS mode .....                                | 103        |
| 8.5       | Delay-off function .....                                         | 113        |
| 8.6       | Starter control .....                                            | 115        |
| <b>9</b>  | <b>Pre-drivers - ExtFET (MOS_Gate1/2/3/4/5) - PRD[1:5] .....</b> | <b>117</b> |
| 9.1       | LS external MOSFET pre-driver .....                              | 117        |
| 9.2       | ON state diagnostic .....                                        | 119        |
| 9.3       | ON/OFF state - Error in status diagnosis .....                   | 119        |
| 9.4       | Error handling .....                                             | 120        |
| 9.5       | OFF state diagnostic .....                                       | 120        |
| <b>10</b> | <b>Pre-drivers - IGNITER IGN[1:6] .....</b>                      | <b>123</b> |
| 10.1      | Ignition pre_drivers diagnosis .....                             | 123        |
| 10.2      | Short to GND .....                                               | 124        |
| 10.3      | Short to BAT .....                                               | 125        |

|           |                                                      |            |
|-----------|------------------------------------------------------|------------|
| 10.4      | Open load .....                                      | 126        |
| 10.5      | Error in driver status .....                         | 126        |
| <b>11</b> | <b>Inductive sensor interface .....</b>              | <b>129</b> |
| 11.1      | VRS interface .....                                  | 129        |
| 11.2      | VRS – Normal mode .....                              | 129        |
| 11.2.1    | VRS normal mode configurations .....                 | 131        |
| 11.2.2    | VRS_A - Manual Hysteresis .....                      | 132        |
| 11.2.3    | VRS_A - Fully Adaptive Hysteresis .....              | 132        |
| 11.2.4    | VRS_A - Adaptive filter time .....                   | 133        |
| 11.2.5    | VRS_B - Manual Hysteresis .....                      | 135        |
| 11.2.6    | VRS_B -Limited Adaptive Hysteresis .....             | 135        |
| 11.2.7    | VRS_B - Fixed Filter Time .....                      | 135        |
| 11.3      | VRS diagnostic mode .....                            | 135        |
| 11.4      | Application circuit .....                            | 136        |
| <b>12</b> | <b>CAN FD interface .....</b>                        | <b>141</b> |
| 12.1      | Functional description .....                         | 142        |
| 12.2      | CAN state diagram .....                              | 143        |
| 12.3      | CAN normal mode .....                                | 144        |
| 12.4      | CAN low-power mode .....                             | 144        |
| 12.5      | CAN Sleep Mode .....                                 | 144        |
| 12.6      | CAN error handling .....                             | 144        |
| 12.6.1    | Dominant CAN_TX time out .....                       | 145        |
| 12.6.2    | CAN permanent recessive .....                        | 145        |
| 12.6.3    | CAN permanent dominant .....                         | 145        |
| 12.6.4    | CAN_RX permanent recessive .....                     | 145        |
| 12.6.5    | Smart reset RSTN effects .....                       | 146        |
| 12.7      | Wake up With U-Chip OFF .....                        | 146        |
| 12.8      | Wake up with U-Chip ON .....                         | 146        |
| 12.8.1    | Wake up options .....                                | 146        |
| 12.8.2    | Normal pattern wake up .....                         | 146        |
| 12.8.3    | No pattern wake up .....                             | 147        |
| 12.9      | Automatic voltage biasing .....                      | 147        |
| 12.10     | CAN reset matrix .....                               | 148        |
| 12.11     | CAN looping mode .....                               | 148        |
| <b>13</b> | <b>LIN/K-LINE interface .....</b>                    | <b>159</b> |
| 13.1      | Functional description .....                         | 159        |
| 13.2      | LIN receive only mode .....                          | 160        |
| 13.3      | LIN thermal shut down .....                          | 160        |
| 13.4      | LIN error handling .....                             | 160        |
| 13.4.1    | LIN dominant TXD timeout .....                       | 160        |
| 13.4.2    | LIN Permanent Dominant .....                         | 160        |
| 13.4.3    | Timing diagram .....                                 | 164        |
| 13.4.4    | LIN permanent recessive .....                        | 167        |
| <b>14</b> | <b>Built in self test .....</b>                      | <b>167</b> |
| 14.1      | Power supply independency and voltage monitors ..... | 167        |
| 14.2      | Analog comparators BIST .....                        | 169        |
| <b>15</b> | <b>Stand-by memory .....</b>                         | <b>171</b> |

|           |                                                                      |            |
|-----------|----------------------------------------------------------------------|------------|
| <b>16</b> | <b>DAC .....</b>                                                     | <b>172</b> |
| <b>17</b> | <b>ADC .....</b>                                                     | <b>173</b> |
| <b>18</b> | <b>Micro second channel (MSC) interface .....</b>                    | <b>174</b> |
| 18.1      | Function description .....                                           | 174        |
| 18.2      | Downstream communication .....                                       | 179        |
| 18.2.1    | Command frame .....                                                  | 179        |
| 18.2.2    | Data frame .....                                                     | 180        |
| 18.3      | Upstream communication .....                                         | 182        |
| 18.4      | Micro Second Channel activity watchdog .....                         | 183        |
| 18.5      | Downstream frame Bit Map .....                                       | 185        |
| 18.5.1    | CONFIG-REG 1 .....                                                   | 189        |
| 18.5.2    | CONFIG-REG 2 .....                                                   | 190        |
| 18.5.3    | CONFIG-REG 3 .....                                                   | 190        |
| 18.5.4    | CONFIG-REG 4 .....                                                   | 191        |
| 18.5.5    | CONFIG-REG 5 .....                                                   | 192        |
| 18.5.6    | CONFIG-REG 6 .....                                                   | 192        |
| 18.5.7    | CONFIG-REG 7 .....                                                   | 193        |
| 18.5.8    | CONFIG-REG 8 .....                                                   | 194        |
| 18.5.9    | CONFIG-REG 9 .....                                                   | 194        |
| 18.5.10   | CONFIG-REG 10 .....                                                  | 195        |
| 18.5.11   | CONFIG-REG 11 .....                                                  | 196        |
| 18.5.12   | CONFIG-REG 12 .....                                                  | 196        |
| 18.5.13   | CONFIG-REG 13 .....                                                  | 197        |
| 18.5.14   | CONFIG-REG 14 .....                                                  | 198        |
| 18.5.15   | CONFIG-REG 15 .....                                                  | 199        |
| 18.5.16   | CONFIG-REG 16 .....                                                  | 199        |
| 18.5.17   | CONFIG-REG 17 .....                                                  | 200        |
| 18.5.18   | CONFIG-REG STBY_NVM .....                                            | 201        |
| 18.5.19   | CONFIG-REG 20 .....                                                  | 202        |
| 18.5.20   | CONFIG-REG 21 .....                                                  | 203        |
| 18.5.21   | Commands .....                                                       | 204        |
| 18.6      | Upstream frame .....                                                 | 206        |
| 18.6.1    | Upstream Bit Map Read1 (CONFIG-REG 1-4) .....                        | 206        |
| 18.6.2    | Upstream Bit Map Read2 (CONFIG-REG 5-8) .....                        | 206        |
| 18.6.3    | Upstream Bit Map Read3 (CONFIG-REG 11-13, 16-0) .....                | 207        |
| 18.6.4    | Upstream Bit Map Read4 (CONFIG-REG 16-1 + Wake up timer_SET) .....   | 207        |
| 18.6.5    | Upstream Bit Map Read5 (CONFIG-REG 17-0 + WAKE_UP_TIMER VALUE) ..... | 208        |
| 18.6.6    | Upstream Bit Map Read6 (DRIVER DIAGNOSIS) .....                      | 208        |
| 18.6.7    | Upstream Bit Map Read7 (DRIVER DIAGNOSIS) .....                      | 209        |
| 18.6.8    | Upstream Bit Map Read8 (DRIVER DIAGNOSIS) .....                      | 209        |
| 18.6.9    | Upstream Bit Map Read9 (DRIVER DIAGNOSIS) .....                      | 210        |
| 18.6.10   | Upstream Bit Map Fast Read10 (SAFETY WDA) .....                      | 210        |
| 18.6.11   | Upstream Bit Map Read11 (SAFETY+BIST+VRS+LIN+ASIC_REV) .....         | 211        |
| 18.6.12   | Upstream Bit Map Read12 (PHOLD+POWER_UP+ADC) .....                   | 214        |
| 18.6.13   | Upstream Bit Map Read13 (SUPPLY UV/OV) .....                         | 215        |
| 18.6.14   | Upstream Bit Map Read14 (THERMAL WARNING AND CAN) .....              | 216        |
| 18.6.15   | Upstream Bit Map Read15 (MEM_STBY_CONFIG+CONFIG_REG20+UCs) .....     | 216        |
| 18.6.16   | Upstream Bit Map Read16 (MEM_STBY_DATA) .....                        | 218        |
| <b>19</b> | <b>Package information. ....</b>                                     | <b>219</b> |
|           | ETQFP package information .....                                      | 219        |
| <b>20</b> | <b>Recommended pad size .....</b>                                    | <b>220</b> |
| <b>21</b> | <b>Appendix A Device behavior .....</b>                              | <b>221</b> |
| 21.1      | Summary of all drivers block .....                                   | 221        |
| 21.2      | Slew rate and on/off delay time .....                                | 222        |
| 21.3      | Power up/down scenarios .....                                        | 223        |



21.4 Bill of material..... 236

**21 Customer Service ..... 238**

**22 Revision history ..... 239**

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## List of tables

|                                                                                                      |    |
|------------------------------------------------------------------------------------------------------|----|
| Table 1-1 Pin function .....                                                                         | 14 |
| Table 2-1 Absolute maximum ratings .....                                                             | 17 |
| Table 2-2 ESD .....                                                                                  | 20 |
| Table 2-3 Pin classify .....                                                                         | 21 |
| Table 2-4 Temperature ranges and thermal data .....                                                  | 21 |
| Table 2-5 Functional range .....                                                                     | 23 |
| Table 3-1 Power up sources and actions summary .....                                                 | 24 |
| Table 3-2 Key pins electrical characteristics .....                                                  | 25 |
| Table 3-3 WK pins electrical characteristics .....                                                   | 26 |
| Table 3-4 VB_STBY pins electrical characteristics .....                                              | 27 |
| Table 3-5 Wake up timer resolution .....                                                             | 31 |
| Table 3-6 Finish wake timer electrical characteristics .....                                         | 35 |
| Table 3-7 SEO timing .....                                                                           | 35 |
| Table 3-8 SEO Circuit electrical characteristic .....                                                | 36 |
| Table 3-9 Smart reset circuit electrical characteristic table 1 .....                                | 39 |
| Table 3-10 Smart reset circuit electrical characteristic table 2 .....                               | 40 |
| Table 3-11 Thermal shutdown .....                                                                    | 41 |
| Table 4-1 Error counter .....                                                                        | 46 |
| Table 4-2 Internal states for WDA_INIT=1 .....                                                       | 48 |
| Table 4-3 Question and answer .....                                                                  | 49 |
| Table 4-4 Watchdog reset behavior .....                                                              | 49 |
| Table 4-5 WDA circuit electrical characteristic .....                                                | 51 |
| Table 4-6 WDA reset circuit electrical characteristic .....                                          | 51 |
| Table 5-1 ON state diagnostic .....                                                                  | 53 |
| Table 5-2 Main relay low-side driver electrical characteristics .....                                | 54 |
| Table 6-1 Pre-boost regulator output electrical characteristics .....                                | 59 |
| Table 6-2 Pre-boost regulator external components electrical characteristics .....                   | 61 |
| Table 6-3 Pre-buck regulator output electrical characteristics .....                                 | 63 |
| Table 6-4 Pre-buck regulator external components electrical characteristics (Vpre application) ..... | 65 |
| Table 6-5 T racking sensor supplies electrical characteristics .....                                 | 65 |
| Table 6-6 VT rack external components characteristics .....                                          | 67 |
| Table 6-7 External VT rack monitor .....                                                             | 68 |
| Table 6-8 VDD5 linear controller pre-driver electrical characteristics .....                         | 69 |
| Table 6-9 Linear controller pre-driver. External components characteristics .....                    | 70 |
| Table 6-10 VDD_IO electrical characteristics .....                                                   | 71 |
| Table 6-11 VB_IN electrical characteristics .....                                                    | 72 |
| Table 6-12 VB_IN_SW electrical characteristics .....                                                 | 73 |
| Table 6-13 Charge pump characteristics .....                                                         | 75 |
| Table 6-14 Clock monitor characteristics .....                                                       | 77 |
| Table 7-1 Driver status diagnosis .....                                                              | 79 |

|                                                                                                    |     |
|----------------------------------------------------------------------------------------------------|-----|
| Table 7-2 STA diagnosis for drivers.....                                                           | 79  |
| Table 7-3 Low-side driver error handling .....                                                     | 83  |
| Table 7-4 Low-side driver - INJECTOR electrical characteristics .....                              | 85  |
| Table 7-5 INJ_ENA electrical characteristics.....                                                  | 87  |
| Table 7-6 Low-side driver - O2 HEATER electrical characteristics .....                             | 89  |
| Table 7-7 Low-side driver - SOLENOID VALVE electrical characteristics .....                        | 92  |
| Table 7-8 Low-side driver - RELAY electrical characteristics.....                                  | 95  |
| Table 7-9 Low-side driver - LED electrical characteristics.....                                    | 99  |
| Table 8-1 High/low-side driver - STARTER electrical characteristics .....                          | 107 |
| Table 8-2 High/low-side driver delay-off function electrical characteristics.....                  | 114 |
| Table 8-3 EN_P and EN_N electrical characteristics .....                                           | 116 |
| Table 9-1 Pre-driver - ExtFET error handling .....                                                 | 120 |
| Table 9-2 Pre-driver - ExtFET electrical characteristics .....                                     | 120 |
| Table 10-1 Pre-driver - IGNITER diagnosis.....                                                     | 124 |
| Table 10-2 Pre-driver - IGNITER electrical characteristics.....                                    | 127 |
| Table 11-1 VRS_A, VRS_B hysteresis and filter time definition .....                                | 131 |
| Table 11-2 VRS_A hysteresis value .....                                                            | 132 |
| Table 11-3 Peak voltage value ranges .....                                                         | 132 |
| Table 11-4 Input peak voltage range .....                                                          | 133 |
| Table 11-5 VRS sensor parameters .....                                                             | 137 |
| Table 11-6 Flying wheel interface function - Diagnosis test.....                                   | 139 |
| Table 12-1 CAN threshold related Low supply voltage flag .....                                     | 149 |
| Table 12-2 CAN communication operating range .....                                                 | 150 |
| Table 12-3 CAN transmit data input: Pin TxDC .....                                                 | 151 |
| Table 12-4 CAN transmit data output: Pin RxDC .....                                                | 151 |
| Table 12-5 CAN transmitter dominant output characteristics.....                                    | 152 |
| Table 12-6 CAN transmitter recessive output characteristics, normal mode .....                     | 153 |
| Table 12-7 CAN transmitter recessive output characteristics, low power mode, biasing active.....   | 153 |
| Table 12-8 CAN transmitter recessive output characteristics, low-power mode, biasing inactive..... | 153 |
| Table 12-9 CAN Receiver input characteristics during normal mode.....                              | 153 |
| Table 12-10 CAN Receiver input characteristics during low power mode, biasing active .....         | 154 |
| Table 12-11 CAN Receiver input characteristics during low power mode, biasing inactive .....       | 154 |
| Table 12-12 CAN Receiver input resistance.....                                                     | 154 |
| Table 12-13 CAN transceiver delay.....                                                             | 155 |
| Table 12-14 Maximum leakage currents on CAN_H and CAN_L, unpowered .....                           | 156 |
| Table 12-15 Biasing control timings.....                                                           | 157 |
| Table 12-16 Standby current consumption (B attery line @RT).....                                   | 157 |
| Table 12-17 Standby current consumption (Battery line @HT).....                                    | 157 |
| Table 12-18 can_auto_bias = 1 and can_wakeup_en = 1.....                                           | 158 |
| Table 12-19 can_auto_bias = 1 and can_wakeup_en = 0.....                                           | 158 |
| Table 13-1 Electrical characteristics of LIN interface .....                                       | 162 |
| Table 16-1 DAC electrical parameters.....                                                          | 172 |

|                                                                       |     |
|-----------------------------------------------------------------------|-----|
| Table 17-1 ADC electrical parameters.....                             | 173 |
| Table 18-1 MSC communication timing parameters .....                  | 174 |
| Table 18-2 MSC electrical characteristics .....                       | 177 |
| Table 18-3 MSC Interface command frame.....                           | 180 |
| Table 18-4 MSC Interface Data frame .....                             | 181 |
| Table 18-5 Control register .....                                     | 181 |
| Table 18-6 MSC Interface Micro Second Channel activity watchdog ..... | 184 |
| Table 18-7 MSC Interface DOWN STREAM FRAME Bit Map .....              | 185 |
| Table 18-8 Commands.....                                              | 204 |
| Table 18-9 MSC Interface Upstream Bit Map Read1 .....                 | 206 |
| Table 18-10 MSC Interface Upstream Bit Map Read2 .....                | 206 |
| Table 18-11 MSC Interface Upstream Bit Map Read3 .....                | 207 |
| Table 18-12 MSC Interface Upstream Bit Map Read4 .....                | 207 |
| Table 18-13 MSC Interface Upstream Bit Map Read5 .....                | 208 |
| Table 18-14 MSC Interface Upstream Bit Map Read6 .....                | 208 |
| Table 18-15 MSC Interface Upstream Bit Map Read7 .....                | 209 |
| Table 18-16 MSC Interface Upstream Bit Map Read8 .....                | 209 |
| Table 18-17 MSC Interface Upstream Bit Map Read9 .....                | 210 |
| Table 18-18 MSC Interface Upstream Bit Map Read10 .....               | 210 |
| Table 18-19 MSC Interface Upstream Bit Map Read11 .....               | 211 |
| Table 18-20 MSC Interface Upstream Bit Map Read12 .....               | 214 |
| Table 18-21 MSC Interface Upstream Bit Map Read13 .....               | 215 |
| Table 18-22 MSC Interface Upstream Bit Map Read14 .....               | 216 |
| Table 18-23 MSC Interface Upstream Bit Map Read15 .....               | 216 |
| Table 18-24 MSC Interface Upstream Bit Map Read16 .....               | 218 |
| Table 21-1 Summary of all drivers block.....                          | 221 |
| Table 21-2 Bill of material .....                                     | 236 |
| Table 22-1 Document revision history .....                            | 239 |

## List of figures

|                                                                                      |     |
|--------------------------------------------------------------------------------------|-----|
| Figure 1-1 Block diagram .....                                                       | 13  |
| Figure 1-2 Pin connection diagram (top view) .....                                   | 14  |
| Figure 2-1 Thermal simulation set-up .....                                           | 22  |
| Figure 3-1 Power up state diagram .....                                              | 24  |
| Figure 3-2 Timing for Wake Up Timer - driver by MSC start/stop commands .....        | 29  |
| Figure 3-3 Timing for Wake Up Timer - activated (started) by KEY negative edge ..... | 29  |
| Figure 3-4 Wake up timer block diagram .....                                         | 31  |
| Figure 3-5 Power hold diagram .....                                                  | 32  |
| Figure 3-6 Impulse timer clear WAKE_IN_DET after time out .....                      | 34  |
| Figure 3-7 Impulse timer disabled before time out .....                              | 34  |
| Figure 3-8 Timer enabled for a new detection .....                                   | 34  |
| Figure 3-9 SEO delay .....                                                           | 36  |
| Figure 3-10 Reset table .....                                                        | 37  |
| Figure 3-11 Reset circuit .....                                                      | 38  |
| Figure 3-12 Reset matrix .....                                                       | 39  |
| Figure 3-13 RSTN/RSTC after run reset timing .....                                   | 39  |
| Figure 4-1 WDA block diagram .....                                                   | 44  |
| Figure 4-2 WDA state flow .....                                                      | 45  |
| Figure 4-3 Monitoring cycle diagram .....                                            | 46  |
| Figure 5-1 Main relay low-side driver stage .....                                    | 52  |
| Figure 6-1 Power supply block diagram with pre-boost regulator permanent VBAT .....  | 57  |
| Figure 6-2 Power supply block diagram with charge pump without pre-boost .....       | 58  |
| Figure 6-3 Pre-boost block diagram .....                                             | 59  |
| Figure 6-4 Boost threshold comparator's schematic .....                              | 62  |
| Figure 6-5 Power up charge pump behavior .....                                       | 74  |
| Figure 6-6 Power down charge pump behavior .....                                     | 75  |
| Figure 6-7 Clock monitor block diagram .....                                         | 76  |
| Figure 7-1 Low-side driver OFF state diagnosis schematic .....                       | 80  |
| Figure 7-2 Low-side driver OFF state diagnosis load thresholds .....                 | 81  |
| Figure 7-3 Low-side driver OFF state fast pull up current behavior .....             | 82  |
| Figure 7-4 Low-side driver OFF state diagnosis timings .....                         | 83  |
| Figure 7-5 INJECTOR LowSide driver stage .....                                       | 84  |
| Figure 7-6 O2 heater LowSide driver stage .....                                      | 88  |
| Figure 7-7 O2 Heater LowSide driver current sense block .....                        | 88  |
| Figure 7-8 Solenoid valves LowSide driver stage .....                                | 92  |
| Figure 7-9 Relay LowSide driver stage .....                                          | 95  |
| Figure 7-10 LED LowSide driver stage .....                                           | 99  |
| Figure 8-1 Configurable high/low -side Driver block schematic .....                  | 102 |
| Figure 8-2 OFF state diagnostic HS config block schematic .....                      | 104 |
| Figure 8-3 OFF state diagnostic HS config .....                                      | 105 |
| Figure 8-4 High-side driver OFF state fast pull down current behavior .....          | 106 |

|                                                                                           |     |
|-------------------------------------------------------------------------------------------|-----|
| Figure 8-5 OFF state diagnostic high-side, timing .....                                   | 107 |
| Figure 8-6 DELAY_OFF timing .....                                                         | 114 |
| Figure 8-7 DELAY_OFF timing terminated by MSC off command .....                           | 114 |
| Figure 8-8 Safety switch off .....                                                        | 116 |
| Figure 9-1 LS External MOSFET Pre-Driver Stage .....                                      | 117 |
| Figure 9-2 PreDriver3 configured for O2H load bit O2H_PDRV_3 = CONFIG_REG_16_1[7] = 1: .. | 118 |
| Figure 9-3 PreDriver1 configured for O2H load bit O2H_PDRV_1 = CONFIG_REG_16_1[6] = 1: .. | 119 |
| Figure 10-1 Ignition pre-drivers block schematic .....                                    | 123 |
| Figure 10-2 IGN pre-drivers diagnosis timing diagram at short to GND .....                | 124 |
| Figure 10-3 IGN pre-drivers diagnosis timing diagram at short to VBAT .....               | 125 |
| Figure 10-4 Low-side drivers ON diagnosis timing diagram at open load .....               | 126 |
| Figure 11-1 VRS interface block diagram .....                                             | 129 |
| Figure 11-2 VRS block diagram - Normal operating mode .....                               | 130 |
| Figure 11-3 Hysteresis application .....                                                  | 131 |
| Figure 11-4 VRS_A fully adaptive hysteresis .....                                         | 133 |
| Figure 11-5 EN_FALLING_FILT = 1 .....                                                     | 134 |
| Figure 11-6 EN_FALLING_FILT = 0 .....                                                     | 134 |
| Figure 11-7 VRS block diagram - Diagnostic operating mode - Current path .....            | 136 |
| Figure 11-8 Sensor sketch .....                                                           | 136 |
| Figure 11-9 Variable reluctance sensor (VRS) .....                                        | 137 |
| Figure 11-10 Hall effect sensor configuration 1 .....                                     | 138 |
| Figure 11-11 Hall effect sensor configuration 2 .....                                     | 138 |
| Figure 12-1 CAN FD interface diagram .....                                                | 141 |
| Figure 12-2 CAN state diagram .....                                                       | 143 |
| Figure 12-3 CAN wake up options .....                                                     | 147 |
| Figure 12-4 CAN_TX input structure .....                                                  | 149 |
| Figure 12-5 CAN transceiver test circuit .....                                            | 149 |
| Figure 13-1 Block diagram of LIN interface .....                                          | 159 |
| Figure 13-2 LIN TimeOut function .....                                                    | 161 |
| Figure 13-3 LIN_TX input structure .....                                                  | 161 |
| Figure 13-4 LIN transmission timing chart .....                                           | 164 |
| Figure 13-5 LIN reception timing chart .....                                              | 165 |
| Figure 13-6 LIN duty cycle timing chart .....                                             | 165 |
| Figure 13-7 LIN slew rate timing chart .....                                              | 166 |
| Figure 13-8 Test circuit for measurement of slew rate .....                               | 166 |
| Figure 14-1 Band-gap supply architecture .....                                            | 168 |
| Figure 14-2 Analog bist implementation .....                                              | 170 |
| Figure 18-1 MSC communication timing diagram .....                                        | 174 |
| Figure 18-2 Communication diagram between $\mu$ C and U-CHIP .....                        | 176 |
| Figure 18-3 MSC voltage levels diagram .....                                              | 177 |
| Figure 18-4 MSC Command Frame bit stream .....                                            | 180 |
| Figure 18-5 MSC Data Frame bit stream .....                                               | 180 |

|                                                                                                     |     |
|-----------------------------------------------------------------------------------------------------|-----|
| Figure 18-6 MSC upstream .....                                                                      | 183 |
| Figure 18-7 MSC upstream commands example .....                                                     | 183 |
| Figure 18-8 MSC activity watchdog time diagram .....                                                | 184 |
| Figure 19-1 ETQFP package outline .....                                                             | 219 |
| Figure 20-1 Recommended pad size .....                                                              | 220 |
| Figure 0-1 Slew rate and on/off delay time .....                                                    | 222 |
| Figure 0-2 Slew rate and on/off delay time .....                                                    | 223 |
| Figure 0-3 Power up permanent battery without boost and cranking .....                              | 224 |
| Figure 0-4 Power up permanent battery with boost and cranking .....                                 | 225 |
| Figure 0-5 Power up with WK_IN and power down with WK_IN in permanent battery condition .....       | 226 |
| Figure 0-6 Power up with WAKE_UP_EOT/CAN and power down with WAKE_UP_EOT/CAN .....                  | 226 |
| Figure 0-7 Power up with WAKE_UP_EOT/CAN and power down with WAKE_UP_EOT/CAN .....                  | 227 |
| Figure 0-8 Main relay driver timeout .....                                                          | 227 |
| Figure 0-9 Power HOLD .....                                                                         | 228 |
| Figure 0-10 Overcurrent OVC removed before VB present with KEY_IN high and Overcurrent<br>OVC ..... | 229 |
| Figure 0-11 Overcurrent OVC permanent after VB present with KEY_IN high .....                       | 229 |
| Figure 0-12 Overcurrent OVC permanent after VB present with KEY_IN high - unlimited retry .....     | 230 |
| Figure 0-13 Overcurrent OVC removed before Tres activation with KEY_IN high and Overcurrent .....   | 231 |
| Figure 0-14 Overcurrent in permanent battery restart conditions during KEY or WK_IN or .....        | 231 |
| Figure 0-15 Overcurrent in permanent battery PHOLD .....                                            | 232 |
| Figure 0-16 Overtemperature in permanent battery restart conditions during KEY or WK_IN or .....    | 232 |
| Figure 0-17 Overtemperature in permanent battery PHOLD .....                                        | 233 |
| Figure 0-18 Driver/predriver section application diagram .....                                      | 234 |
| Figure 0-19 Interface section application diagram .....                                             | 235 |
| Figure 0-20 Multi regulator supply section application diagram .....                                | 235 |

## 1 Block diagram and pin description

### 1.1 Block diagram

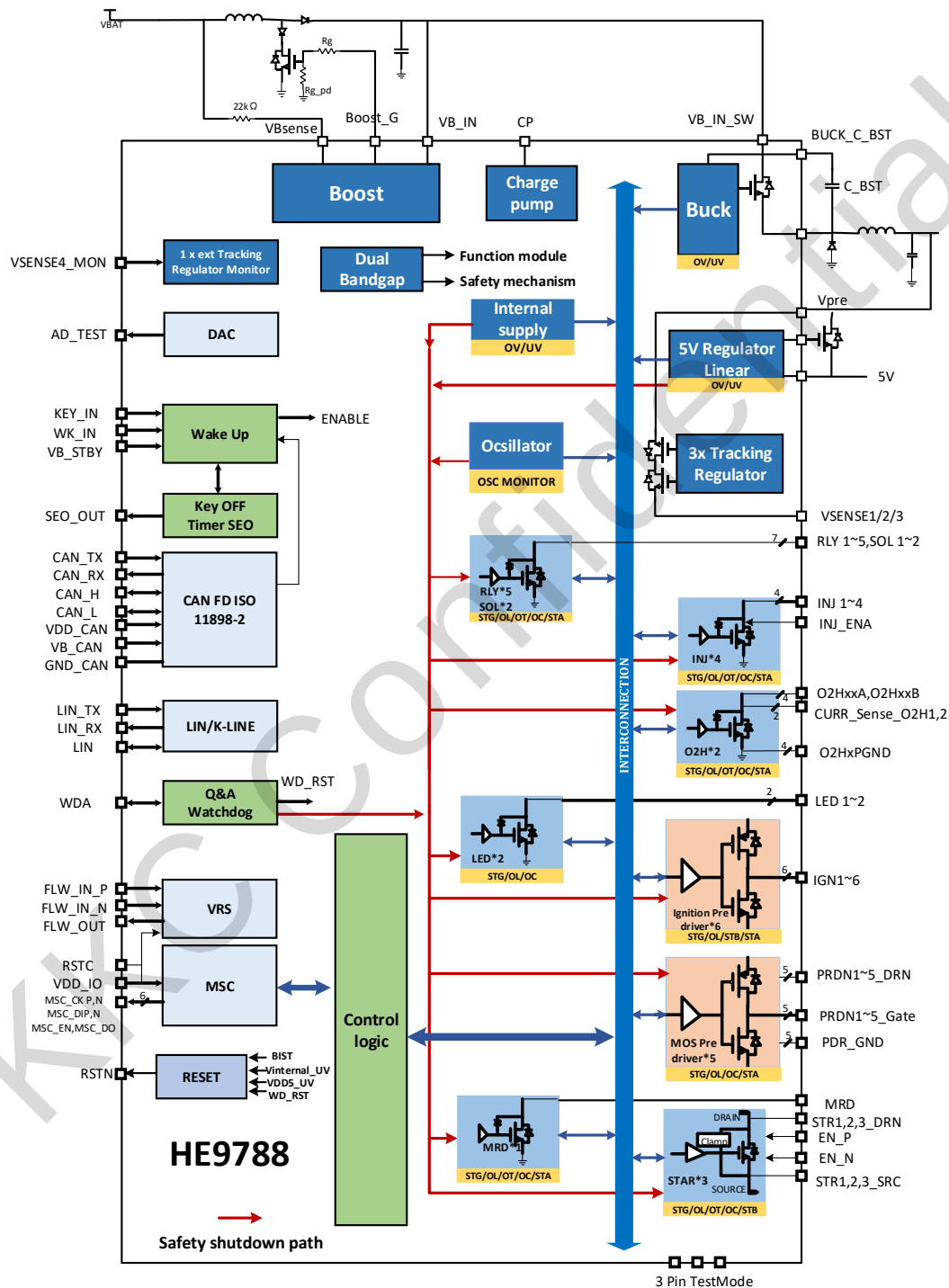


Figure 1-1 Block diagram

## 1.2 Pin description

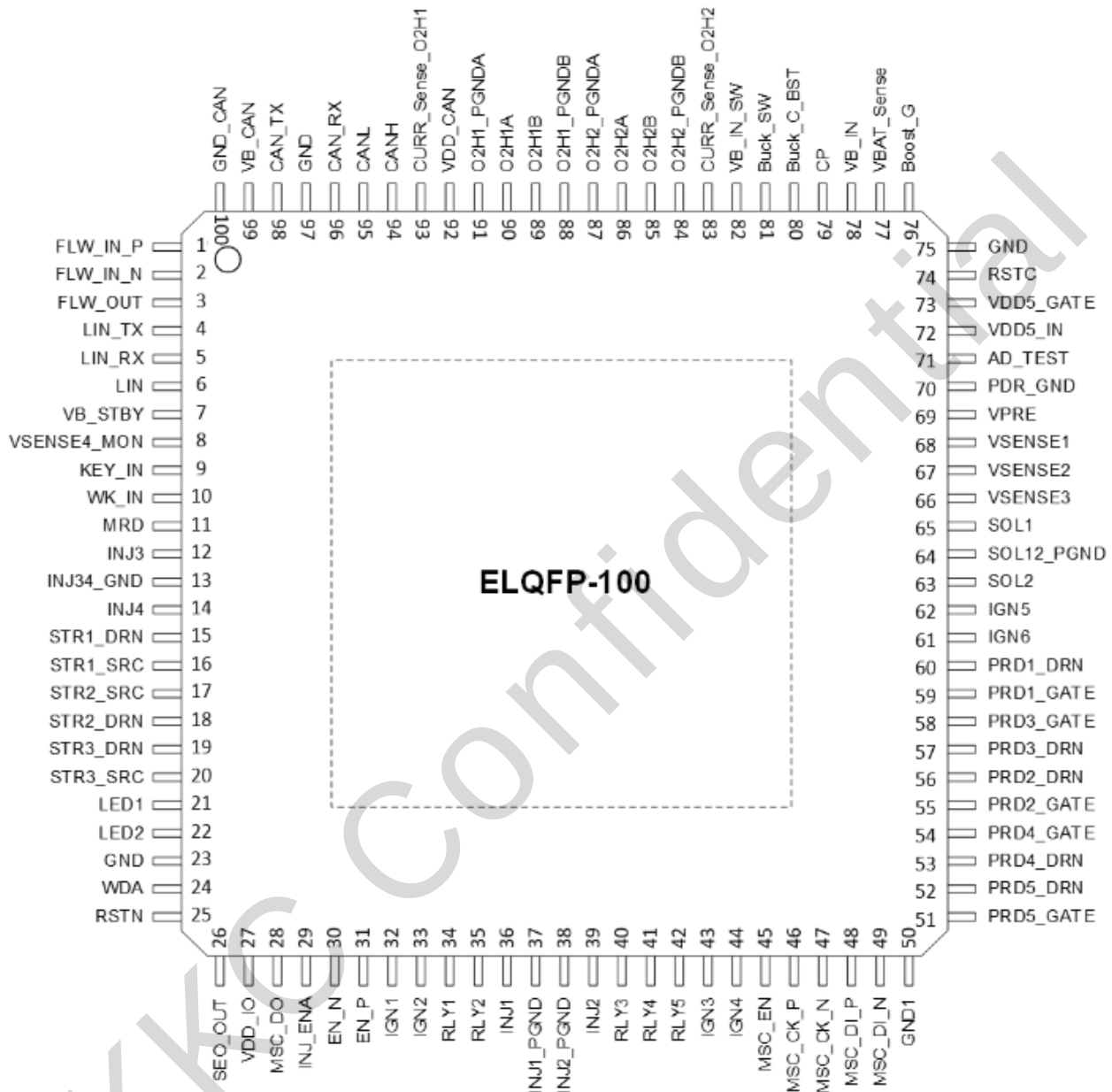


Figure 1-2 Pin connection diagram (top view)

Table 1-1 Pin function

| Pin | Pin name | Function                             | Type        | Note |
|-----|----------|--------------------------------------|-------------|------|
| 1   | FLW_IN_P | Flying wheel inputs voltage positive | Power In    | -    |
| 2   | FLW_IN_N | Flying wheel inputs voltage negative | Power In    | -    |
| 3   | FLW_OUT  | Flying wheel output voltage          | Digital Out | -    |
| 4   | LIN_TX   | LIN data input                       | Digital In  | -    |

# HE9788

Multifunction IC for automotive engine management system



Block diagram and pin description

|    |             |                                                      |                |                           |
|----|-------------|------------------------------------------------------|----------------|---------------------------|
| 5  | LIN_RX      | LIN data output                                      | Digital Out    | -                         |
| 6  | LIN         | LIN driver output                                    | Power In/Out   | -                         |
| 7  | VB_STBY     | Standby function supply voltage                      | Supply         | -                         |
| 8  | VSENSE4_MON | tracking sensor supply Monitor                       | Analog In      | -                         |
| 9  | KEY_IN      | Key signal input                                     | Power In       | -                         |
| 10 | WK_IN       | wake up signal input                                 | Power In       | -                         |
| 11 | MRD         | Main relay drivers output voltage                    | Power Out      | -                         |
| 12 | INJ3        | Injector Driver output voltage                       | Power Out      | -                         |
| 13 | INJ34_PGND  | Ground                                               | GND            | -                         |
| 14 | INJ4        | Injector Driver output voltage                       | Power Out      | -                         |
| 15 | STR1_DRN    | Configurable high/low-side drain voltage             | Power Out      | -                         |
| 16 | STR1_SRC    | Configurable high/low-side source voltage            | Power Out      | -                         |
| 17 | STR2_SRC    | Configurable high/low-side source voltage            | Power Out      | -                         |
| 18 | STR2_DRN    | Configurable high/low-side drain voltage             | Power Out      | -                         |
| 19 | STR3_DRN    | Configurable high/low-side drain voltage             | Power Out      | -                         |
| 20 | STR3_SRC    | Configurable high/low-side source voltage            | Power Out      | -                         |
| 21 | LED1        | LED Driver output voltage                            | Power Out      | -                         |
| 22 | LED2        | LED Driver output voltage                            | Power Out      | -                         |
| 23 | GND         | Ground                                               | GND            | -                         |
| 24 | WDA         | Watchdog output                                      | Digital In/Out | -                         |
| 25 | RSTN        | Reset output for VDD5                                | Digital Out    | -                         |
| 26 | SEO_OUT     | SEO output pin                                       | Digital Out    | -                         |
| 27 | VDD_IO      | Dedicated supply for FLW_OUT, MSC_DO, LIN_RX, CAN_RX | Supply         | -                         |
| 28 | MSC_DO      | MSC digital I/O voltage                              | Digital Out    | -                         |
| 29 | INJ_ENA     | Enable pin for injector driver                       | Digital In     | -                         |
| 30 | EN_N        | enable signals                                       | Digital In     | -                         |
| 31 | EN_P        | enable signals                                       | Digital In     | -                         |
| 32 | IGN1        | Ignition pre-driver output voltage                   | Power Out      | -                         |
| 33 | IGN2        | Ignition pre-driver output voltage                   | Power Out      | -                         |
| 34 | RLY1        | Relay Driver output voltage                          | Power Out      | -                         |
| 35 | RLY2        | Relay Driver output voltage                          | Power Out      | -                         |
| 36 | INJ1        | Injector Driver output voltage                       | Power Out      | -                         |
| 37 | INJ1_PGND   | Ground                                               | GND            | -                         |
| 38 | INJ2_PGND   | Ground                                               | GND            | -                         |
| 39 | INJ2        | Injector Driver output voltage                       | Power Out      | -                         |
| 40 | RLY3        | Relay Driver output voltage                          | Power Out      | -                         |
| 41 | RLY4        | Relay Driver output voltage                          | Power Out      | -                         |
| 42 | RLY5        | Relay Driver output voltage                          | Power Out      | -                         |
| 43 | IGN3        | Ignition pre-driver output voltage                   | Power Out      | -                         |
| 44 | IGN4        | Ignition pre-driver output voltage                   | Power Out      | -                         |
| 45 | MSC_EN      | MSC digital I/O voltage                              | Digital In     | -                         |
| 46 | MSC_CK_P    | MSC digital I/O voltage                              | Digital In     | -                         |
| 47 | MSC_CK_N    | MSC digital I/O voltage                              | Digital In     | -                         |
| 48 | MSC_DI_P    | MSC digital I/O voltage                              | Digital In     | -                         |
| 49 | MSC_DI_N    | MSC digital I/O voltage                              | Digital In     | -                         |
| 50 | GND1        | Power ground                                         | GND            | Shorted to GND by package |
| 51 | PRD5_GATE   | General purpose pre-drivers gate voltage             | Power Out      | -                         |
| 52 | PRD5_DRN    | General purpose pre-drivers feedback voltage         | Power In       | -                         |
| 53 | PRD4_DRN    | General purpose pre-drivers feedback voltage         | Power In       | -                         |
| 54 | PRD4_GATE   | General purpose pre-drivers gate voltage             | Power Out      | -                         |
| 55 | PRD2_GATE   | General purpose pre-drivers gate voltage             | Power Out      | -                         |
| 56 | PRD2_DRN    | General purpose pre-drivers feedback voltage         | Power In       | -                         |
| 57 | PRD3_DRN    | General purpose pre-drivers feedback voltage         | Power In       | -                         |
| 58 | PRD3_GATE   | General purpose pre-drivers gate voltage             | Power Out      | -                         |
| 59 | PRD1_GATE   | General purpose pre-drivers gate voltage             | Power Out      | -                         |
| 60 | PRD1_DRN    | General purpose pre-drivers feedback voltage         | Power In       | -                         |
| 61 | IGN6        | Ignition pre-driver output voltage                   | Power Out      | -                         |

Block diagram and pin description

|     |                     |                                                |                |   |
|-----|---------------------|------------------------------------------------|----------------|---|
| 62  | IGN5                | Ignition pre-driver output voltage             | Power Out      | - |
| 63  | SOL2                | Valve driver output voltage                    | Power Out      | - |
| 64  | SOL12_PGND          | Ground                                         | GND            | - |
| 65  | SOL1                | Valve driver output voltage                    | Power Out      | - |
| 66  | VSENSE3             | 5V tracking sensor supply output voltage       | Power Out      | - |
| 67  | VSENSE2             | 5V tracking sensor supply output voltage       | Supply Out     | - |
| 68  | VSENSE1             | 5V tracking sensor supply output voltage       | Supply Out     | - |
| 69  | VPRE                | Buck output voltage                            | Supply Out     | - |
| 70  | PDR_GND             | Ground pin for predriver supply feedback input | GND            | - |
| 71  | AD_TEST             | AD test pin                                    | Analog Out     | - |
| 72  | VDD5_IN             | 5V regulator feedback voltage                  | Power In       | - |
| 73  | VDD5_GATE           | 5V linear regulator pre-driver output          | Power Out      | - |
| 74  | RST <sub>case</sub> | Reset output for VDDIO supply                  | Digital Out    | - |
| 75  | GND                 | Ground                                         | GND            | - |
| 76  | Boost_G             | Boost LS gate voltage                          | Power Out      | - |
| 77  | VBAT_Sense          | Battery supply voltage                         | Supply In      | - |
| 78  | VB_IN               | Battery supply voltage                         | Supply In      | - |
| 79  | CP                  | Charge pump                                    | Power Out      | - |
| 80  | Buck_C_BST          | Bootstrap capacitor pin                        | Power Out      | - |
| 81  | Buck_SW             | Buck switching pin                             | Power Out      | - |
| 82  | VB_IN_SW            | BUCK supply voltage                            | Power In       | - |
| 83  | CURR_Sense_O2H2     | O2heater drivers output current value          | Analog Out     | - |
| 84  | O2H2_PGND           | Ground                                         | GND            | - |
| 85  | O2H2B               | O2H2B Driver output voltage                    | Power Out      | - |
| 86  | O2H2A               | O2H2A Driver output voltage                    | Power Out      | - |
| 87  | O2H2_PGND           | Ground                                         | GND            | - |
| 88  | O2H1_PGND           | Ground                                         | GND            | - |
| 89  | O2H1B               | O2H1B Driver output voltage                    | Power Out      | - |
| 90  | O2H1A               | O2H1A Driver output voltage                    | Power Out      | - |
| 91  | O2H1_PGND           | Ground                                         | GND            | - |
| 92  | VDD_CAN             | CAN Supply 5 V                                 | Supply In      | - |
| 93  | CURR_Sense_O2H1     | O2heater drivers output current value          | Analog Out     | - |
| 94  | CANH                | CANH output                                    | Power In/Out   | - |
| 95  | CANL                | CANL output                                    | Power In/Out   | - |
| 96  | CAN_RX              | CAN RX data output                             | Digital In/Out | - |
| 97  | GND                 | Ground                                         | GND            | - |
| 98  | CAN_TX              | CAN TX data input                              | Digital In     | - |
| 99  | VB_CAN              | CAN enable signal                              | Supply In      | - |
| 100 | GND_CAN             | Ground CAN                                     | GND            | - |

## 2 Absolute maximum ratings

The component withstands all the following stimuli without any damage or latch-up.

Exceeding any of these values or sustaining it for extended periods may lead to characteristics degradation or component damage.

All voltages are related to analog ground

T<sub>j</sub> -40 to 175 °C unless otherwise specified.

**Table 2-1 Absolute maximum ratings**

| Pin name                  | Parameter                                             | Referred to | Constr.                                              | Min                 | Max | Unit | Pin(1) Direction |
|---------------------------|-------------------------------------------------------|-------------|------------------------------------------------------|---------------------|-----|------|------------------|
| VB_IN                     | Chip supply voltage                                   | GND1        | -                                                    | -0.3                | +40 | V    | S                |
|                           |                                                       |             | 10 ms (ISOpulse1)<br>VB_IN pin shorted with VB_IN_SW | -2                  | -   | V    | S                |
| VB_IN_SW                  | Buck supply voltage                                   | GND1        | -                                                    | -0.3                | +40 | V    | S                |
|                           |                                                       |             | 10 ms (ISOpulse1)                                    | -2                  | -   | V    | S                |
| VBAT_Sense                | Battery supply voltage sense                          | GND1        | -                                                    | -0.3                | +40 | V    | S                |
| VBAT_Sense                | Input current                                         | GND1        | Protect from Reverse battery with 1kΩ resistor       | -16                 | -   | mA   | S                |
| VBAT_Sense                | Input current                                         | GND1        | Protect from ISOPULSE 1 with 1kΩ resistor            | -150                | -   | mA   | S                |
| VB_STBY                   | Standby function supply voltage for EOT               | GND1        | -                                                    | -16                 | +40 | V    | S                |
| Boost_G                   | Boost LS gate voltage                                 | GND1        | -                                                    | -0.3                | +20 | V    | O                |
| VPRE                      | Buck output voltage                                   | GND1        | -                                                    | -0.3                | +20 | V    | S                |
| Buck_SW                   | Buck switching pin                                    | GND1        | -                                                    | -0.3 <sup>(2)</sup> | +40 | V    | S                |
| Buck_C_BST                | Bootstrap capacitor pin                               | GND1        | -                                                    | -0.3                | +45 | V    | I                |
| VDD5_GATE                 | 5 V linear regulator predriver output                 | GND1        | -                                                    | -0.3                | +20 | V    | O                |
| VDD5_IN                   | 5 V regulator feedback voltage                        | GND1        | -                                                    | -0.3                | +20 | V    | S                |
| VDD_IO                    | Dedicated 5 V voltage for FLW_OUT, MSC_DO, TxDC, RxDC | GND1        | -                                                    | -0.3                | +20 | V    | S                |
| VSENSE1, VSENSE2, VSENSE3 | 5 V tracking sensor supply output voltage             | GND1        | -                                                    | -2                  | +40 | V    | S                |
| VSENSE4_MON               | External tracking                                     | GND1        | -                                                    | -2                  | +40 | V    | I                |

|                                                                                                                                          |                                             |      |                                                |         |          |    |     |
|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|------|------------------------------------------------|---------|----------|----|-----|
|                                                                                                                                          | sensor supply Monitor                       |      |                                                |         |          |    |     |
| CP                                                                                                                                       | Charge pump                                 | GND1 | -                                              | -0.3    | VB_IN+5V | V  | S   |
| GND1                                                                                                                                     | Power ground                                | -    | -                                              | -0.3    | +0.3     | V  | O   |
| GND_CAN                                                                                                                                  | Power ground                                | GND1 | -                                              | -0.3    | +0.3     | V  | O   |
| WDA                                                                                                                                      | Digital output (open drain) / Digital input | GND1 | -                                              | -0.3    | +20      | V  | I/O |
| SEO_OUT                                                                                                                                  | Digital output                              | GND1 | -                                              | -0.3    | +20      | V  | O   |
| RSTN, RSTC                                                                                                                               | Digital output / open drain                 | GND1 | -                                              | -0.3    | +20      | V  | I/O |
| AD_TEST                                                                                                                                  | Digital output voltage                      | GND1 | -                                              | -0.3    | +20      | V  | O   |
| INJ_ENA,                                                                                                                                 | Digital input / Digital output              | GND1 | -                                              | -0.3    | +20      | V  | I/O |
| KEY_IN, WK_IN                                                                                                                            | Input voltage                               | GND1 | -                                              | -0.3    | +40      | V  | I   |
| KEY_IN, WK_IN                                                                                                                            | Input current                               | GND1 | Protect from Reverse battery with 1kΩ resistor | -16     | -        | mA | I   |
| KEY_IN, WK_IN                                                                                                                            | Input current                               | GND1 | Protect from ISOPULSE 1 with 1kΩ resistor      | -150    | -        | mA | I   |
| MSC_EN, MSC_CK_P, MSC_CK_N, MSC_DI_P, MSC_DI_N, MSC_DO                                                                                   | MSC digital I/O voltage                     | GND1 | -                                              | -0.3    | +20      | V  | I/O |
| FLW_IN_P, FLW_IN_N                                                                                                                       | Flying wheel inputs voltage                 | GND1 | With 20 mA (Max) inputs current                | -0.3    | Vclamp   | V  | I   |
| FLW_IN_P, FLW_IN_N                                                                                                                       | Flying wheel inputs current                 | -    | -                                              | -20     | +20      | mA | I   |
| FLW_OUT                                                                                                                                  | Flying wheel output voltage (open drain)    | GND1 | -                                              | -0.3    | +20      | V  | O   |
| MRD                                                                                                                                      | Main relay drivers output voltage           | GND1 | -                                              | -16     | Vclamp   | V  | O   |
| INJ1<br>INJ2<br>INJ3<br>INJ4<br>O2H1A<br>O2H1B<br>O2H2A<br>O2H2B<br>SOL1<br>SOL2<br>RLY1<br>RLY2<br>RLY3<br>RLY4<br>RLY5<br>LED1<br>LED2 | Low-side drivers output voltage             | GND1 | -                                              | -1      | Vclamp   | V  | O   |
| STR1_DRN<br>STR2_DRN<br>STR3_DRN                                                                                                         | Configurable high/low-side drain voltage    | GND1 | -                                              | -1      | Vclamp   | V  | O   |
| STR1_SRC<br>STR2_SRC                                                                                                                     | Configurable high/low-side                  | GND1 | -                                              | Vclamp* | +40      | V  | O   |

Absolute maximum ratings

| STR3_SRC                                                                                                              | source voltage                                           |      |                                                                                            |      |      |   |   |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------|--------------------------------------------------------------------------------------------|------|------|---|---|
| IGN1<br>IGN2<br>IGN3<br>IGN4<br>IGN5<br>IGN6                                                                          | Ignition<br>predriver<br>Output voltage                  | GND1 | Negative voltage<br>with limited<br>current at 20 mA,<br>protected by<br>external resistor | -1   | +40  | V | O |
| CURR_Sense_O2H1<br>CURR_Sense_O2H2                                                                                    | O2heater<br>drivers<br>output current<br>value           | GND1 | -                                                                                          | -0.3 | +20  | V | O |
| INJ1_PGND<br>INJ2_PGND<br>INJ34_PGND<br>O2H1_PGND<br>O2H1_PGNDDB<br>O2H2_PGND<br>O2H2_PGNDDB<br>SOL12_PGND<br>PDR_GND | Low-side drivers<br>GND                                  | GND1 | -                                                                                          | -0.3 | +0.3 | V | O |
| PRD1_GATE<br>PRD2_GATE<br>PRD3_GATE<br>PRD4_GATE<br>PRD5_GATE                                                         | General<br>purpose<br>pre-drivers gate<br>voltage        | GND1 | -                                                                                          | -0.3 | +20  | V | O |
| PRD1_DRN<br>PRD2_DRN<br>PRD3_DRN<br>PRD4_DRN<br>PRD5_DRN                                                              | General<br>purpose<br>pre-drivers<br>feedback<br>voltage | GND1 | -                                                                                          | -0.3 | +60  | V | I |
| EN_N<br>EN_P                                                                                                          | Digital input<br>(Cranking<br>drivers enable<br>signals) | GND1 | -                                                                                          | -0.3 | +20  | V | I |
| LIN_TX                                                                                                                | LIN data input                                           | GND1 | -                                                                                          | -0.3 | +20  | V | I |
| LIN_RX                                                                                                                | LIN data output                                          | GND1 | -                                                                                          | -0.3 | +20  | V | O |
| LIN                                                                                                                   | LIN driver<br>output                                     | GND1 | T = -40 °C                                                                                 | -18  | +40  | V | O |
| LIN                                                                                                                   | LIN driver<br>output                                     | GND1 | T = 27 °C, 175 °C                                                                          | -27  | +40  | V | O |
| CAN_TX                                                                                                                | CAN data input                                           | GND1 | -                                                                                          | -0.3 | +20  | V | I |
| CAN_RX                                                                                                                | CAN data<br>output                                       | GND1 | -                                                                                          | -0.3 | +20  | V | O |
| CANH                                                                                                                  | CAN driver<br>output                                     | GND1 | -                                                                                          | -27  | +40  | V | O |
| CANL                                                                                                                  | CAN driver<br>output                                     | GND1 | -                                                                                          | -27  | +40  | V | O |
| VDD_CAN                                                                                                               | CAN Supply 5V                                            | GND1 | -                                                                                          | -0.3 | +20  | V | S |
| VB_CAN                                                                                                                | CAN Supply<br>Battery                                    | GND1 | -                                                                                          | -16  | +40  | V | S |
| VCANH-VCANL                                                                                                           | Differential<br>CAN-bus<br>voltage                       | -    | -                                                                                          | -5   | 10   | V | - |

1. S: supply pin;

I: input pin;

O: output pin;

I/O: input/output pin

2. -2 V allowed during transients.

## 2.1 Latch-up trials

Latch-up tests performed according to JEDEC 78 class 2 Level

## 2.2 ESD

Table 2-2 ESD

| Symbol                | Parameter            | Test condition                                                                                                                                      | Min  | Typ | Max | Unit | Pin    |
|-----------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|-----|------|--------|
| ESD HBM<br>Global     | HBM global pins      | ESD according to Human Body Model (HBM), Q100-002 for global pins; (100pF/1.5kΩ)                                                                    | -4   | -   | 4   | kV   | Global |
| ESD HBM               | HBM local pins       | ESD according to Human Body Model (HBM), Q100-002 for all pins; (100pF/1.5kΩ)                                                                       | -2   | -   | 2   | kV   | ALL    |
| ESD HBM<br>Global CAN | CAN_H,<br>CAN_L pins | ESD according to Human Body Model (HBM), Q100-002 for global pins; (100pF/1.5kΩ) (1)                                                                | -4   | -   | 4   | kV   | Global |
| ESD HBM<br>Global CAN | CAN_H,<br>CAN_L pins | Direct ESD Test according to IEC 61000-4-2 (150 pF, 330Ω) and Hardware Requirements for LIN, CAN and Flexray Interfaces in Automotive Applications" | -4   | -   | 4   | kV   | Global |
| ESD CDM<br>Corner     | CDM corner Pins      | ESD according to Charged Device Model (CDM), Q100-011 Corner pins(Except for pin 1)                                                                 | -750 | -   | 750 | V    | Corner |
| ESD CDM               | CDM all pins         | ESD according to Charged Device Model (CDM), Q100-011 All pins                                                                                      | -500 | -   | 500 | V    | ALL    |

1. HBM with all not zapped pins grounded;
2. ESD CDM Corner pin1 ranges from +-500V;

Table 2-3 Pin classify

| Pin classify                  |                                                                                                                                                                            |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Global pin (ECU connector)    | VB_IN VB_STBY VSENSE1/2 VESNSE3/4_MON KEY_IN WK_IN MRD INJ1/2/3/4 O2H1/2 SOL1/2 RLY1/2/3/4/5 LED1/2 STR1/2/3_DRN STR1/2/3_SRC IGN1/2/3/4/5/6 LIN PRD1/2/3/4/5_DRN          |
| Direction pin (power and I/O) | VDD_IO WDA SOE_OUT RSTN INJ_ENA MSC_EN MSC_CK_P MSC_CK_N MSC_DI_P MSC_DI_N MSC_DO FLW_OUT CURR_Sense_O2H1/2 PRD1/2/3/4/5_GATE EN_P EN_N LIN_TX LIN_RX VDD5_IN CP EN_N EN_P |
| Local pins                    | VB_IN_SW Boost_G VPRE BUCK_SW Buck_C_BST VDD5_GATE FLW_IN_P FLW_IN_N VBAT_Sense                                                                                            |
| GND pins                      | GND1 INJ1/2/3/4_PGND O2H1/2_PGND_A O2H1/2_PGND_B GND                                                                                                                       |

## 2.3 Temperature ranges and thermal data

Table 2-4 Temperature ranges and thermal data

| Pin | Symbol                            | Parameter                              | Test condition                                                                                    | Min | Typ | Max | Unit |
|-----|-----------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------------|-----|-----|-----|------|
| -   | Ta                                | Operating temperature                  | -                                                                                                 | -40 | -   | 125 | °C   |
| -   | Tj                                | Junction temperature                   | -                                                                                                 | -40 | -   | 175 | °C   |
| -   | Tstr                              | Storage temperature                    | -                                                                                                 | -50 | -   | 175 | °C   |
| -   | R <sub>THj-c</sub> <sup>(1)</sup> | Thermal resistance junction to case    | -                                                                                                 | -   | 1   | -   | °C/W |
| -   | R <sub>THj-a</sub> <sup>(1)</sup> | Thermal Resistance junction to ambient | supposing a mounting on board 2s2p + heatsink vias (refer to application note for PCB parameters) | -   | 10  | -   | °C/W |

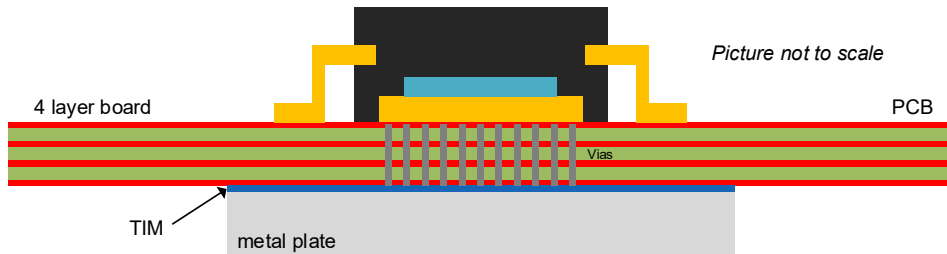
1. With 2s2p PCB thermally enhanced, cold plate as per std Jedec best practice guidelines (JESD51), assuming Pdiss = 5 W dissipated statically and homogeneously, thermal vias under package exposed pad assuming filling ratio 50% and 35 μm vias diameter (157 vias estimation); refer to Figure 2-1 Thermal simulation set-up.

### Note:

All parameters are tested in the temperature range Tj -40 - 150°C; device functionality at high temperature is guaranteed by bench validation, electrical parameters are guaranteed by correlation with ATE tests at reduced temperature and adjusted limits (if needed). Temperature sensor present, please refer to [Section 3.5](#).

## Boundary conditions

- Thermal simulations carried on using typical boundaries for ECU applications:
- 4 layer board with thermal vias
- metal plate in contact with board
- Thermal interface material (TIM) between PCB and metal plate
- $T_{AMB} = 135^{\circ}\text{C}$  / natural convection



With these boundaries and considering uniform power on overall chip surface, package thermal resistance is:  
 **$\text{Theat}_{J-A} = 10^{\circ}\text{C/W}$**

**Figure 2-1 Thermal simulation set-up**

## 2.4 Functional range

Table 2-5 Functional range

| Symbol | Parameter            | Note                                                                                                                                                                                                                                                                                     | Min  | Typ | Max | Unit | Pin   |
|--------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|-----|------|-------|
| -      | Cranking condition   | u-chip in cranking condition, reduced operation range, main relay and delayed off power stages keep the status if delay-off function is triggered. All other driver/regulators /CAN/LIN are in off condition. See description in the Reset Matrix Table, <a href="#">Section 12.10</a> . | 4.25 | -   | 4.8 | V    | VB_IN |
| -      | Low drop condition   | u-chip in low drop operation condition, all functions are guaranteed all parameters are guaranteed unless specified.<br>Vpre and VDD5 are in low drop condition                                                                                                                          | 4.8  | -   | 7   | V    | VB_IN |
| -      | Normal condition     | u-chip in normal operation condition. All functions and parameters are guaranteed, considering the description range of electrical characteristic table and parameters test set up                                                                                                       | 7    | -   | 19  | V    | VB_IN |
| -      | Jump start condition | u-chip in jump start condition, all functions are guaranteed. Parameters are guaranteed unless specified.                                                                                                                                                                                | 19   | -   | 28  | V    | VB_IN |
| -      | Load dump condition  | u-chip in load dump condition. U-chip is no damage. All drivers/regulators/CAN/LIN are in off condition. See description in the Reset Matrix Table, <a href="#">Section 3.3</a> and <a href="#">Section 12.10</a> .                                                                      | 28   | -   | 40  | V    | VB_IN |

### 3 Operation behavior

#### 3.1 Power up/down

##### 3.1.1 Power up/down state diagram chart

The power up/down sequences are shown as state diagram in the following figure:

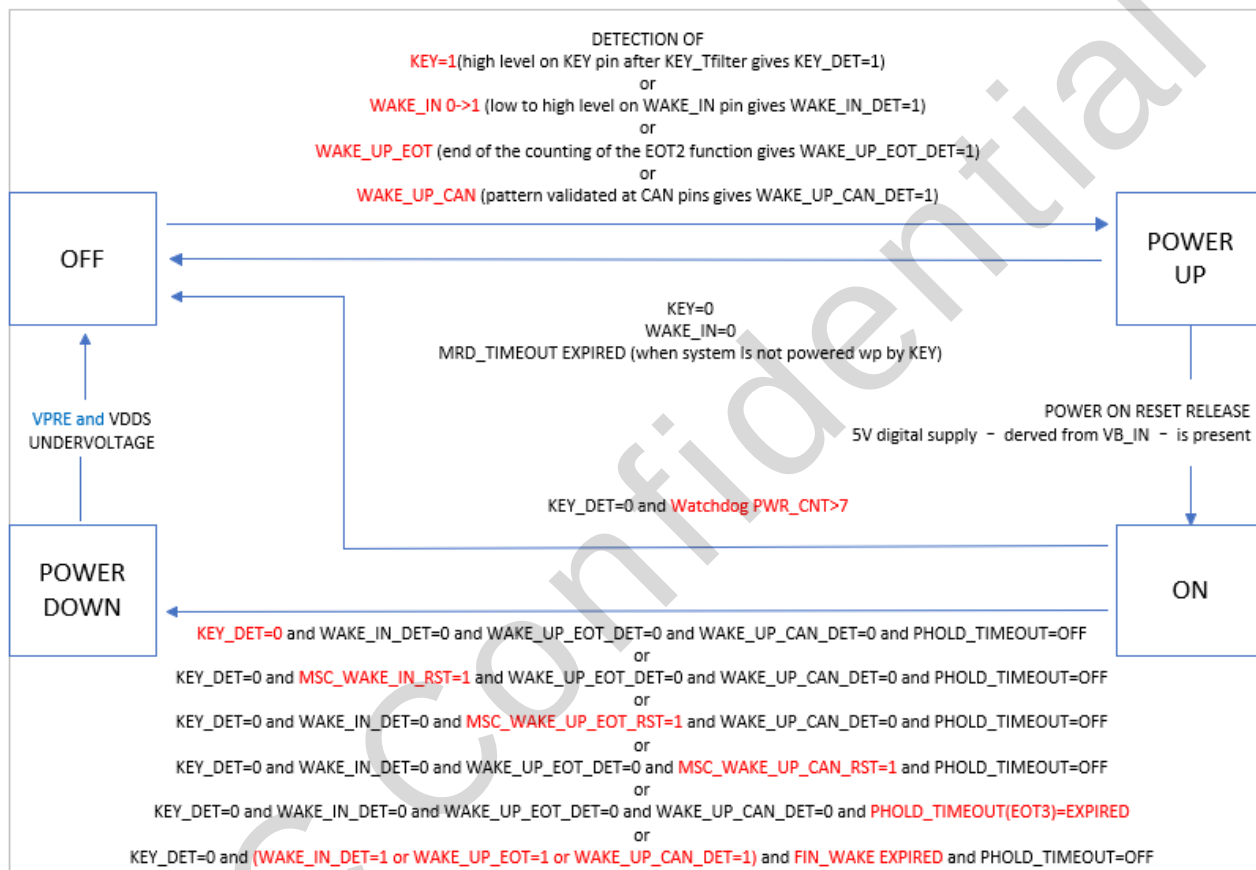


Figure 3-1 Power up state diagram

##### 3.1.2 Power up sources and actions summary

Table 3-1 Power up sources and actions summary

| KEY_IN | WK_IN  | EOT    | CAN_FD | VB_IN     | MRD | Actions                                                   |
|--------|--------|--------|--------|-----------|-----|-----------------------------------------------------------|
| active | -      | -      | -      | permanent | on  | MRD on by KEY_IN pin                                      |
| -      | active | -      | -      | permanent | off | MRD used as normal low-side driver (is controlled by MSC) |
| -      | -      | active | -      | permanent | off | MRD used as normal low-side driver (is controlled by MSC) |
| -      | -      | -      | active | permanent | off | MRD used as normal low-side driver (is controlled by MSC) |

### 3.1.3 Power up from KEY\_IN

The chip is powered up if KEY\_IN signal stays asserted for Key\_Tfilter. There is a MSC bit (*KEY\_IN\_PIN\_FLT* in read command 15 frame 2) to indicate the status of KEY\_IN pin signal after filter time.

### 3.1.4 Power down from KEY\_IN when PHOLD = 0

The chip is powered down if KEY\_IN signal stays de-asserted for Key\_Tfilter if no other power up source is active.

#### Condition:

$4.8 \leq V_{B\_IN} \leq 18 \text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified.

**Table 3-2 Key pins electrical characteristics**

| Symbol      | Parameter                                                           | Test condition             | Min | Typ | Max | Unit          | Pin    |
|-------------|---------------------------------------------------------------------|----------------------------|-----|-----|-----|---------------|--------|
| $V_{IH}$    | Logic input high voltage threshold                                  | -                          | 3.6 | -   | 4.1 | V             | KEY_IN |
| $V_{IL1}$   | Logic input low voltage threshold<br>Main logic in POR not released | -                          | 3.1 | -   | 3.6 | V             | KEY_IN |
| $V_{IHYS}$  | Logic input hysteresis                                              | -                          | 0.1 | -   | 0.8 | V             | KEY_IN |
| Key_Tfilter | Filter time for switching on/off                                    | Guaranteed by test scan(1) | 7.5 | 16  | 24  | ms            | KEY_IN |
| $I_{input}$ | Input current of KEY_IN                                             | KEY_IN $\leq 5 \text{ V}$  | -   | -   | 340 | $\mu\text{A}$ | KEY_IN |
| $R_{PD}$    | Pull-down resistor                                                  | KEY_IN = 1 V               | 170 | 300 | 500 | k $\Omega$    | KEY_IN |

1. Filter time for switching on, is active until KEY logic is out of POR, Filter time for switching off, is active until main logic out of POR.

### 3.1.5 Power up from WK\_IN

The chip is powered up if WK\_IN signal has a rising edge and stays asserted for WK\_Tfilter.

The u-chip will not power up until further rising edge on WK\_IN pin. This detection of a WK\_IN condition flag is named *WK\_IN\_DET* (detection of a WK\_IN condition) and is clear by VDD5/VDDIO under voltage or by the MSC command *WK\_IN\_RST*. In these cases, if no other power-on source is active, a power down sequence is started. When U chip is already in on state (the logic is out of power on reset), the rising edge on pin WK\_IN only sets a flag, with no impact on power supply.

The Main Relay Driver is not switched on by *WAKE\_IN\_DET* rising edge.

There is a MSC bit (*WAKE\_IN\_PIN\_FLT* in read command 15 frame 2) to indicate the status of WK\_IN pin signal after filter time.

A negative glitch on WK\_IN can produce unwanted power up and also prevent the correct power down.

To cover the major cases the falling edge filtering is done both in the VB\_STBY domain and in the VB\_IN domain. Depending on which supply is present the behavior is described in the following cases:

1. If VB\_STBY is present the falling edge filtering is applied also to power up so a negative glitch on WK\_IN will not power up the device.
2. If VB\_IN is present (after the device is already powered up) the falling edge filtering on WK\_IN avoids that a glitch (happening in PHOLD after the MSC has cleared the WK\_IN detection) is recognized as valid detection and the device is kept on.

### 3.1.6 Power down from WK\_IN when PHOLD=0

The chip is powered down if the MSC clears, by the MSC command *WK\_IN\_RST*, the detection of WK\_IN positive edge if no other power up source is active. The power down is actuated even if WK\_IN is still high. The power down is not activated by WK\_IN going low or WK\_IN low level. The detection reset by MSC command, *WK\_IN\_RST* (CONFIG\_REG12[7]), is active at level, so from when MSC writes them at 1 they keep on clearing the detections. If there is further wake detection signal, the detection signal will be cleared, until the MSC writes 0 in the bits.

**Condition:**

$4.8 \leq V_{B\_IN} \leq 18 \text{ V}$ ,  $T_j -40$  to  $175 \text{ }^\circ\text{C}$  unless otherwise specified.

**Table 3-3 WK pins electrical characteristics**

| Symbol       | Parameter                                                              | Test condition               | Min | Typ | Max | Unit          | Pin   |
|--------------|------------------------------------------------------------------------|------------------------------|-----|-----|-----|---------------|-------|
| $V_{IH}$     | Logic input high voltage threshold                                     | -                            | 3.6 | -   | 4.1 | V             | WK_IN |
| $V_{IL}$     | Logic input low voltage threshold                                      | -                            | 3.1 | -   | 3.6 | V             | WK_IN |
| $V_{IHYS}$   | Logic input hysteresis                                                 | -                            | 0.1 | -   | 0.8 | V             | WK_IN |
| $I_{input}$  | Input current of WK_IN                                                 | $W_{K\_IN} \leq 5 \text{ V}$ | -   | -   | 340 | $\mu\text{A}$ | WK_IN |
| WK_Tfilter   | Filter time for switching on/off                                       | Guaranteed by test scan      | 1   | 2   | 3   | ms            | WK_IN |
| $R_{PD}$     | Pull-down resistor                                                     | $W_{K\_IN}=1\text{V}$        | 170 | 300 | 500 | k $\Omega$    | WK_IN |
| Tbattery_det | Main logic POR timeout (used to recognize permanent battery condition) | -                            | 210 | -   | 450 | $\mu\text{s}$ | -     |

### 3.1.7 Power up from WAKE\_UP\_TIMER

The chip is powered up when an end counting is reached inside the wake\_up\_timer block.

The end and enable counting are set by MSC before the chip is powered down.

When U chip is already in on state (the logic is out of power on reset), the wake\_up\_timer end counting only sets a flag, with no impact on power supply. This flag is named *WAKE\_UP\_EOT\_DET* (detection of a wake\_up end of time condition) and is clear by VDD5/VDDIO under voltage or by the MSC command *WAKE\_UP\_EOT\_RST*. If no other power on source is active a power down sequence is started.

The Main Relay Driver is not switched on by *WAKE\_UP\_EOT* if the device is powered in permanent battery.

The MSC sets the programmable time on *Wake up timer\_SET Registers (CONFIG\_REG15-0 to CONFIG\_REG15-2)*. The counting can be started/stopped by MSC command - *CONFIG\_REG1[6:7] = WAKE\_UP\_TIMER\_START\_STOP[0:1] (CONFIG\_REG1 D6;D7)* - or by switching the pin *KEY\_IN* according to a selection set by MSC bit - *CONFIG\_REG20[2] = WAKE\_UP\_TIMER\_EN\_SEL*.

The counter counts time on the *WAKE\_UP\_TIMER\_CNT* Register.

*WAKE\_UP\_TIMER\_CNT* Register and *Wake up timer\_SET* Register are being compared, and when they match, *WAKE\_UP\_EOT\_DET* signal will be set to high level to trigger a power up sequence.

#### Condition:

$4.8\text{ V} \leq \text{VB\_IN} \leq 18\text{ V}$ ,  $3.5\text{ V} \leq \text{VB\_STBY} \leq 18\text{ V}$   $T_j$  -40 to 175 °C unless otherwise specified;

$\text{VB\_STBY} > 3.5\text{ V}$ , EOT can keep the counter value (allow to stop counting),

$5\text{ V} < \text{VB\_STBY} < 18\text{ V}$ , EOT full function, full parameter.

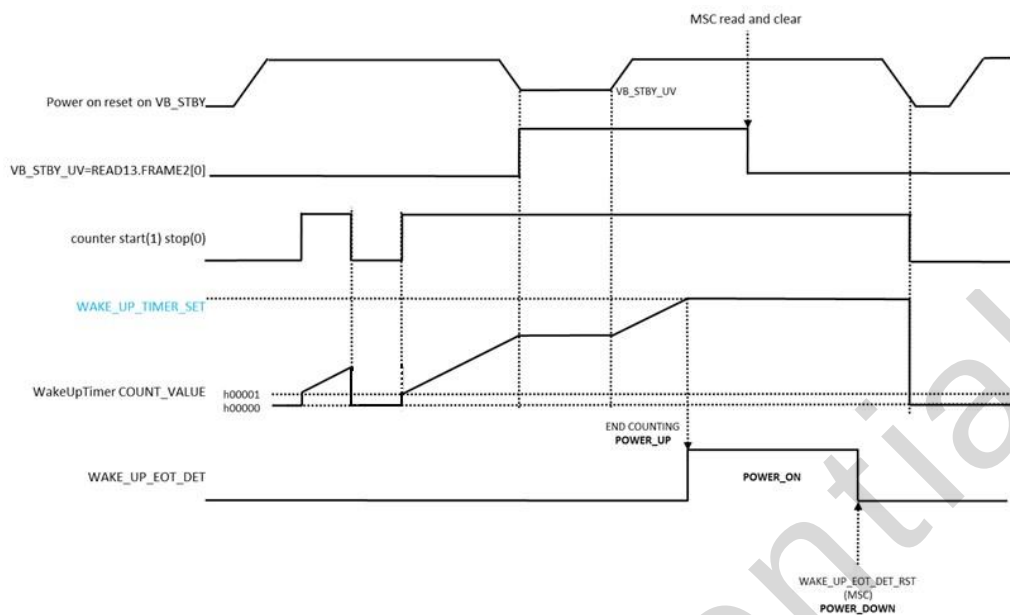
**Table 3-4 VB\_STBY pins electrical characteristics**

| Symbol                  | Parameter                                                               | Test condition                                                                                                           | Min | Typ | Max | Unit | Pin      |
|-------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|------|----------|
| VVB_STBY_T <sub>h</sub> | VB_STBY voltage range that EOT counter keep data or operating correctly | -                                                                                                                        | 3.5 | -   | -   | V    | VB_STB_Y |
| IVB_STBY_EOT_disable    | VB_STBY current with EOT 32 kHz Clock device off                        | $3.5 < \text{VB\_STBY} < 18\text{ V}$ $\text{VB\_CAN} < 0.7\text{ V}$                                                    | -   | -   | 30  | μA   | VB_STB_Y |
| IVB_STBY_EOT_enable     | VB_STBY current with EOT 32 kHz Clock                                   | $3.5 < \text{VB\_STBY} < 18\text{ V}$ $\text{VB\_CAN} < 0.7\text{ V}$<br>$-40\text{ °C} < T_{\text{amb}} < 80\text{ °C}$ | -   | -   | 50  | μA   | VB_STB_Y |

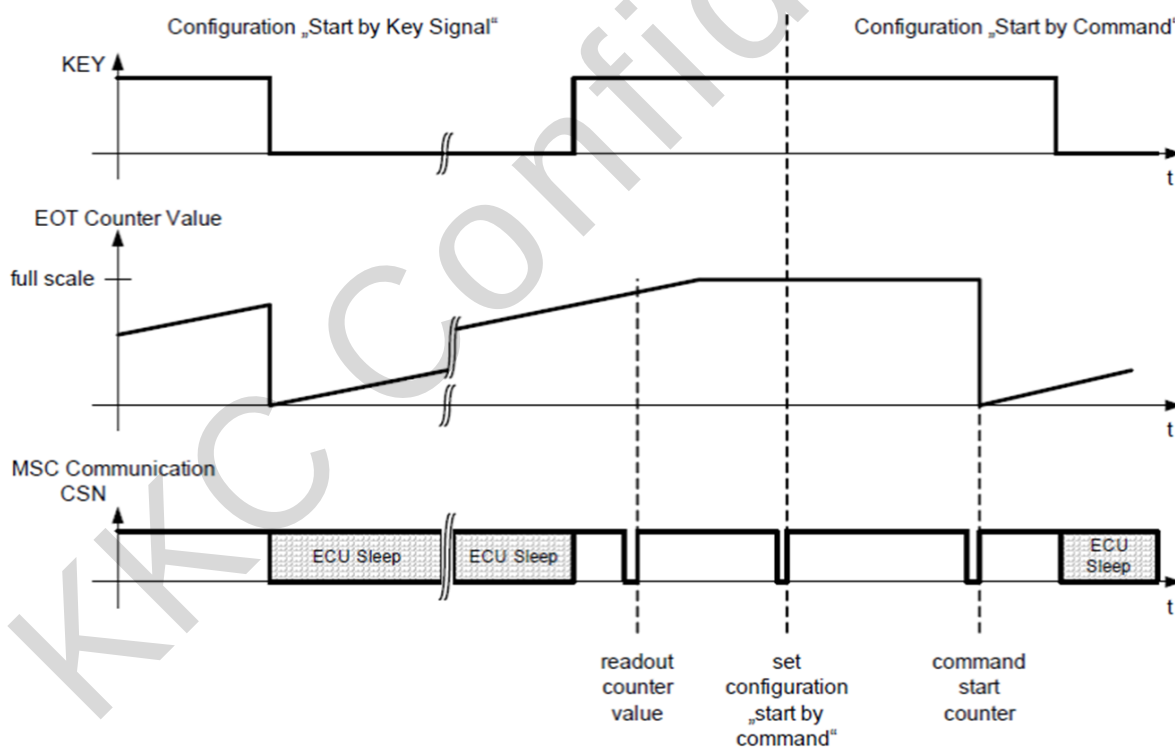
|                     | device off                                                                                      |                                                                                                                                    |     |   |     |               |         |
|---------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----|---|-----|---------------|---------|
| IVB_STBY_EOT_enable | VB_STBY current with EOT 32 kHz Clock device off                                                | $3.5 < VB\_STBY < 18\text{ V}$<br>$VB\_CAN < 0.7\text{ V}$<br>$80\text{ }^{\circ}\text{C} < T_{amb} < 125\text{ }^{\circ}\text{C}$ | -   | - | 70  | $\mu\text{A}$ | VB_STBY |
| EOT_clock_frequency | Freq EOT 32 kHz Clock internal components                                                       | $6\text{ V} < VB\_STBY < 18\text{ V}$<br>$-40\text{ }^{\circ}\text{C} < T_{amb} < 80\text{ }^{\circ}\text{C}$                      | -6% | 1 | +6% | Hz            | VB_STBY |
| EOT_clock_frequency | Freq EOT 32 kHz Clock internal components                                                       | $6\text{ V} < VB\_STBY < 18\text{ V}$<br>$80\text{ }^{\circ}\text{C} < T_{amb} < 125\text{ }^{\circ}\text{C}$                      | -7% | 1 | +7% | Hz            | VB_STBY |
| VB_STBY_UV          | VB_STBY under voltage threshold the 32 kHz clock is active but counter is stopped during crank. | $VB\_STBY > 3.1\text{ V}$                                                                                                          | 3.5 | - | 5   | V             | VB_STBY |

When VB\_STBY\_UV happens, it stops the counter.

The counter will resume when vb\_stby\_uv goes low (no undervoltage).



**Figure 3-2 Timing for Wake Up Timer - driver by MSC start/stop commands**



**Figure 3-3 Timing for Wake Up Timer - activated (started) by KEY negative edge**

#### Reset condition

VB\_STBY under voltage stops (not reset) EOT counter during cranking. The counter is reset in case of power on reset generated on VB\_STBY supply.

### Count up

Wake up Timer shall count up the timer from 1 while the MSC *Wake Up Timer Start* command is given. When *Wake Up Timer Stop* command is given, the Wake Up Timer shall stop counting and the value will clear. While the counter is counting the *EOT\_STATUS* will be 1. If it's not counting *EOT\_STATUS* will be 0.

### Wake Up Timer start/stop commands

Wake Up Timer counts up when msc start command in the MSC *WAKE\_UP\_TIMER\_START\_STOP* register is given. When the msc stop command in the MSC *WAKE\_UP\_TIMER\_START\_STOP* register is given the timer shall stop counting up but the *WAK\_UP\_TIMER\_CNT* register will clear. The bit can be set or reset via MSC.

### WAKE\_UP\_TIMER\_CNT Register

*WAKE\_UP\_TIMER\_CNT* Register consists of 24 bits it's Read Only. The range of *WAKE\_UP\_TIMER\_CNT* is from 0 (h'000000) to 16777215 (h'FFFFFF). *WAKE\_UP\_TIMER\_CNT* Register is set to 0 (h'000000) when power on reset of *VB\_STBY* occurs. *WAKE\_UP\_TIMER\_CNT* Register is set to all "0" via MSC when power on reset period or *VB\_STBY* is open.

### Wake up timer\_SET time register

The *Wake up timer\_SET* register consists of 24 bits. And the register is Read and Write register.

The range of *Wake up timer\_SET* is from 0 (h'000000) to 16777215 (h'FFFFFF). *Wake Up Timer\_SET* Register is set to 0(h'000000) when power on reset of *VB\_STBY* occurs.

### EOT function 1

EOT Function 1 - count "stand" time: After the start of the counter, the counter starts to count up till the value that has to be set before starting it using *Wake\_up\_timer\_SET*. If ECU switches on before the counter reaches the wake up timer\_set value, the  $\mu$ C can read the "stand" time via MSC. If the counter reach the wake up timer\_set value before ECU switch on, the count keeps the wake up timer\_set value.

This function is selected with *MSC CONFIG\_REG6[6]=EOT\_MODE=0*. The EOT function 1 does not set *WAKE\_UP\_EOT\_DET* at 1. *EOT\_MODE* bit can be correctly read only when wake up timer logic is working, else it is read 0.

*VB\_STBY* must be provided to correctly read the *EOT\_MODE* bit. If *VB\_STBY* is removed the register content will be lost.

### EOT function 2

EOT Function 2 - wake-up via *WAKE\_UP\_TIMER\_SET*. Set *WAKE\_UP\_TIMER\_SET* (*CONFIG-REG15-0 / CONFIG-REG15-1/ CONFIG-REG15-2*) to a certain value, then start the counter before shutting down the ECU. The counter counts up until the *WAKE\_UP\_TIMER\_SET* is reached. The reaching of *WAKE\_UP\_TIMER\_SET* activates the wake signal and will start a wake-up sequence. This function is selected with *MSC CONFIG\_REG6[6]=EOT\_MODE=1*. *EOT\_MODE* bit can be correctly read only when wut logic is working, else it is read 0.

*VB\_STBY* supply must be provided to correctly read the *EOT\_MODE* bit.

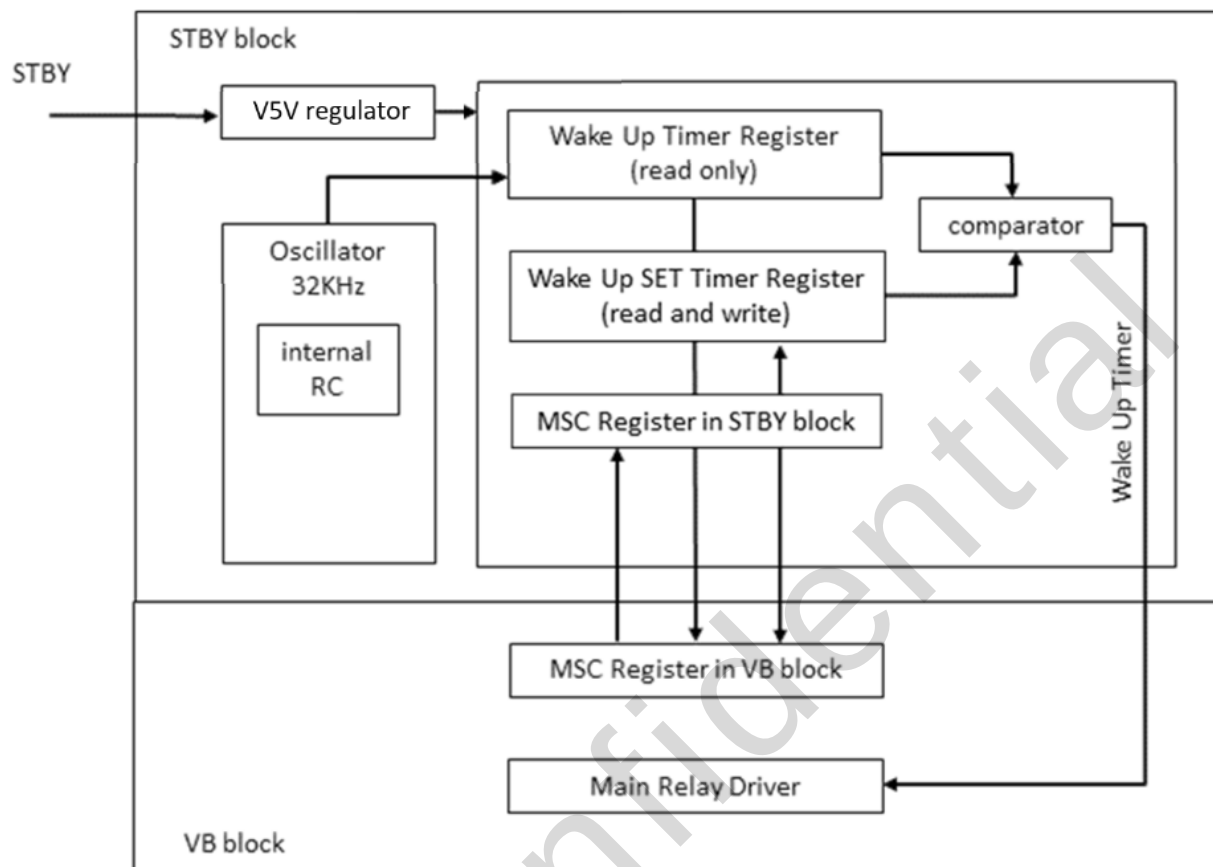


Figure 3-4 Wake up timer block diagram

Table 3-5 Wake up timer resolution

| Range        | resolution |
|--------------|------------|
| 0 - 16777215 | 1 second   |

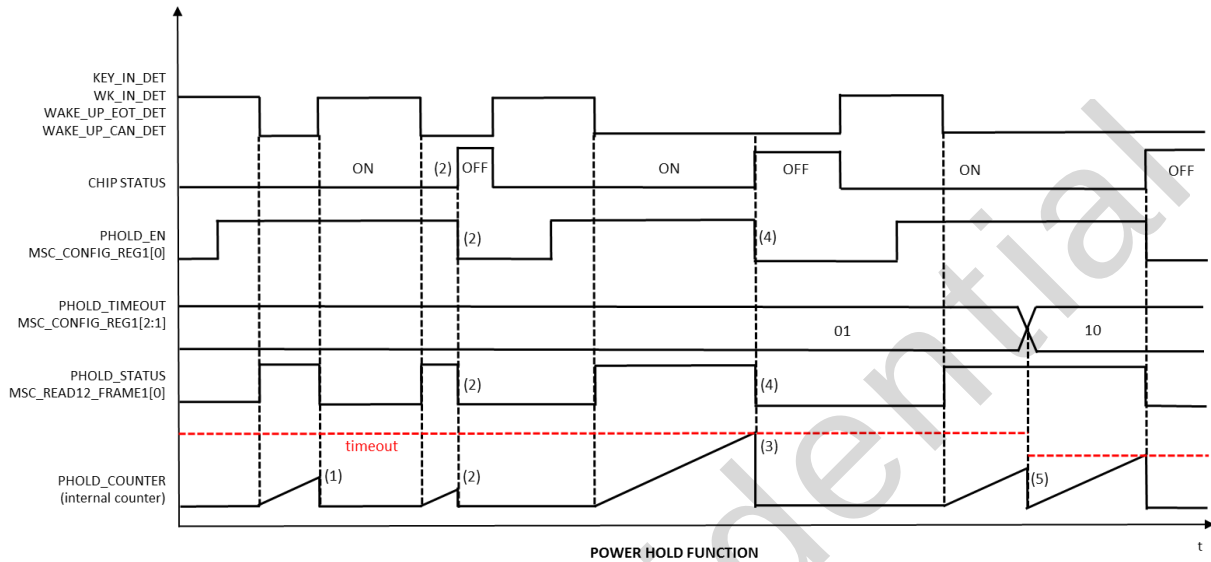
*WAKE\_UP\_TIMER\_CNT* Register can be read via MSC. *Wake up timer\_SET* Register can be read and write via MSC.

### 3.1.8 Power on with PHOLD\_EN (EOT function 3)

The chip is kept switched on by an MSC command (*PHOLD\_EN CONFIG\_REG1[0]*) even though one of the power down sources has been asserted (*KEY\_IN*, *WK\_IN*, *WAKE\_UP\_TIMER*, *WAKE\_UP\_CAN*). The chip is kept on in power hold mode for a time programmed (default is infinity) by MSC (*PHOLD\_TIME[0:1] CONFIG\_REG1 (D1;D2)*). This is also EOT function 3. The power hold mode is enabled by MSC and is cleared by the conditions reported in the reset matrix.

After setting the *PHOLD\_EN=CONFIG\_REG1[0]=1* the PHOLD and the PHOLD timer are enabled when the *KEY\_DET*, *WK\_IN\_DET*, *WAKE\_UP\_EOT\_DET*, *WAKE\_UP\_CAN\_DET*

detections are not active. The  $PHOLD\_TIMER[14:0]=READ12.FRAME[2:1]$  starts counting when the  $KEY\_DET$ ,  $WK\_IN\_DET$ ,  $WAKE\_UP\_EOT\_DET$ ,  $WAKE\_UP\_CAN\_DET$  detections are cleared. When the  $PHOLD\_TIME=CONFIG\_REG1[2:1]$  is configured as 00 (default) the timeout is disabled and the  $PHOLD\_TIMER[14:0]=READ12.FRAME[2:1]$  is read always at 0.



- (1) The timeout counter is cleared on the detection of at least one of the power up sources.
- (2) The timeout counter is cleared if MSC disables the PHOLD function, MRD is switched off, a power down sequence is started.
- (3) The counter has reached the timeout, MRD is switched off, a power down sequence is started.
- (4) The PHOLD\_EN and the PHOLD\_STATUS bits are clear along with the power down sequence.
- (5) A new MSC setting for the timeout makes the timeout counter clear and restart.

**Figure 3-5 Power hold diagram**

### 3.1.9 Power down from WAKE\_UP\_TIMER when PHOLD\_EN=0

The chip is powered down if the MSC clears, by the MSC command  $WAKE\_UP\_EOT\_RST$ ,  $WAKE\_UP\_EOT\_DET$  (The counter counts up until the  $WAKE\_UP\_TIMER\_SET$  is reached) if no other power up source is active and  $PHOLD\_EN = 0$ . The detection reset by MSC command,  $WAKE\_UP\_EOT\_RST=CONFIG\_REG12[6]$ , is active at level not at change, so from when MSC writes them at 1 they keep on clearing the detections. If there is further wake detection signal, the detection signal will be cleared, until the MSC writes 0 in the bits.

### 3.1.10 Power down from PHOLD

The chip is powered down after the power hold timer programmed by MSC has expired or if the PHOLD function is disabled by MSC. The PHOLD timer can be programmed as timeout disabled so that the chip stays powered up.

### 3.1.11 Power up from CAN

The chip can be powered up from CAN according to what reported in the Wake Up (From CAN) chapter. To allow the power up from CAN the pins  $VB\_CAN$ ,  $VB\_STBY$  must be supplied.

When U chip is already in on state (the logic is out of power on reset), the validation of a valid CAN pattern only sets a flag, with no impact on power supply. This flag is named

*WAKE\_UP\_CAN\_DET* (detection of a valid can pattern) and is cleared by VDD5/VDDIO under voltage or by the MSC command *WAKE\_UP\_CAN\_RST*. In this case if no other power on source is active a power down sequence is started.

The Main Relay Driver is not switched on by *WAKE\_UP\_CAN* if the device is powered in permanent battery.

### 3.1.12 Power down from *WAKE\_UP\_CAN* when *PHOLD\_EN*=0

The chip is powered down if the MSC clears, by the MSC command *WAKE\_UP\_CAN\_RST*, *WAKE\_UP\_CAN\_DET* (the detection of *WAKE\_UP\_CAN* high) if no other power up source is active and *PHOLD\_EN* = 0. The detection reset by MSC command, *WAKE\_UP\_CAN\_RST*=*CONFIG\_REG12*[5], is active at level, so from when MSC writes them at 1 they keep on clearing the detections. If there is further wake detection signal, the detection signal will be cleared, until the MSC writes 0 in the bits.

### 3.1.13 Finish wake function

There is a Finish wake function (auto clear mechanism) of *WK\_IN\_DET*, *WAKE\_UP\_EOT\_DET* and *WAKE\_UP\_CAN\_DET*. The purpose is to ensure the system could power down from wake up state after the impulse timer timeout (typ. 1s).

If the software does not stop the impulse timer before time out, the *WK\_IN\_DET*, *WAKE\_UP\_EOT\_DET* and *WAKE\_UP\_CAN\_DET* will be cleared as long as time out. If the software stops the impulse timer (*FIN\_WAKE* = 1) before time out, the *WK\_IN\_DET*, *WAKE\_UP\_EOT\_DET* and *WAKE\_UP\_CAN\_DET* will not be cleared.

The impulse timer is not affected by watchdog(*ERR\_CNT*>4) and EOT. The Finish wake function is not disabled by *RSTN* pulled low externally and *MSC\_SW\_RESET* active. The Finish wake function can be disabled by power supply undervoltage. The reset sources of *FIN\_WAKE* bit are the same as *fin\_wake* function.

Software can decide to clear the *WK\_IN\_DET*, *WAKE\_UP\_EOT\_DET* or *WAKE\_UP\_CAN\_DET* before or after the impulse timer timeout, through set *WK\_IN\_RST*, *WAKE\_UP\_EOT\_RST* or *WAKE\_UP\_CAN\_RST* to 1

*FIN\_WAK* function is stopped by

*FIN\_WAKE* time out,

*FIN\_WAKE* BIT =1,

*WK\_IN\_RST* / *WK\_EOT\_RST* / *WK\_CAN\_RST* of relevant detect signal.

When *WK\_IN\_DET* & *WK\_EOT\_DET* & *WK\_CAN\_DET* = 0, *FIN\_WAK* can be re-triggered by any detection of *WK\_IN\_DET* / *WK\_EOT\_DET* / *WK\_CAN\_DET*. The counter is restarted from 0.

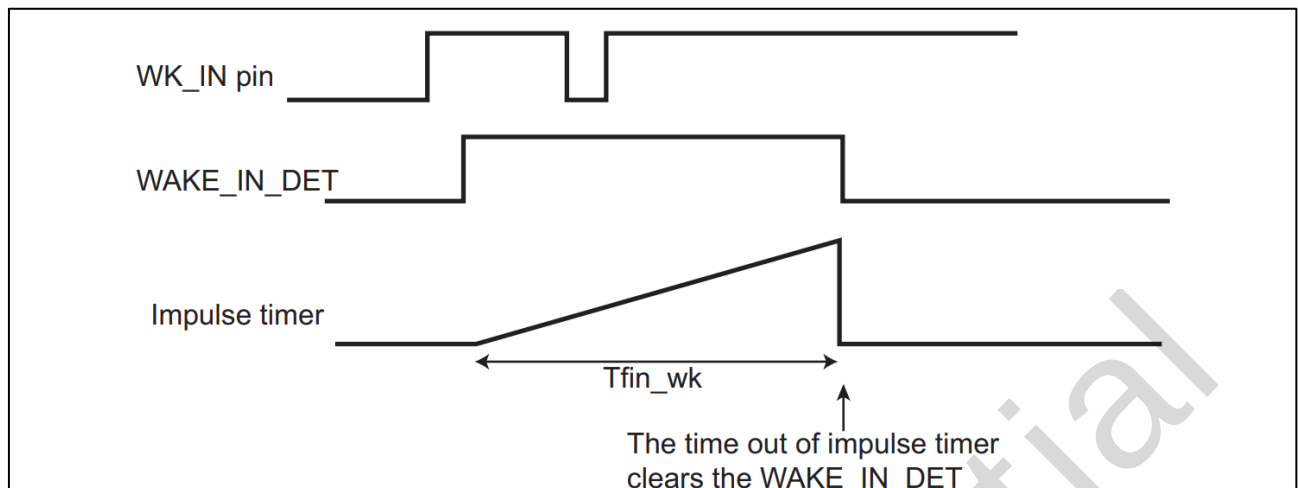


Figure 3-6 Impulse timer clear WAKE\_IN\_DET after time out

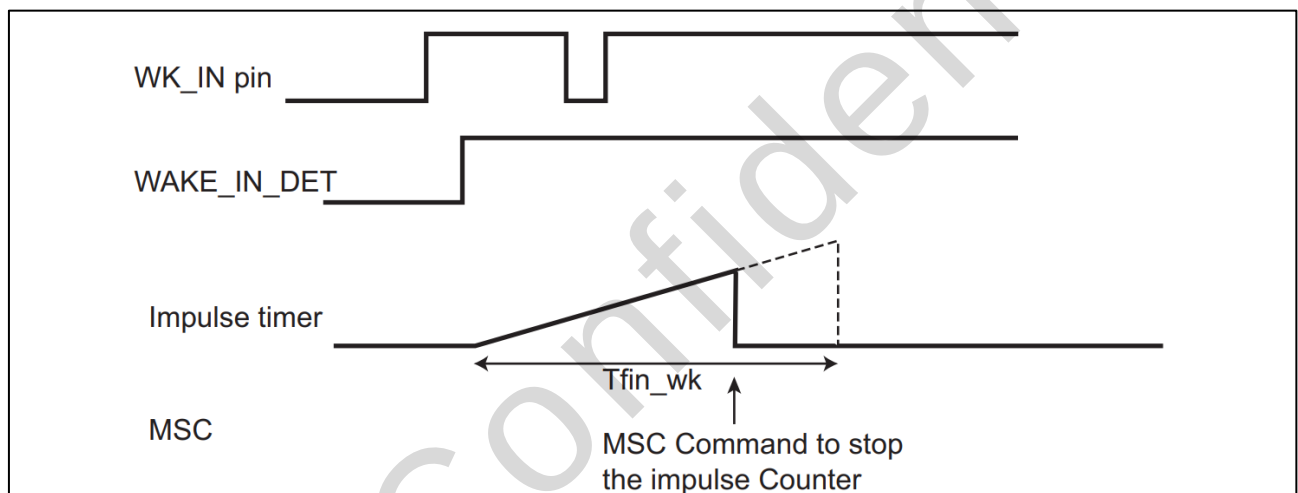


Figure 3-7 Impulse timer disabled before time out

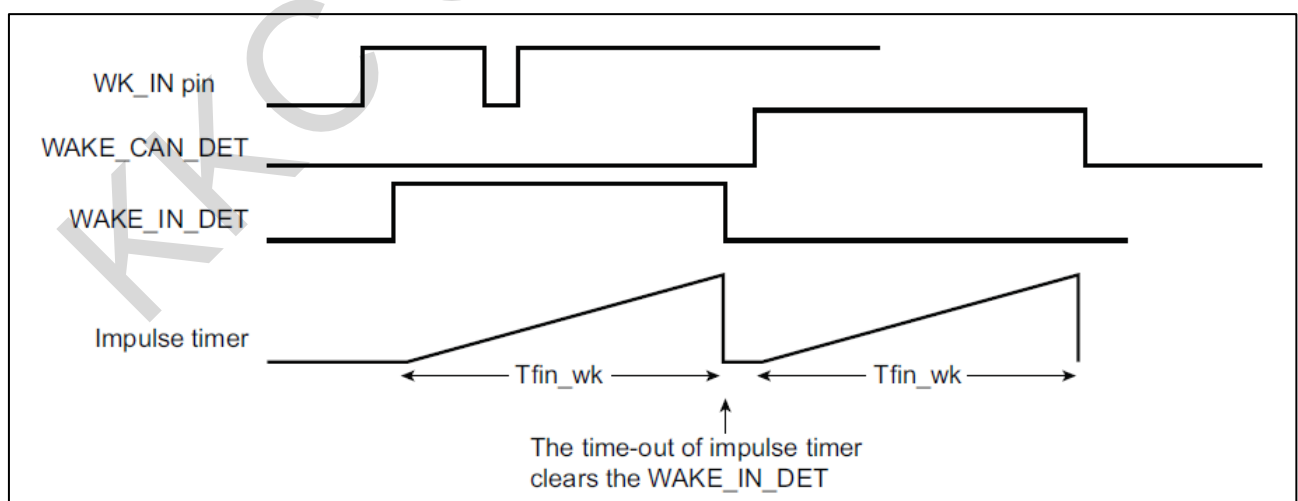


Figure 3-8 Timer enabled for a new detection

Note : Same mechanism applied to WAKE\_IN\_DET, WAKE\_UP\_EOT\_DET and WAKE\_UP\_CAN\_DET.

**Table 3-6 Finish wake timer electrical characteristics**

| Symbol  | Parameter         | Min  | Typ | Max  | Unit |
|---------|-------------------|------|-----|------|------|
| Tfin_wk | Finish wake timer | 0.94 | 1   | 1.06 | s    |

### 3.2 Secure Engine OFF(SEO) function

Secure Engine Off functional switches off, with a certain delay from KEY\_OFF, the engine by a hardware path and without any dependence on external micro and no matter what are the other conditions (e.g. the power hold mode).

After the falling edge of the filtered KEY\_IN a programmable counter is started. The clock of the timer does not depend on any micro clock or function.

SEO\_OUT pin is the output of KEY\_IN after a  $T\_SEO\_DELAY$  (CONFIG\_REG3 [6:7]) delay. Input pin INJ\_ENA enables the internal drivers as listed above (INJ[1:4], is configured), so the configuration of enable/disable of SEO function is realized by hardware in the following way:

For PFI application, SEO\_OUT is connected to INJ\_ENA, when key goes off internal drivers can be switched off.

For GDI application, SEO\_OUT is connected to GDI injector driver enable if present, INJ\_ENA is always enabled. When key off, only the external GDI injector driver is switched off.

During SEO time the external micro can switch on/off the drivers

SEO output is implemented with a push pull output stage, the  $T\_SEO\_DELAY$  can be programmed according to Table 3-7.

**Table 3-7 SEO timing**

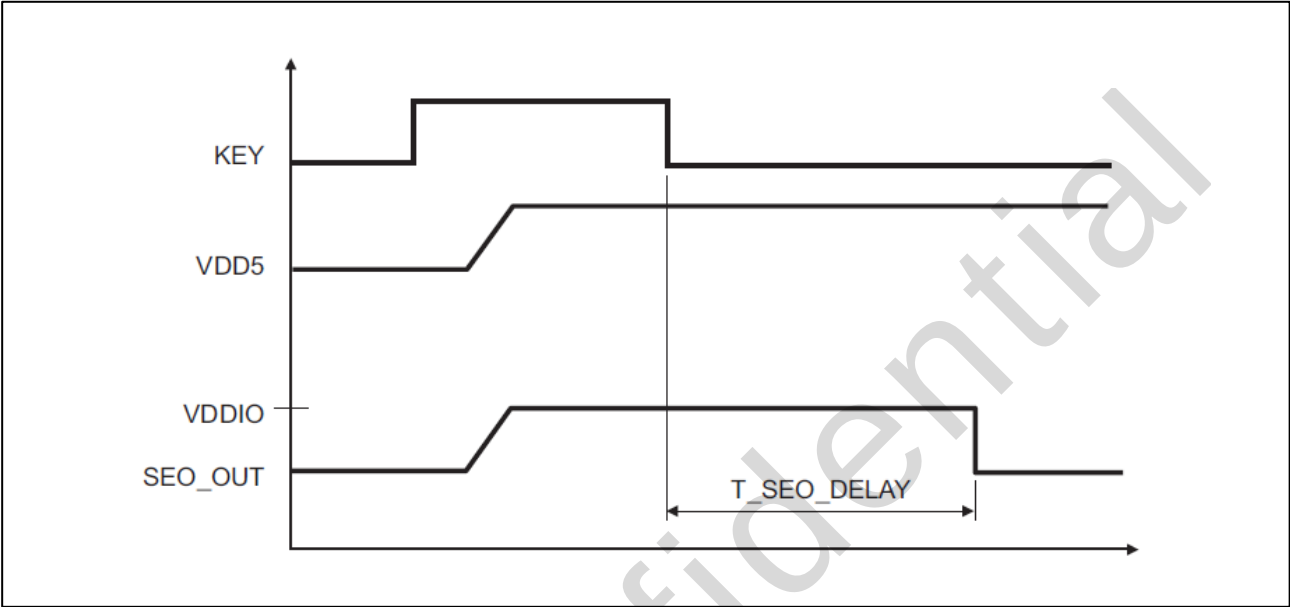
| Symbol      | Parameter | MSC_CONF | Min | Typ | Max | Unit | Pin     |
|-------------|-----------|----------|-----|-----|-----|------|---------|
| T_SEO_DELAY | default   | 00       | 100 | -   | 140 | ms   | SEO_OUT |
| T_SEO_DELAY | -         | 01       | 200 | -   | 250 | ms   | SEO_OUT |
| T_SEO_DELAY | -         | 10       | 400 | -   | 480 | ms   | SEO_OUT |
| T_SEO_DELAY | -         | 11       | 800 | -   | 920 | ms   | SEO_OUT |

**Condition:**

$5.5\text{ V} \leq V_{B\_IN} \leq 18\text{ V}$ ,  $4.75\text{ V} \leq V_{DD\_IO} \leq 5.25\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified.

**Table 3-8 SEO Circuit electrical characteristic**

| Symbol    | Parameter          | Test condition                                          | Min        | Typ | Max | Unit | Pin     |
|-----------|--------------------|---------------------------------------------------------|------------|-----|-----|------|---------|
| SEO_OUT_H | SEO_OUT high level | VDDIO = 5 V or 3.3 V<br>I <sub>sink</sub> current= 2 mA | VDDIO -0.5 | -   | -   | V    | SEO_OUT |



**Figure 3-9 SEO delay**

[illegible]

### Figure 3-10 Reset table

[www.kkchips.com](http://www.kkchips.com)

### 3.4 Smart reset circuit

There are two Reset pins:

1. RSTN: reset pin linked to VDD5 supply;
2. RSTC: reset pin linked to VDDIO supply.

#### 3.4.1 Smart reset RSTN pin

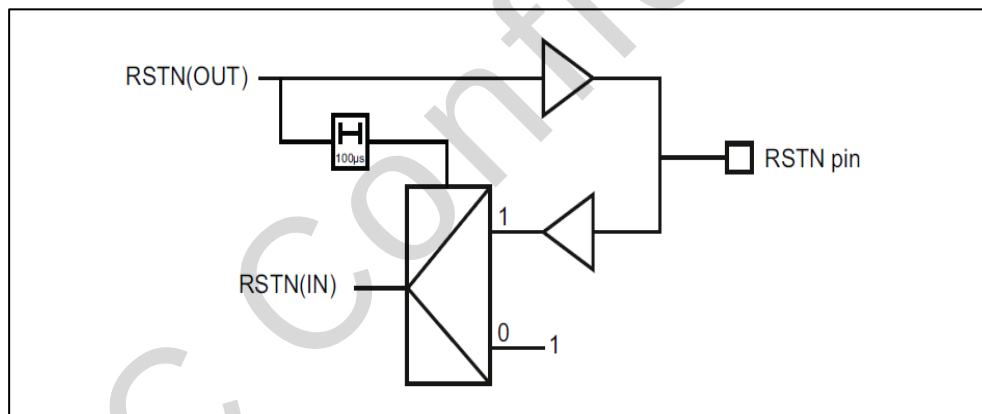
The RSTN pin is an input/output active low. As output pin the Smart Reset circuit takes into account several events in the device in order to generate the proper reset signal at RSTN pin for the microcontroller and for portion of the internal logic as well.

The RSTN in output (RSTN(OUT)) is activated by the contributions described by the reset matrix and is intended as reset for the external microcontroller.

The RSTN in input (RSTN(IN)) is filtered Trstflt\_RST and stops the internal logic as described in the reset matrix.

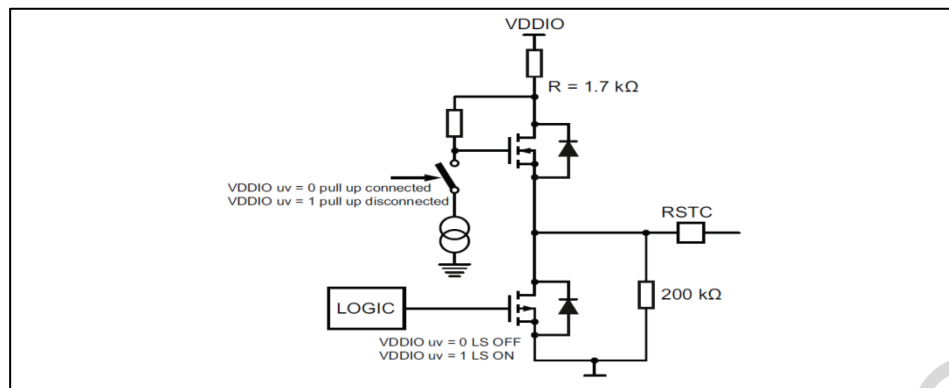
The internal RSTN(OUT) has higher priority respect to the external RSTN(IN). The RSTN(IN) is masked until Tmsk\_RSTN after RSTN(OUT) has been released.

RSTN(OUT) and RSTN(IN) are recombined on the pad according to the scheme.



**Figure 3-11 Reset circuit**

The RSTN activates on the contributions specified by the reset matrix:



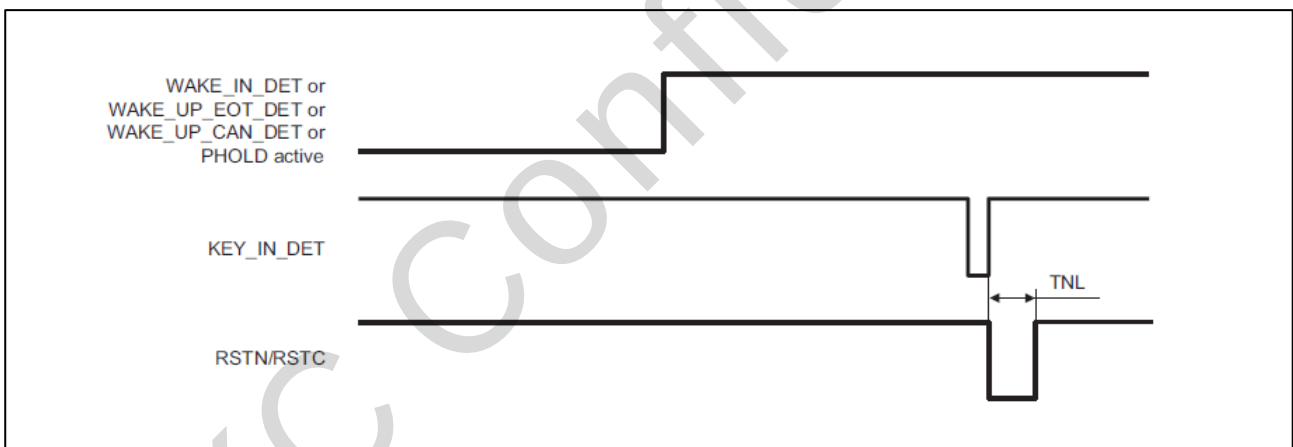
### Figure 3-12 Reset matrix

### 3.4.2 Smart Reset RSTC pin

The RSTC pin is an output active low. As output pin the Smart Reset circuit takes into account several events in the device in order to generate the proper reset signal at RST pin for the microcontroller and for portion of the internal logic as well.

The RSTC activates on the contributions specified by the reset matrix:

### 3.4.3 After run reset



**Figure 3-13 RSTN/RSTC after run reset timing**

As output:

4.8 ≤ V<sub>B IN</sub> ≤ 18 V, T<sub>j</sub> -40 to 175 °C unless otherwise specified;

**Table 3-9 Smart reset circuit electrical characteristic table 1**

| Symbol          | Parameter          | Test condition                                 | Min | Typ | Max | Unit | Pin  |
|-----------------|--------------------|------------------------------------------------|-----|-----|-----|------|------|
| VUV_LO_RSTN     | Output low voltage | $1 < VDD5 < VDD\_UV$                           | -   | -   | 0.6 | V    | RSTN |
| VUV_LO_RSTC     | Output low voltage | $1 < VDDIO < VDDIO\_UV$                        | -   | -   | 0.6 | V    | RSTC |
| IUVres_max_RSTN | Input current      | $VDD5 = VDD\_UV$<br>$V_{UV\_reset} = 0.6V$     | 1   | -   | -   | mA   | RSTN |
| IUVres_max_RSTC | Input current      | $VDDIO = VDDIO\_UV$<br>$V_{IUV\_reset} = 0.6V$ | 1   | -   | -   | mA   | RSTC |

As input:

$4.8 \leq V_{B\_IN} \leq 18 \text{ V}$ ,  $4.5 \leq V_{DD\_IO} \leq 5.5 \text{ V}$ ,  $T_j -40$  to  $175 \text{ }^\circ\text{C}$  unless otherwise specified;

**Table 3-10 Smart reset circuit electrical characteristic table 2**

| Symbol      | Parameter                                          | Test condition           | Min  | Typ | Max                | Unit          | Pin       |
|-------------|----------------------------------------------------|--------------------------|------|-----|--------------------|---------------|-----------|
| RST_L       | RST input low voltage                              | -                        | -0.3 |     | 1.1                | V             | RSTN      |
| RST_H       | RST Input high voltage                             | -                        | 2.3  |     | $V_{DD\_IO} + 0.3$ | V             | RSTN      |
| Trstflt_RST | Reset RST filter time                              | Tested by scan           | 7.5  | 10  | 11.5               | $\mu\text{s}$ | RSTN      |
| RRST_PU     | RST pull up resistor +HS Ron                       | RSTN / RSTC pins to 0V   | 0.95 | -   | 2.5                | k $\Omega$    | RSTN RSTC |
| RRST_PD     | RST pull down resistor                             | RSTN / RSTC pin to VDDIO | 100  | -   | 320                | k $\Omega$    | RSTN RSTC |
| TNL_RST     | After run reset time                               | -                        | 1.4  | 2   | 2.6                | ms            | RSTN RSTC |
| TWDA_RST    | watchdog RST reset time                            | -                        | 1.4  | 2   | 2.6                | ms            | RSTN RSTC |
| Td_UV_RST   | Power on RST reset delay                           | -                        | 17   | -   | 30                 | ms            | RSTN/RSTC |
| Tmsk_RST    | Masking time on RST(IN) after RTN(OUT) is released | -                        | 95   | 100 | 105                | $\mu\text{s}$ | RSTN      |

### 3.5 Thermal shutdown

Table 3-11 Thermal shutdown

| Block           | Overtemperature flags |        |        |        |        |        |        |                        |                          |         |         |         |         |               |         |         |         |
|-----------------|-----------------------|--------|--------|--------|--------|--------|--------|------------------------|--------------------------|---------|---------|---------|---------|---------------|---------|---------|---------|
|                 | OVT[1]                | OVT[2] | OVT[3] | OVT[4] | OVT[5] | OVT[6] | OVT[7] | OVT[8]<br>or<br>OVT[9] | OVT[18]<br>or<br>OVT[19] | OVT[10] | OVT[11] | OVT[12] | OVT[13] | OVT[14]<br>** | OVT[15] | OVT[16] | OVT[17] |
| REGULATOR BUCK  | ▲                     |        |        |        |        |        |        |                        |                          |         |         |         |         |               |         |         |         |
| REGULATOR VDD5, |                       |        |        |        |        |        |        |                        |                          |         |         |         |         |               |         |         |         |
| REGULATOR TRACK |                       |        | ▲      |        |        |        |        |                        |                          |         |         |         |         |               |         |         |         |
| DRIVER INJ1     |                       |        |        | ▲      |        |        |        |                        |                          |         |         |         |         |               |         |         |         |
| DRIVER INJ2     |                       |        |        |        | ▲      |        |        |                        |                          |         |         |         |         |               |         |         |         |
| DRIVER INJ3     |                       |        |        |        |        | ▲      |        |                        |                          |         |         |         |         |               |         |         |         |
| DRIVER INJ4     |                       |        |        |        |        |        | ▲      |                        |                          |         |         |         |         |               |         |         |         |
| DRIVER O2H1     |                       |        |        |        |        |        |        | ▲                      |                          |         |         |         |         |               |         |         |         |
| DRIVER O2H2     |                       |        |        |        |        |        |        |                        | ▲                        |         |         |         |         |               |         |         |         |
| DRIVER SOL1     |                       |        |        |        |        |        |        |                        |                          | ▲       |         |         |         |               |         |         |         |
| DRIVER SOL2     |                       |        |        |        |        |        |        |                        |                          |         | ▲       |         |         |               |         |         |         |
| DRIVER RLY[1:2] |                       |        |        |        |        |        |        |                        |                          |         |         | ▲       |         |               |         |         |         |
| DRIVER RLY[3:5] |                       |        |        |        |        |        |        |                        |                          |         |         |         | ▲       |               |         |         |         |
| DRIVER MRD      |                       |        |        |        |        |        |        |                        |                          |         |         |         |         | ▲             |         |         |         |
| DRIVER STR[1:3] |                       |        |        |        |        |        |        |                        |                          |         |         |         |         |               | ▲       |         |         |
| DRIVER LED[1:2] |                       |        |        |        |        |        |        |                        |                          |         |         |         |         |               |         | ▲       |         |
| LIN             |                       |        |        |        |        |        |        |                        |                          |         |         |         |         |               |         | ▲       |         |
| VRS             |                       |        |        |        |        |        |        |                        |                          |         |         |         |         |               |         |         | ▲       |

**Legenda:**

▲ Shutdown

&lt;blank&gt; Not shutdown

## HE9788

Multifunction IC for automotive engine management system



Operation behavior

OVT[1-16] Dedicated temperature sensor for the power

OVT[3] OVT[3] is used as a common thermal sensor for the three Tracking Regulators

OVT[17] Central Temperature sensor

OVT[14]\*\* Over temperature protection MRD is available only when VB\_IN is present.

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## 4 Watchdog

### 4.1 Basic feature

HE9788 has an internal query and answer (Q&A) watchdog with the aim of monitoring the correctness of microcontroller activity. Via MSC bus a WDA "question" shall be read from a MSC-WDA register. A correct response shall then be written back via MSC in a well defined timing. If response or timing is not correct, then the WDA error counter EC (or ERR\_CNT) is increased. If the error counter is increased to values greater than 4, some output functions are shut off. If the error counter reaches values greater than 7 (overflow), then a RST reset may be generated if this is previously configured via MSC. The first sequence of questions and responses starts from VDD5 higher than the UV threshold.

Other way round, with a RST event also the WDA output pin goes to low.

Note that after startup, reset or an RST\_CNT overflow the initial value of the error counter is 6.

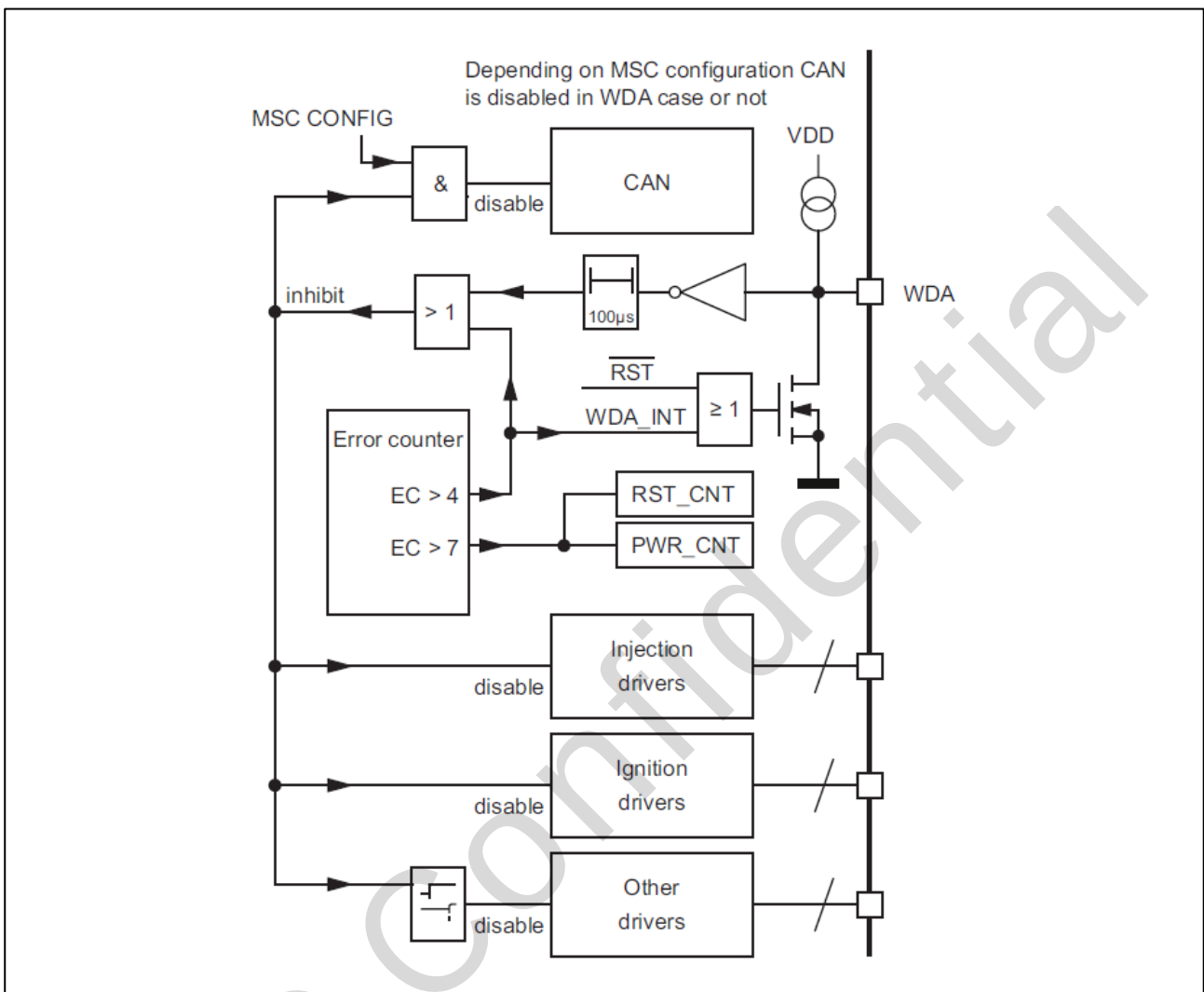
If WDA resets are enabled via MSC (*WDA\_INIT CONFIG-REG 9\_1 D3*): the number of RST events generated by an error counter overflow is limited by the reset counter *RST\_CNT*. If *RST\_CNT* reaches the value of 7, then RST resets via WDA are no longer generated.

With each error counter overflow, the *PWR\_CNT* counter is increased. If it reaches a value greater than 7, then the power latch mode is terminated.

#### Watchdog legenda:

|                     |                                                                                 |
|---------------------|---------------------------------------------------------------------------------|
| <i>ERR_CNT</i> , EC | error 3-bit counter;                                                            |
| <i>RST_CNT</i>      | reset 3-bit counter;                                                            |
| <i>WDA_RST</i>      | internal reset generated by Watchdog;                                           |
| <i>PWR_CNT</i>      | power off 3-bit counter;                                                        |
| <i>PWR_INT</i>      | internal signal indicating <i>PWR_CNT</i> overflow;                             |
| WDA                 | WDA open drain pin (low active);                                                |
| <i>WDA_INT</i>      | WDA internal signal, when high it drives WDA = low, when low it sets WDA= Hi-z; |
| Sequencer-run       | current state of WDA when it is active.                                         |

#### 4.1.1 Monitoring module - WDA Functionality



**Figure 4-1 WDA block diagram**

Note: Delay-off function is not considered in this Block Diagram.

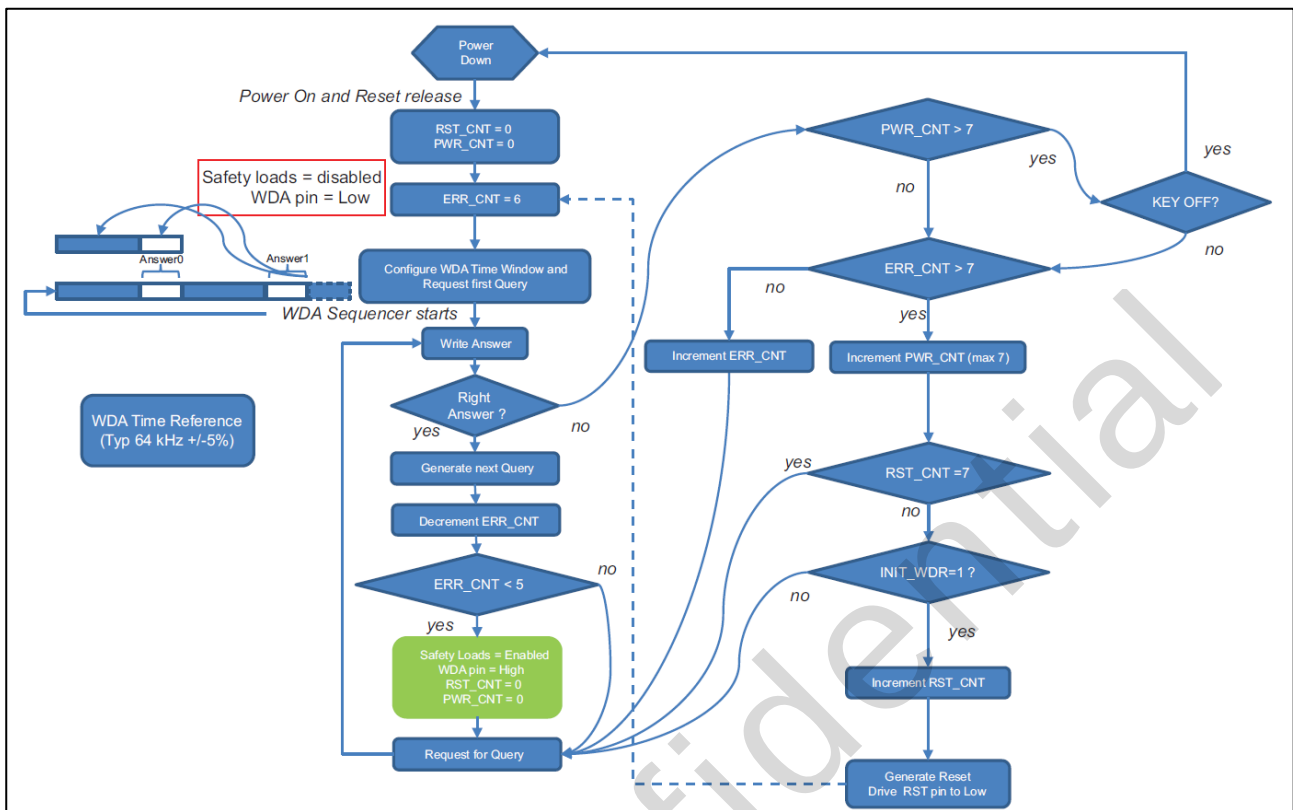


Figure 4-2 WDA state flow

Each time the watchdog error counter is  $EC > 7$  counter  $PWR\_CNT$  counter increases. When this counter is  $PWR\_CNT=7$  and a further error occurs, the power-latch will be terminated if  $KEY\_IN$  is low. The  $PWR\_CNT$ -counter is not cleared when  $EC < 7$ .  $PWR\_CNT$ -counter is cleared when  $EC < 5$  or by  $RST\_UV$ .

The monitoring module works independently of the controller functionality. The monitoring module generates various questions, which the controller must fetch and correctly respond to within a defined time window. The monitoring module checks whether the response is returned in a time window and if the response is fully correct.

The question is a 4-bit word. This 4-bit word can be fetched by the controller using a read access to register *WDA REQULO*. The monitoring module also calculates the expected correct response, which is compared to the actual response from the controller.

The response is a 32-bit word consisting of the 4 bytes *RESP\_BYTE3*, *RESP\_BYTE2*, *RESP\_BYTE1* and *RESP\_BYTE0*. The 4 bytes are sent to the monitoring module via MSC in the order *RESP\_BYTE3* - *RESP\_BYTE2* - *RESP\_BYTE1* - *RESP\_BYTE0* using four times the command *WDA\_RESP* - once for each answer byte.

The monitoring cycle phase is initialized by (the end of) writing of *RESP\_BYTE0* (least significant response byte) or by a write access to the *RESPTIME* register. The cycle starts with a variable wait time (response time, set by register *RESPTIME*), followed by a fixed time window. When a monitoring cycle ends (the end of the fixed time window has been reached) a new monitoring cycle is started automatically.

A correct response within the time window (at a response time  $> 0$  ms) decreases an ERROR COUNTER by one. An incorrect response, a response outside the time window or response time = 0 ms leads to the incrementing of the ERROR COUNTER by one.

“... within the time window” means that the end of writing the last answer byte - i.e.

*RESP\_BYTE0* - falls into the fixed time window mentioned above (see picture below).

Except the last answer byte, the previous answer bytes may also be written earlier than the beginning of the fixed time window.

The question sequence is deterministic. A question will be repeated until it is answered correctly both in value and in time. Then the next question is placed in the sequence.

The ERROR COUNTER (EC) is a 3-bit counter. Various actions are activated depending on the value of the counter.

The result of the comparison of the controller response and the calculated correct response, as well as the next question, are available in registers *REQULO* after receiving the  $\mu$ C response (LSB of *RESP\_BYTE0*) and can be read by the controller.

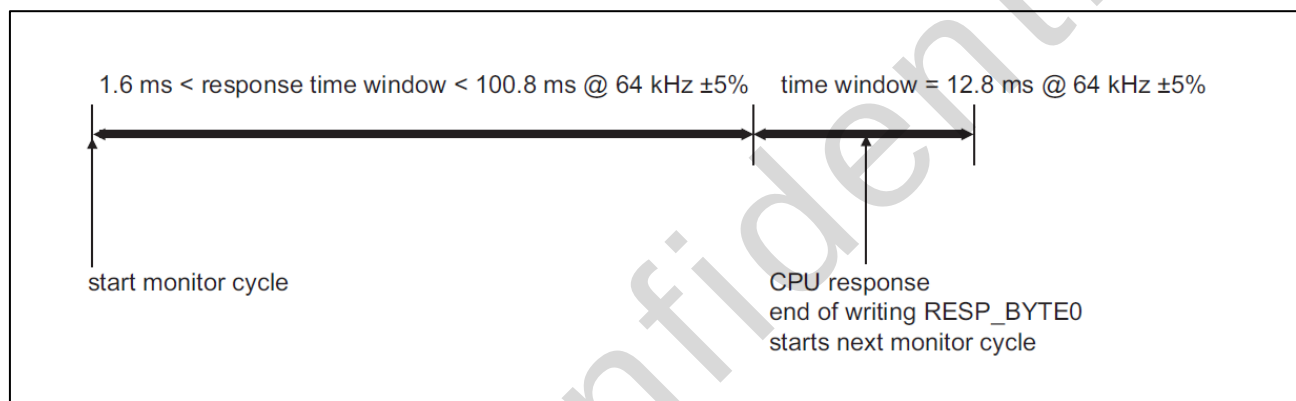


Figure 4-3 Monitoring cycle diagram

#### 4.1.2 ERR\_CNT (EC) and reactions, PWR COUNTER (PWR\_CNT) and generation of the monitoring module reset

Various actions are initiated for specific counter states of the *ERROR COUNTER* EC. The counter reset state is 6.

For *ERROR COUNTER* (EC) > 4, the open drain output WDA is pulled low (active).

Table 4-1 Error counter

| ERR_CNT                 | 0 ... 4          | 5                                                                      | 6 ... 7                                                                | Overflow<br>ERR_CNT > 7                                                                   |
|-------------------------|------------------|------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| WDA                     | Hi-Z             | Low                                                                    | Low                                                                    | Low                                                                                       |
| PWR_CNT                 | 0                | unchanged                                                              | unchanged                                                              | Incremented by 1                                                                          |
| Internal PWR_CNT signal | Low (not active) | If PWR_CNT = 7 and ERR_CNT increase => High (active)<br>Else unchanged | If PWR_CNT = 7 and ERR_CNT increase => High (active)<br>Else unchanged | PWR_CNT < 7 Low (inactive)<br>PWR_CNT 6->7 Low (inactive)<br>PWR_CNT 7 -> 7 High (active) |

If the ERROR COUNTER reaches the value “7” and a further error occurs the *PWR\_CNT* is incremented by one during a sequencer-run.

The state “EC = 7 and a further error occurs” is also called ERROR COUNTER overflow (“EC” > 7).

The counter *PWR\_CNT* is a 3 bit counter.

Behaviour of *PWR\_CNT*:

asynchronous reset to “000” with *RST\_UV*

synchronous reset to “000” IF (EC < 5)

IF (*PWR\_CNT* < 7) AND (sequencer-run AND “EC” 7) THEN *PWR\_CNT* = *PWR\_CNT* + 1  
ELSE unchanged.

The counter cannot be decremented and can be only reset to “000” by an active *RST\_UV* signal (asynchronous) or *<WDA\_INT>* = ‘0’ (synchronous). The signal *PWR\_INT* becomes active ‘1’ when *PWR\_CNT* = “111” and a further error is detected. When *KEY\_IN* = 0, the active *PWR\_INT* signal causes a shut-down of the main relay and the voltage regulators.

This function ensures a secure shutdown of the system in an error state of the  $\mu\text{C}$  in “power-latch”. Signal *PWR\_INT* is set to ‘0’ again only when *<WDA\_INT>* = ‘0’.

#### 4.1.3 Generation of a monitoring module reset

The monitoring module may cause a reset at the pin [RSTN] named “monitoring module reset” in conjunction with the internal signal *WDA\_RST*. The generation of a monitoring module reset depends on the state of the bit *<WDA\_INT>*.

*<WDA\_INT>* = ‘0’ (reset state):

If *<WDA\_INT>* = ‘0’, the signal *<WDA\_RST>* remains always inactive ‘0’ and the monitoring module can never generate a reset. The error counter can only be decremented via correct responses. If *<WDA\_INT>* = ‘0’ the state of the reset counter *<RST\_CNT>* remains unchanged when an ERROR COUNTER overflow occurs (description of the reset counter *<RST\_CNT>* see below).

*<WDA\_INT>* = ‘1’:

If *<WDA\_INT>* = ‘1’, an ERROR COUNTER overflow activates a reset [RST] (signal *<WDA\_RST>* becomes active). The signal *<WDA\_RST>* becomes active (i.e. ‘1’) due to an ERROR COUNTER overflow when the value of the 3 bit reset counter *<RST\_CNT>* (2-0) is 0..6. If the value of *<RST\_CNT>* = “111” and an ERROR COUNTER overflow occurs *<WDA\_RST>* remains inactive (i.e. ‘0’) and no reset is generated. The “reset counter” *<RST\_CNT>* is incremented by one during a sequencer-run due to an ERROR COUNTER overflow when *<WDA\_INT>* = ‘1’ and *<RST\_CNT>* is between 0 and 6. If *<RST\_CNT>* = 7

and an ERROR COUNTER overflow occurs, the counter state remains 7. The counter can not be decremented and can only reset to zero by an active *RST\_UV* signal. The occurrence of a monitoring module reset is indicated via the flag *<WDA\_RST>* = ‘1’.

Reading the flag via MSC clears it automatically. In effect maximum 7 monitoring module resets can be generated between 2 active *RST\_UV* signal. (see also state table for *<WDA\_INT>* = ‘1’ below). The state of the “reset counter” *<RST\_CNT>* can be read via MSC but cannot be changed.

Table 4-2 Internal states for WDA\_INIT=1

| RST_CNT old value | EC>7 and sequencer-run | RST_CNT new value | WDA_RST                              |
|-------------------|------------------------|-------------------|--------------------------------------|
| 000 ... 111       | False                  | = RST_CNT old     | 0, no monitoring module reset        |
| 000 ... 110       | True                   | = RST_CNT old + 1 | 1, monitoring module reset generated |
| 111               | True                   | = 111             | 0, no monitoring module reset        |

#### 4.1.4 Question generation

The generation of the 4-bit question (*REQU[3-0]*) is realized as in the table below. The 4-bit counter only changes to the next state during the sequencer-run when the previous question has been answered correctly in value and in time. The state is changed by a sequencer-run because of a write-access to the RESPTIME register or the expiration of the time window.

When RST\_UV is active the state is set to 0000b

#### 4.1.5 Response comparison

The 2-bit counter <RESP\_CNT (1-0)> counts the received bytes of the 32-bit response and controls the generation of the expected response. Its default value is “11” (corresponds to “waiting for RESP\_BYTE3”).

The <RESP\_ERR> flag is set '1' when a response byte is incorrect. The flag remains '0' if the 32-bit response is correct. The ERROR COUNTER is updated with the flag. The default state of the flag is '0'.

The 2-bit counter <RESP\_CNT (1-0)> and the <RESP\_ERR> flag are reset to their corresponding default values at a sequencer-run. The reset condition of the counter <RESP\_CNT (1-0)> and the <RESP\_ERR> flag are the corresponding default states.

Procedure of the sequential response comparison:

<RESP\_CNT (1-0)> = “11”: switch the expected response for RESP\_BYTE3 to the comparator

Write access: RESP\_BYTE3

Set <RESP\_CNT> to “10”, update <RESP\_ERR> flag

<RESP\_CNT (1-0)> = “10”: switch the expected response for RESP\_BYTE2 to the comparator

Write access: RESP\_BYTE2

set <RESP\_CNT> to “01”, update <RESP\_ERR> flag

<RESP\_CNT (1-0)> = “01”: switch the expected response for RESP\_BYTE1 to the comparator

Write access: RESP\_BYTE1

set <RESP\_CNT> to “00”, update <RESP\_ERR> flag

<RESP\_CNT (1-0)> = “00”: switch the expected response for RESP\_BYTE0 to the comparator

“update <RESP\_ERR> flag” previously described means that it sets <RESP\_ERR> flag to 1 if a response byte is incorrect. Once <RESP\_ERR> flag is set to 1 it maintains its value until next sequencer run.

Write access: RESP\_BYTE0

Start sequencer (SEQU\_START signal), set <RESP\_CNT> to “11”, update <RESP\_ERR> flag, update ERROR COUNTER

Sequencer clears <RESP\_ERR> flag to '0

SEQU\_START = (RESP\_CNT1=0) AND RESP\_CNT0=0) AND “response byte write”

**Table 4-3 Question and answer**

| question REQU (3-0) | RESP_BYTE3 | RESP_BYTE2 | RESP_BYTE1 | RESP_BYTE0 |
|---------------------|------------|------------|------------|------------|
| 0                   | FF         | 0F         | F0         | 00         |
| 1                   | B0         | 40         | BF         | 4F         |
| 2                   | E9         | 19         | E6         | 16         |
| 3                   | A6         | 56         | A9         | 59         |
| 4                   | 75         | 85         | 7A         | 8A         |
| 5                   | 3A         | CA         | 35         | C5         |
| 6                   | 63         | 93         | 6C         | 9C         |
| 7                   | 2C         | DC         | 23         | D3         |
| 8                   | D2         | 22         | DD         | 2D         |
| 9                   | 9D         | 6D         | 92         | 62         |
| A                   | C4         | 34         | CB         | 3B         |
| B                   | 8B         | 7B         | 84         | 74         |
| C                   | 58         | A8         | 57         | A7         |
| D                   | 17         | E7         | 18         | E8         |
| E                   | 4E         | BE         | 41         | B1         |
| F                   | 01         | F1         | 0E         | FE         |

#### 4.1.6 Reset behavior

All monitoring module registers are reset by RST\_UV. The following monitoring module components are also reset by RST\_PRL (internal partial reset).

**Table 4-4 Watchdog reset behavior**

| Component           | Reset Condition          |
|---------------------|--------------------------|
| ERROR COUNTER       | 110b                     |
| Register for “EC>7” | '0'                      |
| Register RESPTIME   | Maximum value:0011 1111b |

RST\_PRL is active when at least one of the following signals is active: RSTN or SW\_RST.

#### 4.1.7 Access during a sequencer-run

A sequencer-run (which means the same as a monitoring cycle) is initiated by the writing of a response (i.e. all answer bytes `<RESP_BYTE3..0>`) or a write to `<RESPTIME>` or by reaching “end of time window”. It’s not interrupted by a new access, i.e. the monitoring module completes the action already started:

- A sequencer-run was initiated by a “response write”: The sequencer completes its task with the data of the previous access and the new data are ignored.
- A sequencer-run was initiated by a “response-time write”: The sequencer uses the response-time of the previous access, the error counter is correspondingly incremented by one and the `<CHRT>` bit (REQUHI register) is set and the new data are ignored. `<CHRT>` will be reset by reading and by the next start of a sequencer run (not reset by the sequencer run that is started by a “response-time write”)
- A sequencer-run was initiated by “end of time window”: The sequencer finishes the started run, the error counter is incremented by one and the new data are ignored.

The writing of a response-time during a sequencer-run will not set the `<CHRT>` bit (REQUHI register). The new response-time value is also not accepted.

At the end of the sequencer run `W_RESP` bit is set to 0 if the response is correct, it is set to 1 if the response is incorrect.

#### 4.1.8 Clock and time references

The monitoring module works independently by the micro-controller clock so that it can monitor the timing of the micro-controller.

This oscillator is integrated in the U-chip and it provides a clock CLK1 for the monitoring module.

Clocked with CLK1, a divider generates the base time of 1.6 ms @ 64 kHz for the response-time and  $8 \times 1.6 \text{ ms} = 12.8 \text{ ms}$  @ 64 kHz for the fixed time window. Accuracy of CLK1 is  $\pm 5\%$ .

The response-time is adjustable by the controller in the range 0ms to about 100ms (register `RESPTIME`).

The response-time can be calculated with the equation

$$\text{Response\_Time} = \text{RESPTIME}[5:0] \times 1.6 \text{ ms}$$

If 39 kHz clock is selected by `wda_win_sel` configuration bit, the base time changes becoming 2.6ms @ 39 kHz. This implies the change of all related timing values:  $8 \times 2.6 \text{ ms} = 20.8 \text{ ms}$  @ 39 kHz for the fixed time window and response time becomes `RESPTIME[5:0] * 2.6 ms`.

The `WDA_RESPTIME (CONFIG_REG9_0)` register is set to '0011 1111'b after a reset. The ERROR COUNTER is incremented by one if the controller changes the response-time. If the response-time is set to 0 ms, then the ERROR COUNTER is incremented by one even if a correct response is received within the time window. The maximum error reaction time is given by: maximum response-time, response at the end of a time-window and ERROR COUNTER 0 is  $5 \times (100.8 \text{ ms} + 12.8 \text{ ms}) = 568 \text{ ms}$ .

In order to assure correct response write, there must be a minimum distance between 2 consecutive writes equal to 5us, this time does not depend on `WDA_WIN_SEL`

(CONFIG\_REG9\_1 D2).

Note: These clock-tolerances have to be taken into account additionally.

As output (open drain):

$5.5\text{ V} \leq \text{VB\_IN} \leq 18\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified;

**Table 4-5 WDA circuit electrical characteristic**

| Symbol          | Parameter             | Test condition                                                                           | Min | Typ | Max | Unit          | Pin |
|-----------------|-----------------------|------------------------------------------------------------------------------------------|-----|-----|-----|---------------|-----|
| WDA_UV_LO       | Output low voltage    | $1.3\text{ V} < \text{VDD5} < \text{VDD\_UV}$<br>$R_{\text{reset}} = 4.7\text{ k}\Omega$ | -   | -   | 0.6 | V             | WDA |
| WDA_IUVres_max  | Input current         | $\text{VDD5} = \text{VDD\_UV}$<br>$V_{\text{UV\_reset}} = 0.6\text{ V}$                  | 5   | -   | -   | mA            | WDA |
| WDA_IlkUV_reset | Input leakage current | $V_{\text{UV\_reset}} > \text{VDD\_UV}$                                                  | -   | -   | 1   | $\mu\text{A}$ | WDA |

As input:

$5.5\text{ V} \leq \text{VB\_IN} \leq 18\text{ V}$ ,  $4.75\text{ V} \leq \text{VDD\_IO} \leq 5.25\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified;

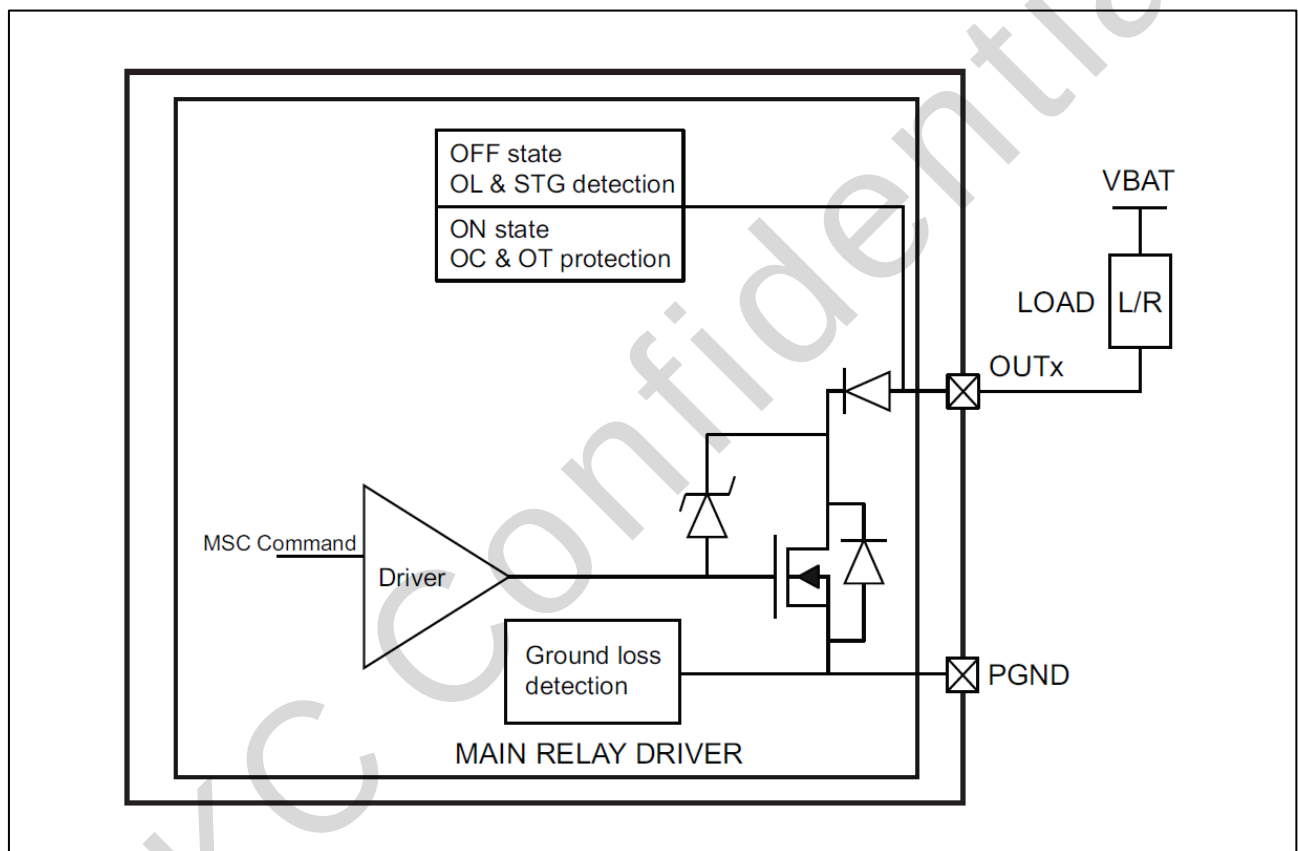
**Table 4-6 WDA reset circuit electrical characteristic**

| Symbol       | Parameter                     | Test condition                         | Min  | Typ | Max                    | Unit             | Pin |
|--------------|-------------------------------|----------------------------------------|------|-----|------------------------|------------------|-----|
| WDA_RST_L    | WDA input low voltage         | -                                      | -0.3 | -   | 1.1                    | V                | WDA |
| WDA_RST_H    | WDA input high voltage        | -                                      | 2.3  | -   | $\text{VDD\_IO} + 0.3$ | V                | WDA |
| WDA_Trst_fit | WDA Reset filter time         | Tested by scan                         | -    | 10  | -                      | $\mu\text{s}$    | WDA |
| -            | WDA pull up external resistor | Application note, parameter not tested | 50   | -   | 250                    | $\text{k}\Omega$ | WDA |

## 5 Main relay driver (low-side driver)

This driver is used to drive main relay only.

The on/off status of the driver depends on the following sources of power on: KEY\_IN WK\_IN WAKE\_UP\_TIMER POWER\_HOLD (MSC), CAN see [Table 4-2](#). The Main Relay Driver can be also switched on/off by a MSC command protected by lock. This is intended to use in the permanent battery supply scheme only if power up was done by wk\_in/EOT/CAN according to [Table 5-1](#). MSC micro commands protected by lock are available for fast on and off diagnosis.



**Figure 5-1 Main relay low-side driver stage**

The low-side (main relay) can work down to  $V_{B\_IN} = V_{OFF\_VB\_IN}$ , during cold crank conditions. The driver has integrated diagnosis, with over-current and overtemperature protection circuit during the driver on. The driver turn on/off slew-rate is internally controlled.

The Main Relay Driver on/off status is controlled by the power up/down modes or by the msc commands according to [Table 5-1](#).

These notes are linked to [Table 5-1](#)

1. The battery status can be determined at power up only.

Main relay driver (low-side driver)

2. The key detection switches on always MRD, Except have sent the MRD\_OFF command under other detection sources (WK\_IN、WAKE\_CAN、WAKE\_EOT、PHOLD) .
3. Once the key detection has switched on MRD, MRD is switched off by one of these events:
  - Clearing of all detections and phold not active (power down),
  - Watchdog power counter timeout,
  - *MSC MRD OFF* command when phold is active (all detections cleared).
4. The msc\_enable\_driver command or trans\_f error or clock monitor error do not affect the MRD even if it is controlled by MSC commands.

## 5.1 ON state diagnostic

Table 5-1 ON state diagnostic

|                                                                                          |                  | MRD             | MSC command |     | OVC                                                           | OVT                                                                                 |
|------------------------------------------------------------------------------------------|------------------|-----------------|-------------|-----|---------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                                                                                          |                  |                 | ON          | OFF |                                                               |                                                                                     |
| KEY_DET=1                                                                                | Permanent Supply | ON              | Y           | Y   | restart by MSC scenario 19 (see <a href="#">Figure 0-14</a> ) | Auto restart without limit of number scenario 21 (see <a href="#">Figure 0-16</a> ) |
| WAKE_IN_DET /<br>WAKE_EOT_DET<br>/<br>WAKE_CAN_DET =1<br>KEY_DET=0                       | Permanent Supply | OFF             | Y           | Y   | restart by MSC scenario 19 (see <a href="#">Figure 0-14</a> ) | Auto restart without limit of number scenario 21 (see <a href="#">Figure 0-16</a> ) |
| PHOLD_EN =1<br>KEY_DET<br>/<br>WAKE_IN_DET<br>/<br>WAKE_EOT_DET<br>/<br>WAKE_CAN_DET = 0 | Permanent Supply | Keep the status | Y           | Y   | restart by MSC scenario 20 (see <a href="#">Figure 0-15</a> ) | Auto restart without limit of number scenario 29 (see <a href="#">Figure 0-17</a> ) |

### 5.1.1 Overcurrent protection

When overcurrent fault is detected the Main Relay Driver MRD is switched off.

The MRD is automatically restarted every T<sub>res</sub> if KEY\_IN is till active in permanent battery supply scheme.

In permanent battery supply scheme in case of power up from WK\_IN or WAKE\_UP\_TIMER or WAKE\_UP\_CAN the MRD is not switched on at power up. In case of permanent battery supply scheme and OVC protection is activated the driver is restarted by a positive transition on the MSC *MRD\_ON* command. It is not restarted by MSC reading.

### 5.1.2 Thermal protection

The overtemperature fault is detected only when VB\_IN is present and the logic is out of power on reset, so when the device is in ON state.

When overtemperature fault is detected the Main Relay Driver is switched off. In permanent battery supply schemes the switching off of the Main Relay Driver does not produce a power down. The MRD is switched on again when temperature decreases to T\_SD\_LOW.

### 5.2 OFF state diagnostic

The OFF diagnosis is performed only in the supply scheme with battery permanently connected.

The off diagnosis is the same as other low-side drivers.

There is a bit *lpupd\_EN* which is controlled by a MSC bit (CONFIG\_REG2 D6), it is used by external  $\mu$ Controller if it is needed to switch on/off both pull up and pull down diagnosis current.

There is a bit *lpupd\_MODE* which is controlled by a MSC bit (CONFIG\_REG2 D7), it is used by external  $\mu$ Controller if it is needed to switch on/off the bigger diagnosis current (I\_LS\_PU1) in fast mode. It is possible to define the maximum number of retry in case of OVC after wake up by KEY. It can be done setting *KEY\_OC\_RETRY\_MAX\_EN* bit in CONFIG-REG 20.

#### Condition:

$5.5\text{ V} \leq \text{VB\_IN} \leq 18\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified;

**Table 5-2 Main relay low-side driver electrical characteristics**

| Symbol              | Parameter                                                            | Test condition                                                                                        | Min | Typ | Max  | Unit       | Pin |
|---------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----|-----|------|------------|-----|
| <b>Main relay</b>   |                                                                      |                                                                                                       |     |     |      |            |     |
| MRD_out_voltage_low | Low-side out voltage                                                 | VB_IN = 4.25 V<br>I <sub>load</sub> = 0.1 A                                                           | -   | -   | 1.35 | V          | MRD |
| MRD_out_voltage_nom | Low-side out voltage                                                 | 5.5 V < VB_IN < 18 V<br>I <sub>load</sub> = 0.3 A                                                     | -   | -   | 1.35 | V          | MRD |
| MRD_lkg             | Output leakage current                                               | Output disabled, diagnostic off                                                                       | -10 | -   | +10  | $\mu$ A    | MRD |
| rev_MRD             | -                                                                    | Reverse battery                                                                                       | -   | -   | -16  | V          | MRD |
| I_load              | Output current                                                       | Application Info                                                                                      | -   | -   | 0.5  | A          | MRD |
| SR_ON_20            | Voltage slew ON State                                                | From 80% to 30% of VOUT VB_IN = 14 V,<br>R <sub>load</sub> = 20 $\Omega$<br>C <sub>load</sub> = 10 nF | 2   | -   | 6    | V/ $\mu$ s | MRD |
| SR_OFF_20           | Voltage slew OFF State                                               | From 30% to 80% of VOUT VB_IN = 14 V,<br>R <sub>load</sub> = 20 $\Omega$<br>C <sub>load</sub> = 10 nF | 2   | -   | 6    | V/ $\mu$ s | MRD |
| Ton_MRD_20          | Propagation Delay from Chip_EN rising edge to 80% output MRD voltage | VB_IN = 14 V,<br>R <sub>load</sub> = 20 $\Omega$<br>C <sub>load</sub> = 10 nF                         | -   | -   | 10   | $\mu$ s    | MRD |

## Main relay driver (low-side driver)

|                             |                                                                               |                                                                                                              |                    |     |                    |    |     |
|-----------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------|-----|--------------------|----|-----|
| Toff_MRD_20                 | Propagation Delay from Chip_EN falling edge to 20% output MRD voltage         | VB_IN=14V,<br>Rload = 20Ω<br>Cload = 10 nF                                                                   | -                  | -   | 10                 | μs | MRD |
| Vclamp_MRD                  | MRD clamping Voltage                                                          | Iload = 0.3 A                                                                                                | 48                 |     | 55                 | V  | MRD |
| Vclamp_MRD_high_curr        | MRD clamping Voltage                                                          | Iload = 1 A<br>Guaranteed by design, provided single pulse energy limits are not violated by clamping action | 48                 | -   | 55                 | V  | MRD |
| IOVC_MRD                    | Over current driver threshold                                                 | -                                                                                                            | 1                  | -   | 2                  | A  | MRD |
| I_LS_ocvflt                 | LS overcurrent filter time                                                    | Guaranteed by scan                                                                                           | 4                  | -   | 7                  | μs | MRD |
| Tft_Res                     | Self retry time after OVC with KEY_IN on or WK_IN active or WKE_UP_EOT active | Guaranteed by scan                                                                                           | 15.2               | 16  | 16.8               | ms | MRD |
| NRes_NN                     | Maximum number of retry when KEY_OC_RETRY_MAX_EN=1                            | Guaranteed by scan                                                                                           | -                  | 32  | -                  | -  | MRD |
| T_SD_HIGH                   | Temperature shut down                                                         | -                                                                                                            | 185                |     | 200                | °C | MRD |
| T_SD_LOW                    | Temperature shut down recover                                                 | -                                                                                                            | 175                | -   | 190                | °C | MRD |
| T_SD_hys                    | Temperature shut down hysteresis                                              | -                                                                                                            | 5                  | -   | 10                 | °C | MRD |
| tmsd_pre_an                 | Thermal shutdown analog filter time                                           | Guaranteed by design                                                                                         | 1.5                | -   | 4                  | μs | MRD |
| t_SD_deglitch               | Digital deglitch filter time on Temperature shut down detection               | Guaranteed by scan                                                                                           | 30                 | -   | -                  | μs | MRD |
| <b>OFF state diagnostic</b> |                                                                               |                                                                                                              |                    |     |                    |    |     |
| VLVT                        | Short to GND threshold voltage                                                | Driver tristate, diag enabled                                                                                | 1.9                | -   | VOU TOPE N - 190mV | V  | MRD |
| V_OL                        | Open load threshold voltage                                                   | Driver tristate, diag enabled                                                                                | VOU TOPE N +160 mV | -   | 3.0                | V  | MRD |
| VOUTOPEN                    | Open load output voltage                                                      | Driver tristate, diag enabled                                                                                | 2.3                | 2.5 | 2.7                | V  | MRD |
| I_LS_PU1                    | Diagnostic pull up current                                                    | In OFF condition, OUTx < VLVT, Ipupd_EN = ENB, Ipupd_MODE = "1"                                              | 2.5                | 3.6 | 4.7                | mA | MRD |
| I_LS_PU2                    | Diagnostic pull up current                                                    | In OFF condition, OUTx < VLVT,                                                                               | 40                 | 70  | 100                | μA | MRD |

Main relay driver (low-side driver)

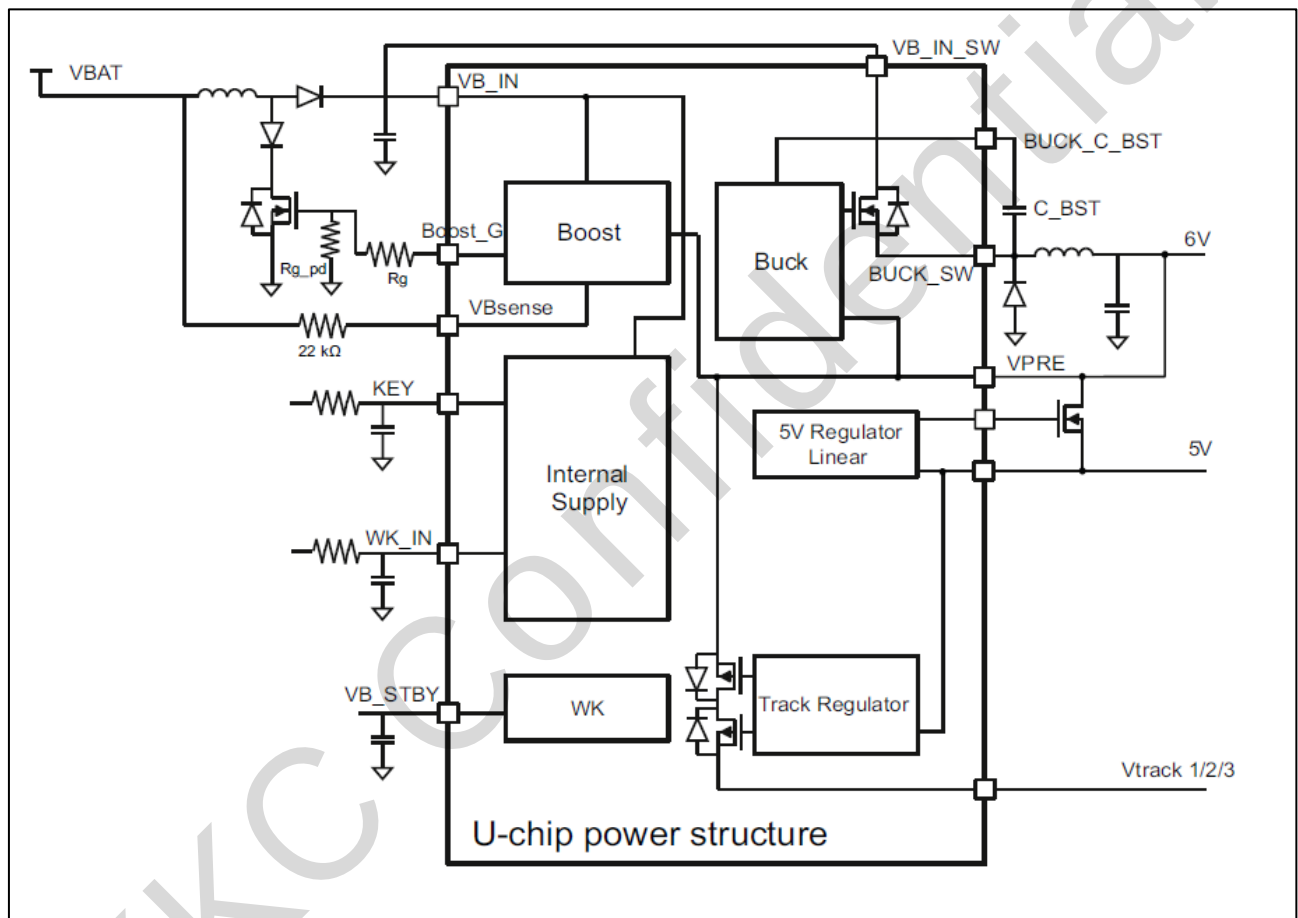
|                                |                                         |                                                               |     |     |     |    |     |
|--------------------------------|-----------------------------------------|---------------------------------------------------------------|-----|-----|-----|----|-----|
|                                |                                         | Ipupd_EN = ENB,<br>Ipupd_MODE = "0"                           |     |     |     |    |     |
| I_LS_PD1                       | Diagnostic pull down current            | In OFF condition,<br>OUTx ><br>VOUTOPEN,<br>Ipupd_EN = ENB    | 60  | 85  | 110 | μA | MRD |
| Tflt_diagoff1                  | DIAG Filter time                        | Filter Mode=0                                                 | 75  | 100 | 125 | μs | MRD |
| Tflt_diagoff2                  | DIAG Filter time                        | Filter Mode=1                                                 | 450 | 600 | 750 | μs | MRD |
| -                              | Minimum OFF time for correct diagnostic | Application note<br>(esd cap < 10 nF)<br>Filter Mode = 0      | 175 | -   | -   | μs | MRD |
| -                              | Fast OFF diagnostic<br>ON→OFF→ON        | Filter Mode = 0                                               | 175 | 200 | 240 | μs | MRD |
| -                              | Fast ON diagnostic<br>OFF→ON→OFF        | -                                                             | 65  | -   | 90  | μs | MRD |
| <b>Driver Reliability Data</b> |                                         |                                                               |     |     |     |    |     |
| EnergyRep MRD                  | Energy Repetitive Pulses                | I_OT_n = 0.3 A<br>Freq = 1 Hz ;<br>4 Miopulses<br>Tj = 150 °C | -   | -   | 6.5 | mJ | MRD |

### 5.3 Error handling

The Main Relay Driver follows the same error handling table as other low-side drivers.

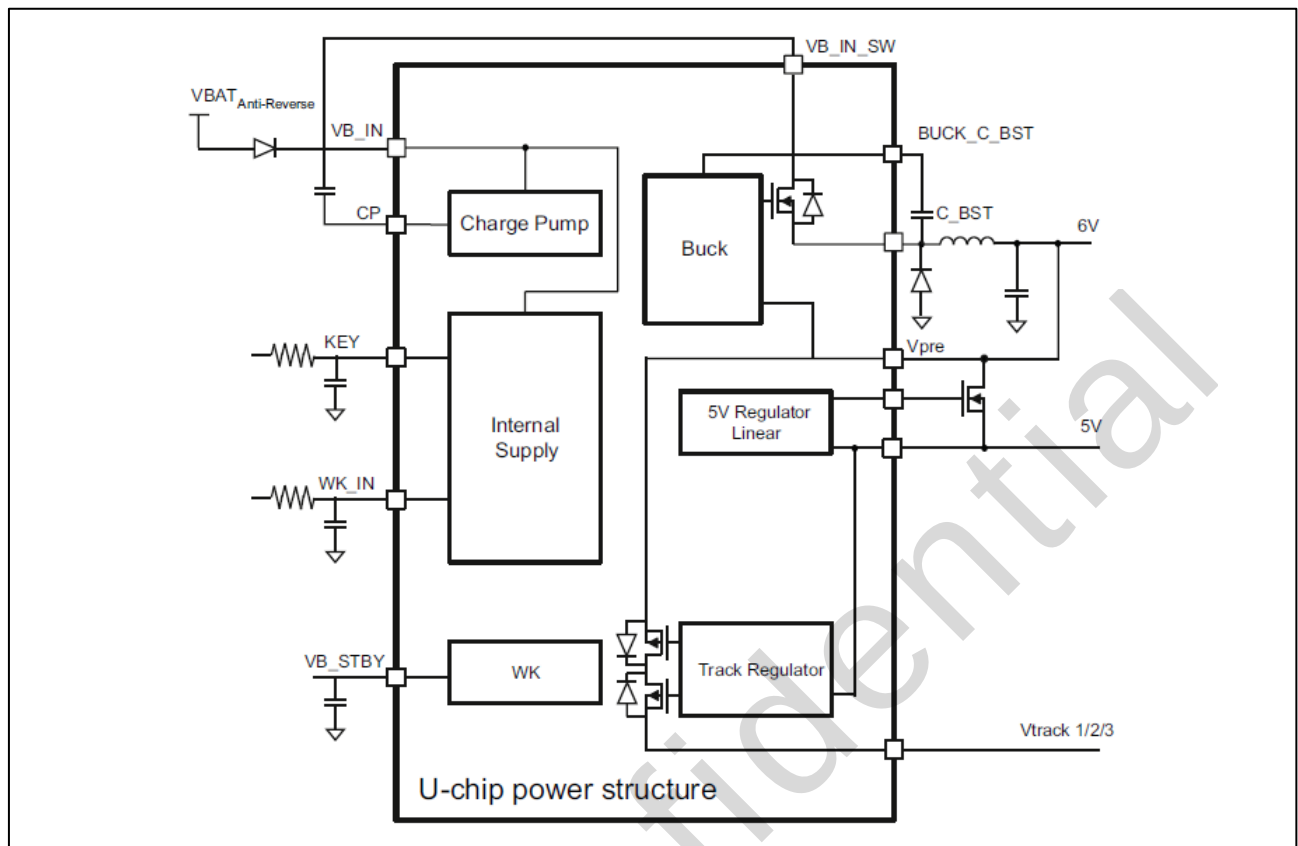
## 6 Multi-voltage regulator supplies

The chip has one pre-boost regulator and one pre-buck regulator, which will supply a 5 V regulator with external MOS. Three 5 V tracking regulators are included. One charge pump is used for HS channel and by the 5 V linear regulator to drive the external MOS. See the diagram in Figure 6-1. The chip becomes to be supplied and the regulators are enabled in the sequence: VB<sub>in</sub> grows till is higher than VB<sub>IN\_uv</sub> threshold, then internal 5V supply, charge pump and buck, pre-boost regulator, 5 V linear regulator, 5 V tracking regulator 1, 2 and 3 are switched on. Then the power on reset is released and the chip is working.



**Figure 6-1** Power supply block diagram with pre-boost regulator permanent VBAT

The pre-boost regulator is an optional choice. It can be disabled (BOOST\_EN (CONFIGREG 1 D5) in case battery voltage is high enough in cranking phase to supply the pre-buck regulator. See the diagram in Figure 6-2.



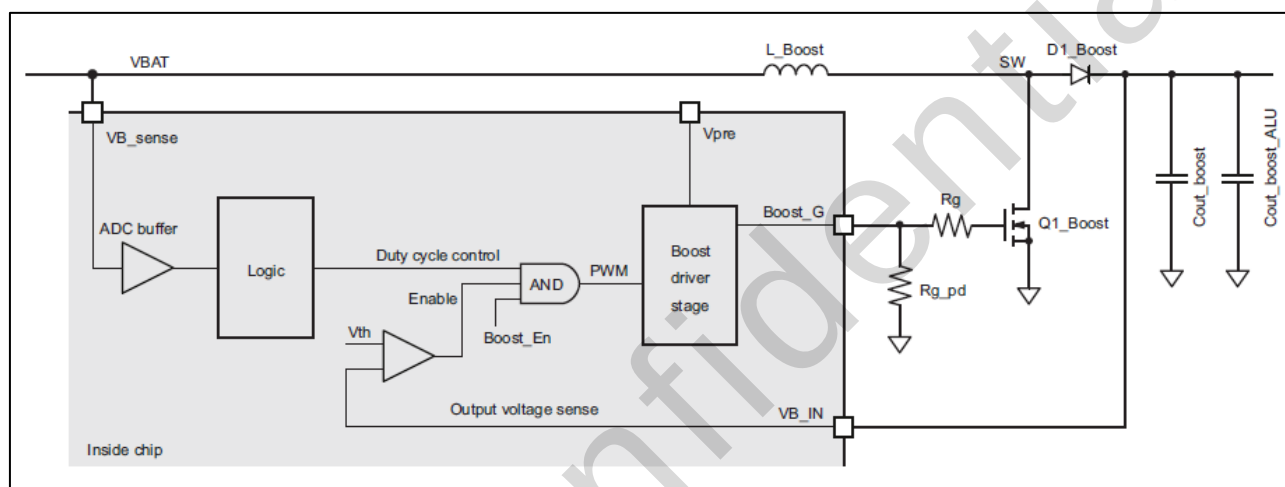
**Figure 6-2** Power supply block diagram with charge pump without pre-boost

## 6.1 Pre-boost regulator

The boost regulator will provide a higher voltage than battery to the buck regulator when battery voltage is too low to maintain 5 V regulator out of reset conditions. Once the output of the pre-boost is higher than VB\_IN\_th (typ 8.5 V), the pre-boost will stop working.

The boost is disabled at the beginning of the power down sequence. When *Boost\_EN* (CONFIG\_REG 1 D5) = 0 the internal driver keeps OFF the external MOS.

The parameters of Pre-boost regulator can be guaranteed when Vbat goes down to 3.0 V during cranking phase. The block diagram of the pre-boost is Figure 6-3.



**Figure 6-3 Pre-boost block diagram**

$3.0\text{ V} \leq \text{VBAT}^{(*)} < 18\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified.

*Note:* (\*)VBAT is the battery line of Figure 6-3 in front of the limiting 22 k  $\Omega$  resistor (not shown in the figure).

**Table 6-1 Pre-boost regulator output electrical characteristics**

| Symbol     | Parameter                                  | Test condition                                                                                                                                               | Min  | Typ | Max  | Unit | Pin   |
|------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|------|------|-------|
| VB_IN_th   | BOOST enable/disable threshold             | When VB_IN < VB_IN_TH, the boost starts for time (ton). When VB_IN ≥ VB_IN_TH, Boost is disabled after ton is completed. ton depends on different duty cycle | 8.07 | 8.5 | 8.93 | V    | VB_IN |
| tdBOOST_EN | BOOST Enable/disable comparator delay time | -                                                                                                                                                            | 0.19 | -   | 2    | μs   | VB_IN |
| Fsw_BOOST  | Switching frequency                        | 43 cycles of 15 MHz main clk, Guaranteed by SCAN                                                                                                             | 334  | 352 | 370  | kHz  | VB_IN |

## Multi-voltage regulator supplies

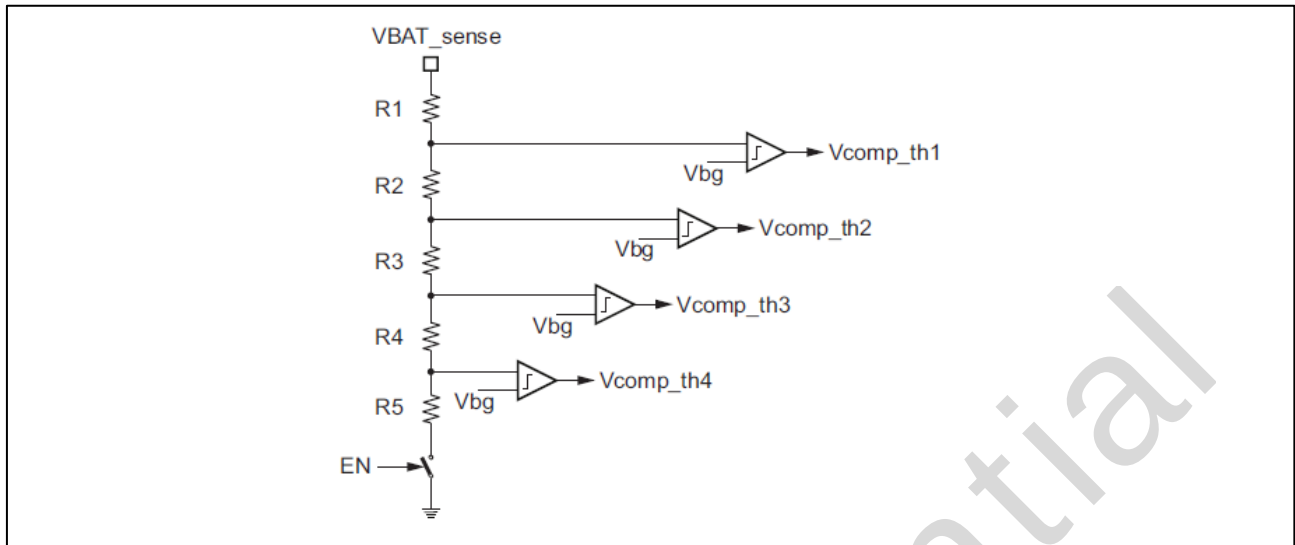
|                   |                                                |                                                                                                                    |      |       |      |    |            |
|-------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------|-------|------|----|------------|
| Boost_DC1         | Boost Duty cycle 1                             | 0V < VBAT_SENSE < Vcomp_th1 33 out of 43 cycles of 15MHz main clk (100% DC is 43 clk cycles)<br>Guaranteed by SCAN | -    | 76.74 | -    | %  | VB_IN      |
| Vcomp_th1         | VBAT_SENSE comparator threshold 1              | -                                                                                                                  | 3.95 | 4.15  | 4.35 | V  | VBAT_SENSE |
| Boost_DC2         | Boost Duty cycle 2                             | Vcomp_th1 < VBAT_SENSE < Vcomp_th2 25 out of 43 clk cycles<br>Guaranteed by SCAN                                   | -    | 58    | -    | %  | VB_IN      |
| Vcomp_th2         | VBAT_SENSE comparator threshold 2              | -                                                                                                                  | 5    | 5.36  | 5.7  | V  | VBAT_SENSE |
| Boost_DC3         | Boost Duty cycle 3                             | Vcomp_th2 < VBAT_SENSE < Vcomp_th3 18 out of 43 clk cycles<br>Guaranteed by SCAN                                   | -    | 41.86 | -    | %  | VB_IN      |
| Vcomp_th3         | VBAT_SENSE comparator threshold 3              | -                                                                                                                  | 6.25 | 6.6   | 6.95 | V  | VBAT_SENSE |
| Boost_DC4         | Boost Duty cycle 4                             | Vcomp_th3 < VBAT_SENSE < Vcomp_th4 10 out of 43 clk cycles<br>Guaranteed by SCAN                                   | -    | 23.85 | -    | %  | VB_IN      |
| Vcomp_th4         | VBAT_SENSE comparator threshold 4              | -                                                                                                                  | 7.4  | 7.8   | 8.2  | V  | VBAT_SENSE |
| Boost_DC5         | Boost Duty cycle 5                             | Vcomp_th4 < VBAT_SENSE 4 out of 43 clk cycles<br>Guaranteed by SCAN                                                | -    | 9.3   | -    | %  | VB_IN      |
| tdBAT_SENS        | VBAT_SENSE comparators delay time              | Vcomp_th1/2/3/4 comparators                                                                                        | 0.19 | -     | 2    | μs | VBAT_SENSE |
| IBOOST_LS_sink    | LS driver sink current                         | BOOST_G = 6 V                                                                                                      | 30   | 50    | 70   | mA | BOOST_G    |
| IBOOST_HS_source  | HS driver source current                       | BOOST_G = 0 V                                                                                                      | 30   | 50    | 70   | mA | BOOST_G    |
| IBOOST_LS_sink    | LS driver sink current                         | BOOST_G = 0.5 V                                                                                                    | 15   | -     | 43   | mA | BOOST_G    |
| IBOOST_HS_source  | HS driver source current                       | BOOST_G = 5.5 V                                                                                                    | 10   | -     | 43   | mA | BOOST_G    |
| BOOST_G_Ileak     | Leakage current                                | From 0 V to max pin voltage                                                                                        | -    | -     | 1.2  | μA | BOOST_G    |
| I_VBAT_Sense_stby | Current leakage on VBAT_sense pin in st-by     | -                                                                                                                  | -    | -     | 1    | μA | VBAT_SENSE |
| I_VBAT_Sense      | Pull down current on VBAT_sense operative mode | -                                                                                                                  | -    | -     | 100  | μA | VBAT_SENSE |
| R_VBAT_sense      | Pull-down resistor from                        | -                                                                                                                  | 438  | 880   | 1320 | kΩ | VBAT_SENSE |

|               |                                 |                         |       |      |       |    |   |
|---------------|---------------------------------|-------------------------|-------|------|-------|----|---|
|               | VBAT_sense to GND               |                         |       |      |       |    |   |
| R1_VBAT_sense | Resistor divider string element | Design info, not tested | 311   | 622  | 933   | kΩ | - |
| R2_VBAT_sense | Resistor divider string element | Design info, not tested | 28.75 | 57.5 | 86.25 | kΩ | - |
| R3_VBAT_sense | Resistor divider string element | Design info, not tested | 18.15 | 36.3 | 54.45 | kΩ | - |
| R4_VBAT_sense | Resistor divider string element | Design info, not tested | 12.5  | 25   | 37.5  | kΩ | - |
| R5_VBAT_sense | Resistor divider string element | Design info, not tested | 67.5  | 135  | 202.5 | kΩ | - |

Boost application information table:

**Table 6-2 Pre-boost regulator external components electrical characteristics**

| Symbol                     | Parameter                            | Min  | Typ | Max                      | Unit | Note                                                                                                                               | Pin        |
|----------------------------|--------------------------------------|------|-----|--------------------------|------|------------------------------------------------------------------------------------------------------------------------------------|------------|
| C <sub>OUT_BOOST</sub>     | Capacitance at VB_IN                 | 36   | 50  | 150                      | μF   | Min 5*10 μF, 10%, 25 V, X7R, 1206, CER CAP;                                                                                        | VB_IN      |
| C <sub>OUT_BOOST_ALU</sub> | Option ALU Capacitance at VB_IN      | 0    | 470 |                          | μF   | ALU 35 V ESR 10 mΩ--3 Ω, 470 μF, 20%, 25 V, SMD, AL, ELEC CAP                                                                      | VB_IN      |
| VBoost_ripple              | -                                    | -    | -   | 1.2<br>1.1<br>1.0<br>0.8 | Vpp  | triangular ripple @ 10 kHz and 25 kHz @ 50 kHz @ 100 kHz @ 350 kHz. All characteristics of all blocks supplied by VB_IN guaranteed | VB_IN      |
| IBOOST                     | Load current at VB_IN                | 0.05 | -   | 2.4                      | A    |                                                                                                                                    | VB_IN      |
| Q1_Boost                   | -                                    | -    | -   | -                        | -    | Choice: STD20NF06L                                                                                                                 | VB_IN      |
| D1_Boost                   | -                                    | -    | -   | -                        | -    | Choice: STPS10H100C                                                                                                                | VB_IN      |
| L_boost                    | BOOST inductance                     | 0.8  | 1   | 1.2                      | μH   | ESR = 10 mΩ                                                                                                                        | VB_IN      |
| L_boost1                   | BOOST inductance                     | -    | 2.5 | -                        | μH   | SR = 25 mΩ, Isat = 23 A                                                                                                            | VB_IN      |
| L_boost1                   | BOOST inductance                     | -    | 3.3 | -                        | μH   | -                                                                                                                                  | VB_IN      |
| Rsense                     | Resistor for VB_sense pin protection | 10   | 22  | 30                       | kΩ   | -                                                                                                                                  | VBAT_Sense |
| Rg                         | -                                    | -    | 2.2 | -                        | Ω    | -                                                                                                                                  | BOOST_G    |
| Rg_pd                      | -                                    | -    | 150 | -                        | kΩ   | -                                                                                                                                  | BOOST_G    |



**Figure 6-4 Boost threshold comparator's schematic**

## 6.2 Pre-buck regulator with internal MOS

Vpre features:

- 6V regulated output voltage (VPRE);
- 2.2 A maximum output current;
- PWM block (ramp generator, comparator and main latch) with one fixed PWM frequency (Fsw\_vpre);
- Internal error amplifier;
- Internal Power FET including Bootstrap;
- dV/dt Control on gate drivers to fulfill EMC requirements;
- Maximum Duty cycle 100%, to allow to charge Bootstrap capacitor;
- Protection by over temperature;
- Soft Start.

The Vpre regulator is a buck regulator which supplies Pre-boost driver/VDD5/Vtrack1.2.3. When the OV of VB\_IN is detected, the Vpre(Buck) will be switched off. This function is selectable by a dedicated MSC bit (*VB\_IN\_OV\_RST\_EN* in CONFIG\_REG12). A bit is also available (*BUCK\_SLOW\_SR* in CONFIG\_REG 21) in order to select slew rate for output voltage; the default value corresponds to fast slew rate, slow slew rate helps to improve emission performance

The Vpre regulator includes a soft start. Range of dV/dt at VPRE is specified in electrical parameters.

Pulse skipping mode: the Vpre regulator can go to pulse skipping mode when the load is too low to avoid that output voltage (VPRE) rises too much. The principle of the pulse skipping mode is to skip some pulses during the switching phase, when the loop can't keep itself the

Multi-voltage regulator supplies

output voltage regulated with a very low load because it is limited by the minimum duty cycle.

VB\_IN\_OV fault can trigger the turn OFF of Buck regulator, as specified into reset matrix. This effect can be masked with bit *VB\_IN\_OV\_RST\_EN* in CONFIG\_REG 12.

**Conditions:**

5.5 V < VB\_IN\_SW < 18 V,  $T_j$  -40 to 175 °C unless otherwise specified.

**Table 6-3 Pre-buck regulator output electrical characteristics**

| Symbol        | Parameter                                                                | Test condition                                                                                                                                                                 | Min           | Typ  | Max           | Unit | Pin  |
|---------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------|---------------|------|------|
| VPREdc        | VPRE regulated voltage (including aging and static line/load regulation) | VB_IN_SW = 7 V to 18 V, $I_{VPRE} = 50$ mA<br>~2.2 A, C = 30 $\mu$ F,<br>L = 22 $\mu$ H                                                                                        | 5.82<br>(-3%) | 6    | 6.18<br>(+3%) | V    | VPRE |
| Vdrift_VPRE   | Total VPRE voltage aging drift                                           | -                                                                                                                                                                              | -0.8          | -    | 0.8           | %    | VPRE |
| Vline_VPRE    | Line regulation voltage                                                  | 7 V < VB_IN_SW < 18V<br>$I_{VPRE} = 50$ mA, 2200 mA<br>C=30 $\mu$ F,<br>L = 22 $\mu$ H, T = 27 °C,<br>175 °C                                                                   | -25           | -    | 25            | mV   | VPRE |
| Vline_VPRE    | Line regulation voltage                                                  | 7 V < VB_IN_SW < 18V<br>$I_{VPRE} = 50$ mA,<br>2200 mA C=30 $\mu$ F,<br>L = 22 $\mu$ H, T = -40 °C                                                                             | -60           | -    | 60            | mV   | VPRE |
| Vload_VPRE    | Load regulation voltage                                                  | VB_IN_SW = 7 V, 18V,<br>50 mA < $I_{VPRE}$ < 2200 mA,<br>C = 30 $\mu$ F, L = 22 $\mu$ H,<br>T = 27 °C, 175 °C                                                                  | -25           | -    | 25            | mV   | VPRE |
| Vload_VPRE    | Load regulation voltage                                                  | VB_IN_SW = 7 V, 18 V,<br>50 mA < $I_{VPRE}$ < 2200 mA,<br>C = 30 $\mu$ F,<br>L = 22 $\mu$ H, T = -40°                                                                          | -60           | -    | 60            | mV   | VPRE |
| Vline_VPRE_tr | Transient line regulation accuracy                                       | VB_IN_SW step 1 2 V to 18 V ; 18 V to 12 V,<br>$dV_{B\_IN\_SW}/dt = 3$ V/ $\mu$ s,<br>$I_{VPRE} = 50$ mA, 1.8 A,<br>C = 30 $\mu$ F, L = 22 $\mu$ H,<br>Guaranteed by design(1) | -8            | VPRE | 8             | %    | VPRE |
| Vload_VPRE_tr | Transient Load regulation accuracy                                       | VB_IN_SW = 13 V,<br>$I_{VPRE} = 50$ mA to 1.8 A<br>$dI/dt = 500$ mA/ $\mu$ s,<br>C = 30 $\mu$ F, L = 22 $\mu$ H,<br>Guaranteed by design(1)                                    | -8            | VPRE | 8             | %    | VPRE |
| ILIMVPRE      | Current limit                                                            | -                                                                                                                                                                              | 3.5           | -    | 5.1           | A    | VPRE |
| VPRE_coupling | VPRE capacitive coupling during                                          | VB_IN_SW from 0 to 13 V in 100 $\mu$ s,                                                                                                                                        | -             | -    | 3             | V    | VPRE |

|                 |                                                                                                             |                                                                           |      |         |      |              |         |
|-----------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|------|---------|------|--------------|---------|
|                 | battery transients                                                                                          | C = 40 $\mu$ F , L = 22 $\mu$ H,<br>device off<br>Design info, not tested |      |         |      |              |         |
| Vuv_vpre_h      | VPRE under voltage hysteresis threshold high; VPRES voltage at Power up to allow Boost activation threshold | Test condition:<br>VB_IN_SW = 5.4 V                                       | 4.38 | -       | 4.65 | V            | VPRE    |
| Vuv_vpre_l      | VPRE under voltage hysteresis threshold low                                                                 | Test condition:<br>VB_IN_SW = 5.4 V                                       | 4.25 | -       | 4.51 | V            | VPRE    |
| Tuv_filter_vpre | VPRE under voltage filter time (2)                                                                          |                                                                           | 10.1 | 12      | 13.3 | $\mu$ s      | VPRE    |
| VPRE_slope      | Slope control                                                                                               | VB_IN_SW = 13 V                                                           | 5    | -       | 40   | V/ms         | VPRE    |
| Tmsd_VPRE_H     | Thermal shutdown_High                                                                                       | Comparator threshold and functionality are tested in production           | 185  | -       | 200  | $^{\circ}$ C | VPRE    |
| Tmsd_VPRE_L     | Thermal shutdown_Low                                                                                        | Comparator threshold and functionality are tested in production           | 175  | -       | 190  | $^{\circ}$ C | VPRE    |
| Tmsd_hyst       | Thermal hysteresis for shutdown                                                                             | -                                                                         | 5    | -       | 10   | $^{\circ}$ C | VPRE    |
| tmsd_pre_an     | Thermal shutdown analog filter time                                                                         | -                                                                         | 1.5  | -       | 4    | $\mu$ s      | VPRE    |
| tmsd_pre_dig    | Thermal shutdown digital filter time                                                                        | -                                                                         | 10   | 20      | 30   | $\mu$ s      | VPRE    |
| VPRE_rpd        | VPRE pulldown resistor                                                                                      | To keep VPRES below 10 V with Buck switch leakage in OFF state            | 0.2  | -       | 1.05 | M $\Omega$   | VPRE    |
| Fsw_VPRE        | Switching frequency                                                                                         | Fosc from main 15 MHz oscillator, typical 462.5 kHz                       | 440  | Fosc/32 | 500  | kHz          | BUCK_SW |
| DCVPRE(BUCK)    | Duty cycle range                                                                                            | -                                                                         | 3%   | -       | 100% | -            | BUCK_SW |
| RVPRE           | VPRE (Buck_POWER MOS) transistor RDSON                                                                      | -                                                                         | -    | -       | 250  | m $\Omega$   | BUCK_SW |
| TRVPRE          | Rise Time (10-90%)                                                                                          | BUCK_SLOW_SR=0                                                            | 5    | -       | 25   | ns           | BUCK_SW |
| TRVPRE_slow     | Rise Time (10-90%)                                                                                          | BUCK_SLOW_SR=1                                                            | 5    | -       | 30   | ns           | BUCK_SW |
| TFVPRE          | Fall Time (10-90%)                                                                                          | Iload = 1 A                                                               | 5    | -       | 27   | ns           | BUCK_SW |
| VPRE_SW_Ileak   | Leakage current                                                                                             | From 0 V to max pin voltage                                               | -    | -       | 1    | $\mu$ A      | BUCK_SW |

1. Fault detection during power up sequence is masked until RSTN release to avoid false triggering at ramp-up phase.
2. No VDD5 UV/OV violation is expected during specified transient.

**Note:** When  $18 < VB\_IN\_SW < VOV\_VB\_IN\_L$ , the function of Vpre is guaranteed and no reset happens.

**Table 6-4 Pre-buck regulator external components electrical characteristics (Vpre application)**

| Symbol      | Parameter                                                  | Min | Typ | Max | Unit | Note                                                                          | Pin        |
|-------------|------------------------------------------------------------|-----|-----|-----|------|-------------------------------------------------------------------------------|------------|
| CVPRE       | External capacitor on Vbuck supply                         | 15  | 40  | 150 | μF   | Min 3*10 μF Ceramic X7R 16 V in parallel<br>CER,X7R,10 μF , 10%,16 V,1206, Sn | VPRE       |
| LVPRE       | Switching regulator external inductor                      | 15  | 22  | 29  | μH   |                                                                               | VPRE       |
| CBST        | Bootstrap capacitor                                        | -   | 47  | -   | nF   | Ceramic X7R 16V<br>47nF, 10%, 16 V, X7R, 0603,<br>TINNED, CER CAP/            | BUCK_C_BST |
| DVBUCK_CTRL | Switching regulator external diode (Schottky or ultrafast) | -   | -   | -   | -    | Schottky Diode STPS5L60-Y                                                     | -          |

### 6.3 Tracking regulator for sensors supply

There are three 5V tracking regulators with self protection from over-current, over-voltage and short to battery. STB/OVC/OV don't disable tracking, but set a flag readable by MSC. Those three tracking regulators track VDD5\_in voltage with 150mA output. There is one thermal sensor, shared between the three tracking regulators, to provide the thermal shut down protection for those three tracking sensor supplies. The switching off of one internal tracking occurs if that tracking is also in OVC.

Once the thermal shut down has happened, the tracking sensor in over-current condition will be off until temperature decreases to thermal shut down threshold low. Those three tracking regulators can be singularly disabled by dedicated MSC bit (*TRK\_EN* CONFIG\_REG7 [3:5] in CONFIG\_REG 7). VB\_IN\_OV fault can trigger the turn OFF of Tracking regulators, as specified into reset matrix. This effect can be masked with bit VB\_IN\_OV\_RST\_EN in CONFIG\_REG 12.

#### Conditions:

5.5 V < VB\_IN\_SW ≤ 18 V, T<sub>j</sub> -40 to 175 °C unless otherwise specified.

**Table 6-5 Tracking sensor supplies electrical characteristics**

| Symbol               | Parameter                                                                                                        | Test condition                              | Min           | Typ | Max                 | Unit | Pin         |
|----------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------|-----|---------------------|------|-------------|
| ΔVTRK                | Output voltage tracking error (including voltage aging and static load/line transient regulation) <sup>(1)</sup> | Vpre = 5.6 - 18 V<br>1 mA < IVSENS < 150 mA | VDD5<br>-15mV |     | VDD5<br>+20mV       | V    | VSENSE1/2/3 |
| I <sub>MAX_VS</sub>  | Output current limitation                                                                                        | VSENS = -1 V                                | 150           |     | 300                 | mA   | VSENSE1/2/3 |
| I <sub>TH_VS</sub>   | Output over-current threshold                                                                                    | Vpre = 5.6 - 18 V                           | 150           |     | I <sub>max_vs</sub> | mA   | VSENSE1/2/3 |
| V <sub>line_VS</sub> | Line regulation                                                                                                  | Vpre = 5.6 - 18 V                           | -15           |     | 30                  | mV   | VSENSE1/2/3 |

Multi-voltage regulator supplies

|                                   |                                                    |                                                                                                                                                |                  |      |                       |    |             |
|-----------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------|-----------------------|----|-------------|
|                                   | voltage                                            | IVSENS=50mA<br>Ctrk=470nF (2)                                                                                                                  |                  |      |                       |    |             |
| V <sub>load_VS</sub>              | Load regulation voltage                            | V <sub>pre</sub> = 6 V<br>1 mA<IVSENS<150 mA<br>Ctrk = 470 nF Note(2)                                                                          | -15              |      | 30                    | mV | VSENSE1/2/3 |
| V <sub>load_tran</sub>            | Transient load regulation                          | V <sub>pre</sub> = 6 V IVSENS =1 mA to 150 mA and viceversa<br>dI/dt = 150 mA/μs<br>Ctrk = 470 nF Guaranteed by design(3)                      | -15              |      | +15                   | %  | VSENSE1/2/3 |
| V <sub>line_tran</sub>            | Transient line regulation                          | V <sub>PRE</sub> step 5.6V to 7V and Viceversa<br>dV <sub>PRE</sub> /dt=3V/μs<br>IVSENS=1mA,150mA<br>Ctrk=470nF<br>Guaranteed by design(4),(3) | -15              |      | +15                   | %  | VSENSE1/2/3 |
| I <sub>sink_VS</sub>              | Short circuit reverse current                      | Output shorted to V <sub>B_IN</sub> +2V                                                                                                        | -                | -    | -4                    | mA | VSENSE1/2/3 |
| V <sub>SCB_TH_VS</sub>            | Short to battery threshold                         | -                                                                                                                                              | V <sub>pre</sub> |      | V <sub>pre</sub> +0.2 | V  | VSENSE1/2/3 |
| V <sub>UV_VS_l</sub>              | VSENSE1/2/3 undervoltage hysteresis threshold low  | -                                                                                                                                              | 4.5              | 4.65 | 4.75                  | V  | VSENSE1/2/3 |
| V <sub>UV_VS_h</sub>              | VSENSE1/2/3 undervoltage hysteresis threshold high | -                                                                                                                                              | 4.7              | 4.8  | 4.9                   | V  | VSENSE1/2/3 |
| V <sub>OV_VS</sub>                | VSENSE1/2/3 overvoltage threshold                  | -                                                                                                                                              | 5.3              | 5.4  | 5.5                   | V  | VSENSE1/2/3 |
| PSRR <sub>VS</sub>                | Supply voltage tracking rejection                  | f =10 kHz                                                                                                                                      |                  |      | -35                   | dB | VSENSE1/2/3 |
| V <sub>OS_VS</sub>                | Over shoot during power up                         | -                                                                                                                                              |                  |      | 5.5                   | V  | VSENSE1/2/3 |
| T <sub>UV_filter_VS_running</sub> | Undervoltage filter time(2)                        | cover by SCAN TEST                                                                                                                             | 16               | 24   | 32                    | μs | VSENSE1/2/3 |
| T <sub>OV_filter_VS_running</sub> | Overvoltage filter time(2)                         | cover by SCAN TEST                                                                                                                             | 16               | 24   | 32                    | μs | VSENSE1/2/3 |
| T <sub>OC_filter_VS_running</sub> | Overcurrent filter time                            | cover by SCAN TEST                                                                                                                             | 16               | 24   | 32                    | μs | VSENSE1/2/3 |
| T <sub>msd_VPRE_H</sub>           | Thermal shutdown_High                              | Comparator threshold and functionality are tested in production                                                                                | 185              |      | 200                   | °C | VSENSE1,2,3 |
| T <sub>msd_VPRE_L</sub>           | Thermal shutdown_Low                               | Comparator threshold and functionality are tested in production                                                                                | 175              |      | 190                   | °C | VSENSE1,2,3 |
| T <sub>msd_hyst</sub>             | Thermal hysteresis for shutdown                    | -                                                                                                                                              | 5                |      | 10                    | °C | VSENSE1,2,3 |
| t <sub>msd_pre_an</sub>           | Thermal shutdown analog filter time                | Guaranteed by design                                                                                                                           | 1.5              |      | 4                     | μs | VSENSE1,2,3 |
| t <sub>msd_pre_dig</sub>          | Thermal shutdown digital filter time               | Implemented in VHDL                                                                                                                            | 10               | 20   | 30                    | μs | VSENSE1,2,3 |

1. No oscillation expected with 0mA load current.
2. line and load regulation are guaranteed until the loop is able to regulate after they are defined by low drop parameters;  $ITH\_VS < IMAX\_VS$  which is guaranteed by design; Over shoot during power up should not be triggered over voltage warning. When Vtrk output is higher than VSCB\_TH\_VS, The Vtrk output structure will be switched off automatically in order to block the big inrush current into chip through Vtrk circuit. When Vtrk output is higher than VOV\_VS, an over voltage flag will be set. UV/OV fault detection during power up sequence is masked until RSTN release to avoid false triggering at ramp-up phase.
3. No UV/OV violation is expected during specified transient.
4. Specification valid until Tracking regulator is in regulation range, or, in other words, until Vsense X output voltage is determined by the effect of voltage regulation loop and not by secondary effects on other regulators.

**Table 6-6 VT rack external components characteristics**

| Symbol   | Parameter                 | Min           | Typ  | Max | Unit | Note                                                             | Pin             |
|----------|---------------------------|---------------|------|-----|------|------------------------------------------------------------------|-----------------|
| Cvs1/2/3 | VSENSE external capacitor | 470nF<br>-20% | 0.47 | 20  | μF   | ESR_max<20 mΩ<br>Suggest part numbers:<br>GCM188R71C105KA4<br>9D | VSENSE<br>1/2/3 |

## 6.4 External tracking regulator monitor

Vsense4\_mon is an input pin for monitoring an external tracking regulator output voltage.

### Conditions:

5.5 V < VB\_IN\_SW ≤ 18 V, T<sub>j</sub> -40 to 175 °C unless otherwise specified.

**Table 6-7 External VT rack monitor**

| Symbol                                   | Parameter                           | Test condition       | Min  | Typ  | Max  | Unit | Pin          |
|------------------------------------------|-------------------------------------|----------------------|------|------|------|------|--------------|
| V <sub>UV_VS_L</sub>                     | VSENSE4 undervoltage threshold low  | -                    | 4.55 | 4.65 | 4.75 | V    | VSENSE 4_MON |
| V <sub>UV_VS_H</sub>                     | VSENSE4 undervoltage threshold high | -                    | 4.7  | 4.8  | 4.95 | V    | VSENSE 4_MON |
| V <sub>OV_VS</sub>                       | VSENSE4 overvoltage threshold       | -                    | 5.3  | 5.4  | 5.55 | V    | VSENSE 4_MON |
| T <sub>UV_filter_VS_running</sub><br>(1) | Undervoltage filter time            | Covered by SCAN TEST | 16   | 24   | 32   | μs   | VSENSE 4_MON |
| T <sub>OV_filter_VS_running</sub><br>(1) | Overvoltage filter time             | Covered by SCAN TEST | 16   | 24   | 32   | μs   | VSENSE 4_MON |

1. Fault detection during power up sequence is masked until RSTN release to avoid false triggering at ramp-up phase.

## 6.5 VDD5 linear regulator with external MOSFET

VDD5 voltage is generated by a linear controller with internal pre-driver and external MOSFET.

Electrical characteristics are available in [Table 6-8](#) and [Table 6-9](#). VB\_IN\_OV fault can trigger the turn OFF of VDD5 regulator, as specified into reset matrix. This effect can be masked with bit *VB\_IN\_OV\_RST\_EN* in CONFIG\_REG 12.

VB\_IN\_UV fault triggers the turn OFF of VDD5 regulator, as specified into reset matrix. This effect can be masked with bit *VDD5\_OFF\_SEL* in CONFIG\_REG 11.

VDD5\_OV fault has effects into the device specified into reset matrix. These effects can be masked with bit *VDD5\_OV\_RST\_EN* in CONFIG\_REG 12.

### Conditions:

5.5 V < VB\_IN\_SW ≤ 18 V, 5.5 V < VB\_IN ≤ 18 V, T<sub>j</sub> -40 to 175°C unless otherwise specified.

**Table 6-8 VDD5 linear controller pre-driver electrical characteristics**

| Symbol                      | Parameter                                                     | Test condition                                                                                                                      | Min | Typ | Max | Unit | Pin     |
|-----------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|------|---------|
| V <sub>VDD5</sub>           | 5 V regulated voltage including static load / line regulation | I <sub>load</sub> = 10 mA, 1.5 A                                                                                                    | 4.9 | 5   | 5.1 | V    | VDD5_IN |
| SR <sub>p-up_VDD5</sub>     | Output voltage slew rate at power-up                          | Cvdd5 = 10 μF; from VDD5*10% to VDD5*90%                                                                                            | 2   | -   | 25  | V/ms | VDD5_IN |
| I <sub>VDD5</sub>           | Controller stability over VDD5 output current range           | -                                                                                                                                   | 1.5 | -   | -   | A    | VDD5_IN |
| V <sub>line_VDD5</sub>      | Line regulation voltage                                       | 5.6 V < VPre < 18 V<br>I <sub>load</sub> = 10mA, 1.5A<br>Cvdd5 = 20 μF (1)                                                          | -25 | -   | 25  | mV   | VDD5_IN |
| V <sub>load_VDD5</sub>      | Load regulation voltage                                       | VPre 5.6 V, 18 V<br>10 mA < I <sub>load</sub> < 1.5 A<br>Cvdd5 = 20 μF (1)                                                          | -25 | -   | 25  | mV   | VDD5_IN |
| V <sub>line_VDD5_tran</sub> | Line transient regulation voltage                             | Vpre from 5.6 V to 7 V by 1 μs<br>I <sub>load</sub> = 10 mA, 1200 mA<br>Cvdd5 = 20μF Ext MOS STD20NF06L<br>Guaranteed by design (1) | -   | -   | 300 | mV   | VDD5_IN |
| V <sub>load_VDD5_tran</sub> | Load transient regulation voltage                             | VPre = 6 V I <sub>load</sub> from 10 mA to 1200 mA by 1.5μs<br>Cvdd5=20μF Ext MOS STD20NF06L<br>Guaranteed by design (1)            | -   | -   | 280 | mV   | VDD5_IN |

Multi-voltage regulator supplies

|                               |                                                   |                                                                                                                         |      |      |      |               |               |
|-------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------|------|------|---------------|---------------|
| $V_{\text{drift\_Vdd5}}$      | Total VDD5 voltage aging drift                    | -                                                                                                                       | -0.8 | -    | 0.8  | %             | VDD5_IN       |
| $\text{PSRR}_{\text{VDD5}}$   | Supply voltage rejection ratio                    | $C_{\text{out}} = 10 \mu\text{F}$ , $\text{ESR} = 0.1\Omega$<br>$I_{\text{load}} = 600 \text{ mA}$ $f = 10 \text{ kHz}$ | -    | -    | -35  | dB            | VDD5_IN       |
| $V_{\text{OS\_VDD5}}$         | VDD5 voltage with overshoot admitted at switch on | -                                                                                                                       | 0    | 0.1  | 0.2  | V             | VDD5_IN       |
| $T_{\text{UV\_filter\_VDD5}}$ | Undervoltage filter time(1)                       | -                                                                                                                       | 16   | 24   | 32   | $\mu\text{s}$ | VDD5_IN       |
| $T_{\text{OV\_filter\_VDD5}}$ | Overvoltage filter time(1)                        | -                                                                                                                       | 16   | 24   | 32   | $\mu\text{s}$ | VDD5_IN       |
| $V_{\text{UV\_VDD5\_L}}$      | VDD5 undervoltage hysteresis threshold low        | -                                                                                                                       | 4.5  | 4.65 | 4.75 | V             | VDD5_IN       |
| $V_{\text{UV\_VDD5\_H}}$      | VDD5 undervoltage hysteresis threshold high       | -                                                                                                                       | 4.7  | 4.8  | 4.9  | V             | VDD5_IN       |
| $V_{\text{OV\_VDD5}}$         | VDD5 overvoltage threshold                        | -                                                                                                                       | 5.25 | 5.4  | 5.5  | V             | VDD5_IN       |
| $V_{\text{VDD5\_GATE\_Min}}$  | Pre-driver gate voltage                           | $V_{\text{Pre}} = 4.8 \text{ V}$ to force regulator in low drop condition                                               | 8    | -    | -    | V             | VDD5_G<br>ATE |
| $V_{\text{VDD5\_GATE\_MAX}}$  | Pre-driver gate voltage                           | Force 5 mA into VDD5_GATE                                                                                               | -    | -    | 15   | V             | VDD5_G<br>ATE |
| $V_{\text{GS OFF condition}}$ | VDD5_GATEVDD5_IN                                  | Device off                                                                                                              | -    | -    | 0.5  | V             | VDD5_G<br>ATE |

1. line and load regulation are guaranteed until the loop is able to regulate after they are defined by low drop parameters; Over shoot during power up should not be triggered over voltage warning. UV/OV fault detection during power up sequence is masked until RSTN release to avoid false triggering at ramp-up phase. With external MOS different from STD20NF06L regulated voltage could violate UV threshold but for a time shorter than digital deglitch, so it would not be able to trigger UV/OV fault. With STD20NF06L MOS no UV/OV violation is expected on this and other regulators.

Table 6-9 Linear controller pre-driver. External components characteristics

| Symbol              | Parameter                         | Note                                                                                        | Min | Typ | Max  | Unit             | Pin                      |
|---------------------|-----------------------------------|---------------------------------------------------------------------------------------------|-----|-----|------|------------------|--------------------------|
| $C_{\text{VDD5}}$   | External VDD5 capacitor           | 20 $\mu\text{F}$ to guarantee $\text{ESR} = 50 \text{ m}\Omega$ for load transient response | 5   | 30  | 60   | $\mu\text{F}$    | VDD5_IN                  |
| -                   | External n-MOS                    | -                                                                                           | -   | -   | -    | -                | VDD5_IN<br>VDD5_G<br>ATE |
| -                   | $R_{\text{gate\_source}}$         | -                                                                                           | -   | 1   | -    | $\text{M}\Omega$ | -                        |
| $g_{\text{fs}}$     | Forward Transconductance          | $V_{\text{DS}} = 25 \text{ V}$ , $I_{\text{D}} = 10 \text{ A}$                              | 4.5 | -   | -    | S                | -                        |
| $V_{\text{GS(th)}}$ | Gate Threshold Voltage            | $V_{\text{DS}} = V_{\text{GS}}$ , $I_{\text{D}} = 250 \mu\text{A}$                          | 2.0 | -   | 4.0  | V                | -                        |
| $C_{\text{iss}}$    | input capacitance of external FET | -                                                                                           | -   | 370 | 1020 | pF               | VDD5_G<br>ATE            |

## 6.6 VDD\_IO

VDD\_IO pin is the external supply pin for internal I/O circuit.

**Table 6-10 VDD\_IO electrical characteristics**

| Symbol                   | Parameter                                | Test condition                              | Min | Typ | Max | Unit | Pin    |
|--------------------------|------------------------------------------|---------------------------------------------|-----|-----|-----|------|--------|
| I <sub>VDD_IO</sub>      | Input current of VDD_IO                  | VB=14V, all VDD_IO related pin without load | 2   | 4.8 | 8   | mA   | VDD_IO |
| V <sub>VDD_IO</sub>      | Operation range                          | -                                           | 3.1 | -   | 5.5 | V    | VDD_IO |
| VDDIO_UV_LOW             | Under voltage threshold on pin VDDIO     | -                                           | 2.9 | 3.0 | 3.1 | V    | VDD_IO |
| T <sub>f_VDD_IO_UV</sub> | Under voltage filter time <sup>(1)</sup> | Tested by scan                              | 3   | -   | 10  | μs   | VDD_IO |

1. Fault detection during power up sequence is masked until RSTN release to avoid false triggering at ramp-up phase.

## 6.7 VB\_IN and VB\_IN\_SW SUPPLY

VB\_IN pin is the power supply pin for internal circuit.

During cranking phase even the VB\_IN is = 4.25 V the chip can keep the status, the internal POR is not triggered and also the internal supply under voltage is not triggered (internal 5V regulator undervoltage), the PHOLD timer value is also kept.

**Table 6-11 VB\_IN electrical characteristics**

| Symbol                 | Parameter                                                    | Test condition                                      | Min  | Typ | Max  | Unit    | Pin          |
|------------------------|--------------------------------------------------------------|-----------------------------------------------------|------|-----|------|---------|--------------|
| $I_{VB\_IN}$           | VB_IN current in functional mode                             | -                                                   | -    | -   | 80   | mA      | VB_IN        |
| $I_{VB\_IN\_STBY}$     | VB_IN current in OFF state.                                  | -                                                   | -    | -   | 13   | $\mu$ A | VB_IN        |
| $I_{VB\_IN\_SW}$       | VB_IN_SW current in functional mode                          | VB_IN=VB_IN_S<br>W=13.5V, $V_{pre}$<br>load = 500mA | 200  | -   | 350  | mA      | VB_IN_S<br>W |
| $V_{OFF\_VB\_IN}$      | Min VB_IN voltage to keep the MRD, start relay status        | -                                                   | -    | -   | 4.25 | V       | VB_IN        |
| $V_{UV\_VB\_IN\_L}$    | VB_IN undervoltage hysteresis threshold low                  | -                                                   | 4.7  | 4.8 | 4.9  | V       | VB_IN        |
| $V_{UV\_VB\_H}$        | VB_IN undervoltage hysteresis threshold high                 | -                                                   | 4.75 | 4.9 | 5.05 | V       | VB_IN        |
| $V_{UV\_VB\_IN\_HYST}$ | -                                                            | -                                                   | -    | 0.1 | -    | V       | VB_IN        |
| $V_{OV\_VB\_IN\_L}$    | VB_IN over voltage threshold low                             | -                                                   | 28   | -   | -    | V       | VB_IN        |
| $V_{OV\_VB\_IN\_H}$    | VB_IN over voltage threshold high                            | -                                                   | -    | -   | 32   | V       | VB_IN        |
| $V_{OV\_VB\_IN\_HYST}$ | -                                                            | -                                                   | -    | 2   | -    | V       | VB_IN        |
| $t_{VBOV1}$            | Filter time1 for VB_IN overvoltage(1)                        | Tested by scan                                      | 70   | 85  | 100  | $\mu$ s | VB_IN        |
| $t_{VBOV2}$            | Filter time2 for VB_IN overvoltage(1)                        | Tested by scan                                      | 11   | 15  | 19   | ms      | VB_IN        |
| $t_{VBUV1}$            | Filter time 1 for VB_IN undervoltage $V_{UV\_VB\_IN\_L}$ (1) | Tested by scan                                      | 5    | -   | 15   | $\mu$ s | VB_IN        |
| $t_{VBUV2}$            | Filter time 2 for VB_IN undervoltage $V_{UV\_VB\_IN\_H}$ (1) | Tested by scan                                      | 5    | -   | 15   | $\mu$ s | VB_IN        |
| $V_{UV\_VB\_CP\_ON}$   | VB_IN threshold for main                                     | Charge Pump ON                                      | 3.75 | 4   | 4.15 | V       | VB_IN        |

|                           |                                            |                 |     |     |     |   |       |
|---------------------------|--------------------------------------------|-----------------|-----|-----|-----|---|-------|
|                           | logic reset                                |                 |     |     |     |   |       |
| V <sub>UV_VB_CP_OFF</sub> | VB_IN<br>threshold for main<br>logic reset | Charge Pump OFF | 5.4 | 5.5 | 5.7 | V | VB_IN |

1. Fault detection during power up sequence is masked until RSTN release to avoid false triggering at ramp-up phase.

When VB\_IN is higher than 4.25 V (V<sub>off\_vb\_in</sub>), the starter relay and MRD can hold the state.

VB\_IN\_SW pin is the dedicated supply pin for pre-buck regulator.

*Note:* VB\_IN\_SW must be shorted to VB\_IN in PCB layout.

**Conditions:**

4.8 V ≤ VB\_IN ≤ 18 V; T<sub>j</sub> -40 to 175° C unless otherwise specified;

**Table 6-12 VB\_IN\_SW electrical characteristics**

| Symbol                     | Parameter                            | Test condition | Min | Typ | Max | Unit | Pin      |
|----------------------------|--------------------------------------|----------------|-----|-----|-----|------|----------|
| I <sub>VB_IN_sw_STBY</sub> | VB_IN_SW<br>current in OFF<br>state. | -              | -   | -   | 10  | μA   | VB_IN_SW |

## 6.8 Charge pump

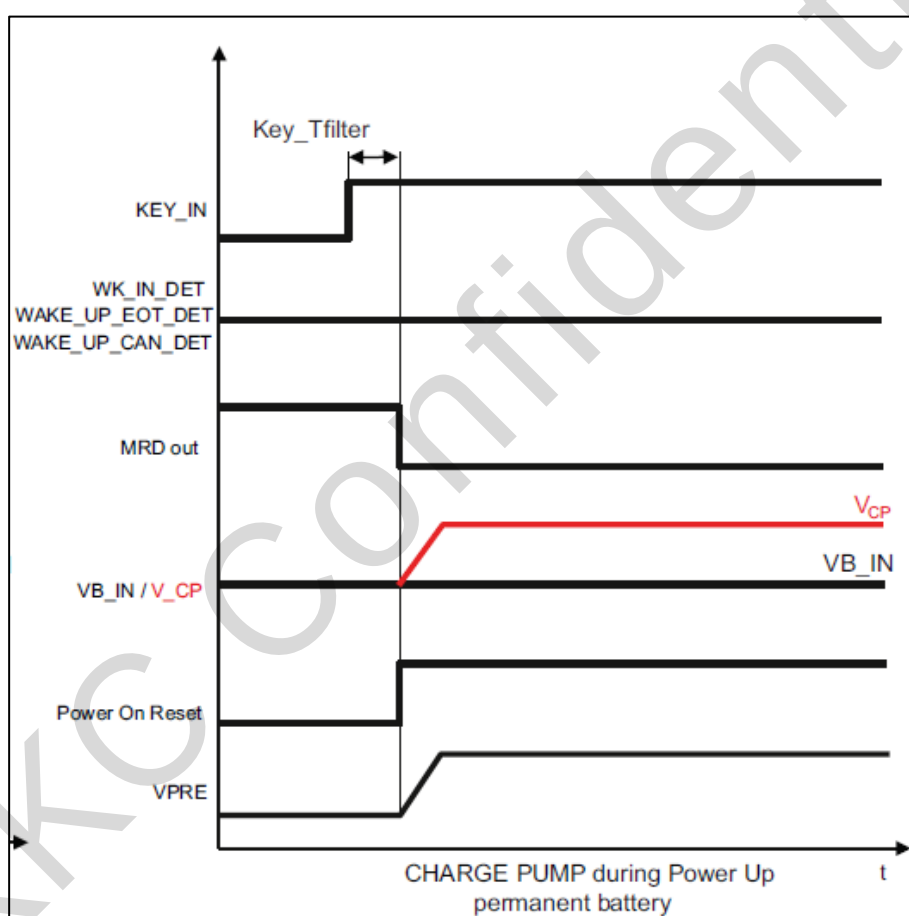
Charge pump provides a permanent voltage of at least 4.8 V above VB\_IN when VB\_IN is higher than 5.5 V, the VDD5 linear regulator, H-side switch and so on.

Once VB\_IN overvoltage is detected ( $VB\_IN > VOV\_VB\_IN\_H$  for  $t > tVBOV2$ ), the charge pump will be switched off automatically.

Charge pump can provide voltage of 4V above VB\_IN when VB\_IN in low power condition ( $4.25V < VB\_IN < 5.5V$ )

This charge pump is for internal use only.

*Note: It is forbidden to use this charge pump pin for external load.*



**Figure 6-5** Power up charge pump behavior

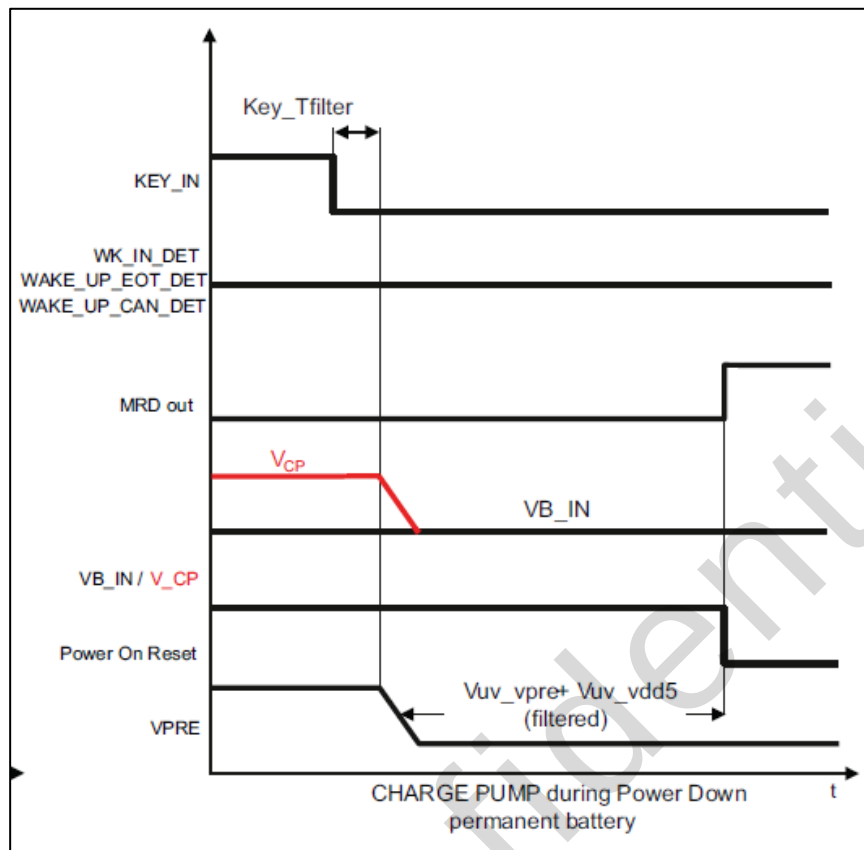


Figure 6-6 Power down charge pump behavior

**Conditions:**

$4.8\text{ V} \leq \text{VB\_IN} \leq 18\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified;

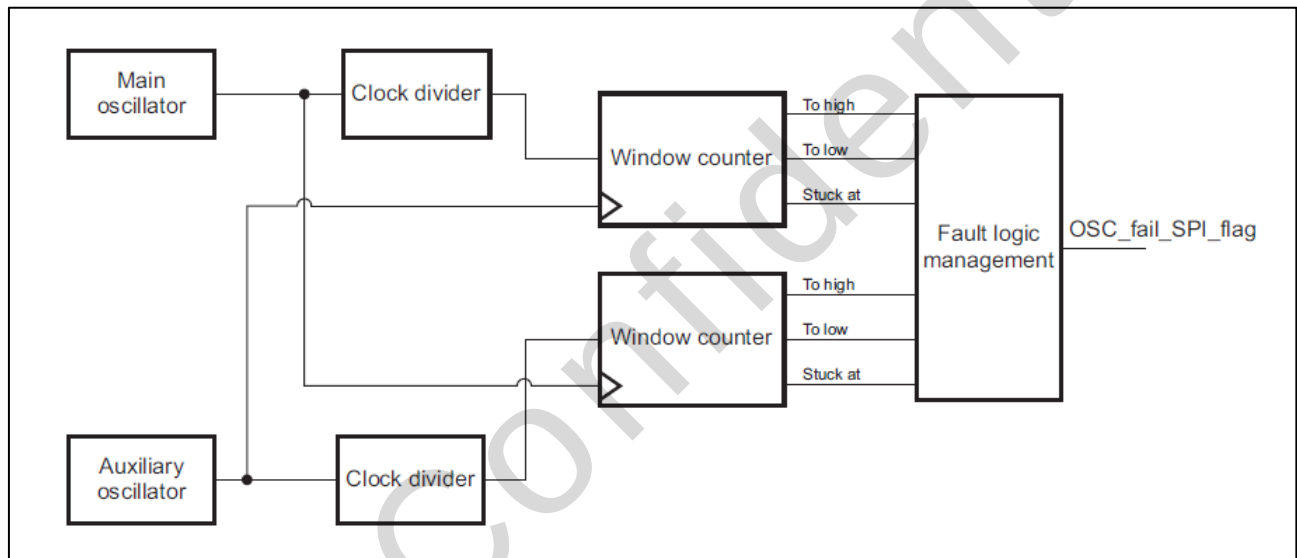
Table 6-13 Charge pump characteristics

| Symbol      | Parameter                      | Test condition                                                                                 | Min                | Typ | Max          | Unit | Pin |
|-------------|--------------------------------|------------------------------------------------------------------------------------------------|--------------------|-----|--------------|------|-----|
| CP          | External charge Pump capacitor | -                                                                                              | -20%               | 100 | +20%         | nF   | CP  |
| Vcp         | Output voltage                 | VB_IN = 5.5 V-18 V<br>Driver Starter 1 & 2 & 3<br>configured as HS and<br>switching            | VB_I<br>N+4.8<br>V | -   | VB_I<br>N+8V | V    | CP  |
|             |                                | VB_IN_OFF(4.25V)<V<br>B_IN<5.<br>5 V with 2 Driver Starter<br>configured as HS and<br>fully ON | VB_I<br>N+4V       | -   | VB_I<br>N+8V | V    | CP  |
| Icp_leakage | Leakage current in off state   | CP = VB_IN; chip off                                                                           | -3                 | -   | 3            | μA   | CP  |

## 6.9 Main oscillator

The device uses an internal oscillator, called MAIN Oscillator, to synchronize internal logic and internal charge pump. A second oscillator is used to check internally if the Main oscillator is working correctly. The Main Oscillator output is continuously verified by the Auxiliary Oscillator output. The two clocks frequency are measured and compared so that out of range frequency values can be detected. In case of frequency error greater than 20% a flag “*freq\_err*” (Upstream Read11) is set and read and cleared by MSC.

In case of frequency error greater than 30% another flag “*clk\_mon\_fault*” (Upstream Read11) is set and read and cleared by MSC, drivers are disabled when this error occurs. The monitor circuit is able also to detect a stuck condition in the reference clock, moreover the check is done on both the main and the auxiliary oscillator so that a failure in any of the two oscillators is able to set the fault flag.



**Figure 6-7 Clock monitor block diagram**

The Main and Auxiliary oscillators are kept independent using redundant circuits, biased by independent current sources, and layouted in different floor plan regions kept isolated thanks to deep trench usage. The only common point between the oscillators is the digital supply that is however monitored by the independent mechanisms.

Aux oscillator is used just for safety purpose and connected to a small digital portion (no impact on EMC performance). The main oscillator has spread spectrum solution applied by default to reduce emissions; it means that oscillator frequency has an average value equal to MainOSCIL with a triangular modulation applied, so final frequency is  $\text{MainOSCIL} \pm 3.5\%$  and a triangular variation of about 111 kHz frequency.

### Conditions:

$4.8 \leq V_{B\_IN} \leq 18V$ ,  $T_j -40$  to  $175\text{ }^{\circ}\text{C}$  unless otherwise specified;

**Table 6-14 Clock monitor characteristics**

| Symbol    | Parameter                               | Test condition                                                        | Min   | Typ | Max   | Unit | Pin |
|-----------|-----------------------------------------|-----------------------------------------------------------------------|-------|-----|-------|------|-----|
| MainOSCIL | Internal MAIN oscillator frequency      | VB_IN = 4.8 V-18 V                                                    | 14.25 | 15  | 15.75 | MHz  | -   |
| MainOSCIL | Internal redundant oscillator frequency | VB_IN = 4.8 V-18 V                                                    | -     | 15  | -     | MHz  | -   |
| -         | Delta 1 Oscillator CHECK                | VB_IN = 4.8 V-18 V<br>"freq_err= 1"                                   | 20    | -   | -     | %    | -   |
| -         | Delta 2 Oscillator CHECK                | VB_IN=4.8V-18V<br>"FREQ_ERR=1"&<br>"clk_mon_fault=1"(drive<br>rs OFF) | 30    | -   | -     | %    | -   |

## 7 Low-side drivers

Low-side drivers are used for power stages driving and they include:

- 4-channel Low-side Injectors INJ[1:4];
- 2-channel Low-side O2 Heaters O2H[1:2];
- 2-channel Low-side Solenoids SOL[1:2];
- 5-channel Low-side Relays RLY[1:5];
- 2-channel Low-side LED[1:2].

All drivers support the same failure diagnosis.

### 7.1 Diagnosis

The driver has the following fault diagnosis:

Overcurrent (Short To Vbat) protection in On Phase (OVC);

Open Load in Off phase (DIAGOL);

Short to Ground in Off Phase (DIAGLV);

Over Temperature Protection in On phase (OT).

When an overcurrent fault is detected the driver switches off with higher slew rate (FASTSR) to reduce the power dissipation.

#### 7.1.1 ON state - overcurrent protection

An overcurrent protection is present for LS transistors of each driver channel. Once the overcurrent fault is detected the faulted power MOS is switched off and a fault bit is set. This bit can be reset by every Read Diag Communication to re-activate the power stage in this two condition:

- The MSC command still High, when the Fault OVC Bit is read by MSC communication.
- The MSC command becomes Low and then again High.

To increase EMC robustness and avoid unwanted fault triggering a small deglitch filter is applied.

In order to avoid false OVC detection during the turn on slew rate phase, an analog blanking time filtering strategy is implemented. This means that the OVC comparator output is masked until the driver gate voltage has reached a threshold value which guarantees that the slew rate phase is ended.

#### 7.1.2 ON state - thermal protection

To protect power stages from temperature overheat (high battery range of operation, soft short conditions, etc.) a dedicated thermal sensor placed close to power MOS is present in the layout sensing FET temperature; One threshold is implemented (T\_SD\_H) with Hysteresis (T\_SD\_hys). When the T\_SD\_H threshold is detected the Driver switches OFF.

Low-side drivers

Once the power stage has been switched-off for over-temperature detection, it will be able to switch on again when temperature is decreased below thermal shut down threshold plus hysteresis value to avoid high frequency on-off cycling.

### 7.1.3 ON/OFF state - Error in on status diagnosis

To avoid unwanted driving of power stages the consistency of gate driving signal with digital command is always monitored inside the device. Each time a turn-on/turn-off signal is output by the logic core its effectiveness is verified comparing it with the status of the output voltage.

The main logic can check the consistence between inner logic and actual signal at output pin reading the OpenLoad comparator or Short-To-Ground Comparator according to [Table 7-2](#).

The driver status diagnosis (STA) is activated by the *MSC on command*, the relative comparator is filtered for Tsta filter time to validate the diagnosis. In case of MOS pre-driver the selected blanking time Tblank is applied before filtering.

**Table 7-1 Driver status diagnosis**

| Symbol | Parameter                           | Test condition | Min | Typ  | Max | Unit | Pin |
|--------|-------------------------------------|----------------|-----|------|-----|------|-----|
| Tsta   | Driver status diagnosis filter time | Tested by scan | 30  | 32.5 | 35  | μs   | -   |

**Table 7-2 STA diagnosis for drivers**

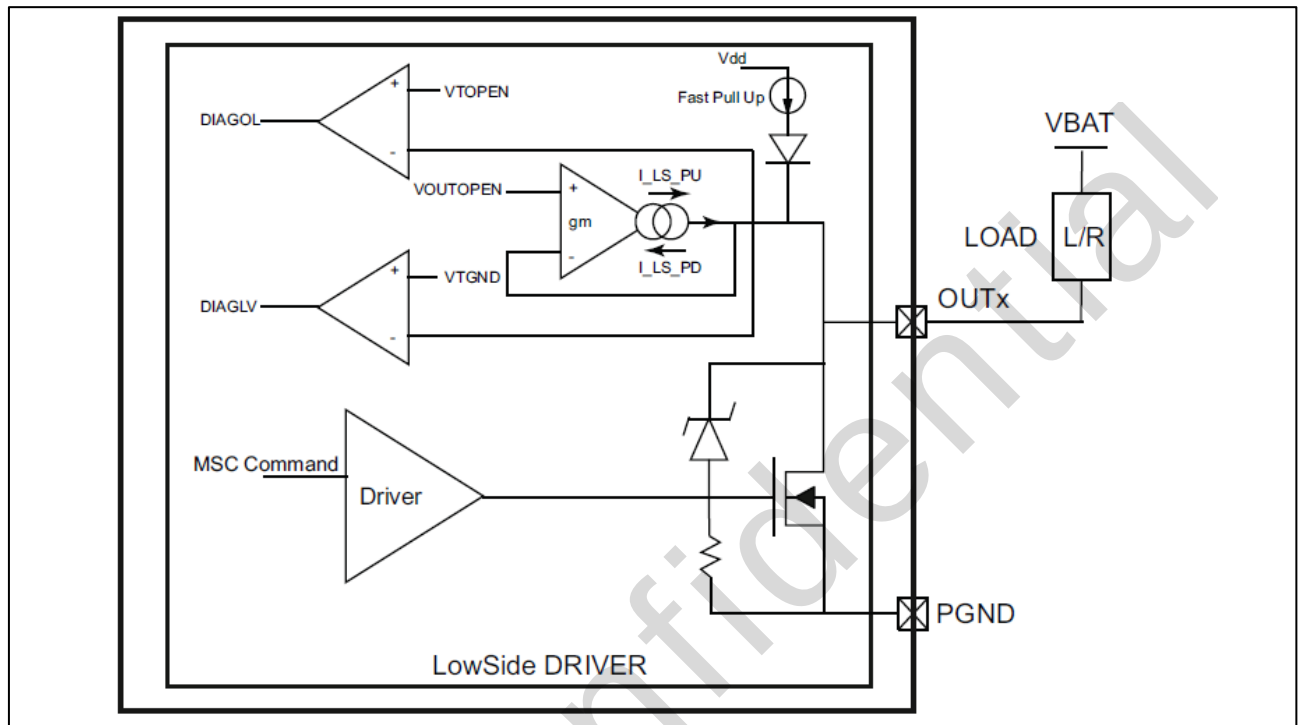
| STA diagnosis for drivers                           |                                                                                                                                                        |                                                                                                        |                                                                                                         |                              |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------|
| -                                                   | INJ,SOL,O2H, RLY,STR, in Low-side configuration                                                                                                        | STR in high-side configuration.                                                                        | MOS pre-driver                                                                                          | IGN                          |
| ON state                                            | Detected by STG comparator, if the output pin voltage is higher than VLVT, STA fault is detected                                                       | Detected by Open load comparator, if the output pin voltage is lower than V_OL, STA fault is detected. | Detected by STG comparator, if the MOS_DRN pin voltage is higher than VLVT, STA fault is detected.      | Same information as STG bit. |
| OFF state (for LS/HS/PDRV)<br>LS_ON state (for IGN) | Detected by Open load comparator or Short To Ground comparator. If the output pin voltage is lower than V_OL or lower than VLVT STA fault is detected. | Detected by STG comparator, if the output pin voltage is higher than VLVT, STA fault is detected.      | Detected by Open load comparator, if the MOS_DRN pin voltage is lower than V_OL, STA fault is detected. | Not able to detect.          |
| STA processing                                      | filtering                                                                                                                                              | filtering                                                                                              | blanking+filtering                                                                                      | filtering                    |

**Note:** In case of activation of OVT[17] the STA diagnosis is not available.

In case of activation of relative driver OVC the STA diagnosis is available.

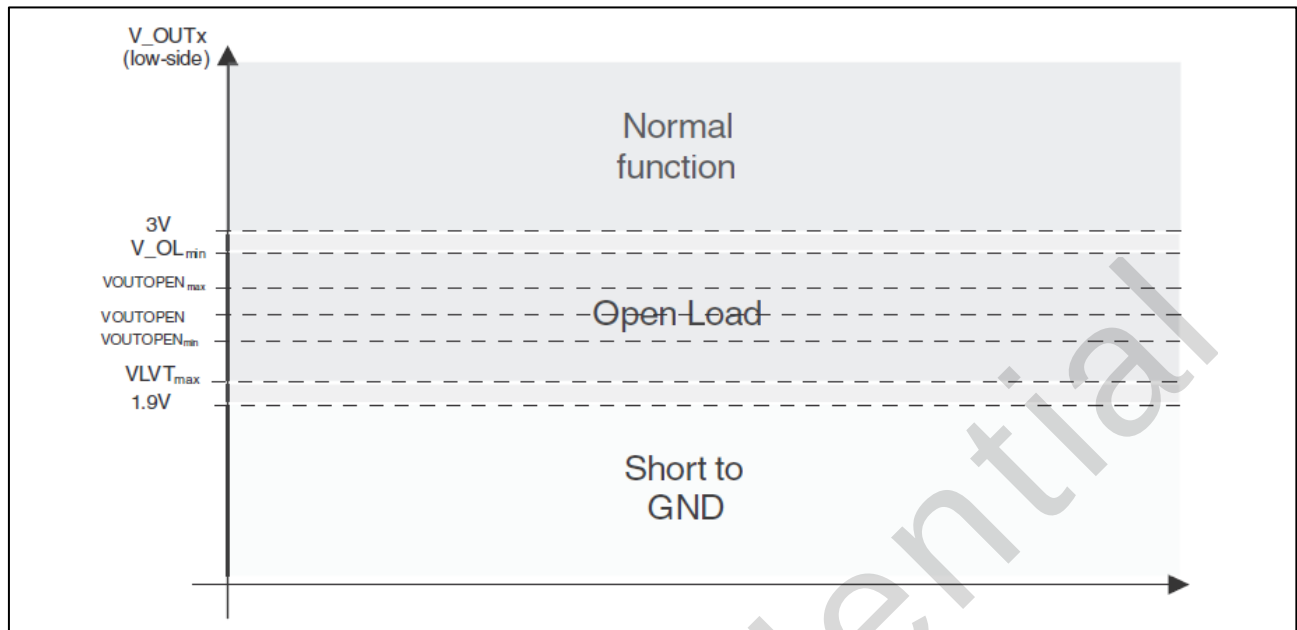
#### 7.1.4 OFF state - short load and open load

The device provides off-state diagnosis for each driver channel. Simplified schematic of implemented diagnosis is shown in [Figure 7-1](#).



### Figure 7-1 Low-side driver OFF state diagnosis schematic

In the Low-Side Driver we can have three different load conditions as shown in [Figure 7-2](#):



**Figure 7-2 Low-side driver OFF state diagnosis load thresholds**

Normal Load: it means output driver connected to the load, No Fault Present,  $V_{OUT} \geq V_{OL}$  Threshold

Open Load: It means output driver disconnected from the load, Open Load Fault Present,  $VLVT \leq V_{OUT} \leq V_{OL}$  Threshold

Short To Ground: it means output driver shorted to GND voltage, Short To Gnd Fault Present,  $V_{OUT} \leq VLVT$  Threshold There is a bit *IPUPD\_EN* in *CONFIG\_REG2* which is controlled by a MSC bit, that it is used

by external  $\mu$ Controller if it is needed to switch on/off both pull up and pull down diagnosis current.

The Diagnostic in Off state is done by the following blocks:

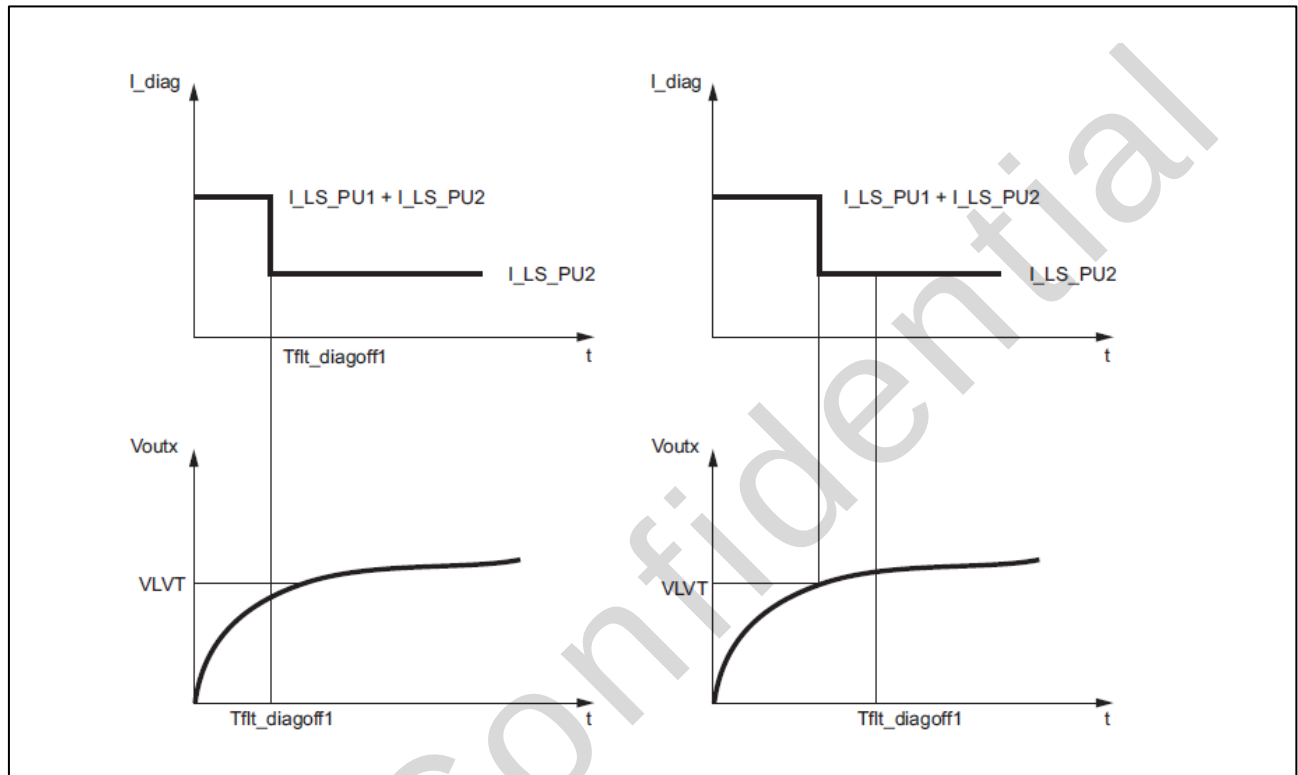
- An internal Buffer using the pull up/down currents is able to force the *VOUTOPEN* voltage on the Low-Side Driver Output when the driver is in Open Load Condition.
- A comparator to detect the Open load Condition. The Open Load fault is detected when the Low-Side output pin is  $VLVT \leq V_{OUT} \leq V_{OL}$  Threshold for a time longer than Diag Filter Time.
- A comparator to detect the Short to GND Condition. The Short to GND fault is detected when the Low-Side output pin is  $V_{OUT} \leq VLVT$  Threshold for a time longer than Diag Filter Time.

There is a bit *IPUPD\_MODE* in *CONFIG\_REG2* which is controlled by a MSC bit, it is used by external

$\mu$ Controller if it is needed to switch on/off the bigger diagnosis current (*I\_LS\_PU1*) in fast mode.

The Fast pull up current is used to avoid the false short to tGND diagnosis when the driver

Low-side drivers switches off in Open load condition. This current switches on after the falling edge of MSC command (driver in off state) and remains on until the Diag Filter Time is expired or until the VOUT is > VLVT Threshold. (note: if fast pull up current is selected, the filter time Tflt\_diagoff1 is suggested. Otherwise, the filter time Tflt\_diagoff2 is suggested)



**Figure 7-3 Low-side driver OFF state fast pull up current behavior**

Despite the fast mode of all drivers in off diagnostic condition (*lpupd\_MODE* bit), for Injector and Solenoid drivers there is a dedicated MSC bit (*IDIAG\_HIGH\_SOL/IDIAG\_HIGH\_INJ* in *CONFIG\_REG13*) for each kind of driver to select off diagnostic high or low pull up/down current

By default diagnostic pull up/down currents are disabled and comparator outputs are masked by internal logic, to enable OFF state diagnostic the channel must be put first in tristate condition and then (if not already done with previous MSC frames) diagnostic must be enabled.

A Filter Time (Diag Filter Time) is implemented in order to avoid to detect false diagnosis as shown in [Figure 7-4](#).

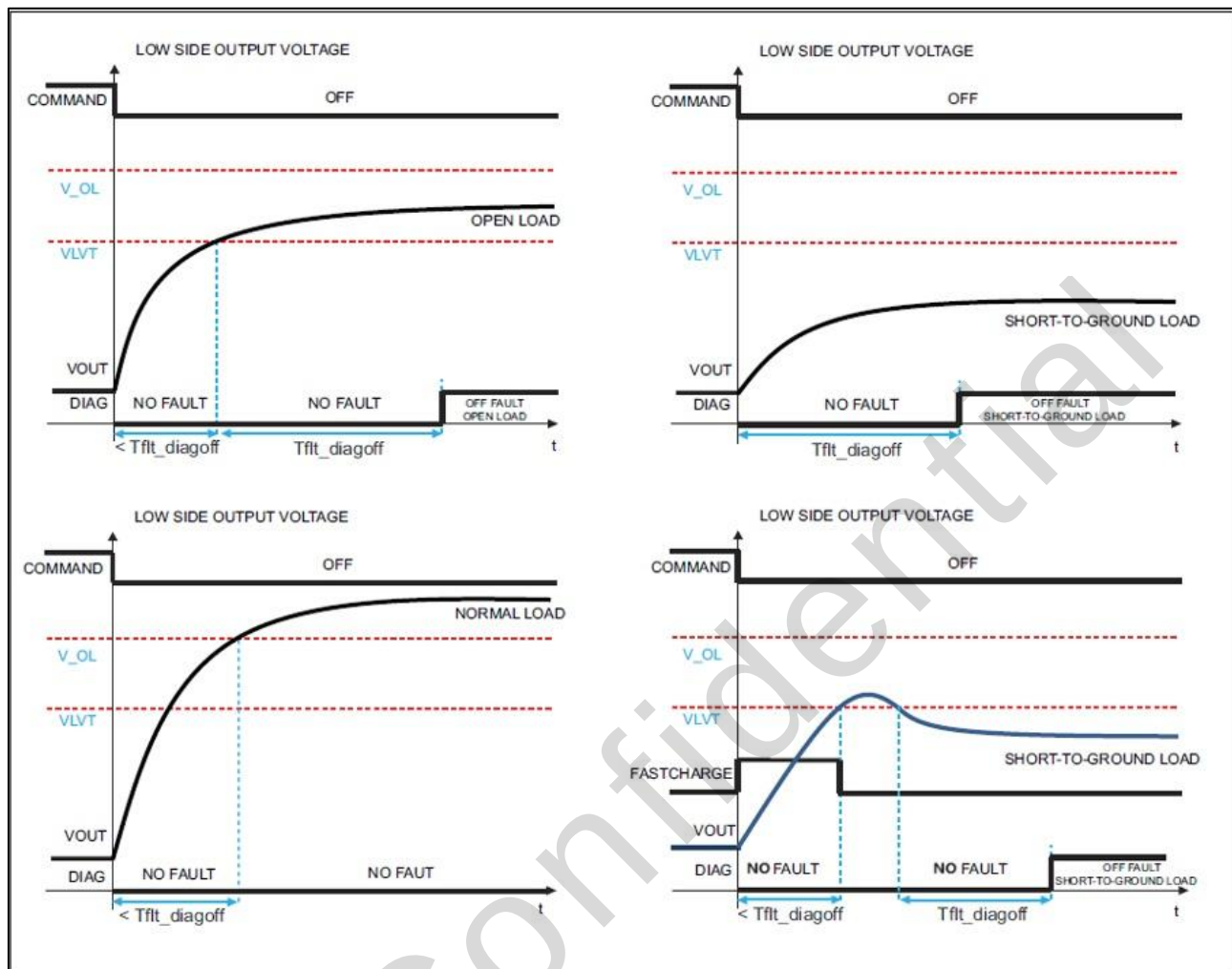


Figure 7-4 Low-side driver OFF state diagnosis timings

If DiagOL or DiagLV signals remain high for a time higher than Diag Filter Time the fault bit is setted and can be read by Read Diag Communication.

This bit is reset by every Read Diag Communication.

### 7.1.5 Error handling

Table 7-3 Low-side driver error handling

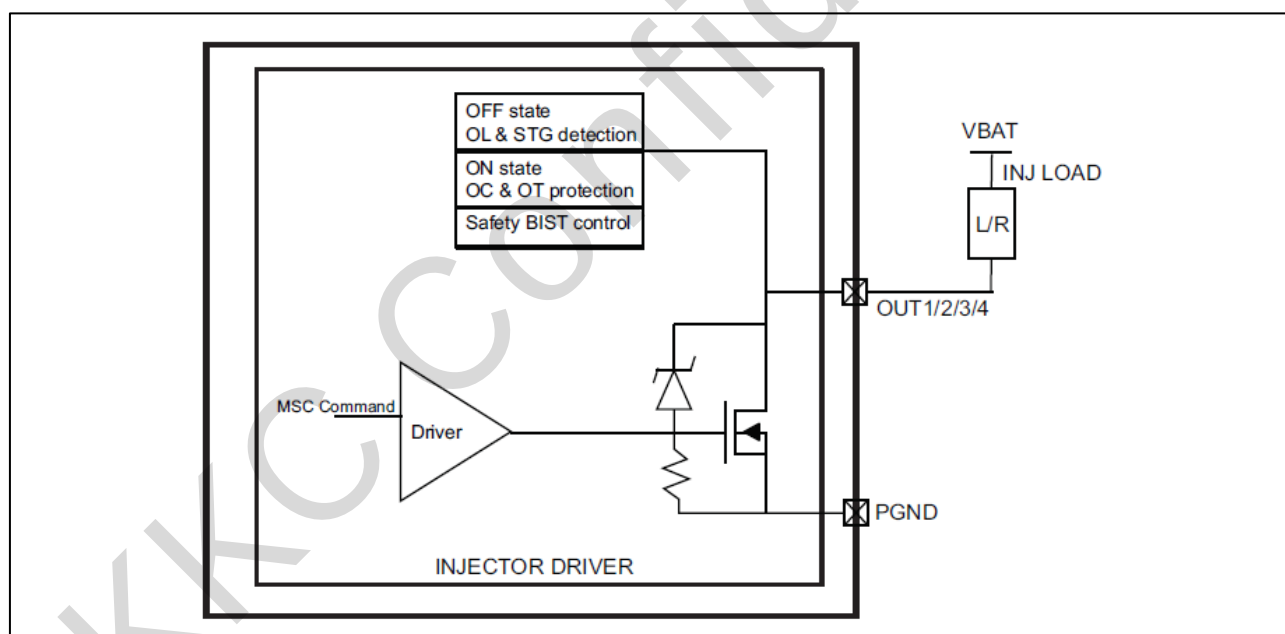
| Type of error            | Detection condition | Action                                                                      | Clear MSC flag | Restart condition                                                                                     |
|--------------------------|---------------------|-----------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------|
| <b>Driver Diagnostic</b> |                     |                                                                             |                |                                                                                                       |
| Overcurrent (OVC)        | Driver on           | Driver is put in off state. The fault is latched in MSC diagnosis register. | On read        | At MSC read of diagnosis register if MSC command is still high. If MSC command is low and high again. |
| Short to ground          | Driver off          | The fault is latched                                                        | On read        | -                                                                                                     |

Low-side drivers

|                                 |                                                                                      |                                                                             |         |                                                                                      |
|---------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------|
| (STG)                           |                                                                                      | in MSC diagnosis register.                                                  |         |                                                                                      |
| Open load (OPL)                 | Driver off                                                                           | The fault is latched in MSC diagnosis register.                             | On read | -                                                                                    |
| Driver Status Error (STA)       | Driver on/off<br>When the driver output level is not aligned with the command on/off | The fault is latched in MSC diagnosis register.                             | On read | -                                                                                    |
| Overttemperature (OVT) Shutdown | Driver on                                                                            | Driver is put in off state. The fault is latched in MSC diagnosis register. | On read | The driver restarts when the temperature decreases. The fault is cleared at MSC read |

## 7.2 Low-side driver - INJECTOR INJ[1:4]

These 4 low-side drivers are designed to driver Injector Load. They are driven by *MSC command*. The output voltage is clamped to voltage limit by internal clamp circuit.



**Figure 7-5 INJECTOR LowSide driver stage**

### Conditions:

$5.5\text{ V} \leq \text{VB\_IN} \leq 18\text{ V}$ ;  $T_j$  -40 to 175 °C unless otherwise specified;

Table 7-4 Low-side driver - INJECTOR electrical characteristics

| Symbol                | Parameter                                                             | Test condition                                                                                                                           | Min  | Typ  | Max  | Unit               | Pin        |
|-----------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|--------------------|------------|
| <b>INJECTOR</b>       |                                                                       |                                                                                                                                          |      |      |      |                    |            |
| LS_RdsON              | Low-side RdsON                                                        | $T_j = 150\text{ }^{\circ}\text{C}$ , $I_{\text{LOAD}} = 3\text{ A}$                                                                     | 0.3  | 0.44 | 0.6  | $\Omega$           | INJ1/2/3/4 |
| LS_RdsON              | -                                                                     | $T_j = 25\text{ }^{\circ}\text{C}$ , $I_{\text{LOAD}} = 3\text{ A}$                                                                      | 0.22 | 0.3  | 0.45 | $\Omega$           | INJ1/2/3/4 |
| LS_RdsON              | -                                                                     | $T_j = -40\text{ }^{\circ}\text{C}$ , $I_{\text{LOAD}} = 3\text{ A}$                                                                     | 0.15 | 0.2  | 0.3  | $\Omega$           | INJ1/2/3/4 |
| OUTx_ikg              | Output leakage current                                                | Output disabled, diagnostic off                                                                                                          | -10  | -    | 10   | $\mu\text{A}$      | INJ1/2/3/4 |
| Irev_OUTx             | Body Diode reverse current voltage drop                               | $I_{\text{LOAD}} = -2\text{ A}$                                                                                                          | -    | -    | 2    | V                  | INJ1/2/3/4 |
| SR_ON                 | Voltage slew ON State                                                 | From 80% to 30% of $V_{\text{OUT}}$ , $V_{\text{B\_IN}} = 14\text{ V}$ , $R_{\text{load}} = 15\Omega$ , $C_{\text{load}} = 10\text{ nF}$ | 0.6  | -    | 1.84 | V/ $\mu\text{s}$   | INJ1/2/3/4 |
| SR_OFF                | Voltage slew OFF State                                                | From 80% to 30% of $V_{\text{OUT}}$ , $V_{\text{B\_IN}} = 14\text{ V}$ , $R_{\text{load}} = 15\Omega$ , $C_{\text{load}} = 10\text{ nF}$ | 0.6  | -    | 1.84 | V/ $\mu\text{s}$   | INJ1/2/3/4 |
| S/RGkill_LSH          | FAST VS/R off when and OVC fault happens                              | From 30% to 80% of $V_{\text{OUT}}$ , $V_{\text{B\_IN}} = 14\text{ V}$ , $R_{\text{load}} = 15\Omega$ , $C_{\text{load}} = 10\text{ nF}$ | 5    | -    | 20   | V/ $\mu\text{s}$   | INJ1/2/3/4 |
| Ton_OUTx              | Propagation Delay from MSC_EN rising edge to 80% output OUTx voltage  | $V_{\text{B\_IN}} = 14\text{ V}$ , $R_{\text{load}} = 15\Omega$ , $C_{\text{load}} = 10\text{ nF}$                                       | -    | -    | 8    | $\mu\text{s}$      | INJ1/2/3/4 |
| Toff_OUTx             | Propagation Delay from MSC_EN falling edge to 20% output OUTx voltage | $V_{\text{B\_IN}} = 14\text{ V}$ , $R_{\text{load}} = 15\Omega$ , $C_{\text{load}} = 10\text{ nF}$                                       | -    | -    | 8    | $\mu\text{s}$      | INJ1/2/3/4 |
| Vclamp_OUTx           | OUTx clamping Voltage                                                 | $I_{\text{LOAD}} = 1.3\text{ A}$                                                                                                         | 50   | 55   | 60   | V                  | INJ1/2/3/4 |
| Vclamp_OUTx_high_curr | OUTx clamping Voltage                                                 | $I_{\text{LOAD}} = 3\text{ A}$<br>Guaranteed by design, provided single pulse energy limits are not violated by clamping action          | 50   | 55   | 60   | V                  | INJ1/2/3/4 |
| I_OVC_INJ             | Over Current Driver Threshold                                         | -                                                                                                                                        | 3    | -    | 6    | A                  | INJ1/2/3/4 |
| I_LS_ocv_fit          | LS overcurrent filter time                                            | Guaranteed by scan                                                                                                                       | 4    | -    | 7    | $\mu\text{s}$      | INJ1/2/3/4 |
| T_SD_HIGH             | Temperature shut down                                                 | -                                                                                                                                        | 185  | -    | 200  | $^{\circ}\text{C}$ | INJ1/2/3/4 |
| T_SD_LOW              | Temperature shut down recover                                         | -                                                                                                                                        | 175  | -    | 190  | $^{\circ}\text{C}$ | INJ1/2/3/4 |
| T_SD_hys              | Temperature shut down hysteresis                                      | -                                                                                                                                        | 5    | -    | 10   | $^{\circ}\text{C}$ | INJ1/2/3/4 |
| tmsd_pre_an           | Thermal shutdown analog filter time                                   | Guaranteed by design                                                                                                                     | 1.5  | -    | 4    | $\mu\text{s}$      | INJ1/2/3/4 |
| t_SD_deglitch         | Digital deglitch                                                      | Guaranteed by design                                                                                                                     | -    | 10   | -    | $\mu\text{s}$      | INJ1/2/3/4 |

## Low-side drivers

|                                                 |                                                      |                                                                                                                                          |                                        |     |                                        |    |                |
|-------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-----|----------------------------------------|----|----------------|
|                                                 | filter time on<br>Temperature shut<br>down detection |                                                                                                                                          |                                        |     |                                        |    | 4              |
| <b>Driver Reliability Data</b>                  |                                                      |                                                                                                                                          |                                        |     |                                        |    |                |
| EnergyRep INJ                                   | Energy repetitive<br>pulse                           | T <sub>case</sub> = 30 °C;<br>I <sub>OUT_n</sub> = 1.8 A,<br>18 Miopulses                                                                | -                                      | -   | 7.5                                    | mJ | INJ1/2/3/<br>4 |
| EnergyRep INJ                                   | Energy repetitive<br>pulse                           | T <sub>case</sub> = 115 °C;<br>I <sub>OUT_n</sub> = 1.4 A<br>648 Miopulses                                                               | -                                      | -   | 4                                      | mJ | INJ1/2/3/<br>4 |
| EnergyRep INJ                                   | Energy repetitive<br>pulse                           | T <sub>case</sub> = 130 °C;<br>I <sub>OUT_n</sub> = 1 A<br>96 Miopulses                                                                  | -                                      | -   | 3                                      | mJ | INJ1/2/3/<br>4 |
| EnergyRep INJ                                   | Energy repetitive<br>pulse                           | T <sub>case</sub> = 140 °C;<br>I <sub>OUT_n</sub> = 1 A<br>4 Miopulses                                                                   | -                                      | -   | 3                                      | mJ | INJ1/2/3/<br>4 |
| <b>Driver Reliability Data Generator Defect</b> |                                                      |                                                                                                                                          |                                        |     |                                        |    |                |
| EnergyRep INJ                                   | Energy repetitive<br>pulse                           | T <sub>case</sub> = 25 °C ;<br>I <sub>OUT_n</sub> = 2 A<br>0.5 Miopulses                                                                 | -                                      | -   | 9                                      | mJ | INJ1/2/3/<br>4 |
| EnergyRep INJ                                   | Energy repetitive<br>pulse                           | T <sub>case</sub> = 135 °C;<br>I <sub>OUT_n</sub> = 1.5 A<br>0.5 Miopulses                                                               | -                                      | -   | 8                                      | mJ | INJ1/2/3/<br>4 |
| <b>Driver Reliability Data jump start</b>       |                                                      |                                                                                                                                          |                                        |     |                                        |    |                |
| EnergyRep INJ                                   | Energy repetitive<br>pulse                           | T <sub>case</sub> = 25 °C;<br>I <sub>OT_n</sub> = 3.0 A<br>MAX.0.021Mio cycles<br>10 jumps starts over<br>lifetime, each start <<br>2min | -                                      | -   | 17.5                                   | mJ | INJ1/2/3/<br>4 |
| EnergyRep INJ                                   | Energy repetitive<br>pulse                           | T <sub>c</sub> = 75 °C ;<br>I <sub>OT_n</sub> = 2.3A<br>MAX.0.021Mio cycles<br>10jumps starts over<br>lifetime,each start<2min           | -                                      | -   | 10                                     | mJ | INJ1/2/3/<br>4 |
| <b>OFF state diagnostic</b>                     |                                                      |                                                                                                                                          |                                        |     |                                        |    |                |
| VLVT                                            | Short to GND<br>threshold voltage                    | Driver tristate, diag<br>enabled                                                                                                         | 1.9                                    | -   | V <sub>OUT</sub><br>OPEN<br>-180<br>mV | V  | INJ1/2/3/<br>4 |
| V_OL                                            | Open load<br>threshold voltage                       | Driver tristate, diag<br>enabled                                                                                                         | V <sub>OUT</sub><br>OPEN<br>+160<br>mV | -   | 3                                      | V  | INJ1/2/3/<br>4 |
| VOUTOPEN                                        | Open load output<br>voltage                          | Driver tristate, diag<br>enabled                                                                                                         | 2.3                                    | 2.5 | 2.7                                    | V  | INJ1/2/3/<br>4 |
| I_LS_PU1                                        | Diagnostic pull up<br>current                        | In OFF condition, OUTx<br>< VLVT, Ipupd_EN<br>= ENB,<br>Ipupd_MODE = "1"                                                                 | 2.5                                    | 3.6 | 4.7                                    | mA | INJ1/2/3/<br>4 |
| I_LS_PU2                                        | Diagnostic pull up<br>low current                    | In OFF condition,<br>OUTx < VLVT,<br>Ipupd_EN = ENB,<br>Ipupd_MODE = "0" and<br>IDIAG_HIGH_INJ = "0"                                     | 40                                     | 70  | 100                                    | μA | INJ1/2/3/<br>4 |
| I_LS_PU3                                        | Diagnostic pull up<br>high current                   | In OFF condition,<br>OUTx < VLVT,<br>Ipupd_EN = ENB,<br>Ipupd_MODE = "0" and<br>IDIAG_HIGH_INJ = "1"                                     | 100                                    | -   | 200                                    | μA | INJ1/2/3/<br>4 |
| I_LS_PD1                                        | Diagnostic pull                                      | In OFF condition,                                                                                                                        | 60                                     | 85  | 110                                    | μA | INJ1/2/3/      |

Low-side drivers

|               |                                               |                                                                                 |     |     |     |    |                |
|---------------|-----------------------------------------------|---------------------------------------------------------------------------------|-----|-----|-----|----|----------------|
|               | down high current                             | OUTx > VOUTOPEN,<br>lpupd_EN = ENB<br>IDIAG_HIGH_INJ = "0"                      |     |     |     |    | 4              |
| I_LS_PD2      | Diagnostic pull<br>down high current          | In OFF condition,<br>OUTx > VOUTOPEN,<br>lpupd_EN = ENB<br>IDIAG_HIGH_INJ = "1" | 325 | -   | 550 | μA | INJ1/2/3/<br>4 |
| Tflt_diagoff1 | DIAG Filter time                              | Filter Mode = 0                                                                 | 75  | 100 | 125 | μs | INJ1/2/3/<br>4 |
| Tflt_diagoff2 | DIAG Filter time                              | Filter Mode = 1                                                                 | 450 | 600 | 750 | μs | INJ1/2/3/<br>4 |
| -             | Minimum OFF<br>time for correct<br>diagnostic | Application note (esd<br>cap < 10nF)<br>Filter Mode=0                           | 175 | -   | -   | μs | ESVx           |

### 7.2.1 Enable pin INJ\_ENA

The Pin INJ\_ENA is a digital input pin that is used as enable of Injector driver. If this pin is at low digital level the internal logic drives all injector drivers in OFF state. If this pin is at high digital, all injector drivers can be switched on by *dedicated MSC bit*. This pin can be used also as a SEO function for injector driver. If the application doesn't use this enable function, the INJ\_ENA pin have to be pulled up externally to VDD5 by a resistor of 10 kΩ

#### Conditions:

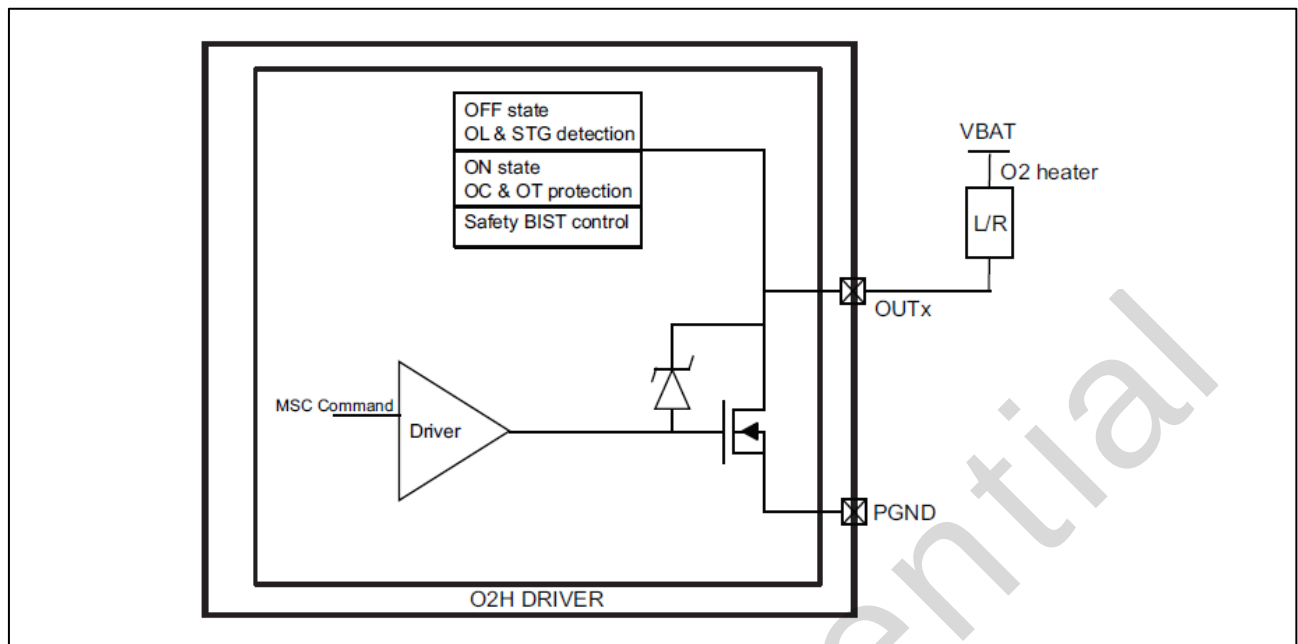
5.5 V ≤ VB\_IN ≤ 18 V, T<sub>j</sub> -40 to 175 °C unless otherwise specified.

**Table 7-5 INJ\_ENA electrical characteristics**

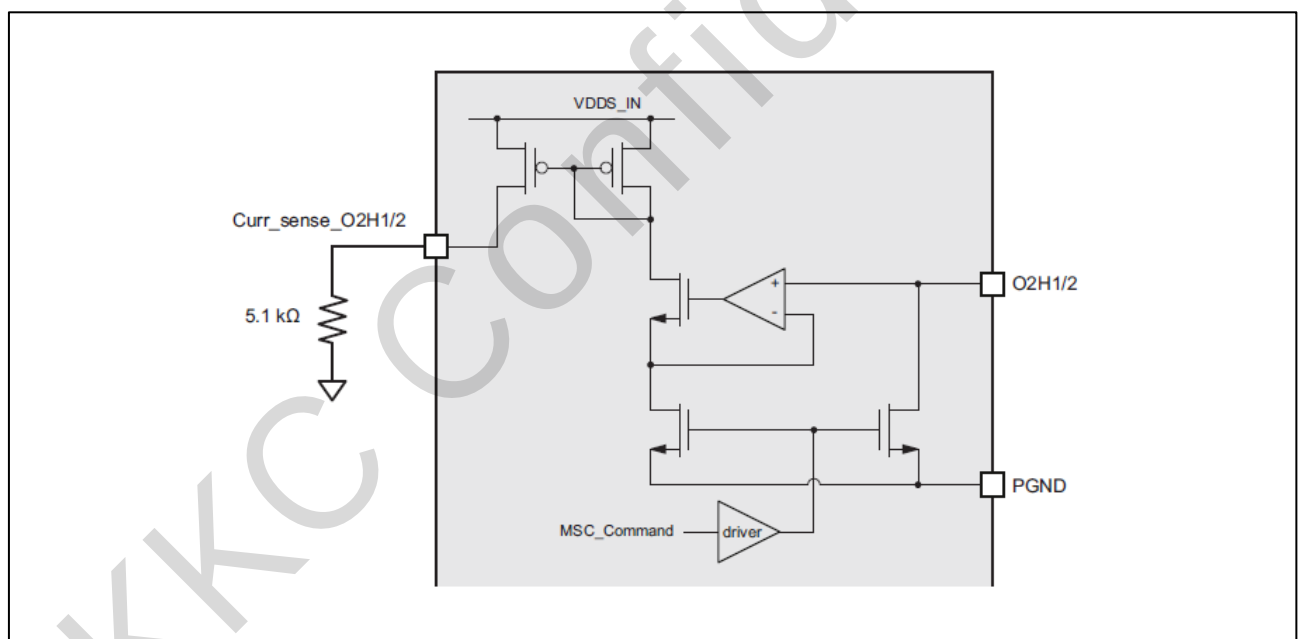
| Symbol   | Parameter                   | Test condition | Min  | Typ | Max  | Unit | Pin     |
|----------|-----------------------------|----------------|------|-----|------|------|---------|
| VIH_Tx   | HIGH_level input<br>voltage | -              | 1.75 | -   | -    | V    | INJ_ENA |
| VIL_Tx   | LOW_level input<br>voltage  | -              | -    | -   | 0.75 | V    | INJ_ENA |
| VHYS_Tx  | Input voltage<br>hysteresis | -              | 0.1  | 0.5 | -    | V    | INJ_ENA |
| PDOWN_Tx | Pull-down<br>resistance     | -              | 50   | 100 | 200  | kΩ   | INJ_ENA |

### 7.3 Low-side driver - O2 HEATER O2H[1:2]

These 2 LowSide driver are designed to driver O2 heater Load. They are driven by *MSC command*. The Output voltage is clamped to voltage limit by internal clamp circuit. O2 channel can be configured as valve driver with faster slew rate and lower over current threshold by MSC commands. Current sense generates on Curr\_sense\_O2H pin a reduced copy of the current flowing through O2H Power. Current sense current generator is done with a P-channel current mirror connected to VDD5\_IN, for this reason voltage on Curr\_sense\_O2H1 and Curr\_sense\_O2H2 pins can never exceed VDD5\_IN.



**Figure 7-6** O2 heater LowSide driver stage



**Figure 7-7** O2 Heater LowSide driver current sense block

**Conditions:**

$5.5\text{ V} \leq \text{VB\_IN} \leq 18\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified.

Table 7-6 Low-side driver - O2 HEATER electrical characteristics

| Symbol                | Parameter                                                             | Test condition                                                                                                                                                             | Min  | Typ  | Max  | Unit             | Pin    |
|-----------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|------------------|--------|
| <b>O2H</b>            |                                                                       |                                                                                                                                                                            |      |      |      |                  |        |
| LS_RdsON              | Low-side RdsON                                                        | $T_j = 150\text{ }^{\circ}\text{C}, I_{\text{LOAD}} = 3\text{ A}$                                                                                                          | 0.04 | 0.16 | 0.23 | $\Omega$         | O2H1/2 |
|                       |                                                                       | $T_j = 25\text{ }^{\circ}\text{C}, I_{\text{LOAD}} = 3\text{ A}$                                                                                                           | 0.04 | 0.12 | 0.2  | $\Omega$         | O2H1/2 |
|                       |                                                                       | $T_j = -40\text{ }^{\circ}\text{C}, I_{\text{LOAD}} = 3\text{ A}$                                                                                                          | 0.03 | 0.09 | 0.13 | $\Omega$         | O2H1/2 |
| OUTx_lkg              | Output leakage current                                                | Output disabled, diagnostic off $V_{\text{pin}} = 13.5\text{ V}$                                                                                                           | -10  | -    | +10  | $\mu\text{A}$    | O2H1/2 |
| Irev_OUTx             | Body diode reverse current voltage drop                               | $I_{\text{LOAD}} = -2\text{ A}$                                                                                                                                            | -    | -    | 2    | V                | O2H1/2 |
| SR_ON                 | Voltage slew ON State                                                 | From 80% to 30% of VOUT<br>$V_{\text{B\_IN}} = 14\text{ V}$ , $R_{\text{load}} = 3\text{ }\Omega$ ,<br>$C_{\text{load}} = 10\text{ nF}$ ; configure as O2                  | 0.2  | -    | 0.8  | V/ $\mu\text{s}$ | O2H1/2 |
| SR_ON_fast            | Fast voltage slew ON State                                            | From 80% to 30% of VOUT<br>$V_{\text{B\_IN}} = 14\text{ V}$ , $R_{\text{load}} = 3\text{ }\Omega$ ,<br>$C_{\text{load}} = 10\text{ nF}$ ; configure as sol, fast slew rate | 2    | -    | 6    | V/ $\mu\text{s}$ | O2H1/2 |
| SR_OFF                | Voltage slew OFF state                                                | From 30% to 80% of VOUT<br>$V_{\text{B\_IN}} = 14\text{ V}$ , $R_{\text{load}} = 3\text{ }\Omega$ ,<br>$C_{\text{load}} = 10\text{ nF}$ ; configure as O2                  | 0.2  | -    | 0.8  | V/ $\mu\text{s}$ | O2H1/2 |
| SR_OFF_fast           | Fast Voltage slew OFF State                                           | From 30% to 80% of VOUT<br>$V_{\text{B\_IN}} = 14\text{ V}$ , $R_{\text{load}} = 3\text{ }\Omega$ ,<br>$C_{\text{load}} = 10\text{ nF}$ ; configure as sol, fast slew rate | 2    | -    | 6    | V/ $\mu\text{s}$ | O2H1/2 |
| S/RGkill_LSH          | FAST VS/R off when an OVC fault happens                               | From 30% to 80% of VOUT<br>$V_{\text{B\_IN}} = 14\text{ V}$ , $R_{\text{load}} = 3\text{ }\Omega$ , $C_{\text{load}} = 10\text{ nF}$                                       | 5    | -    | 20   | V/ $\mu\text{s}$ | O2H1/2 |
| Ton_OUTx              | Propagation Delay from MSC_EN rising edge to 80% output OUTx voltage  | $V_{\text{B\_IN}} = 14\text{ V}$ , $R_{\text{load}} = 3\text{ }\Omega$ , $C_{\text{load}} = 10\text{ nF}$                                                                  | -    | -    | 12   | $\mu\text{s}$    | O2H1/2 |
| Toff_OUTx             | Propagation Delay from MSC_EN falling edge to 20% output OUTx voltage | $V_{\text{B\_IN}} = 14\text{ V}$ , $R_{\text{load}} = 3\text{ }\Omega$ , $C_{\text{load}} = 10\text{ nF}$                                                                  | -    | -    | 10   | $\mu\text{s}$    | O2H1/2 |
| Vclamp_OUTx           | OUTx clamping Voltage                                                 | $I_{\text{LOAD}} = 1.3\text{ A}$                                                                                                                                           | 45   | 50   | 55   | V                | O2H1/2 |
| Vclamp_OUTx_high_curr | OUTx clamping Voltage                                                 | $I_{\text{LOAD}} = 1.3\text{ A}$<br>Guaranteed by design, provided single pulse energy limits are not violated by clamping action                                          | 45   | 50   | 55   | V                | O2H1/2 |
| I_OVC1_O2H            | Over current driver                                                   | configure as sol                                                                                                                                                           | 3    | 4.5  | 6    | A                | O2H1/2 |

## Low-side drivers

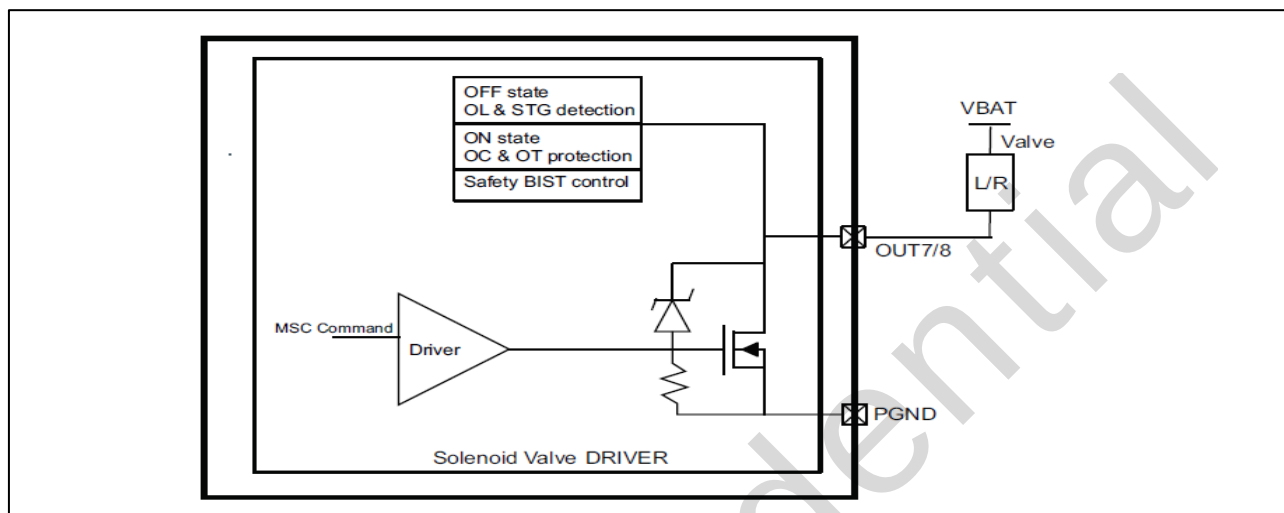
|                                                 |                                                                 |                                                                                                                    |                    |     |                    |    |        |
|-------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------|-----|--------------------|----|--------|
|                                                 | threshold 1                                                     |                                                                                                                    |                    |     |                    |    |        |
| I_OVC2_O2H                                      | Over current driver threshold 2                                 | T <sub>j</sub> = -40 °C configured as O2                                                                           | 8.6                | -   | 12.4               | A  | O2H1/2 |
| I_OVC2_O2H                                      | Over current driver threshold 2                                 | T <sub>j</sub> = +25 °C configured as O2                                                                           | 8.0                | -   | 11.2               | A  | O2H1/2 |
| I_OVC2_O2H                                      | Over current driver threshold 2                                 | T <sub>j</sub> = +150 °C configured as O2                                                                          | 7.8                | -   | 10.5               | A  | O2H1/2 |
| I_LS_ocvflt                                     | LS overcurrent filter time                                      | Guaranteed by scan                                                                                                 | 4                  | -   | 7                  | μs | O2H1/2 |
| T_SD_HIGH                                       | Temperature shut down                                           | -                                                                                                                  | 185                | -   | 200                | °C | O2H1/2 |
| T_SD_LOW                                        | Temperature shut down recover                                   | -                                                                                                                  | 175                | -   | 190                | °C | O2H1/2 |
| T_SD_hys                                        | Temperature shut down hysteresis                                | -                                                                                                                  | 5                  | -   | 10                 | °C | O2H1/2 |
| tmsd_pre_an                                     | Thermal shutdown analog filter time                             | Guaranteed by design                                                                                               | 1.5                | -   | 4                  | μs | O2H1/2 |
| t_SD_deglitch                                   | Digital deglitch filter time on Temperature shut down detection | Guaranteed by scan                                                                                                 | -                  | 10  | -                  | μs | O2H1/2 |
| <b>Driver reliability data</b>                  |                                                                 |                                                                                                                    |                    |     |                    |    |        |
| EnergyRep O2H                                   | Energy repetitive pulses                                        | T <sub>case</sub> = 25 °C ; I <sub>OT_n</sub> = 1.5A 18 Miopulses                                                  | -                  | -   | 15                 | mJ | O2H1/2 |
| EnergyRep O2H                                   | Energy repetitive pulses                                        | T <sub>case</sub> = 115 °C ; I <sub>OT_n</sub> = 1.2A 648 Miopulses                                                | -                  | -   | 10                 | mJ | O2H1/2 |
| EnergyRep O2H                                   | Energy repetitive pulses                                        | T <sub>case</sub> = 130 °C ; I <sub>OT_n</sub> = 1.0A 96 Miopulses                                                 | -                  | -   | 9                  | mJ | O2H1/2 |
| EnergyRep O2H                                   | Energy repetitive pulses                                        | T <sub>case</sub> = 140 °C ; I <sub>OT_n</sub> = 1.0A 4 Miopulses                                                  | -                  | -   | 9                  | mJ | O2H1/2 |
| <b>Driver Reliability Data jump start</b>       |                                                                 |                                                                                                                    |                    |     |                    |    |        |
| EnergyRep O2H                                   | Energy repetitive pulses                                        | T <sub>case</sub> = 25 °C ; I <sub>OT_n</sub> = 2.2A 0.021 Miopulses 10 jumps starts over lifetime, each start < 2 | -                  | -   | 30                 | mJ | O2H1/2 |
| EnergyRep O2H                                   | Energy repetitive pulses                                        | T <sub>case</sub> = 75 °C ; I <sub>OT_n</sub> = 1.8A 0.021 Miopulses 10 jumps starts over lifetime, each start < 2 | -                  | -   | 18                 | mJ | O2H1/2 |
| <b>Driver reliability data generator defect</b> |                                                                 |                                                                                                                    |                    |     |                    |    |        |
| EnergyRep O2H                                   | Energy repetitive pulses                                        | T <sub>case</sub> = 25 °C ; I <sub>OT_n</sub> = 1.6A 0.5 Miopulses                                                 | -                  | -   | 17.5               | mJ | O2H1/2 |
| EnergyRep O2H                                   | Energy repetitive pulses                                        | T <sub>case</sub> = 135 °C ; I <sub>OT_n</sub> = 1.1 A 0.5 Miopulses                                               | -                  | -   | 10                 | mJ | O2H1/2 |
| <b>OFF state diagnostic</b>                     |                                                                 |                                                                                                                    |                    |     |                    |    |        |
| VLVT                                            | Short to GND threshold voltage                                  | Driver tristate, diag enabled                                                                                      | 1.9                | 2.1 | VOU TOPE N -180 mV | V  | O2H1/2 |
| V_OL                                            | Open load threshold voltage                                     | Driver tristate, diag enabled                                                                                      | VOU TOPE N +160 mV | 2.9 | 3.0                | V  | O2H1/2 |
| VOUTOPEN                                        | Open load output voltage                                        | Driver tristate, diag enabled                                                                                      | 2.3                | 2.5 | 2.7                | V  | O2H1/2 |
| I_LS_PU1                                        | Diagnostic pull up current                                      | In OFF condition, OUTx < VLVT,                                                                                     | 2.5                | 3.6 | 4.7                | mA | O2H1/2 |

## Low-side drivers

|                                        |                                         |                                                                          |      |     |      |     |                      |
|----------------------------------------|-----------------------------------------|--------------------------------------------------------------------------|------|-----|------|-----|----------------------|
|                                        |                                         | Ipupd_EN = ENB,<br>Ipupd_MODE = "1"                                      |      |     |      |     |                      |
| I_LS_PU2                               | Diagnostic pull up current              | In OFF condition,<br>OUTx < VLVT,<br>Ipupd_EN = ENB,<br>Ipupd_MODE = "0" | 40   | 70  | 100  | μA  | O2H1/2               |
| I_LS_PD1                               | Diagnostic pull down current            | In OFF condition,<br>OUTx > VOUTOPEN,<br>Ipupd_EN = ENB                  | 60   | 85  | 105  | μA  | O2H1/2               |
| Tflt_diagoff1                          | DIAG Filter time                        | Filter Mode = 0                                                          | 75   | 100 | 125  | μs  | O2H1/2               |
| Tflt_diagoff2                          | DIAG Filter time                        | Filter Mode = 1                                                          | 450  | 600 | 750  | μs  | O2H1/2               |
| -                                      | Minimum OFF time for correct diagnostic | Application note (esd cap < 10 nF,) Filter Mode = 0                      | 175  | -   | -    | μs  | O2H1/2               |
| <b>Current sensing characteristics</b> |                                         |                                                                          |      |     |      |     |                      |
| Gain_sense                             | Transresistance sense gain              | -                                                                        |      | 0.8 |      | V/A | Curr_Sense<br>O2H1/2 |
| V_OUT_0p5A                             | Output voltage AD for Ipower = 0.5 A    | Rext = 5.1 kΩ                                                            | 0.3  | 0.4 | 0.55 | V   | Curr_Sense<br>O2H1/2 |
| V_OUT_1A                               | Output voltage AD for Ipower = 1 A      | Rext = 5.1 kΩ                                                            | 0.64 | 0.8 | 0.96 | V   | Curr_Sense<br>O2H1/2 |
| V_OUT_2A                               | Output voltage AD for Ipower = 2 A      | Rext = 5.1 kΩ                                                            | 1.44 | 1.6 | 1.76 | V   | Curr_Sense<br>O2H1/2 |
| V_OUT_3A                               | Output voltage AD for Ipower = 3 A      | Rext = 5.1 kΩ                                                            | 2.16 | 2.4 | 2.64 | V   | Curr_Sense<br>O2H1/2 |
| V_OUT_4A                               | Output voltage AD for Ipower = 4 A      | Rext = 5.1 kΩ                                                            | 2.88 | 3.2 | 3.52 | V   | Curr_Sense<br>O2H1/2 |

## 7.4 Low-side driver - SOLENOID SOL [1:2]

These 2 LowSide drivers are designed to driver Solenoid Valves Load. They are driven by *MSC command*. The Output voltage is clamped to voltage limit by internal clamp circuit.



**Figure 7-8 Solenoid valves LowSide driver stage**

### Conditions:

$5.5\text{ V} \leq \text{VB\_IN} \leq 18\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified.

**Table 7-7 Low-side driver - SOLENOID VALVE electrical characteristics**

| Symbol                 | Parameter                  | Test condition                                                                                                                        | Min  | Typ  | Max  | Unit                   | Pin    |
|------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------|------|------|------------------------|--------|
| <b>Solenoid valves</b> |                            |                                                                                                                                       |      |      |      |                        |        |
| LS_RdsON               | Low-side RdsON             | $T_j = 150\text{ }^{\circ}\text{C}$ , $I_{\text{LOAD}} = 3\text{ A}$                                                                  | 0.24 | 0.35 | 0.47 | $\Omega$               | SOL1/2 |
|                        |                            | $T_j = 25\text{ }^{\circ}\text{C}$ , $I_{\text{LOAD}} = 3\text{ A}$                                                                   | 0.17 | 0.22 | 0.35 | $\Omega$               | SOL1/2 |
|                        |                            | $T_j = -40\text{ }^{\circ}\text{C}$ , $I_{\text{LOAD}} = 3\text{ A}$                                                                  | 0.12 | 0.17 | 0.23 | $\Omega$               | SOL1/2 |
| OUTx_lkg               | Output leakage current     | Output disabled, diagnostic off $V_{\text{pin}} = 13.5\text{ V}$                                                                      | -10  | -    | +10  | $\mu\text{A}$          | SOL1/2 |
| Irev_OUTx              | Body diode reverse current | $I_{\text{LOAD}} = -2\text{ A}$                                                                                                       | -    | -    | 2    | V                      | SOL1/2 |
| SR_ON                  | Voltage slew ON State      | From 80% to 30% of VOUT<br>$\text{VB\_IN} = 14\text{ V}$ , $R_{\text{load}} = 15\text{ }\Omega$ ,<br>$C_{\text{load}} = 10\text{ nF}$ | 0.55 | -    | 1.75 | $\text{V}/\mu\text{s}$ | SOL1/2 |
| SR_OFF                 | Voltage slew OFF state     | From 30% to 80% of VOUT<br>$\text{VB\_IN} = 14\text{ V}$ , $R_{\text{load}} = 15\text{ }\Omega$ ,<br>$C_{\text{load}} = 10\text{ nF}$ | 0.6  | -    | 1.75 | $\text{V}/\mu\text{s}$ | SOL1/2 |
| S/RGkill_LSH           | FAST VS/R off              | From 30% to 80% of                                                                                                                    | 5    | -    | 20   | $\text{V}/\mu\text{s}$ | SOL1/2 |

## Low-side drivers

|                                                 |                                                                       |                                                                                                              |     |     |      |    |        |
|-------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----|-----|------|----|--------|
|                                                 | when an OVC fault happens                                             | VOUT<br>VB_IN = 14 V, Rload = 15Ω, Cload = 10 nF                                                             |     |     |      |    |        |
| Ton_OUTx                                        | Propagation Delay from MSC_EN rising edge to 80% output OUTx voltage  | VB_IN = 14 V, Rload = 15Ω, Cload = 10 nF                                                                     | -   | -   | 8    | μs | SOL1/2 |
| Toff_OUTx                                       | Propagation Delay from MSC_EN falling edge to 20% output OUTx voltage | VB_IN = 14 V, Rload = 15Ω, Cload = 10 nF                                                                     | -   | -   | 8    | μs | SOL1/2 |
| Vclamp_OUTx                                     | OUTx clamping Voltage                                                 | ILOAD = 1.3 A                                                                                                | 50  | 55  | 60   | V  | SOL1/2 |
| Vclamp_OUTx_high_curr                           | OUTx clamping Voltage                                                 | ILOAD = 3 A<br>Guaranteed by design, provided single pulse energy limits are not violated by clamping action | 50  | 55  | 60   | V  | SOL1/2 |
| I_OVC_SOL                                       | Over current driver threshold                                         | -                                                                                                            | 3   | 4.5 | 6    | A  | SOL1/2 |
| I_LS_ocvflt                                     | LS overcurrent filter time                                            | Guaranteed by scan                                                                                           | 4   | -   | 7    | μs | SOL1/2 |
| T_SD_HIGH                                       | Temperature shut down                                                 | -                                                                                                            | 185 | -   | 200  | °C | SOL1/2 |
| T_SD_LOW                                        | Temperature shut down recover                                         | -                                                                                                            | 175 | -   | 190  | °C | SOL1/2 |
| T_SD_hys                                        | Temperature shut down hysteresis                                      | -                                                                                                            | 5   | -   | 10   | °C | SOL1/2 |
| -                                               | Thermal shutdown analog filter time                                   | Guaranteed by design                                                                                         | 1.5 | -   | 4    | μs | SOL1/2 |
| t_SD_deglitch                                   | Digital deglitch filter time on Temperature shut down detection       | Guaranteed by scan                                                                                           | -   | 10  | -    | μs | SOL1/2 |
| <b>Driver reliability data</b>                  |                                                                       |                                                                                                              |     |     |      |    |        |
| Energy_Rep_SOL                                  | Energy repetitive pulses                                              | Tcase = 25 °C ; I_OT_n = 1.5A 18 Miopulses                                                                   | -   | -   | 15   | mJ | SOL1/2 |
| Energy_Rep_SOL                                  | Energy repetitive pulses                                              | Tcase = 115 °C ; I_OT_n = 1.2A 648 Miopulses                                                                 | -   | -   | 10   | mJ | SOL1/2 |
| Energy_Rep_SOL                                  | Energy repetitive pulses                                              | Tcase = 130 °C ; I_OT_n = 1.0A 96 Miopulses                                                                  | -   | -   | 9    | mJ | SOL1/2 |
| Energy_Rep_SOL                                  | Energy repetitive pulses                                              | Tcase = 140 °C ; I_OT_n = 1.0A 4 Miopulses                                                                   | -   | -   | 9    | mJ | SOL1/2 |
| <b>Driver Reliability Data jump start</b>       |                                                                       |                                                                                                              |     |     |      |    |        |
| Energy_Rep_SOL                                  | Energy repetitive pulses                                              | Tcase = 25 °C ; I_OT_n = 2.2A 0.021 Miopulses<br>10 jumps starts over lifetime, each start < 2 min           | -   | -   | 30   | mJ | SOL1/2 |
| Energy_Rep_SOL                                  | Energy repetitive pulses                                              | Tcase = 75 °C ; I_OT_n = 1.8A 0.021 Miopulses<br>10 jumps starts over lifetime, each start < 2 min           | -   | -   | 18   | mJ | SOL1/2 |
| <b>Driver reliability data generator defect</b> |                                                                       |                                                                                                              |     |     |      |    |        |
| EnergyRep SOL                                   | Energy repetitive pulses                                              | Tcase = 25 °C ; I_OT_n = 1.6A 0.5 Miopulses                                                                  | -   | -   | 17.5 | mJ | SOL1/2 |

Low-side drivers

|                             |                                         |                                                                                          |                    |     |                    |    |        |
|-----------------------------|-----------------------------------------|------------------------------------------------------------------------------------------|--------------------|-----|--------------------|----|--------|
| EnergyRep SOL               | Energy repetitive pulses                | T <sub>case</sub> = 135 °C ; I <sub>OT_n</sub> = 1.1 A 0.5 Miopulses                     | -                  | -   | 10                 | mJ | SOL1/2 |
| <b>OFF state diagnostic</b> |                                         |                                                                                          |                    |     |                    |    |        |
| VLVT                        | Short to GND threshold voltage          | Driver tristate, diag enabled                                                            | 1.9                | 2.1 | VOU TOPE N - 180mV | V  | SOL1/2 |
| V_OL                        | Open load threshold voltage             | Driver tristate, diag enabled                                                            | VOU TOPE N +160 mV | 2.9 | 3.0                | V  | SOL1/2 |
| VOUTOPEN                    | Open load output voltage                | Driver tristate, diag enabled                                                            | 2.3                | 2.5 | 2.7                | V  | SOL1/2 |
| I_LS_PU1                    | Diagnostic pull up current              | In OFF condition, OUTx < VLVT, lpupd_EN = ENB, lpupd_MODE = "1"                          | 2.5                | 3.6 | 4.7                | mA | SOL1/2 |
| I_LS_PU2                    | Diagnostic pull up current              | In OFF condition, OUTx < VLVT, lpupd_EN = ENB, lpupd_MODE = "0" and IDIAG_HIGH_SOL = "0" | 40                 | 70  | 100                | µA | SOL1/2 |
| I_LS_PU3                    | Diagnostic pull up high current         | In OFF condition, OUTx < VLVT, lpupd_EN=ENB,lpupd_MO DE="0" and IDIAG_HIGH_SOL = "1"     | 100                |     | 200                | µA | SOL1/2 |
| I_LS_PD1                    | Diagnostic pull down current            | In OFF condition, OUTx > VOUTOPEN, lpupd_EN = ENB and IDIAG_HIGH_SOL = "0"               | 60                 | 85  | 105                | µA | SOL1/2 |
| I_LS_PD2                    |                                         | In OFF condition, OUTx > VOUTOPEN, lpupd_EN = ENB IDIAG_HIGH_SOL = "1"                   | 325                | -   | 550                | µA | SOL1/2 |
| Tflt_diagoff1               | DIAG Filter time                        | Filter Mode = 0                                                                          | 75                 | 100 | 125                | µs | SOL1/2 |
| Tflt_diagoff2               | DIAG Filter time                        | Filter Mode = 1                                                                          | 450                | 600 | 750                | µs | SOL1/2 |
| -                           | Minimum OFF time for correct diagnostic | Application note (esd cap < 10 nF,) Filter Mode = 0                                      | 175                | -   | -                  | µs | SOL1/2 |

## 7.5 Low-side driver - RELAY RLY [1:5]

These 5 LowSide drivers are optimized for relay or generic low current loads driving. They are driven by MSC command. The Output voltage is clamped to voltage limit by internal clamp circuit.

Each relay driver is separately put in on or off state by the *MSC CONFIG-REG 14 [0:4]* just for the time required to perform on or off diagnosis (Fast Diagnosis). At the end of the diagnosis it is automatically put in the state determined by the current value of MSC data frame.

The micro command is activated by writing 1 from 0 in the MSC registers (both for ON and OFF).

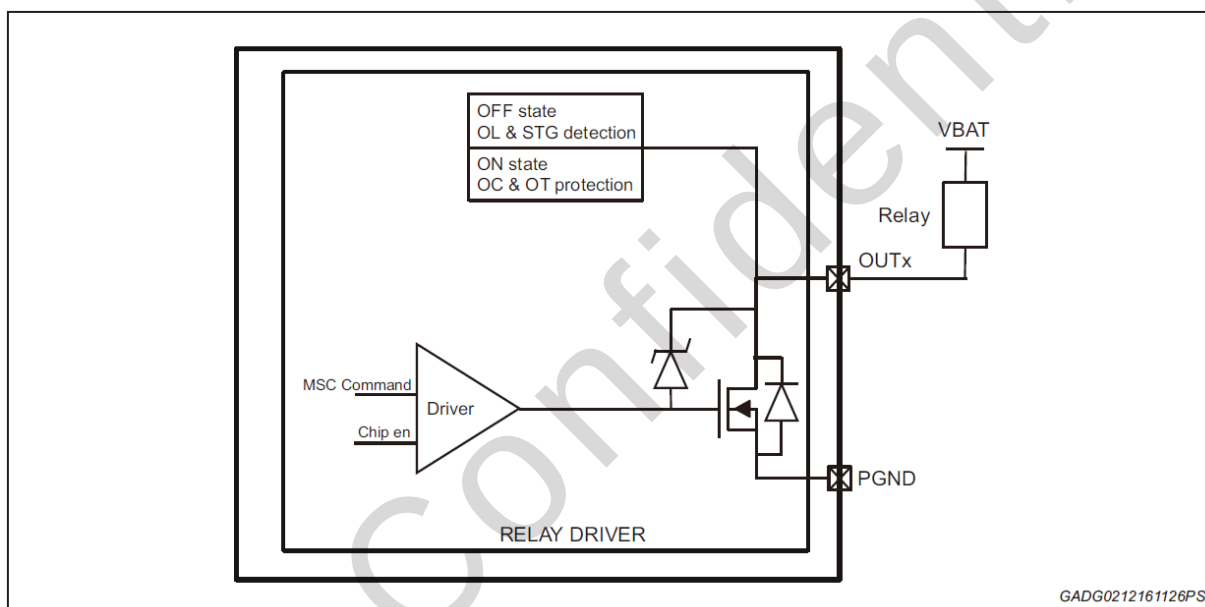


Figure 7-9 Relay LowSide driver stage

### Conditions:

$5.5\text{ V} \leq \text{VB\_IN} \leq 18\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified; the RLY4 can be associated to DELAY\_OFF function if the *MSC CONFIG\_REG\_3 [0] = RLY4\_DLY\_OFF\_EN* is set. In this case the RLY4 is configured as starter and can work down to  $\text{VB\_IN} = 4.25\text{ V}$  for THOLD.

Table 7-8 Low-side driver - RELAY electrical characteristics

| Symbol       | Parameter      | Test condition                                         | Min | Typ | Max  | Unit     | Pin           |
|--------------|----------------|--------------------------------------------------------|-----|-----|------|----------|---------------|
| <b>Relay</b> |                |                                                        |     |     |      |          |               |
| LS_RdsON     | Low-side RdsON | $T_j = 150\text{ °C}$ , $I_{\text{LOAD}} = 1\text{ A}$ | -   | -   | 1.5  | $\Omega$ | RLY/1/2/3/4/5 |
|              |                | $T_j = 25\text{ °C}$ , $I_{\text{LOAD}} = 1\text{ A}$  | -   | -   | 0.82 | $\Omega$ | RLY/1/2/      |

## Low-side drivers

|                       |                                                                       |                                                                                                                                              |     |     |      |                    |                   |
|-----------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|------|--------------------|-------------------|
|                       |                                                                       | $T_j = -40\text{ }^{\circ}\text{C}$ , $I_{\text{LOAD}} = 1\text{ A}$                                                                         | -   | -   | 0.63 | $\Omega$           | 3/4/5             |
| LS_RdsON_RL4          | Low-side RdsON                                                        | $V_{\text{B\_IN}} = 4.25\text{ V}$ ,<br>$I_{\text{LOAD}} = 0.6\text{ A}$                                                                     | -   | -   | 3    | $\Omega$           | RLY/1/2/<br>3/4/5 |
| OUTx_Ikg              | Output leakage current                                                | Output disabled,<br>diagnostic off                                                                                                           | -10 | -   | +10  | $\mu\text{A}$      | RLY/1/2/<br>3/4/5 |
| Irev_OUTx             | Body diode reverse current voltage drop                               | $I_{\text{LOAD}} = -0.6\text{ A}$                                                                                                            | -   | -   | 2    | V                  | RLY/1/2/<br>3/4/5 |
| SR_ON                 | Voltage slew ON State                                                 | From 80% to 30% of $V_{\text{OUT}}$ , $V_{\text{B\_IN}} = 14\text{ V}$ ,<br>$R_{\text{load}} = 68\Omega$<br>$C_{\text{load}} = 10\text{ nF}$ | 0.4 | -   | 2.1  | V/ $\mu\text{s}$   | RLY/1/2/<br>3/4/5 |
| SR_OFF                | Voltage slew OFF State                                                | From 30% to 80% of $V_{\text{OUT}}$ , $V_{\text{B\_IN}} = 14\text{ V}$ ,<br>$R_{\text{load}} = 68\Omega$<br>$C_{\text{load}} = 10\text{ nF}$ | 0.4 | -   | 2.1  | V/ $\mu\text{s}$   | RLY/1/2/<br>3/4/5 |
| S/RGkill              | FAST VS/R off when an OVC fault happens                               | From 30% to 80% of $V_{\text{OUT}}$ , $V_{\text{B\_IN}} = 14\text{ V}$ ,<br>$R_{\text{load}} = 68\Omega$<br>$C_{\text{load}} = 10\text{ nF}$ | 5   | -   | 20   | V/ $\mu\text{s}$   | RLY/1/2/<br>3/4/5 |
| Ton_OUTx              | Propagation Delay from MSC_EN rising edge to 80% output OUTx voltage  | $V_{\text{OUT}}$ , $V_{\text{B\_IN}} = 14\text{ V}$ ,<br>$R_{\text{load}} = 68\Omega$<br>$C_{\text{load}} = 10\text{ nF}$                    | -   | -   | 10   | $\mu\text{s}$      | RLY/1/2/<br>3/4/5 |
| Toff_OUTx             | Propagation Delay from MSC_EN falling edge to 20% output OUTx voltage | $V_{\text{B\_IN}} = 14\text{ V}$ ,<br>$R_{\text{load}} = 68\Omega$<br>$C_{\text{load}} = 10\text{ nF}$                                       | -   | -   | 10   | $\mu\text{s}$      | RLY/1/2/<br>3/4/5 |
| Vclamp_OUTx           | OUTx clamping Voltage                                                 | $I_{\text{LOAD}} = 0.6\text{ A}$                                                                                                             | 45  | -   | 55   | V                  | RLY/1/2/<br>3/4/5 |
| Vclamp_OUTx_high_curr | OUTx clamping Voltage                                                 | $I_{\text{LOAD}} = 1\text{ A}$<br>Guaranteed by design, provided single pulse energy limits are not violated by clamping action              | 45  | -   | 55   | V                  | RLY/1/2/<br>3/4/5 |
| I_OVC_RLY             | Over current driver threshold                                         | -                                                                                                                                            | 1   | 1.5 | 2    | A                  | RLY/1/2/<br>3/4/5 |
| I_LS_ocvflt           | LS overcurrent filter time                                            | Guaranteed by scan                                                                                                                           | 4   | -   | 7    | $\mu\text{s}$      | RLY/1/2/<br>3/4/5 |
| T_SD_HIGH             | Temperature shut down                                                 | -                                                                                                                                            | 185 | -   | 200  | $^{\circ}\text{C}$ | RLY/1/2/<br>3/4/5 |
| T_SD_LOW              | Temperature shut down recover                                         | -                                                                                                                                            | 175 | -   | 190  | $^{\circ}\text{C}$ | RLY/1/2/<br>3/4/5 |
| T_SD_hys              | Temperature shut down hysteresis                                      | -                                                                                                                                            | 5   | -   | 10   | $^{\circ}\text{C}$ | RLY/1/2/<br>3/4/5 |
| -                     | Thermal shutdown analog filter time                                   | Guaranteed by design                                                                                                                         | 1.5 |     | 4    | $\mu\text{s}$      | RLY/1/2/<br>3/4/5 |
| t_SD_deglitch         | Digital deglitch filter time on Temperature shut down detection       | Guaranteed by scan                                                                                                                           | -   | 10  | -    | $\mu\text{s}$      | RLY/1/2/<br>3/4/5 |

## Driver reliability data

## Low-side drivers

|                                                 |                                   |                                                                                                                                         |                                |     |                                    |    |                   |
|-------------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----|------------------------------------|----|-------------------|
| EnergyRep RLY                                   | Energy repetitive pulses          | T <sub>case</sub> = 25 °C;<br>I <sub>OT_n</sub> = 0.45 A<br>1.1 Miopulses                                                               | -                              | -   | 9                                  | mJ | RLY/1/2/<br>3/4/5 |
| EnergyRep RLY                                   | Energy repetitive pulses          | T <sub>case</sub> = 115 °C;<br>I <sub>OT_n</sub> = 0.3 A<br>40 Miopulses                                                                | -                              | -   | 6.5                                | mJ | RLY/1/2/<br>3/4/5 |
| EnergyRep RLY                                   | Energy repetitive pulses          | T <sub>case</sub> = 130 °C;<br>I <sub>OT_n</sub> = 0.3 A<br>9 Miopulses                                                                 | -                              | -   | 6.5                                | mJ | RLY/1/2/<br>3/4/5 |
| EnergyRep RLY                                   | Energy repetitive pulses          | T <sub>case</sub> = 140 °C;<br>I <sub>OT_n</sub> = 0.3 A<br>1 Miopulses                                                                 | -                              | -   | 6.5                                | mJ | RLY/1/2/<br>3/4/5 |
| <b>Driver reliability data generator defect</b> |                                   |                                                                                                                                         |                                |     |                                    |    |                   |
| EnergyRep RLY                                   | Energy repetitive pulses          | T <sub>case</sub> = 25 °C;<br>I <sub>OT_n</sub> = 0.5 A<br>0.02 Miopulses                                                               | -                              | -   | 11                                 | mJ | RLY/1/2/<br>3/4/5 |
| EnergyRep RLY                                   | Energy repetitive pulses          | T <sub>case</sub> = 135 °C;<br>I <sub>OT_n</sub> = 0.35 A<br>0.02 Miopulses                                                             | -                              | -   | 8                                  | mJ | RLY/1/2/<br>3/4/5 |
| <b>Driver reliability data JUMP START</b>       |                                   |                                                                                                                                         |                                |     |                                    |    |                   |
| EnergyRep RLY                                   | Energy repetitive pulses          | T <sub>case</sub> = 25 °C;<br>I <sub>OT_n</sub> = 0.75 A<br>0.001 Miopulses,<br>10 jumps starts over<br>lifetime,<br>each start < 2 min | -                              | -   | 25                                 | mJ | RLY/1/2/<br>3/4/5 |
| EnergyRep RLY                                   | Energy repetitive pulses          | T <sub>c</sub> = 75 °C;<br>I <sub>OT_n</sub> = 0.5 A<br>0.001 Miopulses,<br>10 jumps starts over<br>lifetime,<br>each start < 2 min     | -                              | -   | 17                                 | mJ | RLY/1/2/<br>3/4/5 |
| <b>OFF state diagnostic</b>                     |                                   |                                                                                                                                         |                                |     |                                    |    |                   |
| VLVT                                            | Short to GND<br>threshold voltage | Driver tristate, diag<br>enabled                                                                                                        | 1.9                            |     | VOU<br>TOPE<br>N<br>-<br>180m<br>V | V  | RLY/1/2/<br>3/4/5 |
| V_OL                                            | Open load<br>threshold<br>voltage | Driver tristate, diag<br>enabled                                                                                                        | VOU<br>TOPE<br>N<br>+160<br>mV |     | 3.0                                | V  | RLY/1/2/<br>3/4/5 |
| VOUTOPEN                                        | Open load output<br>voltage       | Driver tristate, diag<br>enabled                                                                                                        | 2.3                            | 2.5 | 2.7                                | V  | RLY/1/2/<br>3/4/5 |
| I_LS_PU1                                        | Diagnostic pull up<br>current     | In OFF condition,<br>OUTx < VLVT,<br>Ipupd_EN = ENB,<br>Ipupd_MODE = "1"                                                                | 2.3                            | 3.6 | 4.7                                | mA | RLY/1/2/<br>3/4/5 |
| I_LS_PU2                                        | Diagnostic pull up<br>current     | In OFF condition,<br>OUTx < VLVT,<br>Ipupd_EN = ENB,<br>Ipupd_MODE = "0"                                                                | 40                             | 70  | 100                                | μA | RLY/1/2/<br>3/4/5 |
| I_LS_PD1                                        | Diagnostic pull<br>down current   | In OFF condition,<br>OUTx > VOUTOPEN,<br>Ipupd_EN = ENB                                                                                 | 60                             | 85  | 105                                | μA | RLY/1/2/<br>3/4/5 |
| Tflt_diagoff1                                   | DIAG Filter time                  | Filter Mode=0                                                                                                                           | 75                             | 100 | 125                                | μs | RLY/1/2/<br>3/4/5 |
| Tflt_diagoff2                                   | DIAG Filter time                  | Filter Mode=1                                                                                                                           | 450                            | 600 | 750                                | μs | RLY/1/2/<br>3/4/5 |

## HE9788

Multifunction IC for automotive engine management system



Low-side drivers

|   |                                         |                                                     |     |     |     |    |               |
|---|-----------------------------------------|-----------------------------------------------------|-----|-----|-----|----|---------------|
| - | Minimum OFF time for correct diagnostic | Application note (esd cap < 10 nF,) Filter Mode = 0 | 175 | -   | -   | μs | RLY/1/2/3/4/5 |
| - | Fast OFF diagnostic ON→OFF→ON           | Filter Mode=0                                       | 175 | 200 | 225 | μs | RLY/1/2/3/4/5 |
| - | Fast ON diagnostic OFF→ON→OFF           | -                                                   | 72  |     | 90  | μs | RLY/1/2/3/4/5 |

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## 7.6 Low-side driver - LED[1:2]

These 2 LowSide drivers are designed to drive LED Load. They are driven by *MSC command*. The Output voltage is clamped to voltage limit by internal clamp circuit.

The pull down diagnostic current of LED driver is configurable by MSC (*LEDx\_PD\_EN CONFIG\_REG1 D3;D4*). Overcurrent and Short to Ground diagnostic function are available, open load diagnostic function is not available when pull down diagnostic current is disabled.

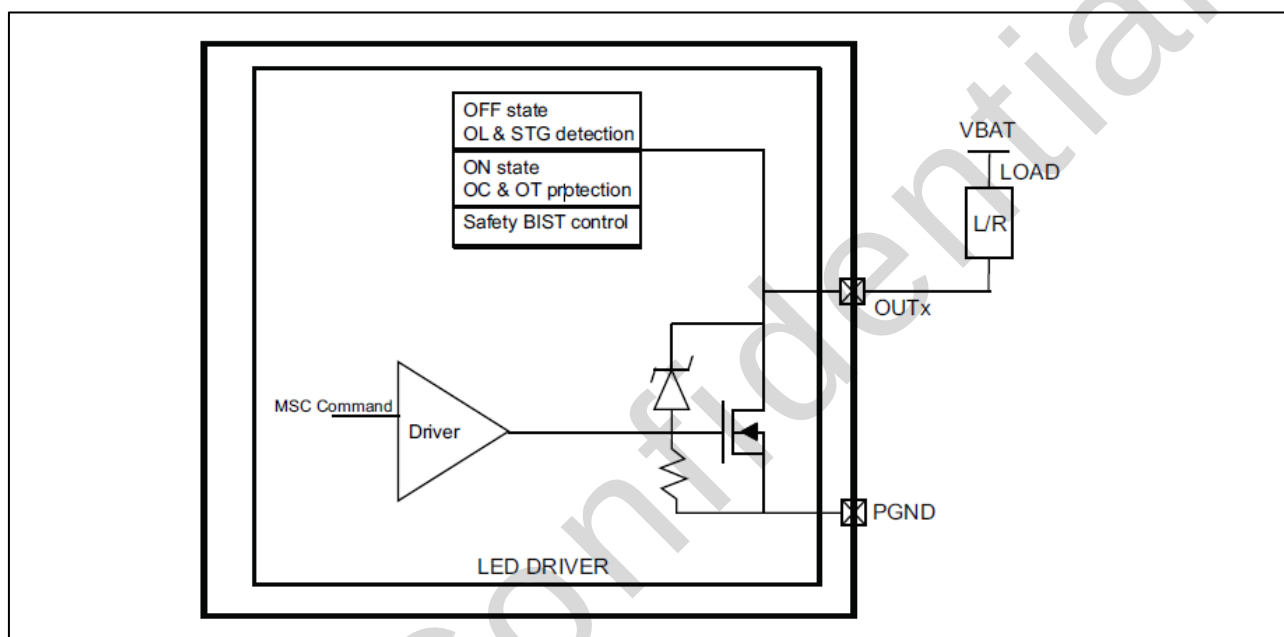


Figure 7-10 LED LowSide driver stage

### Conditions:

$5.5\text{ V} \leq V_{B\_IN} \leq 18\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified;

Table 7-9 Low-side driver - LED electrical characteristics

| Symbol     | Parameter                  | Test condition                                                  | Min | Typ | Max | Unit          | Pin    |
|------------|----------------------------|-----------------------------------------------------------------|-----|-----|-----|---------------|--------|
| <b>LED</b> |                            |                                                                 |     |     |     |               |        |
| LS_RdsON   | Low-side RdsON             | $T_j = 150\text{ }^{\circ}\text{C}$ , $I_{LOAD} = 70\text{ mA}$ | 10  | 14  | 20  | $\Omega$      | LED1/2 |
|            |                            | $T_j = 25\text{ }^{\circ}\text{C}$ , $I_{LOAD} = 70\text{ mA}$  | 6   | 8   | 12  | $\Omega$      | LED1/2 |
|            |                            | $T_j = -40\text{ }^{\circ}\text{C}$ , $I_{LOAD} = 70\text{ mA}$ | 4.9 | 6   | 9   | $\Omega$      | LED1/2 |
| OUTx_lkg   | Output leakage current     | Output disabled, diagnostic off                                 | -10 | -   | +10 | $\mu\text{A}$ | LED1/2 |
| Irev_OUTx  | Body diode reverse current | $I_{LOAD} = -50\text{ mA}$                                      | -   | -   | 2   | V             | LED1/2 |

## Low-side drivers

|                       | voltage drop                                                          |                                                                                                                |                    |     |                    |      |        |
|-----------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------|-----|--------------------|------|--------|
| SR_ON                 | Voltage slew ON state                                                 | From 80% to 30% of VOUT<br>VB_IN = 14 V, Rload = 270Ω,                                                         | 10                 | -   | 25                 | V/μs | LED1/2 |
| SR_OFF                | Voltage slew OFF state                                                | From 30% to 80% of VOUT<br>VB_IN = 14 V, Rload = 270Ω                                                          | 13                 | -   | 21                 | V/μs | LED1/2 |
| Ton_OUTx              | Propagation Delay from MSC_EN rising edge to 80% output OUTx voltage  | VB_IN = 14 V, Rload=270Ω,                                                                                      | -                  | -   | 5                  | μs   | LED1/2 |
| Toff_OUTx             | Propagation Delay from MSC_EN falling edge to 20% output OUTx voltage | VB_IN = 14 V, Rload=270Ω,                                                                                      | -                  | -   | 5                  | μs   | LED1/2 |
| Vclamp_OUTx           | OUTx clamping Voltage                                                 | ILOAD = 50 mA                                                                                                  | 40                 | 45  | 50                 | V    | LED1/2 |
| Vclamp_OUTx_high_curr | OUTx clamping Voltage                                                 | ILOAD = 70 mA<br>Guaranteed by design, provided single pulse energy limits are not violated by clamping action | 40                 | 45  | 50                 | V    | LED1/2 |
| I_OVC_LED             | Over current driver threshold                                         | -                                                                                                              | 70                 | -   | 110                | mA   | LED1/2 |
| I_LS_ocvflt           | LS overcurrent filter time                                            | Guaranteed by scan                                                                                             | 4                  | -   | 7                  | μs   | LED1/2 |
| VLVT                  | Short to GND threshold voltage                                        | Driver tristate, diag enabled                                                                                  | 1.9                | -   | VOU TOPE N - 180mV | V    | LED1/2 |
| V_OL                  | Open load threshold voltage                                           | Driver tristate, diag enabled                                                                                  | VOU TOPE N +160 mV | -   | 3                  | V    | LED1/2 |
| VOUTOPEN              | Open load output voltage                                              | Driver tristate, diag enabled                                                                                  | 2.3                | 2.5 | 2.7                | V    | LED1/2 |
| I_LS_PU1              | Diagnostic pull up current                                            | In OFF condition, OUTx < VLVT, Ipupd_EN = ENB, Ipupd_MODE = "1"                                                | 2.5                | 3.6 | 4.7                | mA   | LED1/2 |
| I_LS_PU2              | Diagnostic pull up current                                            | In OFF condition, OUTx < VLVT, Ipupd_EN = ENB, Ipupd_MODE = "0"                                                | 40                 | 70  | 100                | μA   | LED1/2 |
| I_LS_PD1              | Diagnostic pull down current                                          | In OFF condition, OUTx > VOUTOPEN, Ipupd_EN = ENB LEDx_PD_EN = 1                                               | 60                 | 85  | 105                | μA   | LED1/2 |
| Tflt_diagoff1         | DIAG filter time                                                      | Filter Mode = 0                                                                                                | 75                 | 100 | 125                | μs   | LED1/2 |
| Tflt_diagoff2         | DIAG filter time                                                      | Filter Mode = 1                                                                                                | 450                | 600 | 750                | μs   | LED1/2 |

## HE9788

Multifunction IC for automotive engine management system



Low-side drivers

|   |                                               |                                                        |     |   |   |    |        |
|---|-----------------------------------------------|--------------------------------------------------------|-----|---|---|----|--------|
| - | Minimum OFF<br>time for correct<br>diagnostic | Application note<br>(esd cap < 10nF,)<br>Filter Mode=0 | 175 | - | - | μs | LED1/2 |
|---|-----------------------------------------------|--------------------------------------------------------|-----|---|---|----|--------|

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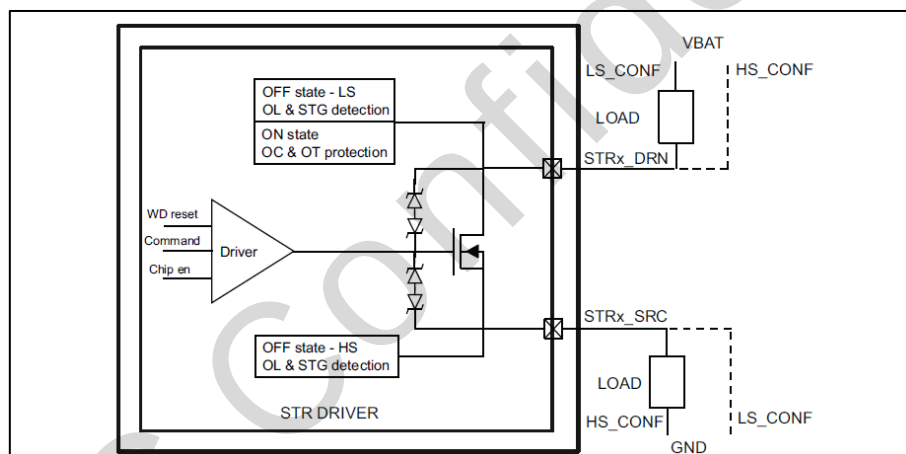
## 8 High/low-side drivers – STARTER STR[1:3]

These 3 high/low-side drivers with diagnosis and overcurrent protection have a floating architecture and can be used in high-side or low-side mode.

The configuration is defined by MSC bit (*LSD/HSD\_DRVx-CFG CONFIG\_REG3 D1~D3*).

The driver is optimized for relay and low-current loads, and it can be associated to smart starter functional block.

They are driven by MSC command. The Output voltage is clamped to voltage limit by internal clamp circuit. In order to guarantee a negligible output current in case of LED application an internal dedicated comparator is present. This comparator, in case of highside configuration, Diagoff disabled and channel OFF, determines the disabling of internal driver in case of output voltage (STR\_SRC) pin in the range of  $-0.4V_{typ} < STR\_SRC < 1.8V_{typ}$ . This implementation guarantees a negligible output current down to 0V on STR\_SRC pin (where internal compensation is no more effective) and at the same time the correct reenabling of the driver when the output voltage goes negative, in order to guarantee the correct clamping functionality during recirculation. In any other condition (Low-side configuration, or Diagoff enabled, or channel ON) the effect of this comparator is masked and an output current will be observed.



**Figure 8-1 Configurable high/low-side Driver block schematic**

The driver has the following detection diagnosis in low-side mode:

- Overcurrent (Short to Vbat) protection in On Phase (OVC);
- Open Load in Off phase (DIAGOL);
- Short to Ground in Off Phase (DIAGLV);
- Over Temperature Protection in On phase (OT);
- The driver has the following detection diagnosis in high-side mode:
- Overcurrent (Short To GND) protection in On Phase (OVC);
- Open Load in Off phase (DIAGOL);
- Short to battery in Off Phase (DIAGHV);
- Over Temperature Protection in On phase (OT);

High/low-side drivers - STARTER STR[1:3]

In low-side configuration, if they are configured as starter `CONFIG_REG3[4]=STR2_EN=1` and `CONFIG_REG3[5]=STR3_EN=1`, the driver STR2 and STR3 can work down to VB\_IN = 4.25 V. for THOLD.

In high-side configuration, if they are configured as starter `CONFIG_REG3[4]=STR2_EN=1` and `CONFIG_REG3[5]=STR3_EN=1`, the driver STR2 and STR3 can work down to VB\_IN = 4.25 V for THOLD.

Furthermore, the driver can latch its ON state for THOLD even if the main microcontroller is temporarily not working due to low-battery conditions.

HS/LS are intended to be used as starter relay driver. One for high-side, one for low-side. STR2/3 should keep the status till VOFF\_VB\_IN.(4.25 V).

Each starter driver is separately put in on or off state by the *MSC CONFIG-REG 14[5:7]* just for the time required to perform on or off diagnosis (Fast Diagnosis). At the end of the diagnosis it is automatically put in the state determined by the current value of *MSC data frame*.

The micro command is activated by writing 1 from 0 in the MSC registers (both for ON and OFF).

## **8.1 ON state diagnosis**

The ON state diagnosis has the same features as the low-side driver ON state diagnosis described in chapter [Section 7.1.1](#) and [Section 7.1.2](#).

## **8.2 ON/OFF state - Error in on status diagnosis**

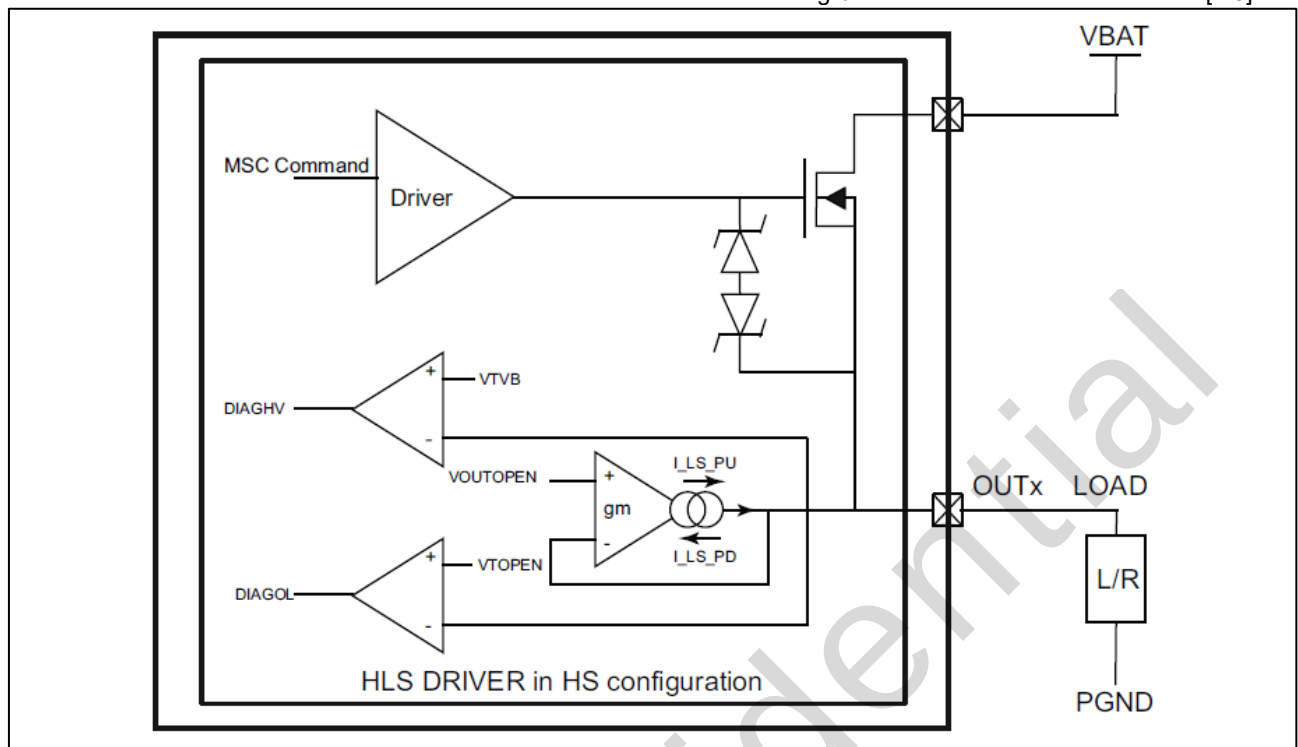
Refer to paragraph [Section 7.1.3](#)

## **8.3 OFF state diagnosis LS mode**

The OFF state diagnosis in LS mode has the same features as the low-side driver OFF state diagnosis described in paragraph [Section 7.1.4](#).

## **8.4 OFF state diagnosis HS mode**

The device provides off-state diagnostic for HS mode, simplified schematic of implemented diagnostic is shown in [Figure 8-2](#).



**Figure 8-2 OFF state diagnostic HS config block schematic**

In the HS mode we can have three different load conditions:

- Normal Load, it means output driver connected to the load, No Fault Present,  $V_{OUT} \leq V_{OL}$  Threshold;
- Open Load, It means output driver disconnected to the load, Open Load Fault Present,  $V_{HVT} \geq V_{OUT} \geq V_{OL}$  Threshold;
- Short To VB, it means output driver shorted to VB voltage, Short To VB Fault Present,  $V_{OUT} > V_{HVT}$ .

There is a bit *IPUPD\_EN\_STRx* in CONFIG\_REG13 which is controlled by a MSC bit, it is used by external  $\mu$ Controller if it is needed to switch on/off both pull up and pull down diagnosis current.

There is a bit *IPUPD\_MODE* in CONFIG\_REG2 which is controlled by a MSC bit, it is used by external

$\mu$ Controller if it is needed to switch on/off the bigger diagnosis current( $I_{HS\_PD1}$ ) in fast mode.

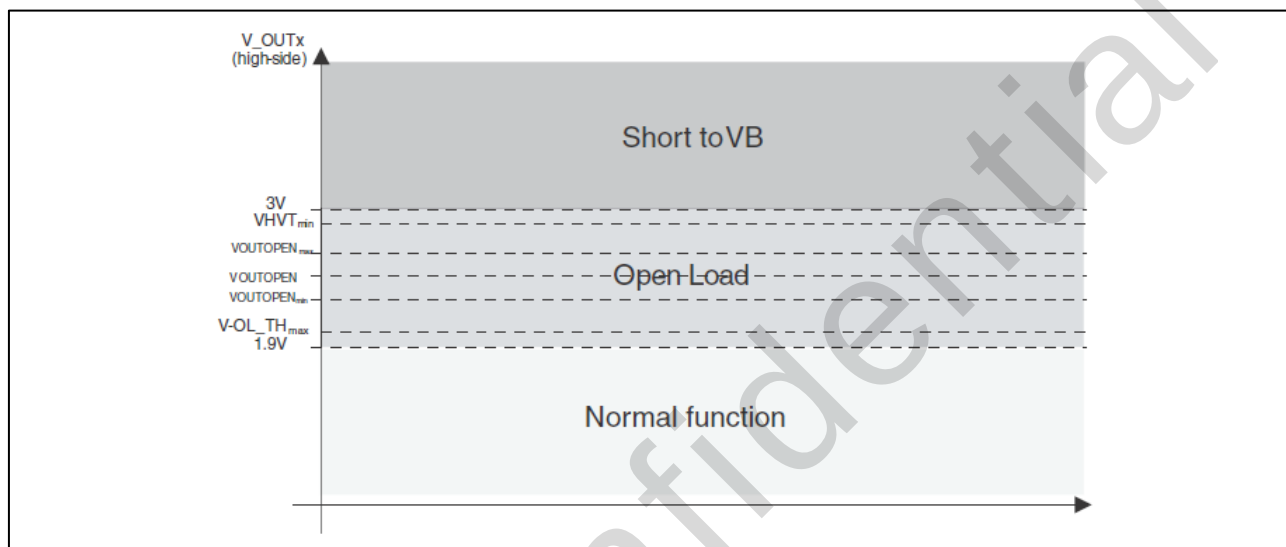
The Diagnostic in Off state is done by the following block:

- An internal Buffer using the pull up/down currents is able to force the  $V_{OUTOPEN}$  voltage on the High-Side Driver Output when the driver is in Open Load Condition.
- A comparator to detect the Open load Condition. The Open Load fault is detected when the  $OUTx$  pin is  $V_{HVT} \geq V_{OUT} \geq V_{OL}$  Threshold for a time longer than Diag Filter Time.
- A comparator to detect the Short to VB Condition. The Short To VB fault is detected

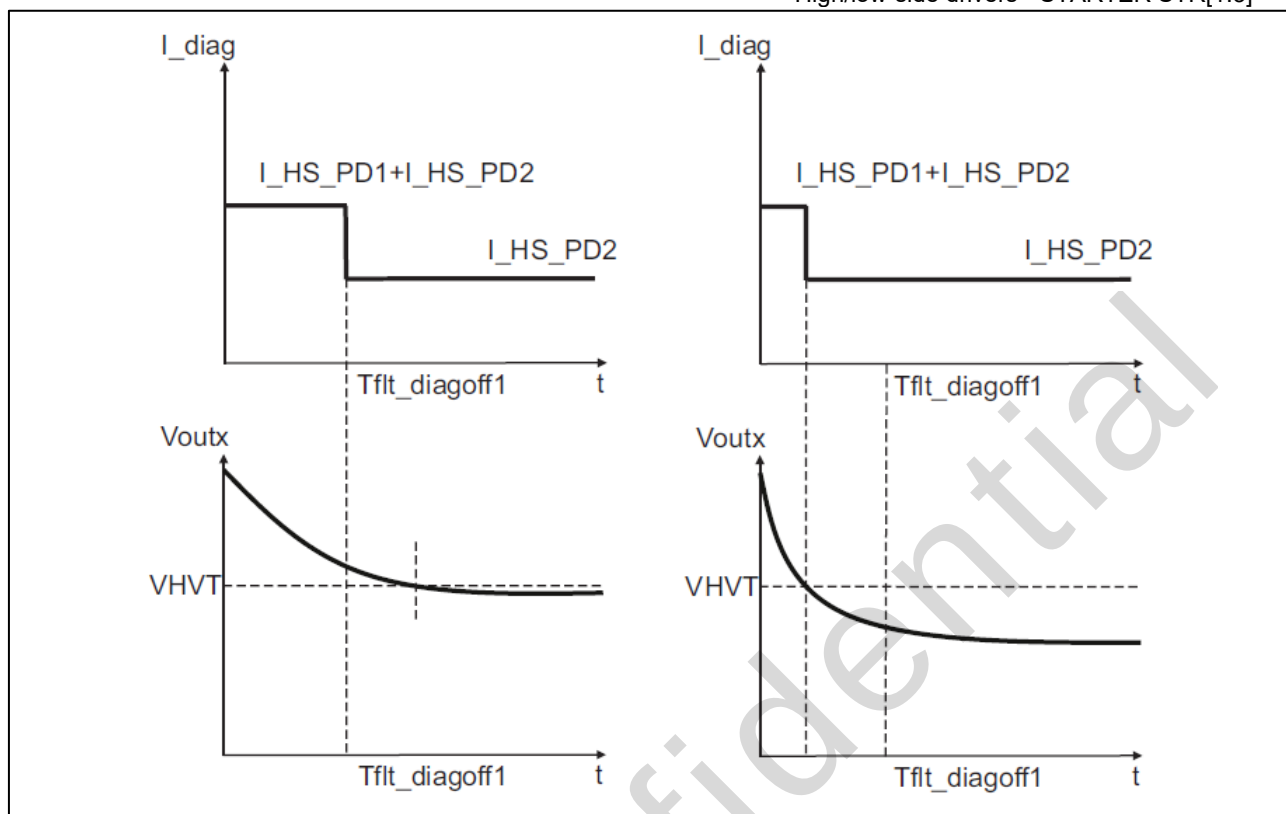
High/low-side drivers - STARTER STR[1:3]

when the OUTx pin is  $V_{OUT} \geq V_{HVT}$  Threshold for a time longer than Diag Filter Time.

- Fast pull down current, this current is used to avoid the false short to VB diagnosis when the driver switches off in Open load condition. This current switches on after the falling edge of *MSC command* (driver in off state) and remain on until the Diag Filter Time is expired or until the  $V_{OUT}$  is  $< V_{HVT}$  Threshold. (Note: if fast pull down current is selected, the filter time  $T_{flt\_diagoff1}$  is suggested. Otherwise, the filter time  $T_{flt\_diagoff2}$  is suggested).



**Figure 8-3** OFF state diagnostic HS config



**Figure 8-4 High-side driver OFF state fast pull down current behavior**

By default diagnostic pull up/down currents are disabled and comparator outputs are masked by internal logic, to enable OFF state diagnostic the channel must be put first in tristate condition and then (if not already done with previous MSC frames) diagnostic must be enabled. Once the desired channel is put in tristate a false diagnostic can be sensed due to the time needed to discharge output voltage through the load.

A Filter Time (Diag Filter Time) is implemented in order to avoid to detect false diagnosis as shown in [Figure 8-5](#).

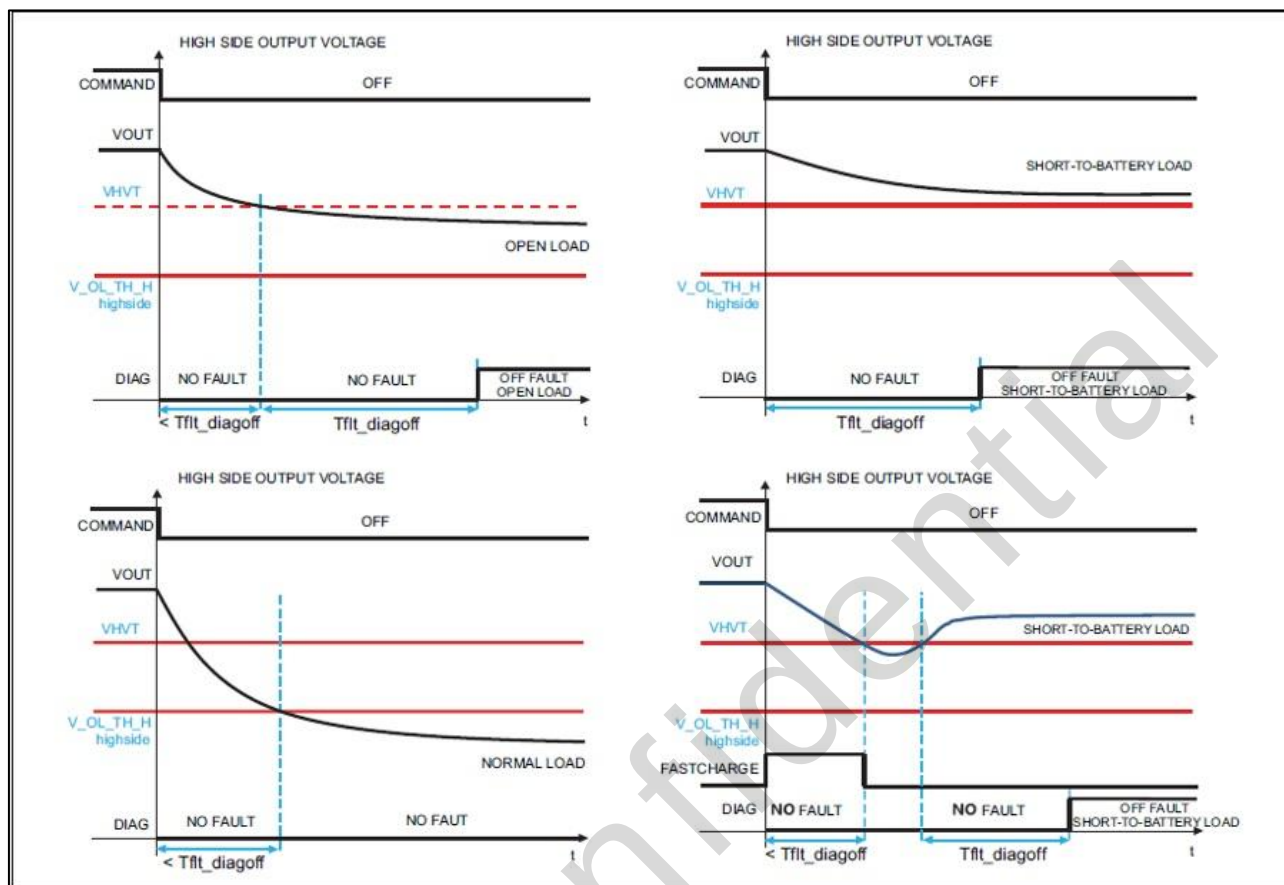


Figure 8-5 OFF state diagnostic high-side, timing

If DiagOL or DiagHV signal remains high for a time higher than Diag Filter Time the fault bit is set and can be read by Read Diag Communication.

This bit can be reset by every Read Diag Communication.

#### Conditions:

$5.5\text{ V} \leq \text{VB\_IN} \leq 18\text{ V}$ ,  $T_j -40$  to  $175\text{ }^\circ\text{C}$  unless otherwise specified;

Table 8-1 High/low-side driver - STARTER electrical characteristics

| Symbol                        | Parameter                               | Test condition                                                                         | Min | Typ | Max  | Unit | Pin      |
|-------------------------------|-----------------------------------------|----------------------------------------------------------------------------------------|-----|-----|------|------|----------|
| Irev_OUTx                     | Body diode reverse current voltage drop | I <sub>LOAD</sub> = -0.6 A                                                             | -   | -   | 1.5  | V    | STR1/2/3 |
| <b>Relay HS configuration</b> |                                         |                                                                                        |     |     |      |      |          |
| RdsON                         | Drain-source resistance                 | HS configuration<br>VB_IN = 13.5 V;<br>T <sub>j</sub> = 150 °C, I <sub>load</sub> = 1A | -   | -   | 1.5  | Ω    | STR1/2/3 |
| RdsON                         | Drain-source resistance                 | HS configuration<br>VB_IN = 13.5 V;<br>T <sub>j</sub> = 25 °C, I <sub>load</sub> = 1 A | -   | -   | 0.81 | Ω    | STR1/2/3 |

High/low-side drivers - STARTER STR[1:3]

|                               |                                                                                      |                                                                                                                                             |      |     |      |      |          |
|-------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------|-----|------|------|----------|
| RdsON                         | Drain-source resistance                                                              | HS configuration<br>VB_IN = 13.5 V;<br>T <sub>j</sub> = -40 °C, I <sub>load</sub> = 1 A                                                     | -    | -   | 0.63 | Ω    | STR1/2/3 |
| RdsON                         | Drain-source resistance                                                              | HS configuration<br>VB_IN = 4.25 V<br>I <sub>load</sub> = 1 A                                                                               | -    | -   | 3    | Ω    | STR1/2/3 |
| STR_SRC_lkg                   | Output leakage current                                                               | Driver disabled,<br>diagnostic off ; HSide<br>configuration                                                                                 | -10  | -   | +10  | μA   | STR1/2/3 |
| SR_ON                         | HS configuration<br>Voltage slew ON<br>State                                         | From 30% to 80% of<br>VOUT VB_IN = 14 V,<br>R <sub>load</sub> = 68Ω                                                                         | 0.6  | -   | 2.1  | V/μs | STR1/2/3 |
| SR_OFF                        | HS configuration<br>Voltage slew<br>OFF State                                        | From 80% to 30% of<br>VOUT VB_IN = 14 V,<br>R <sub>load</sub> = 68 Ω                                                                        | 0.7  | -   | 2.1  | V/μs | STR1/2/3 |
| S/RGkill_HS                   | FAST VS/R off<br>when an OVC<br>fault happens<br>high-side<br>configuration          | From 80% to 30% of<br>VOUT VB_IN = 14 V,<br>R <sub>load</sub> = 68Ω                                                                         | 5    | -   | 20   | V/μs | STR1/2/3 |
| Ton_OUTx_Hside                | Propagation<br>Delay from<br>MSC_EN rising<br>edge to 20%<br>output OUTx<br>voltage  | VB_IN = 14 V,<br>R <sub>load</sub> = 68Ω                                                                                                    | -    | -   | 10   | μs   | STR1/2/3 |
| Toff_OUTx_Hside               | Propagation<br>Delay from<br>MSC_EN falling<br>edge to 80%<br>output OUTx<br>voltage | VB_IN = 14 V,<br>R <sub>load</sub> = 68Ω                                                                                                    | -    | -   | 10   | μs   | STR1/2/3 |
| Vclamp_OUTx_HS                | OUTx clamping<br>voltage HS<br>configuration                                         | I <sub>LOAD</sub> = 0.6 A                                                                                                                   | -4.1 | -   | -2.5 | V    | STR1/2/3 |
| I_OVC_STR                     | Over current<br>driver threshold<br>HS configuration                                 | -                                                                                                                                           | 1    | 1.5 | 2    | A    | STR1/2/3 |
| I_HS_ocvflt                   | HS overcurrent<br>filter time                                                        | Guaranteed by scan                                                                                                                          | 4    | -   | 7    | μs   | STR1/2/3 |
| <b>Relay LS configuration</b> |                                                                                      |                                                                                                                                             |      |     |      |      |          |
| RdsON                         | Drain-source resistance                                                              | LS configuration<br>VB_IN = 13.5 V;<br>T <sub>j</sub> = 150 °C, I <sub>load</sub> = 1A                                                      | -    | -   | 1.5  | Ω    | STR1/2/3 |
| RdsON                         | Drain-source resistance                                                              | LS configuration<br>VB_IN = 13.5 V;<br>T <sub>j</sub> = 25 °C, I <sub>load</sub> = 1 A                                                      | -    | -   | 0.81 | Ω    | STR1/2/3 |
| RdsON                         | Drain-source resistance                                                              | LS configuration<br>VB_IN = 13.5 V;<br>T <sub>j</sub> = -40 °C, I <sub>load</sub> = 1<br>A                                                  | -    | -   | 0.63 | Ω    | STR1/2/3 |
| RdsON                         | Drain-source resistance                                                              | LS configuration<br>VB_IN = 4.25 V<br>I <sub>load</sub> = 1 A                                                                               | -    | -   | 3    | Ω    | STR1/2/3 |
| STR_DRN_lkg                   | Output leakage current                                                               | Output disabled,<br>diagnostic off ;<br>Low-Side configuration,<br>DRN1=DRN2=DRN3=1<br>8 V, total current from 3<br>pins, T = -40 °C, 27 °C | -10  | -   | +18  | μA   | STR1/2/3 |
| STR_DRN_lkg                   | Output leakage                                                                       | Output disabled,                                                                                                                            | -10  | -   | +48  | μA   | STR1/2/3 |

High/low-side drivers - STARTER STR[1:3]

|                                             |                                                                                      |                                                                                                                          |     |     |      |      |          |
|---------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----|-----|------|------|----------|
|                                             | current                                                                              | diagnostic off ;<br>Low-Side configuration,<br>DRN1=DRN2=DRN3=1<br>8 V, total current from 3<br>pins, T = 175 °C         |     |     |      |      |          |
| SR_ON                                       | LS configuration<br>voltage slew ON<br>State                                         | From 80% to 30% of<br>VOUT VB_IN = 14 V,<br>Rload = 68Ω                                                                  | 0.8 | -   | 2.5  | V/μs | STR1/2/3 |
| SR_OFF                                      | LS configuration<br>voltage slew<br>OFF State                                        | From 30% to 80% of<br>VOUT VB_IN = 14 V,<br>Rload = 68Ω,                                                                 | 0.8 | -   | 2.5  | V/μs | STR1/2/3 |
| S/RGkill_LS                                 | FAST VS/R off<br>when an OVC<br>fault happens in<br>low-side<br>configuration        | From 30% to 80% of<br>VOUT VB_IN = 14 V,<br>Rload = 68Ω,                                                                 | 5   | -   | 20   | V/μs | STR1/2/3 |
| Ton_OUTx_Lside                              | Propagation<br>Delay from<br>MSC_EN rising<br>edge to 80%<br>output OUTx<br>voltage  | VB_IN = 14 V,<br>Rload = 68Ω                                                                                             | -   | -   | 10   | μs   | STR1/2/3 |
| Toff_OUTx_Lside                             | Propagation<br>Delay from<br>MSC_EN falling<br>edge to 20%<br>output OUTx<br>voltage | VB_IN = 14 V,<br>Rload = 68Ω                                                                                             | -   | -   | 10   | μs   | STR1/2/3 |
| Vclamp_OUTx_LS                              | OUTx clamping<br>voltage LS<br>configuration                                         | ILOAD = 0.6 A                                                                                                            | 40  | -   | 50   | V    | STR1/2/3 |
| Vclamp_OUTx_LS_<br>high_curr                | OUTx clamping<br>voltage LS<br>configuration                                         | ILOAD = 1 A<br>Guaranteed by design,<br>provided single pulse<br>energy limits are not<br>violated by clamping<br>action | 40  | -   | 50   | V    | STR1/2/3 |
| I_OVC_STR                                   | Over current<br>driver threshold<br>LS configuration                                 | -                                                                                                                        | 1   | 1.5 | 2    | A    | STR1/2/3 |
| I_LS_ocv_fit                                | LS overcurrent<br>filter time                                                        | Guaranteed by scan                                                                                                       | 4   | -   | 7    | μs   | STR1/2/3 |
| <b>Driver reliability data as Low-Side</b>  |                                                                                      |                                                                                                                          |     |     |      |      |          |
| EnergyRep                                   | Energy<br>repetitive pulses                                                          | Tcase = 25 °C;<br>I_OT_n = 0.45 A<br>1.1 Miopulses                                                                       | -   | -   | 9    | mJ   | STR1/2/3 |
| EnergyRep                                   | Energy<br>repetitive pulses                                                          | Tc = 115 °C;<br>I_OT_n = 0.3 A<br>40 Miopulses                                                                           | -   | -   | 6.5  | mJ   | STR1/2/3 |
| EnergyRep                                   | Energy<br>repetitive pulses                                                          | Tc = 130 °C;<br>I_OT_n = 0.3 A<br>9 Miopulses                                                                            | -   | -   | 6.5  | mJ   | STR1/2/3 |
| EnergyRep                                   | Energy<br>repetitive pulses                                                          | Tc = 140 °C;<br>I_OT_n = 0.3 A<br>1 Miopulses                                                                            | -   | -   | 6.5  | mJ   | STR1/2/3 |
| <b>Driver reliability data as High-Side</b> |                                                                                      |                                                                                                                          |     |     |      |      |          |
| EnergyRep                                   | Energy<br>repetitive pulses                                                          | Tcase = 25 °C;<br>I_OT_n = 0.45 A<br>1.1 Miopulses                                                                       | -   | -   | 12.5 | mJ   | STR1/2/3 |
| EnergyRep                                   | Energy<br>repetitive pulses                                                          | Tc = 115 °C;<br>I_OT_n = 0.3 A                                                                                           | -   | -   | 9    | mJ   | STR1/2/3 |

High/low-side drivers - STARTER STR[1:3]

|                                                              |                                     |                                                                                                                                      |                                |     |                                    |    |          |
|--------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----|------------------------------------|----|----------|
|                                                              |                                     | 40 Miopulses                                                                                                                         |                                |     |                                    |    |          |
| EnergyRep                                                    | Energy repetitive pulses            | T <sub>c</sub> = 130 °C;<br>I <sub>OT_n</sub> = 0.3 A<br>9 Miopulses                                                                 | -                              | -   | 9                                  | mJ | STR1/2/3 |
| EnergyRep                                                    | Energy repetitive pulses            | T <sub>c</sub> = 140 °C;<br>I <sub>OT_n</sub> = 0.3 A<br>1 Miopulses                                                                 | -                              | -   | 9                                  | mJ | STR1/2/3 |
| <b>Driver reliability data generator defect as Low-Side</b>  |                                     |                                                                                                                                      |                                |     |                                    |    |          |
| EnergyRep STR                                                | Energy repetitive pulses            | T <sub>c</sub> = 25 °C ;<br>I <sub>OT_n</sub> = 0.5 A<br>0.02 Miopulses                                                              | -                              | -   | 11                                 | mJ | STR1/2/3 |
| EnergyRep STR                                                | Energy repetitive pulses            | T <sub>c</sub> = 135 °C ;<br>I <sub>OT_n</sub> = 0.35 A<br>0.02 Miopulses                                                            | -                              | -   | 8                                  | mJ | STR1/2/3 |
| <b>Driver reliability data generator defect as High-Side</b> |                                     |                                                                                                                                      |                                |     |                                    |    |          |
| EnergyRep STR                                                | Energy repetitive pulses            | T <sub>c</sub> = 25 °C ;<br>I <sub>OT_n</sub> = 0.5 A<br>0.02 Miopulses                                                              | -                              | -   | 15                                 | mJ | STR1/2/3 |
| EnergyRep STR                                                | Energy repetitive pulses            | T <sub>c</sub> = 135 °C ;<br>I <sub>OT_n</sub> = 0.35 A<br>0.02 Miopulses                                                            | -                              | -   | 11                                 | mJ | STR1/2/3 |
| <b>Driver reliability data JUMP START as Low-Side</b>        |                                     |                                                                                                                                      |                                |     |                                    |    |          |
| EnergyRep STR                                                | Energy repetitive pulses            | T <sub>c</sub> = 25 °C ;<br>I <sub>OT_n</sub> = 0.75A<br>0.001 Miopulses,<br>10 jumps starts over<br>lifetime, each start < 2<br>min | -                              | -   | 25                                 | mJ | STR1/2/3 |
| EnergyRep STR                                                | Energy repetitive pulses            | T <sub>c</sub> = 75 °C ;<br>I <sub>OT_n</sub> = 0.5 A<br>0.001 Miopulses,<br>10 jumps starts over<br>lifetime, each start < 2<br>min | -                              | -   | 17                                 | mJ | STR1/2/3 |
| <b>Driver reliability data JUMP START as High-Side</b>       |                                     |                                                                                                                                      |                                |     |                                    |    |          |
| EnergyRep STR                                                | Energy repetitive pulses            | T <sub>c</sub> = 25 °C ;<br>I <sub>OT_n</sub> = 0.75A<br>0.001 Miopulses,<br>10 jumps starts over<br>lifetime, each start < 2<br>min | -                              | -   | 31                                 | mJ | STR1/2/3 |
| EnergyRep STR                                                | Energy repetitive pulses            | T <sub>c</sub> = 75 °C ;<br>I <sub>OT_n</sub> = 0.5 A<br>0.001 Miopulses,<br>10 jumps starts over<br>lifetime, each start < 2<br>min | -                              | -   | 21                                 | mJ | STR1/2/3 |
| <b>OFF state diagnostic - HS configuration</b>               |                                     |                                                                                                                                      |                                |     |                                    |    |          |
| VHVT                                                         | Short to VB<br>threshold<br>voltage | HS configuration                                                                                                                     | VOU<br>TOPE<br>N<br>+160<br>mV | -   | 3.0                                | V  | STR1/2/3 |
| V_OL_TH_Highside                                             | Open load<br>threshold<br>voltage   | HS configuration                                                                                                                     | 1.9                            | -   | VOU<br>TOPE<br>N<br>-<br>180m<br>V | V  | STR1/2/3 |
| VOUTOPEN_High<br>side                                        | Open load output<br>voltage         | HS configuration                                                                                                                     | 2.3                            | 2.5 | 2.7                                | V  | STR1/2/3 |

High/low-side drivers - STARTER STR[1:3]

|                                                |                                                                                                                                                                            |                                                                                           |      |      |            |    |          |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------|------|------------|----|----------|
| I_HS_PD1                                       | Diagnostic pull down current in high-side configuration                                                                                                                    | In OFF condition, OUTx STRx_SRC > VHVT, Ipupd_EN IPUPD_EN_STRx = ENB, Ipupd_MODE = "1"    | 2.5  | 3.6  | 4.7        | mA | STR1/2/3 |
| I_HS_PD2                                       |                                                                                                                                                                            | In OFF condition, OUTx STRx_SRC > VHVT, Ipupd_EN IPUPD_EN_STRx = ENB, Ipupd_MODE = "0"    | 40   | 70   | 100        | μA | STR1/2/3 |
| I_HS_PU1                                       | Diagnostic pull up current in high-side configuration                                                                                                                      | In OFF condition, OUTx STRx_SRC < VOUTOPEN, Ipupd_EN IPUPD_EN_STRx = ENB STR_SRCx = 1.9 V | 35   | -    | 75         | μA | STR1/2/3 |
| I_HS_PU2                                       |                                                                                                                                                                            | In OFF condition, OUTx STRx_SRC < VOUTOPEN, Ipupd_EN IPUPD_EN_STRx = ENB STR_SRCx = 0 V   | 100  | -    | 180        | μA | STR1/2/3 |
| I_HS_OUT_Dis1                                  | Output current in high-side configuration (measured on STR_SRC)                                                                                                            | OFF condition, Diagoff disabled, STR_SRC > or = 0 V                                       | -15  | -    | +15        | μA | STR1/2/3 |
| I_HS_OUT_Dis2                                  |                                                                                                                                                                            | OFF condition, Diagoff disabled, STR_SRC < 0 V                                            | -190 | -    | +10        | μA | STR1/2/3 |
| V_negative_threshold                           | Output voltage of STR_SRC below which the output current is measured again                                                                                                 | OFF condition, Diagoff disabled<br>Guaranteed by design                                   | -    | -    | -0.2       | V  | STR1/2/3 |
| V_positive_threshold                           | Positive threshold of internal comparator responsible of remove output current on STR_SRC pin (over this voltage negligible current is guaranteed by compensation circuit) | OFF condition, Diagoff disabled<br>Guaranteed by design                                   | -    | 1.8  | -          | V  | STR1/2/3 |
| Tflt_diagoff2                                  | -                                                                                                                                                                          | Guaranteed by scan<br>Cload = 15 nF                                                       | 1000 | 1200 | 1500       | us | STR1/2/3 |
| Tflt_diagoff1                                  | -                                                                                                                                                                          | Guaranteed by scan<br>Cload = 15 nF                                                       | 130  | 160  | 200        | us | STR1/2/3 |
| <b>OFF state diagnostic - LS configuration</b> |                                                                                                                                                                            |                                                                                           |      |      |            |    |          |
| VLVT                                           | Short to GND threshold voltage                                                                                                                                             | LS configuration                                                                          | 1.9  | -    | VOU TOPE N | V  | STR1/2/3 |

High/low-side drivers - STARTER STR[1:3]

|                                                |                                                                                            |                                                                        |                                |     |       |    |          |
|------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------|-----|-------|----|----------|
|                                                |                                                                                            |                                                                        |                                |     | 180mV |    |          |
| V_OL_TH_Lowside                                | Open load threshold voltage                                                                | LS configuration                                                       | VOU<br>TOPE<br>N<br>+160<br>mV | -   | 3.0   | V  | STR1/2/3 |
| VOUTOPEN_Lowside                               | Open load output voltage                                                                   | LS configuration                                                       | 2.3                            | 2.5 | 2.7   | V  | STR1/2/3 |
| I_LS_PU1                                       | Diagnostic pull up current in Ls configuration                                             | In OFF condition, STRx_DRN < VLVT, IPUPD_EN_STRx =ENB, Ipupd_MODE ="1" | 2.5                            | 3.6 | 4.7   | mA | STR1/2/3 |
| I_LS_PU2                                       |                                                                                            | In OFF condition, STRx_DRN < VLVT, IPUPD_EN_STRx =ENB, Ipupd_MODE ="0" | 40                             | 70  | 100   | μA | STR1/2/3 |
| I_LS_PD1                                       | Diagnostic pull down current in Ls configuration                                           | In OFF condition, STRx_DRN > VOUTOPEN, IPUPD_EN_STRx =ENB              | 55                             | 85  | 100   | μA | STR1/2/3 |
| Tflt_diagoff2                                  | DIAG Filter time in Ls configuration Filter Mode = 1                                       | Guaranteed by scan Cload = 15 nF                                       | 450                            | 600 | 750   | μs | STR1/2/3 |
| Tflt_diagoff1                                  | DIAG Filter time in Ls configuration fast discharge enable Filter Mode = 0 (fast diag off) | Guaranteed by scan Cload = 15 nF                                       | 75                             | 100 | 125   | μs | STR1/2/3 |
| <b>Common diagnostic - LS/HS configuration</b> |                                                                                            |                                                                        |                                |     |       |    |          |
| T_SD_HIGH                                      | Temperature shut down                                                                      | -                                                                      | 185                            | -   | 200   | °C | STR1/2/3 |
| T_SD_LOW                                       | Temperature shut down recover                                                              | -                                                                      | 175                            | -   | 190   | °C | STR1/2/3 |
| T_SD_hys                                       | Temperature shut down hysteresis                                                           | -                                                                      | 5                              | -   | 10    | °C | STR1/2/3 |
| -                                              | Thermal shutdown analog filter time                                                        | Guaranteed by design                                                   | 1.5                            | -   | 4     | μs | STR1/2/3 |
| t_SD_deglitch                                  | Digital deglitch filter time on Temperature shut down detection                            | Guaranteed by scan                                                     | -                              | 10  | -     | μs | STR1/2/3 |
| -                                              | Fast OFF diagnostic ON->OFF->ON                                                            | Filter Mode=0                                                          | 220                            | 245 | 270   | μs | STR1/2/3 |
| -                                              | Fast ON diagnostic OFF->ON->OFF                                                            | -                                                                      | 65                             | -   | 90    | μs | STR1/2/3 |

## 8.5 Delay-off function

The Delay-off function involves the RLY4 and STR[2:3] drivers when they are configured as starters by the bit `MSC_CONFIG_REG3[0]=RLY4_DELAY_OFF_EN=1`, `CONFIG_REG3[4]=STR2_EN=1`, `CONFIG_REG3[5]=STR3_EN=1`. These drivers follow the reset matrix as other drivers when they are not configured as starters.

If the drivers is configured as starter and it is in on state its delay-off timer starts when one of the following events occurs:

- under-voltage of the main supply VDD5 is detected,
- over-voltage of the main supply VDD5 is detected,
- the MSC time out occurs,
- an active signal ("0") at pin WDA(IN).
- an active signal ("0") at pin RSTN(IN).

During the delay off time THOLD

- The driver is ON regardless of whether the ON/OFF command is reset.
- The starter configuration is held.
- The HSD/LSD configuration bit is held.
- The driver is ON even in case of Watchdog `ERR_CNT>4` and `ERR_CNT>7`
- The driver is ON even in case of `VDDIO_UV`.
- Once it has started, the delay\_off is not restarted by a new triggering event before THOLD has expired.

The delay-off timer can be terminated by one of the following events:

THOLD time expires.

Driver is switched off by *MSC command off*.

After DELAY\_OFF time the RLY4, STR[2:3] take the status of the on/off data command. If before the delay off time expires, the above triggering sources are recovered, the delay-off function remains valid.

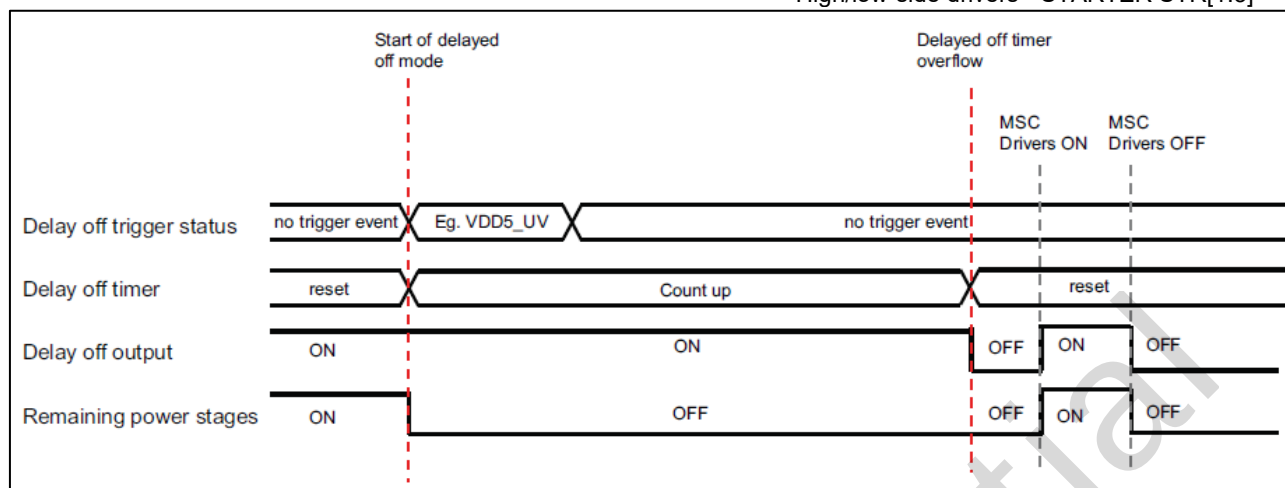


Figure 8-6 DELAY\_OFF timing

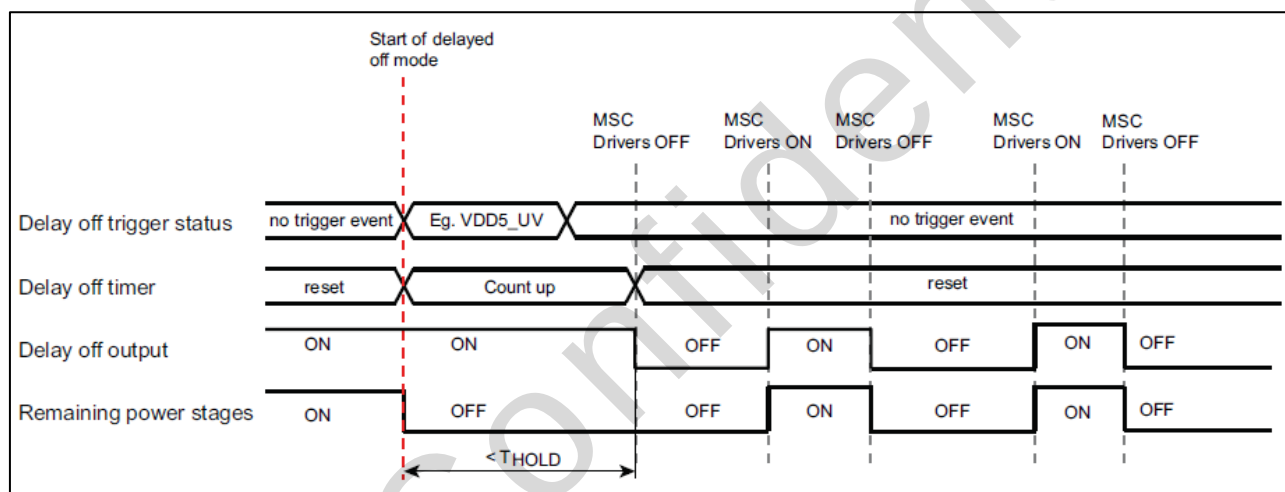


Figure 8-7 DELAY\_OFF timing terminated by MSC off command

**Conditions:**
 $5.5\text{ V} \leq V_{B\_IN} \leq 18\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified;

Table 8-2 High/low-side driver delay-off function electrical characteristics

| Symbol | Parameter      | Test condition | Min | Typ | Max | Unit | Pin         |
|--------|----------------|----------------|-----|-----|-----|------|-------------|
| THOLD  | Delay off time | -              | 400 | -   | 800 | ms   | RLY4 STR2/3 |

## 8.6 Starter control

To prevent unintended activation of starter relays at the same time a dual control mechanism is implemented inside the logic by means of independent MSC commands for on/off for each starter (*RLY4\_ON RLY4\_OFF STR2\_ON STR2\_OFF STR3\_ON STR3\_OFF*). If the RLY4, STR2, STR3 are not configured as starter they are controlled by data frame on/off as the other drivers

The starter control function comes with a dedicated failsafe management circuit to always provide a safety switch off path able to disconnect the main supply in case of fault conditions. Starter switch off path is implemented with dual polarity command acting on drivers STR[2:3] and RLY4: if the EN\_P pin is high or the EN\_N pin is high or if the FAULT\_WARN is activated. RLY4, STR2, STR3 are switched off by the safety switched off path independently if they are configured as starter.

Safety switch off path can be triggered both by the micro-controller through EN pins or by the HE9788 detection of an internal fault as described in the [Figure 8-8](#). The functionality of the switch off path is guaranteed through physical isolation from the rest of the device: the circuit is kept layouted in a dedicated area surrounded by trench isolations, avoiding top routing over this cell and keeping a safe distance respect to the nearby functional top routing to ensure complete independency.

The switch off path generates locally its supply voltage to drive the safety turn-off switch in order to avoid interactions with the rest of the HE9788; the supply voltage is created starting directly from battery line. All the circuits used inside the switch-off path are able to sustain high voltage to prevent unwanted functionality in case of local failure, moreover no external reference is routed to this area and the EN\_N and EN\_P control logic is located inside this area as well to complete isolation and remove any dependency from the rest of the HE9788.

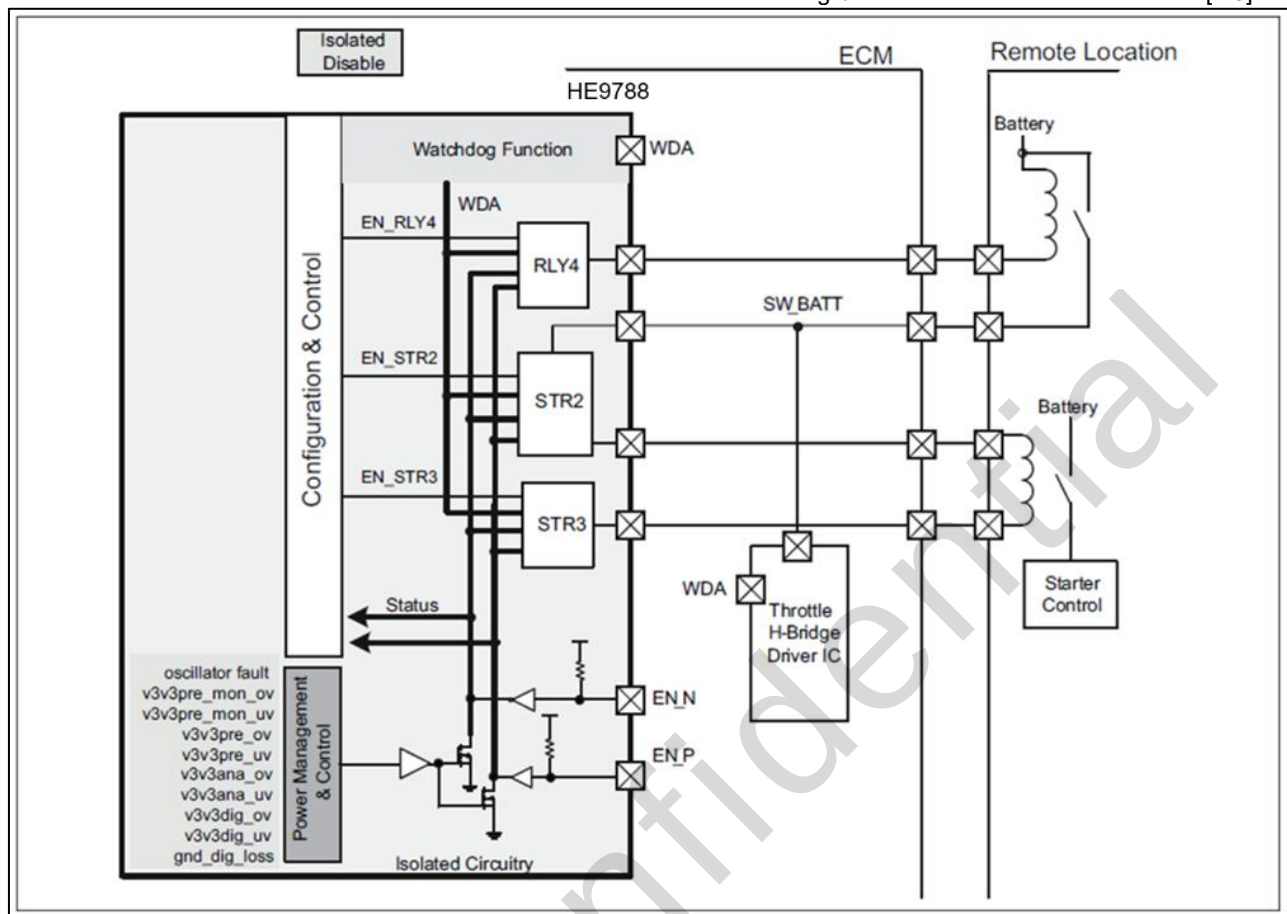


Figure 8-8 Safety switch off

**Conditions:**

$5.5\text{ V} \leq V_{B\_IN} \leq 18\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified;

**Table 8-3 EN\_P and EN\_N electrical characteristics**

| Symbol  | Parameter                | Test condition | Min  | Typ | Max  | Unit | Pin  |
|---------|--------------------------|----------------|------|-----|------|------|------|
| VIH_TX  | HIGH_level input voltage | -              | 1.75 | -   | -    | V    | EN_P |
| VIL_TX  | LOW_level input voltage  | -              | -    | -   | 0.75 | V    | EN_P |
| VHYS_TX | Input voltage hysteresis | -              | 0.06 | 0.5 | -    | V    | EN_P |
| PULL_TX | Pull-up resistance       | -              | 50   | 100 | 200  | kΩ   | EN_P |
| VIH_TX  | HIGH_level input voltage | -              | 1.75 | -   | -    | V    | EN_N |
| VIL_TX  | LOW_level input voltage  | -              | -    | -   | 0.75 | V    | EN_N |
| VHYS_TX | Input voltage hysteresis | -              | 0.06 | 0.5 | -    | V    | EN_N |
| PULL_TX | Pull-up resistance       | -              | 50   | 100 | 200  | kΩ   | EN_N |

*Note: For this application the function is implemented, but kept disabled using external resistor to pull EN\_N and EN\_P to low.*

## 9 Pre-drivers - ExtFET (MOS\_Gate1/2/3/4/5) - PRD[1:5]

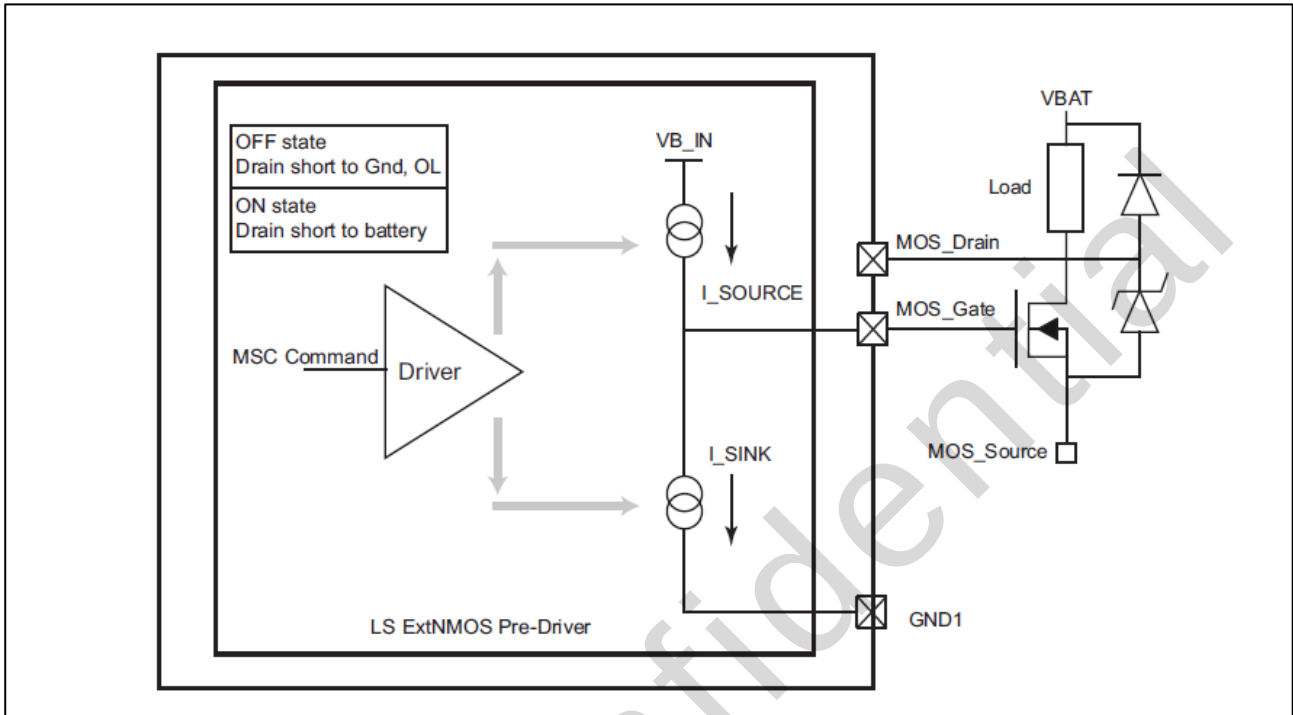


Figure 9-1 LS External MOSFET Pre-Driver Stage

### 9.1 LS external MOSFET pre-driver

These 5 pre-drivers are designed to drive LS external MOSFET Load. They are driven by *MSC command*.

The push-pull stage is made up of limited source current and sink current. The voltage of MOS\_Drain is monitored for diagnosis.

The driver has the following detection diagnosis:

- \_ Drain short to battery @ pre-driver = On state
- \_ Drain short to ground @ pre-driver = Off state
- \_ Drain open load @ pre-driver = Off state

Predriver1 & Predriver3 can be used to drive ExtMos for O2H Load.

For these two predrivers there is the possibility to select a lower gate current and sink current (typ = 250  $\mu$ A).

A dedicated MSC bit is used to select the predriver O2H function (Bit *O2H\_PDRV* = *CONFIG\_REG\_16\_1[7:6]*). Considering that this sink/source current is very low the logic will switch to the selected higher sink/source current after the cmd delay time (typ 150  $\mu$ s for *CONFIG\_REG20[7]=PDRV\_O2H\_DLY=0* or 300  $\mu$ s

Pre-drivers - ExtFET (MOS\_Gate1/2/3/4/5) - PRD[1:5]

$CONFIG\_REG20[7]=PDRV\_O2H\_DLY=1$ ). After the CMD delay time the current used is the one selected by  $CONFIG\_REG4=PRDx\_IDRV[1:0]$ .

When the Predriver is used for O2H load the overcurrent detection can't be done using a VDS comparator due to slow voltage slew-rate, for this reason has to be used and external Rshunt.

When the Predriver is used for O2H load no blanking time is applied to OVC detection. The OVC detection is done reading the voltage drop on this external Rshunt by the VDS comparator of another predriver (not used).

This means that when the Predriver 1 and Predriver 3 are used for O2H load the Predriver 2 and Predriver 4 can't be used.

Bit  $O2H\_PDRV = CONFIG\_REG\_16\_1[7:6] = 00$ : all 5 predriver channels can be used

Bit  $O2H\_PDRV = CONFIG\_REG\_16\_1[7:6] = 11$ :

predriver 2 & Predriver 4 can't be used;

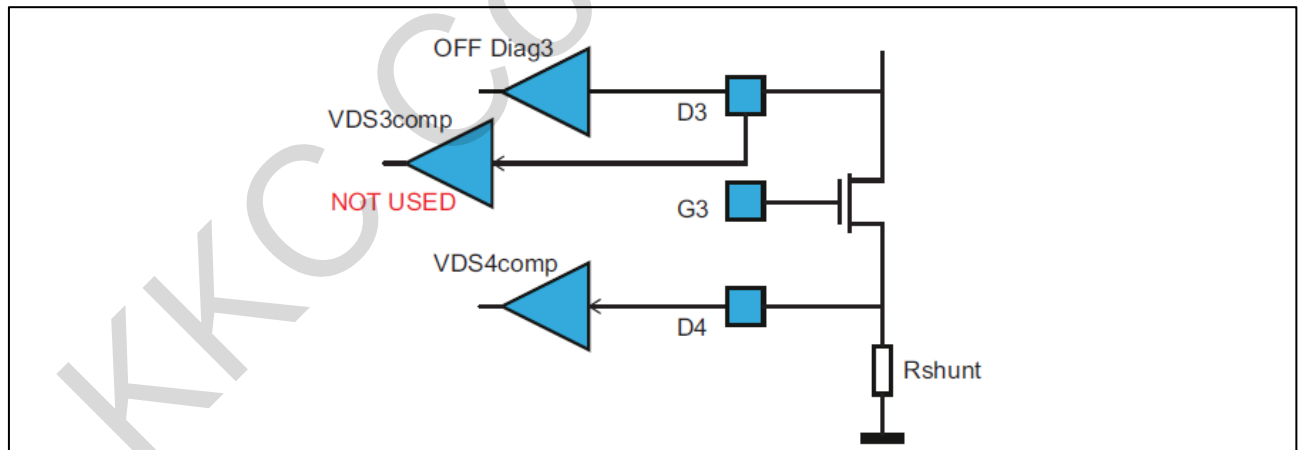
predriver 1 & predriver3 work with low gate current and sink current (typ = 250 uA);

it is possible read the Overcurrent on external Rshunt of Predriver1&3 using the Vds comparator of Drain2 and Drain4;

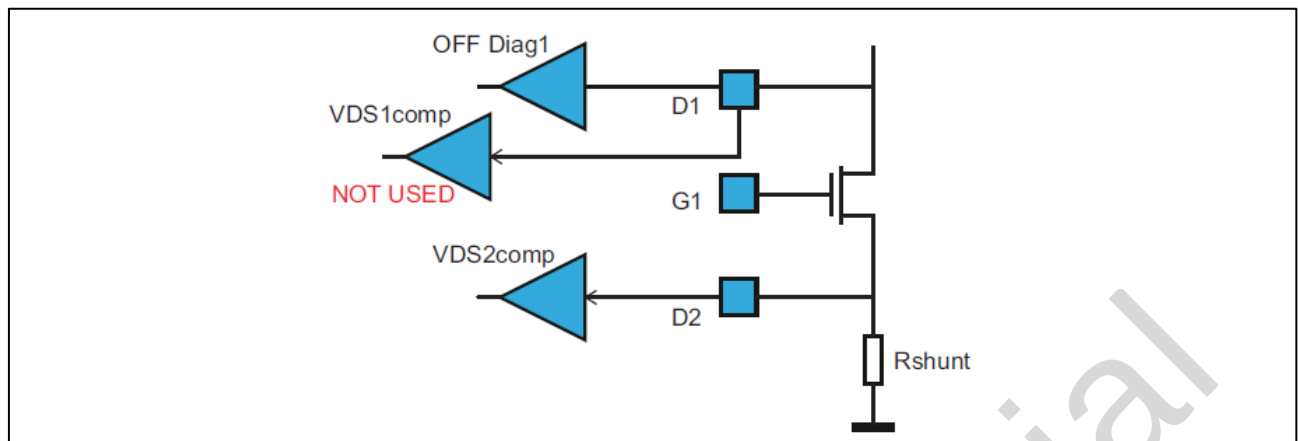
Predriver 5 works as in  $O2H\_PDRV = CONFIG\_REG\_16\_1[7:6] = 00$ .

There is a dedicated ground pin PDR\_GND used for source pin of External Mos or ground connection of External shunt resistor. The internal voltage reference for VDS comparator threshold ( $V_{th\_VDS\_xxx}$  parameter) is connected to this ground pin PDR\_GND. Application has to connect this PDR\_GND pin to the source of external MOS or ground of external shunt resistor in order to obtain a good accuracy of overcurrent threshold.

Pre-driver after power up is OFF, output LOW.



**Figure 9-2** PreDriver3 configured for O2H load bit  $O2H\_PDRV\_3 = CONFIG\_REG\_16\_1[7] = 1$ :



**Figure 9-3** PreDriver1 configured for O2H load bit O2H\_PDRV\_1 = CONFIG\_REG\_16\_1[6] = 1:

## 9.2 ON state diagnostic

In the On state of pre-driver, the diagnosis will compare the Drain-source voltage with the threshold generated internal. If  $MOS\_DRAIN\_X > V_{th\_DS\_XXX}$  ( $PRDx\_VDS[0:2]$  *CONFIG\_REG5* / *CONFIG\_REG6* / *CONFIG\_REG7*) for a filter time, the Drain short to battery fault will be confirmed. The internal comparator reads the voltage on drain pin versus a dedicated GND sense pin (PIN X) that at application should be connected on the source of external mos. Starting from the condition ON state a programmable blanking time (*MSC CONFIG\_REG16=PRDx\_BLK[1:0]*) is applied to prevent the detection of the short-to battery fault.

The blanking time is not applied when the driver is already ON. The pre-driver will switch OFF and the fault will be latched until the Read\_Diag\_CMD action. This bit can be reset by every Read Diag Communication. To re-activate ON the power stage in these two following conditions:

The MSC command still High, when the Fault OVC Bit is read by MSC communication.

The MSC command becomes Low and then again High.

When Bit MSC\_O2HPredr\_1 = 1 the VDS comparator of predriver 1 is ignored.

When Bit MSC\_O2HPredr\_3 = 1 the VDS comparator of predriver 3 is ignored.

When Bit *MSC\_O2HPredr* = 1 the VDS comparator of predriver 2 (or 4) is enabled immediately with the turn ON command of Predriver (and checked by logic after the blanking time), while when Bit *MSC\_O2HPredr* = 0 the VDS comparator of corresponding channel is enabled after selected blanking time.

The over current protection can be done reading the voltage on external Rshunt using the VDS comparator of predriver 2&4.

### 9.3 ON/OFF state - Error in status diagnosis

Refer to [Section 7.1.3](#)

## 9.4 Error handling

**Table 9-1 Pre-driver - ExtFET error handling**

| Type of error             | Detection condition                                                           | Action                                                                        | Clear MSC flag | Restart condition                                                                                     |
|---------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------|
| <b>Driver diagnostic</b>  |                                                                               |                                                                               |                |                                                                                                       |
| Overcurrent (OVC)         | Driver on                                                                     | Driver is put in off state.<br>The fault is latched in MSC diagnosis register | On read        | At MSC read of diagnosis register if MSC command is still high. If MSC command is low and high again. |
| Short to ground (STG)     | Driver off                                                                    | The fault is latched in MSC diagnosis register.                               | On read        |                                                                                                       |
| Open load (OPL)           | Driver off                                                                    | The fault is latched in MSC diagnosis register.                               | On read        |                                                                                                       |
| Driver Status Error (STA) | Driver on/off When the driver output level is not aligned with the cmd on/off | The fault is latched in MSC diagnosis register                                | On read        |                                                                                                       |

## 9.5 OFF state diagnostic

The Predriver follows the same off diagnosis as low-side driver. But despite the fast mode of all drivers in off diagnostic condition (*lpupd\_MODE* bit), for all pre\_Driver, there is a dedicated MSC bit (*IDIAG\_HIGH\_PDRV*) to select off diagnostic high or low pull up/down current

### Conditions:

$5.5\text{ V} \leq \text{VB\_IN} \leq 18\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified;

**Table 9-2 Pre-driver - ExtFET electrical characteristics**

| Symbol            | Parameter                                         | Test condition                 | Min | Typ | Max | Unit | Pin                |
|-------------------|---------------------------------------------------|--------------------------------|-----|-----|-----|------|--------------------|
| <b>Pre-driver</b> |                                                   |                                |     |     |     |      |                    |
| Vgate_on          | On state MOS_Gate voltage                         | I_source_00                    | 5   | -   | 6   | V    | MOS_Gate 1/2/3/4/5 |
| Vgate_off         | Off state MOS_Gate voltage                        | I_sink_00                      | 0   | -   | 0.1 | V    | MOS_Gate 1/2/3/4/5 |
| Ileak_off         | Leakage current of MOS_Gate @ driver in tri-state |                                | -1  | -   | 1   | μA   | MOS_Gate 1/2/3/4/5 |
| I_source_00       | Pre-driver source current 1                       | MOS_Gate1/2/3/4/5 short to gnd | 14  | 20  | 26  | mA   | MOS_Gate 1/2/3/4/5 |
| I_source_01       | Pre-driver source current 2                       | MOS_Gate1/2/3/4/5 short to gnd | 7   | 10  | 13  | mA   | MOS_Gate 1/2/3/4/5 |
| I_source_10       | Pre-driver                                        | MOS_Gate1/2/3/4/5              | 2.8 | 4   | 5.2 | mA   | MOS_Gate           |

Pre-drivers - ExtFET (MOS\_Gate1/2/3/4/5) - PRD[1:5]

|                                |                                               |                                                                                                 |                 |       |                 |    |                     |
|--------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------|-------|-----------------|----|---------------------|
|                                | source current 3                              | short to gnd                                                                                    |                 |       |                 |    | 1/2/3/4/5           |
| I_source_11                    | Pre-driver source current 4                   | MOS_Gate1/2/3/4/5 short to gnd                                                                  | 1.3             | 2     | 2.6             | mA | MOS_Gate 1/2/3/4/5  |
| I_source_O2H                   | Pre-driver source current 5                   | MOS_Gate1/3/ short to gnd                                                                       | 0.155           | 0.25  | 0.355           | mA | MOS_Gate 1/3        |
| I_sink_00                      | Pre-driver sink current 1                     | MOS_Gate1/2/3/4/5 short to 5 V                                                                  | -26             | -20   | -14             | mA | MOS_Gate 1/2/3/4/5  |
| I_sink_01                      | Pre-driver sink current 2                     | MOS_Gate1/2/3/4/5 short to 5 V                                                                  | -13             | -10   | -7              | mA | MOS_Gate 1/2/3/4/5  |
| I_sink_10                      | Pre-driver sink current 3                     | MOS_Gate1/2/3/4/5 short to 5 V                                                                  | -5.2            | -4    | -2.8            | mA | MOS_Gate 1/2/3/4/5  |
| I_sink_11                      | Pre-driver sink current 4                     | MOS_Gate1/2/3/4/5 short to 5 V                                                                  | -2.6            | -2    | -1.4            | mA | MOS_Gate 1/2/3/4/5  |
| I_sink_O2H                     | Pre-driver sink current 5                     | MOS_Gate1/3 short to 5V                                                                         | -0.355          | -0.25 | -0.155          | mA | MOS_Gate 1/3        |
| <b>ON/OFF state diagnostic</b> |                                               |                                                                                                 |                 |       |                 |    |                     |
| Vth_DS_000                     | Diagnosis Vds threshold 1                     |                                                                                                 | 120             | 150   | 190             | mV | MOS_Drai n1/2/3/4/5 |
| Vth_DS_001                     | Diagnosis Vds threshold 2                     |                                                                                                 | 220             | 245   | 285             | mV | MOS_Drai n1/2/3/4/5 |
| Vth_DS_010                     | Diagnosis Vds threshold 3                     |                                                                                                 | 300             | 325   | 360             | mV | MOS_Drai n1/2/3/4/5 |
| Vth_DS_011                     | Diagnosis Vds threshold 4                     |                                                                                                 | 380             | 405   | 445             | mV | MOS_Drai n1/2/3/4/5 |
| Vth_DS_100                     | Diagnosis Vds threshold 5                     |                                                                                                 | 500             | 525   | 560             | mV | MOS_Drai n1/2/3/4/5 |
| Vth_DS_101                     | Diagnosis Vds threshold 6                     |                                                                                                 | 620             | 660   | 710             | mV | MOS_Drai n1/2/3/4/5 |
| Vth_DS_110                     | Diagnosis Vds threshold 7                     |                                                                                                 | 900             | 950   | 1000            | mV | MOS_Drai n1/2/3/4/5 |
| T_filter_ovc                   | Over current filtering time                   | Tested by scan                                                                                  | 5               | -     | 7               | μs | MOS_Drai n1/2/3/4/5 |
| V_OL                           | Open load threshold voltage                   | Driver tristate, diag enabled                                                                   | VOUTOPEN +160mV | -     | 3               | V  | MOS_Drai n1/2/3/4/5 |
| VOUTOPEN                       | Open load output voltage                      | Driver tristate, diag enabled                                                                   | 2.3             | 2.5   | 2.7             | V  | MOS_Drai n1/2/3/4/5 |
| VLVT                           | Output short circuit to GND Voltage threshold | Driver tristate, diag enabled                                                                   | 1.9             | -     | VOUTOPEN -180mV | V  | MOS_Drai n1/2/3/4/5 |
| I_LS_PU1                       | Diagnostic pull up current                    | In OFF condition, MOS_DRAINx < VLVT, lpupd_EN = ENB, lpupd_MODE = "1"                           | 2.5             | 3.6   | 4.7             | mA | MOS_Drai n1/2/3/4/5 |
| I_LS_PU2                       | Diagnostic pull up current                    | In OFF condition, MOS_DRAINx < VLVT, lpupd_EN = ENB, lpupd_MODE = "0" and IDIAG_HIGH_PDRV = "0" | 40              | 70    | 100             | μA | MOS_Drai n1/2/3/4/5 |
| I_LS_PU3                       | Diagnostic pull up high current               | In OFF condition, MOS_Drainx < VLVT, lpupd_EN = ENB,                                            | 100             | -     | 200             | μA | MOS_Drai n1/2/3/4/5 |

|               |                                                                    |                                                                                                     |      |     |      |    |                        |
|---------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------|-----|------|----|------------------------|
|               |                                                                    | lpupd_MODE = "0"<br>and<br>IDIAG_HIGH_PDRV<br>= "1"                                                 |      |     |      |    |                        |
| I_LS_PD1      | Diagnostic pull<br>down current                                    | In OFF condition,<br>MOS_DRAINx<br>> VOUTOPEN,<br>lpupd_EN = ENB<br>and<br>IDIAG_HIGH_PDRV<br>= "0" | 60   | 85  | 105  | μA | MOS_Drai<br>n1/2/3/4/5 |
| I_LS_PD2      | Diagnostic pull<br>down high<br>current                            | In OFF condition,<br>OUT MOS_Drainx<br>> VOUTOPEN,<br>lpupd_EN = ENB<br>IDIAG_HIGH_PDRV<br>= "1"    | 350  | -   | 550  | μA | MOS_Drai<br>n1/2/3/4/5 |
| I_leakage     | MOS_DRAIN<br>leakage current                                       | Output disable<br>diagnostic<br>OFF condition                                                       | -10  |     | 10   | μA | MOS_Drai<br>n1/2/3/4/5 |
| Tflt_diagoff1 | DIAG Filter time                                                   | Filter Mode=0                                                                                       | 75   | 100 | 125  | μs | MOS_Drai<br>n1/2/3/4/5 |
| Tflt_diagoff2 | DIAG Filter time                                                   | Filter Mode=1                                                                                       | 450  | 600 | 750  | μs | MOS_Drai<br>n1/2/3/4/5 |
| Tblank_00     | Blanking time<br>on<br>ON diag                                     | PRD1_BLK[0:1]=00                                                                                    | 5.1  | 6   | 6.7  | μs | MOS_Drai<br>n1/2/3/4/5 |
| Tblank_01     | Blanking time<br>on<br>ON diag                                     | PRD1_BLK[0:1]=01                                                                                    | 11.1 | 12  | 13.5 | μs | MOS_Drai<br>n1/2/3/4/5 |
| Tblank_10     | Blanking time<br>on<br>ON diag                                     | PRD1_BLK[0:1]=10                                                                                    | 17.1 | 18  | 20   | μs | MOS_Drai<br>n1/2/3/4/5 |
| Tblank_11     | Blanking time<br>on<br>ON diag                                     | PRD1_BLK[0:1]=11                                                                                    | 23.1 | 24  | 26.7 | μs | MOS_Drai<br>n1/2/3/4/5 |
| cmd_dly_0     | Command delay<br>for<br>MSC_CONFIG_<br>REG20[7]=PDR<br>V_O2H_DLY=0 | Tested by scan                                                                                      | 140  | 150 | 160  | μs | MOS_Drai<br>n1/3       |
| cmd_dly_1     | Command delay<br>for<br>MSC_CONFIG_<br>REG20[7]=PDR<br>V_O2H_DLY=1 | Tested by scan                                                                                      | 280  | 300 | 320  | μs | MOS_Drai<br>n1/3       |

## 10 Pre-drivers – IGNITER IGN[1:6]

These 6 Ignition pre-drivers are designed to drive ignition load. They are driven by *MSC command*. The push-pull stage is made up of a high-side current generator and a low-side driver.

When an overcurrent fault is detected the driver switches off with higher slew rate (FASTSR) to reduce the power dissipation.

There is a bit selection for disabling the LSD in case of external IGBT. It is suggested to configure the MSC bit (*IGN\_LSD\_DIS CONFIG\_REG6 D7*) disabling the LSD of igniter before the *msc\_driver\_enable* command.

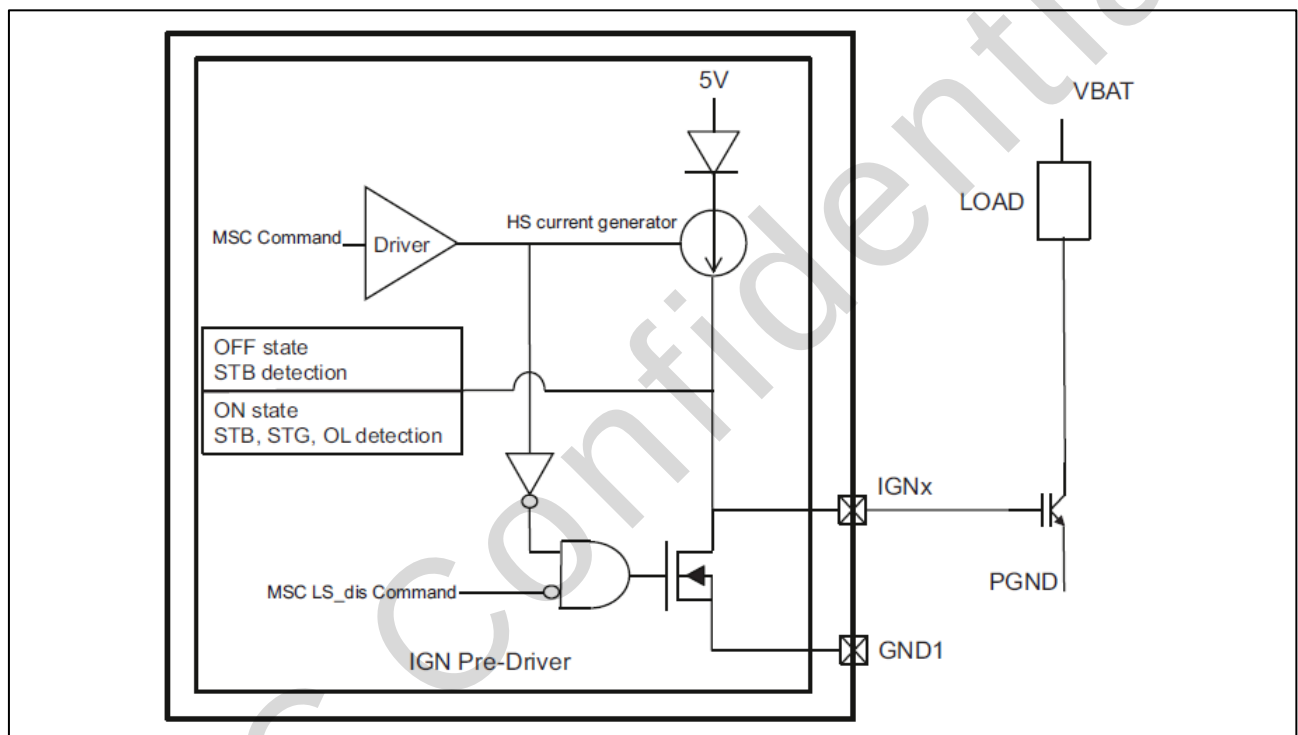


Figure 10-1 Ignition pre-drivers block schematic

### 10.1 Ignition pre\_drivers diagnosis

The driver has the following detection diagnosis:

- Short To Vbat protection by checking the IGNx pin voltage higher than  $V_{th\_STB\_IGN}$ , when the driver is enabled high (high-side on) or low (low-side on);
- Open Load when the driver is enabled high (high-side on) (DIAGOL);
- Short to Ground when the driver is enabled high (high-side on) (DIAGLV).

Table 10-1 Pre-driver - IGNITER diagnosis

| Controlled load                                               | Diagnosis & protection type                                                    | Detected faults @ pre-driver enabled HIGH                       | Detected faults @ pre-driver enabled LOW |
|---------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------|
| Internal IGBT/smart IGBT<br>(IGN_DIAG (CONFIG_REG2 D5) = "0") | The controlled device is module internal. No diagnosis and protection required | -                                                               | -                                        |
| External IGBT Load<br>(IGN_DIAG = "1")                        | IGNx voltage and current monitoring                                            | _ IGNx short to GND<br>_ IGNx open load<br>_ IGNx short to VBAT | _ IGNx short to VBAT                     |

## 10.2 Short to GND

When IGNx pin voltage is lower than  $V_{th\_STG\_IGN}$  for a filter time at pre-driver is enabled high, short to ground fault is detected. So the fault is latched until Read\_Diag\_CMD action.

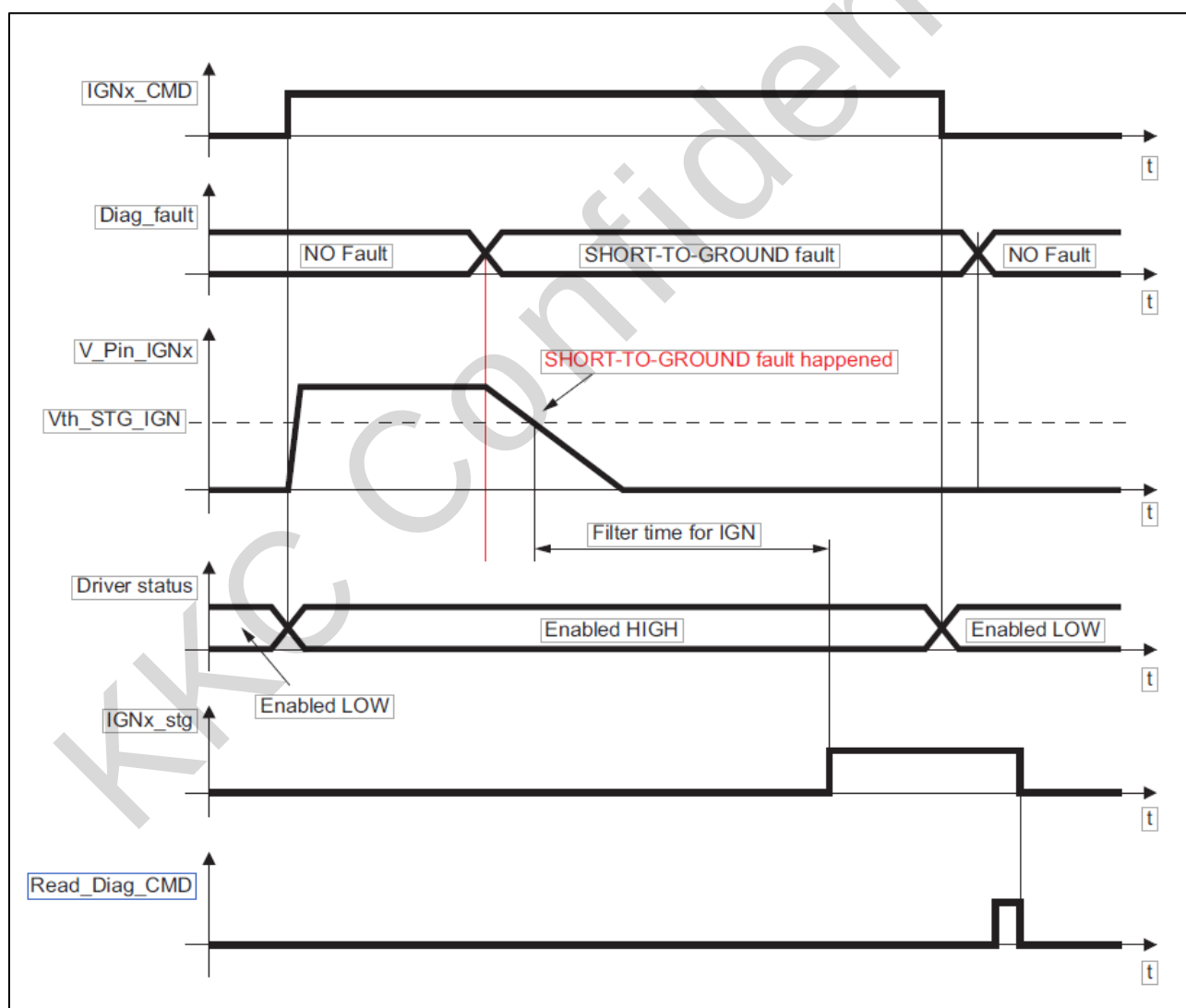


Figure 10-2 IGN pre-drivers diagnosis timing diagram at short to GND

### 10.3 Short to BAT

When IGNx pin voltage is higher than  $V_{th\_STB\_IGN}$  for a filter time at pre-driver is enabled high or enabled low, short to battery fault is detected and the driver is disabled. So the fault is latched.

During Short-to-Bat fault present the driver is auto-restart with these modalities:

- The high-side restart during OVC is on read diagnostic or  $IGNx\_CMD$  goes to low and high again.
- The low-side restart during OVC is on read diagnostic.

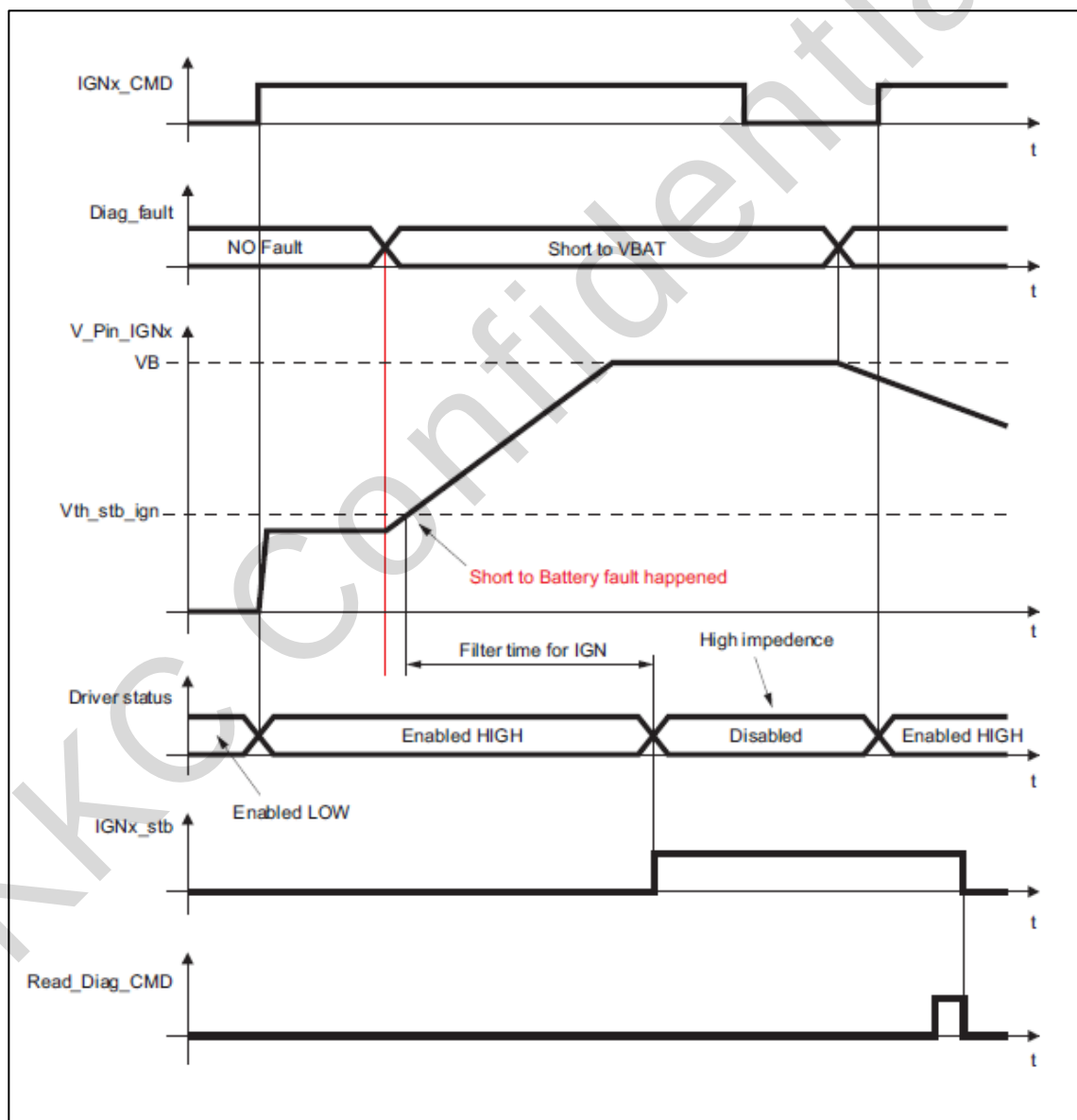


Figure 10-3 IGN pre-drivers diagnosis timing diagram at short to VBAT

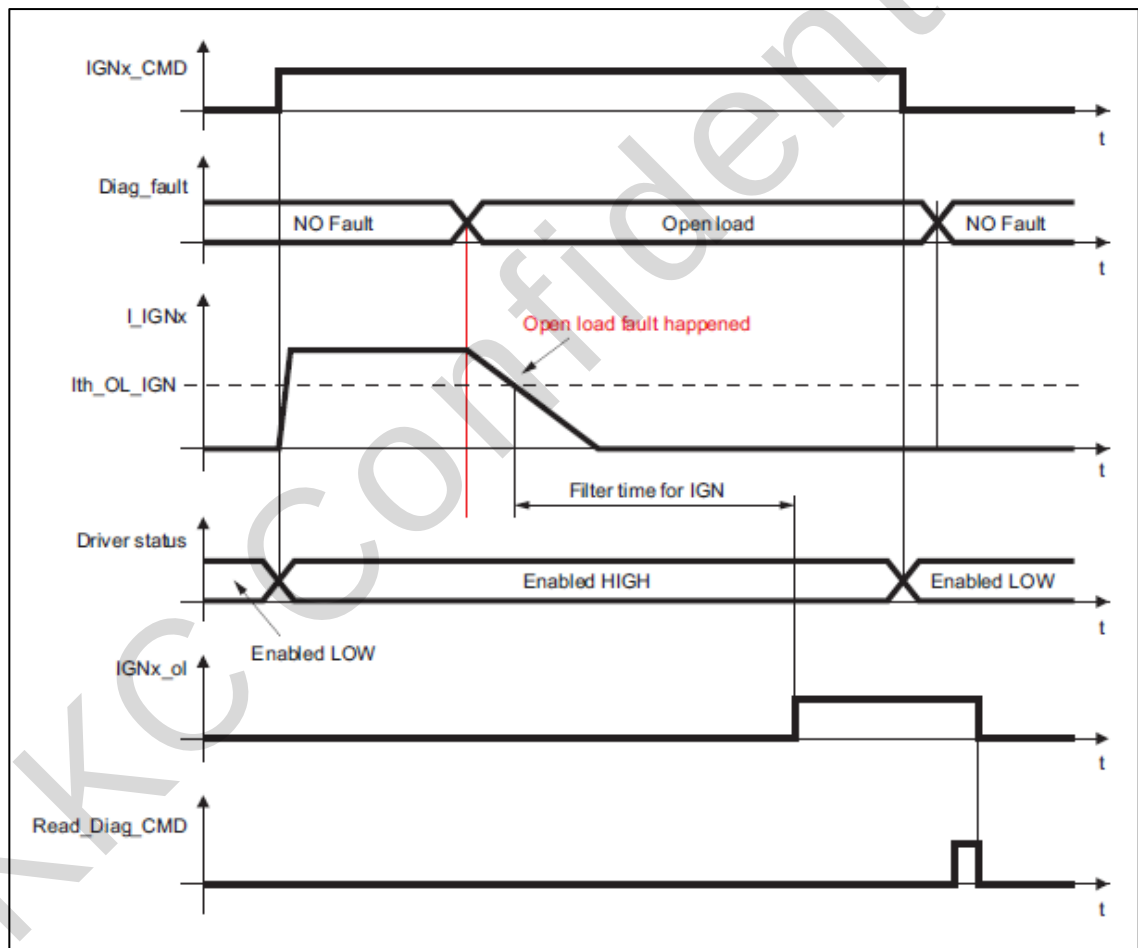
## 10.4 Open load

When IGNx pre-driver is enabled HIGH, if IGNx current is lower than OL current detection  $I_{th\_OL\_IGN}$  for a filter time, open load fault is detected and the driver is not disabled. Open load function can be disabled by a MSC bit. Open Load fault is latched until Read\_Diag\_cmd action. OPL threshold is selectable between 2 values, the default (between 0.4 and 0.8mA) and a lower one.

OPL threshold reduction is available just in low current limitation condition, so when  $OL\_RED$  and  $IGN\_CURRENT\_CFG$  ( $CONFIG\_REG2$  D4) bit are both high.

## 10.5 Error in driver status

Refer to [Section 7.1.3](#)



**Figure 10-4** Low-side drivers ON diagnosis timing diagram at open load

### Conditions:

$5.5\text{ V} \leq V_{B\_IN} \leq 18\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified;

Table 10-2 Pre-driver - IGNITER electrical characteristics

| Symbol                                        | Parameter                                 | Test condition                                                                                | Min          | Typ | Max  | Unit | Pin                |
|-----------------------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------|--------------|-----|------|------|--------------------|
| <b>High-side current generator parameters</b> |                                           |                                                                                               |              |     |      |      |                    |
| IOH1_HS_IGN                                   | High-side output current 1                | VIGNX = 4 V<br>VDD5 = 5 V<br>MSC Bit<br>IGN_CURRENT<br>CFG = 1,<br>OL_RED = 0                 | 4.7          | -   | 15.3 | mA   | IGN1/2/3/<br>4/5/6 |
| IOH1_HS_IGN_OL_RED                            | High-side output current 1                | VIGNX = 4 V<br>VDD5 = 5 V<br>MSC Bit<br>IGN_CURRENT<br>CFG = 1,<br>OL_RED = 1                 | 2            | -   | 15   | mA   | IGN1/2/3/<br>4/5/6 |
| IOH2_HS_IGN                                   | High-side output current 2                | VIGNX = 4 V<br>VDD5 = 5 V<br>MSC Bit<br>IGN_CURRENT<br>CFG = 0 (Default Value),<br>OL_RED = 0 | 14.7         | -   | 30.3 | mA   | IGN1/2/3/<br>4/5/6 |
| VOH1_HS_IGN                                   | Output voltage 1 @ high-side on           | Isource = 5 mA<br>MSC Bit<br>IGN_CURRENT<br>CONFIG = 1,<br>OL_RED = 0                         | VDD5<br>-0.4 | -   | VDD5 | V    | IGN1/2/3/<br>4/5/6 |
| VOH1_HS_IGN_OL_RED                            | Output voltage 1 @ high-side on           | Isource = 2 mA<br>MSC Bit<br>IGN_CURRENT<br>CONFIG = 1,<br>OL_RED = 1                         | VDD5<br>-0.4 | -   | VDD5 | V    | IGN1/2/3/<br>4/5/6 |
| VOH2_HS_IGN                                   | Output voltage 2 @ high-side on           | Isource = 15mA<br>MSC Bit<br>IGN_CURRENT<br>CONFIG = 0 (Default Value)                        | VDD5<br>-0.6 | -   | VDD5 | V    | IGN1/2/3/<br>4/5/6 |
| IOFF_LK_IGN                                   | High-side leakage current @ off condition | IGNx = 2.5 V in off                                                                           | -            | -   | 10   | μA   | IGN1/2/3/<br>4/5/6 |
| <b>Low-side MOSFET parameters</b>             |                                           |                                                                                               |              |     |      |      |                    |
| RDS-on_LS_IGN                                 | Drain-source resistance                   | I <sub>load</sub> = 0.1A<br>@ T = -40 °C                                                      | 1.8          | 2.2 | 3.1  | Ω    | IGN1/2/3/<br>4/5/6 |
| RDS-on_LS_IGN                                 | Drain-source resistance                   | I <sub>load</sub> = 0.1 A<br>@ T = 27 °C                                                      | 2.4          | 3.0 | 3.91 | Ω    | IGN1/2/3/<br>4/5/6 |
| RDS-on_LS_IGN                                 | Drain-source resistance                   | I <sub>load</sub> = 0.1 A<br>@ T = 150 °C                                                     | 3.8          | 5.0 | 7.0  | Ω    | IGN1/2/3/<br>4/5/6 |
| VOL_LS_IGN                                    | Output voltage @ LS low-side on           | IOL = 20 μA<br>sunked                                                                         | -            | -   | 0.1  | V    | IGN1/2/3/<br>4/5/6 |
| VOL_LS_IGN                                    | Output voltage @ LS low-side on           | IOL= 100 mA<br>sunked                                                                         | -            | -   | 0.7  | V    | IGN1/2/3/<br>4/5/6 |
| Irev_LS_IGN                                   | Body diode reverse current voltage drop   | I <sub>load</sub> = 0.3 mA<br>sourced                                                         | -            | -   | 1    | V    | IGN1/2/3/<br>4/5/6 |
| <b>General IGN block parameters</b>           |                                           |                                                                                               |              |     |      |      |                    |
| TDON_IGN                                      | Turn-on delay time                        | CLOAD = 10 nF<br>Isource = 10 mA                                                              | -            | -   | 10   | μs   | IGN1/2/3/<br>4/5/6 |
| TDOFF_IGN                                     | Turn-off delay time                       | C LOAD = 10 nF                                                                                | -            | -   | 10   | μs   | IGN1/2/3/<br>4/5/6 |

|                  |                                                         |                                 |      |   |      |    |                    |
|------------------|---------------------------------------------------------|---------------------------------|------|---|------|----|--------------------|
| TR_IGN           | Output rise time                                        | CLOAD =10 nF<br>Isource = 10 mA | -    | - | 9    | μs | IGN1/2/3/<br>4/5/6 |
| TF_IGN           | Output fall time                                        | C LOAD = 10 nF                  | 0.05 | - | 2    | μs | IGN1/2/3/<br>4/5/6 |
| IOUT_LK_IGN      | Output leakage current                                  | Vpin = 13.5 V                   | -    | - | 10   | μA | IGN1/2/3/<br>4/5/6 |
| Vth_STB_IGN      | Short circuit to VBAT voltage threshold                 | IGNx = VBAT                     | 5.3  | - | 6.5  | V  | IGN1/2/3/<br>4/5/6 |
| Vth_STG_IGN      | Short circuit to GND threshold                          | -                               | 2.1  | - | 2.7  | V  | IGN1/2/3/<br>4/5/6 |
| Ileak_STG_IGN    | Leakage current @ short circuit to GND, IGN in tristate | IGNx = GND                      | -    | - | 10   | μA | IGN1/2/3/<br>4/5/6 |
| Ileak_STB_IGN    | Leakage current @ short circuit to VBAT,IGN in tristate | IGNx = VBAT                     | -1   | - | 1    | mA | IGN1/2/3/<br>4/5/6 |
| Ileak_STB_IGN    | Leakage current @ short circuit to VBAT                 | GND = VB_IN = 0V<br>IGNx = 18 V | -    | - | 10   | μA | IGN1/2/3/<br>4/5/6 |
| TFLT_SC GOL IGN  | Open load and short circuit to GND filter time          | -                               | 72   | - | 90   | μs | IGN1/2/3/<br>4/5/6 |
| TFLT_SCB IGN     | Short circuit to VBAT switch-off delay                  | -                               | 5.9  | - | 7.2  | μs | IGN1/2/3/<br>4/5/6 |
| Ith_OL_IGN       | Open load current threshold                             | -                               | 0.18 | - | 1.25 | mA | IGN1/2/3/<br>4/5/6 |
| Ith_OL_IGN_OL_RE | Open load current threshold                             | OL_RED = 1                      | 0.03 | - | 0.2  | mA | IGN1/2/3/<br>4/5/6 |

## 11 Inductive sensor interface

The interface handles signals coming from magnetic pick-up sensors or Hall Effect sensors.

The interface feeds the digital signal to microcontroller that extracts flying wheel rotational position, angular speed and acceleration.

### 11.1 VRS interface

VRS interface runs either in normal mode to convert the input differential voltage or in diagnostic mode to detect eventual short to ground, short to battery or open condition at sensor pins. Figure 11-1 shows the main parts inside VRS block and, in red, the internal connection in case of diagnostic mode active.

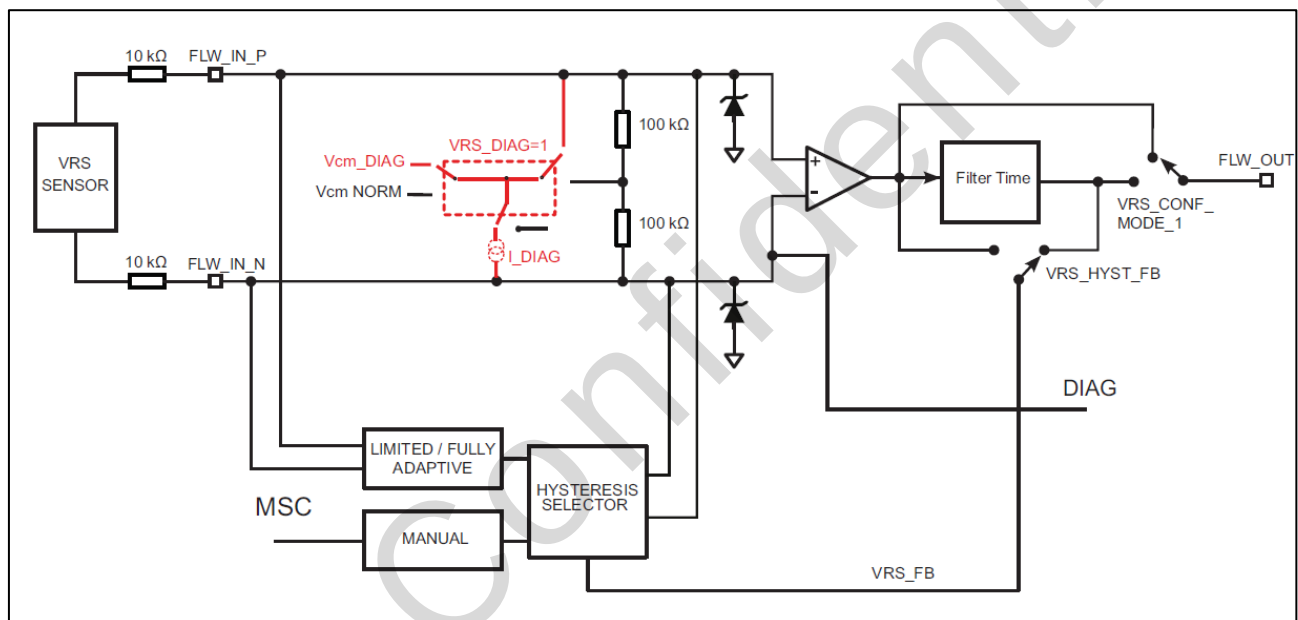


Figure 11-1 VRS interface block diagram

Operating mode is defined in VRS register through *VRS\_DIAG* in *CONFIG\_REG8* bit: when *VRS\_DIAG*=0, VRS block is set in normal mode; when *VRS\_DIAG*=1 the VRS diagnosis mode is activated.

If HE9788 is supplied and VRS is running, in case *FLW\_IN\_P* or *FLW\_IN\_N* voltage rises their values are clamped at *VclpH*.

In case of activation of clamp both on *FLW\_IN\_P* and *FLW\_IN\_N*, it is guaranteed by design that *FLW\_IN\_P* voltage is higher than *FLW\_IN\_N*.

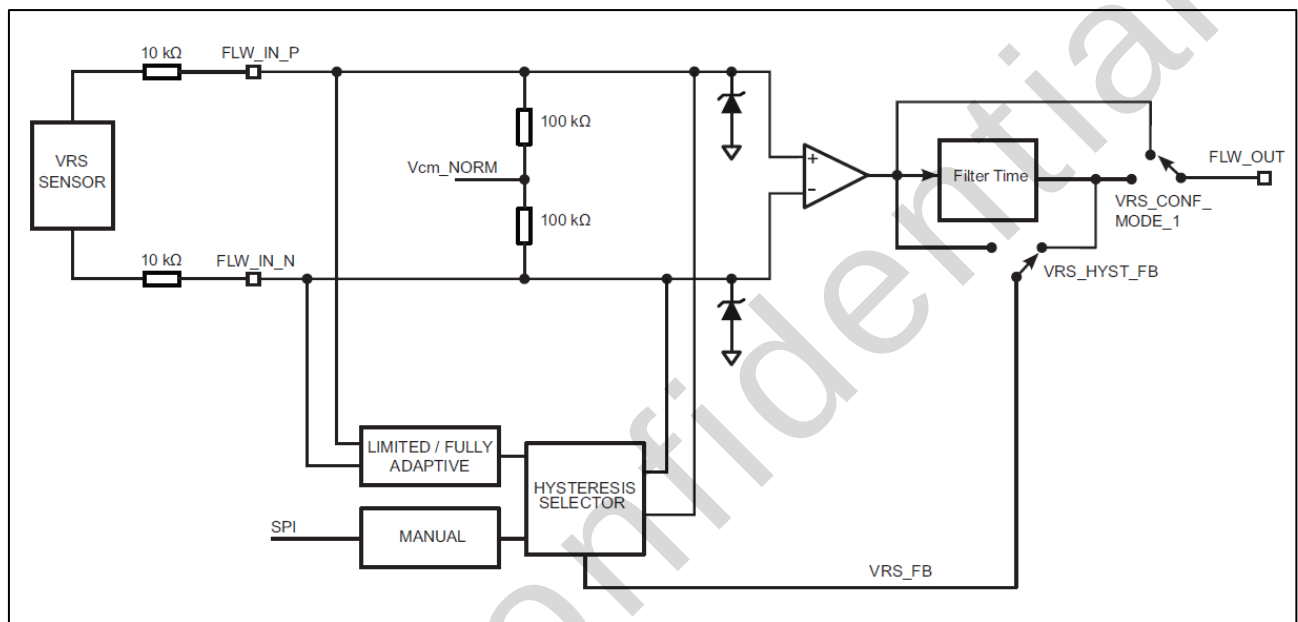
### 11.2 VRS – Normal mode

The VRS normal mode is set with *VR\_DIAG*=0

In Normal mode, the circuit is configured as the one reported in Figure 11-2. It allows decoding the VRS signal while the wheel is in rotation.

Due to the high variability of the input signal ( $\pm 200$  V) the input is clamped in the range  $[V_{clpH}:V_{clpL}]$  in order to allow the analog circuitry to process the signal itself. Moreover, the sensor input pins have an input common mode, VCM.

The preconditioned input signal is then processed by a zero-crossing comparator, which toggles at each transition of the input signal.



**Figure 11-2 VRS block diagram - Normal operating mode**

To avoid spurious commutations of the zero crossing comparator, a hysteresis mechanism is implemented. HE9788 is able to sink a hysteresis current which generates a voltage drop across the external resistors. The voltage levels related to the hysteresis function shown hereafter are calculated considering an external series resistance of 10 kΩ on FLW\_IN\_P and 10 kΩ on FLW\_IN\_N pins.

As reported in Figure 11-3, the  $V_{diff}$  ( $FLW\_IN\_P - FLW\_IN\_N$ ) input differential signal exhibits

some steps at each zero crossing:

when the output of the zero crossing comparator is high, the hysteresis current is kept OFF;

when the output of the zero crossing comparator is low, the hysteresis current is switched ON.

This approach applies the hysteresis current only on the transition L-H of the VRS\_FB signal.

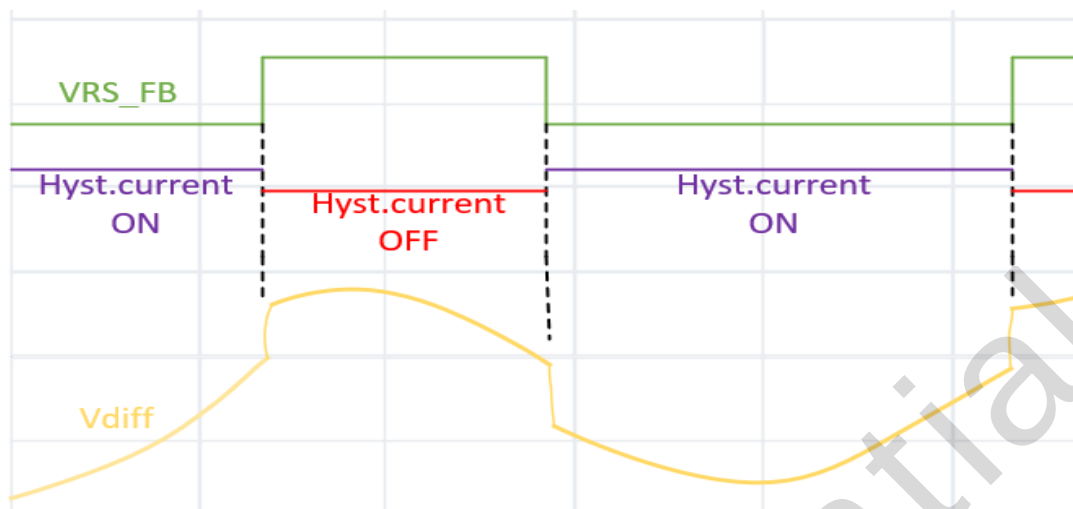


Figure 11-3 Hysteresis application

The output of the zero crossing comparator can be further processed by a filtering circuit or directly routed to FLW\_OUT.

### 11.2.1 VRS normal mode configurations

HE9788 integrates two main configurable architectures: VRS\_A and VRS\_B. These architectures are selected in VRS register, *VRS\_MODE\_SEL* bit in CONFIG\_REG 2.

Once VRS\_A (*VRS\_MODE\_SEL* =1) or VRS\_B (*VRS\_MODE\_SEL* =0) has been configured, hysteresis and filtering strategy are defined through *VRS\_MODE[1:0]* bit in the same CONFIG\_REG 8 register:

*VRS\_MODE[1]* defines filtering function (OFF/ON and if ON, its time value)

*VRS\_MODE[0]* defines the hysteresis (manual or adaptive)

Table 11-1 summarizes the parameters of VRS\_A and VRS\_B architecture; the next paragraphs detail the two configurations.

Table 11-1 VRS\_A, VRS\_B hysteresis and filter time definition

| <i>VRS_MO<br/>DE_SEL</i> | <i>VRS<br/>_MODE[<br/>1:0]</i> | Filter |           | Hyst.            |                                            |
|--------------------------|--------------------------------|--------|-----------|------------------|--------------------------------------------|
| VRS_A configuration      |                                |        |           |                  |                                            |
| 1                        | 00                             | OFF    | 0 μs      | Manual           | Ref to VRS_A - Manual Hysteresis           |
| 1                        | 01                             | OFF    | 0 μs      | Full adaptive    | Ref. to VRS_A - Fully Adaptive Hysteresis  |
| 1                        | 10                             | ON     | T(n-1)/32 | Manual           | Ref to VRS_A - Manual Hysteresis           |
| 1                        | 11                             | ON     | T(n-1)/32 | Full adaptive    | Ref. to VRS_A - Fully Adaptive Hysteresis  |
| VRS_B configuration      |                                |        |           |                  |                                            |
| 0                        | 00                             | OFF    | 0 μs      | Manual           | Ref. to VRS_B - Manual Hysteresis          |
| 0                        | 01                             | OFF    | 0 μs      | Limited adaptive | Ref. to VRS_B -Limited Adaptive Hysteresis |
| 0                        | 10                             | ON     | 4 μs      | Manual           | Ref. to VRS_B - Manual Hysteresis          |
| 0                        | 11                             | ON     | 4 us      | Limited adaptive | Ref. to VRS_B -Limited Adaptive Hvsteresis |

In case a change of *VRS\_MODE\_SEL* bit within the normal operating mode occurs (1->0 or 0->1) with hysteresis current active, this leads to the change of the hysteresis (to HI1 or HI3, according the new selection programmed) not synchronized with any *VRS\_FB* zero crossing.

### 11.2.2 VRS\_A - Manual Hysteresis

To set the manual hysteresis on *VRS\_A* configuration, bit *VRS\_MODE[0]* has to be configured at '0'. Hysteresis value is manually set through *VRS\_HYST[2:0]* of CONFIGREG 8 according to *Table 11-2*. Such hysteresis is fixed until a new SPI programming occurs.

Default hysteresis current after exiting reset is HI3.

New MSC current value is updated during HYST CURRENT OFF phase that means the output comparator is high.

**Table 11-2 VRS\_A hysteresis value**

| Hysteresis current [H] | Min | Typ | Max  | Unit    | Correspondent value on 20 k $\Omega$ ext. resistor | Unit |
|------------------------|-----|-----|------|---------|----------------------------------------------------|------|
| HI1                    | 3   | 5   | 7    | $\mu$ A | 100                                                | mV   |
| HI2                    | 7.5 | 10  | 13.5 | $\mu$ A | 200                                                | mV   |
| HI3                    | 13  | 17  | 23   | $\mu$ A | 347                                                | mV   |
| HI4                    | 23  | 32  | 40   | $\mu$ A | 644                                                | mV   |
| HI5                    | 35  | 51  | 60   | $\mu$ A | 1020                                               | mV   |
| No Hyst                | -   | -   | -    | -       | -                                                  | -    |

### 11.2.3 VRS\_A - Fully Adaptive Hysteresis

To set the adaptive hysteresis on *VRS\_A* configuration, bit *VRS\_MODE[0]* has to be set to '1'. In this configuration, *VRS* input differential signal is fed into a peak detector circuit and then quantized on 5 different voltage levels, based on 4 *PVi* thresholds (see *Table 11-3*). Default hysteresis current after exiting reset is HI3.

**Table 11-3 Peak voltage value ranges**

| Peak voltage [ <i>PVi</i> ] | Min  | Typ  | Max  | Unit |
|-----------------------------|------|------|------|------|
| PV1                         | 600  | 930  | 1300 | mV   |
| PV2                         | 1200 | 1600 | 1950 | mV   |
| PV3                         | 2000 | 2300 | 2650 | mV   |
| PV4                         | 2600 | 3000 | 3300 | mV   |

The quantized output is sent to a logic block (Hysteresis Selection Table) that chooses the proper hysteresis value (HIi) depending on the input peak voltage (*PVi*), see *Table 11-4*.

Table 11-4 Input peak voltage range

| Input peak voltage range | Selected hysteresis (Hii) |
|--------------------------|---------------------------|
| 0 - PV1                  | HI1                       |
| PV1 PV2                  | HI2                       |
| PV2 PV3                  | HI3                       |
| PV3 PV4                  | HI4                       |
| > PV4                    | HI5                       |

Peak detector and Hysteresis Selection Table circuits are enabled by VRS\_FB signal according to *HYS\_FB\_SEL* in CONFIG-REG 8 bit value that establishes if the feedback signal is before or after the filter time.

VRS input differential voltage is continuously acquired: its max value, reached during timeframe VRS\_FB signal is asserted (hysteresis current is off), is latched through a peak detector; such a peak defines a specific value of hysteresis current, turned on as soon as the VRS\_FB falls to zero and switches OFF when next rising edge occurs.

Based on the hysteresis current, the signal is processed by a squaring circuit which processes the output signal of the comparator, see Figure 11-4.

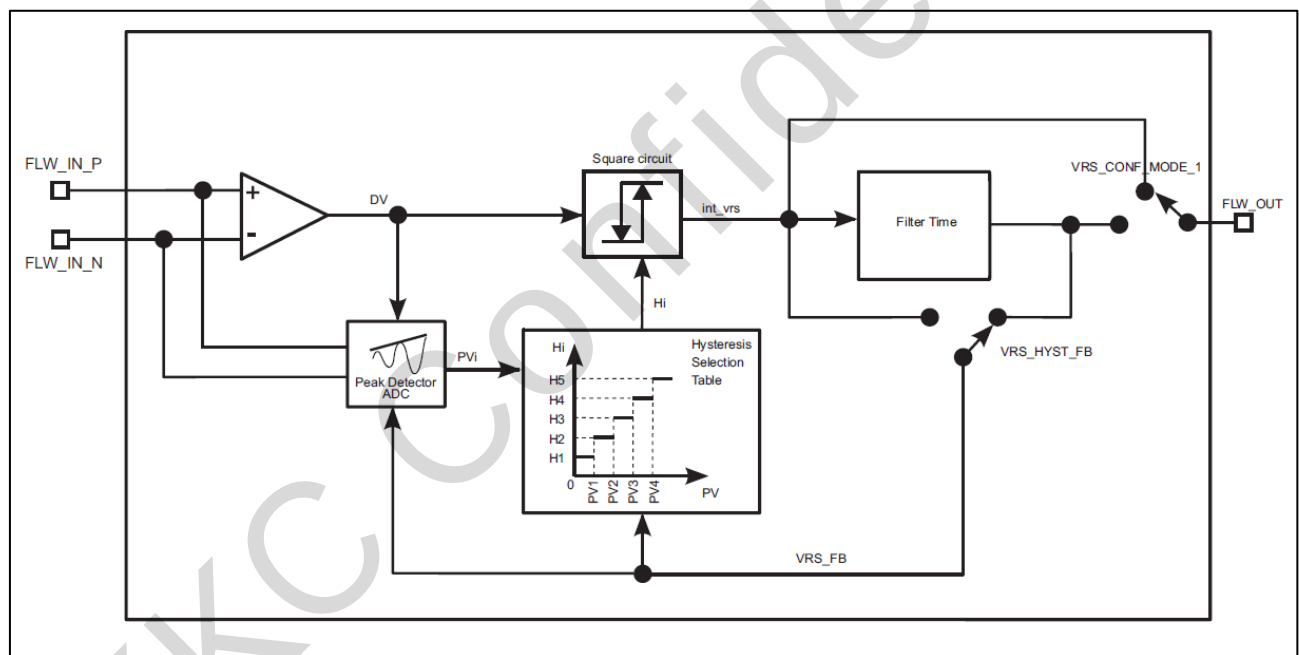


Figure 11-4 VRS\_A fully adaptive hysteresis

#### 11.2.4 VRS\_A - Adaptive filter time

In VRS\_A mode, it is possible to enable the filter time on the output of the zero crossing comparator through the bit *VRS\_MODE[1]* of CONFIG-REG 8.

Once enabled, the most suitable internal filter based on the input signal frequency is determined.

According to VRS previous output period, filter time value is updated as follows:

$$T_{\text{filter}(n)} = \frac{T_{\text{period}(n-1)}}{32}$$

If the value of the previous period is lower than 128  $\mu\text{s}$ , the filter time would be saturated at 4  $\mu\text{s}$  fixed value. After reset  $T_{\text{filter}} = 200 \mu\text{s}$  (TYP)

Through *EN\_FALLING\_FILT* CONFIG-REG8 bit in VRS register, it is possible to configure two different strategies for the filtering algorithm.

VRS\_OUT rising edge: the transition depends on the hysteresis crossing of differential signal  $V_{\text{diff}}$ ; VRS output is set if  $V_{\text{diff}}$  remains asserted and stable for a period longer than  $T_{\text{filter}}$ .

VRS\_OUT falling edge: the transition depends on the zero crossing of differential signal  $V_{\text{diff}}$ :

*EN\_FALLING\_FILT* = 1: VRS\_OUT is deasserted when the signal is low and remains stable for at least  $T_{\text{filter}}$ ; see Figure 11-5.

*EN\_FALLING\_FILT* = 0: VRS\_OUT is de-asserted at first zero crossing transition of differential signal and next eventual commutations are ignored for  $T_{\text{filter}}$  time. See Figure 11-6.

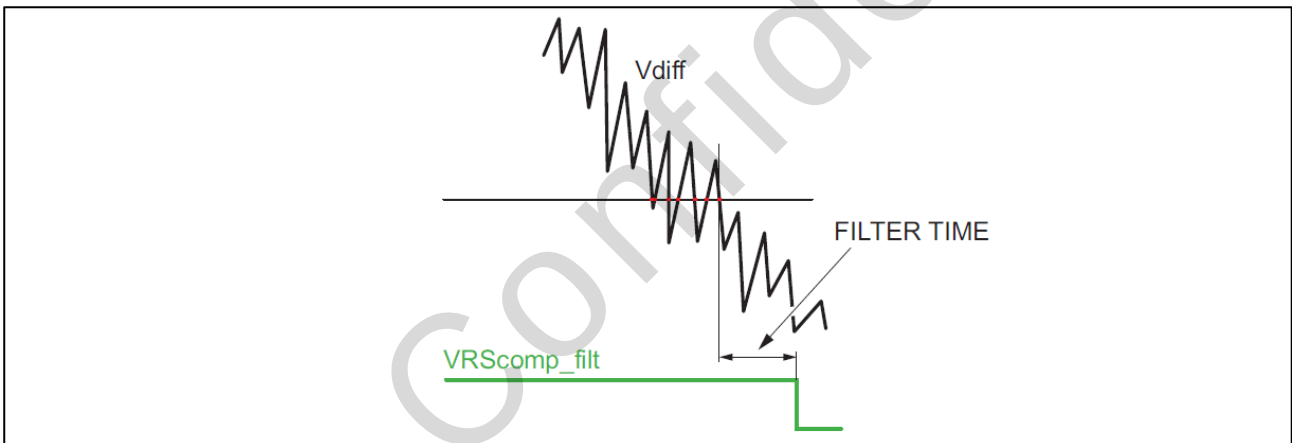


Figure 11-5 *EN\_FALLING\_FILT* = 1

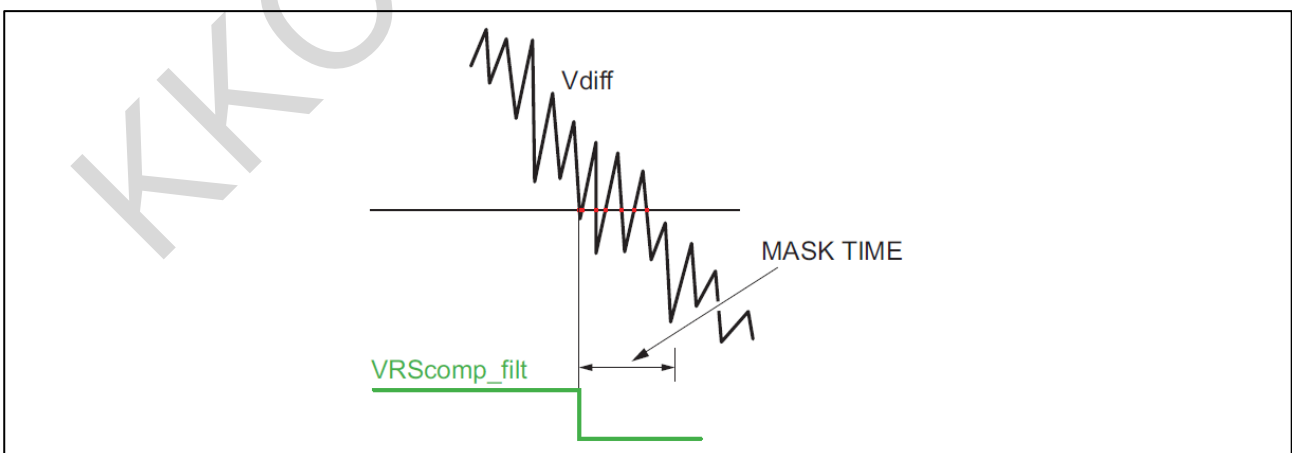


Figure 11-6 *EN\_FALLING\_FILT* = 0

### 11.2.5 VRS\_B - Manual Hysteresis

To set the manual hysteresis on VRS\_B configuration, bit *VRS\_MODE* [0] has to be set to '0'. Hysteresis value is manually set through *VRS\_HYST* [2:0] of CONFIG-REG 8 according to [Table 11-2](#) Such hysteresis is fixed until a new SPI programming occurs.

Once a new value is defined, new hysteresis threshold is applied after the second VRS\_FB H-L transition and until the next rising edge of the VRS input differential voltage occurs.

### 11.2.6 VRS\_B -Limited Adaptive Hysteresis

To set the limited adaptive hysteresis on VRS\_B configuration, bit *VRS\_MODE* [0] has to be set to '1'. In this mode, user programs a hysteresis threshold through *VRS\_HYST* [2:0] bit in VRS register and the internal logic selects a hysteresis based on input signal peak value: the maximum of these two values is actually applied.

Once a new value is defined, new hysteresis threshold is applied after the second VRS\_FB H-L transition and until the next rising edge of the VRS input differential voltage occurs.

### 11.2.7 VRS\_B - Fixed Filter Time

In VRS\_B configuration, it is possible to enable the filter time on the output of the zero crossing comparator through the bit *VRS\_MODE*[1] of VRS register. This configuration allows defining the internal filter time at a fixed value of 4  $\mu$ s, active on both rising and falling edges of VRS output.

As per VRS\_A architecture, *EN\_FALLING\_FILT* allows configuring the same two different strategies for the filtering algorithm.

## 11.3 VRS diagnostic mode

The diagnostic mode is selected through *VRS\_DIAG* = '1' in CONFIG-REG8. This mode provides feedback to detect faulty conditions either on FLW\_IN\_P or FLW\_IN\_N.

To be noted that diagnostic results are not reliable while the flying wheel is rotating. If a fault is detected in DIAG mode, VRS correct functionality is not guaranteed. Fault bit *VRS\_DIAG* of VRS register is consequently set.

[Figure 11-7](#) shows the circuit used in Diagnostic mode. When VRS diagnostic mode is activated, FLW\_IN\_P is fixed at Vcm\_DIAG and IDIAG current generator is enabled, the current path is that one in green in [Figure 11-7](#).

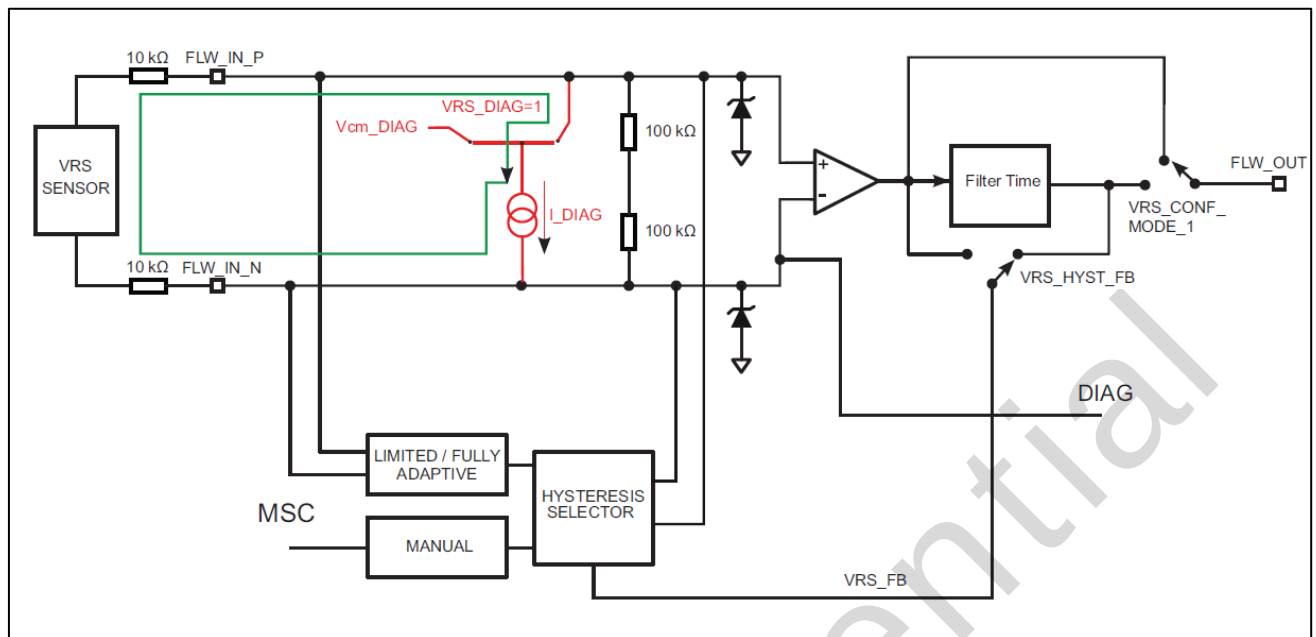


Figure 11-7 VRS block diagram - Diagnostic operating mode - Current path

## 11.4 Application circuit

Sensor sketch and parameters are reported in Figure 11-8 and Table 11-5 VRS sensor parameters.

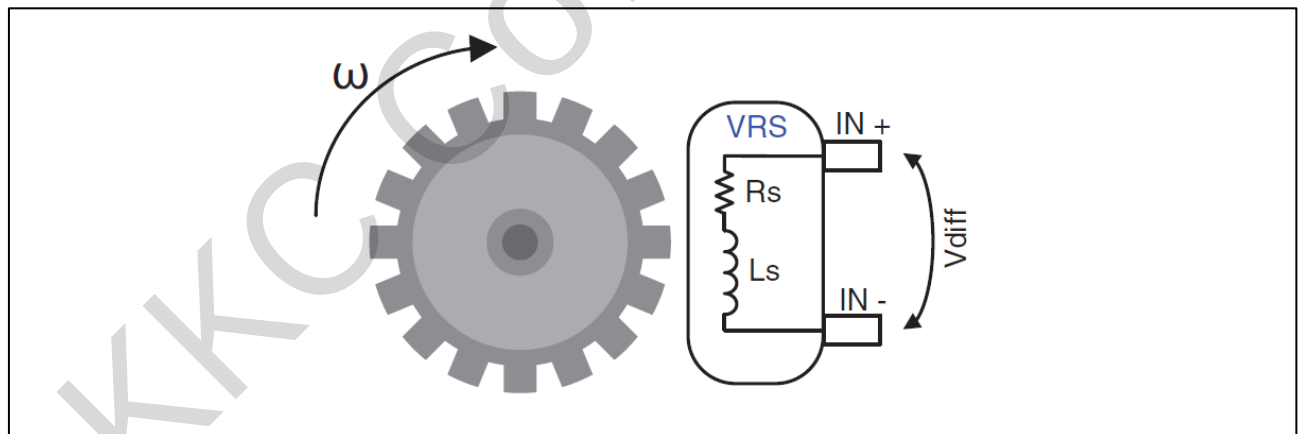


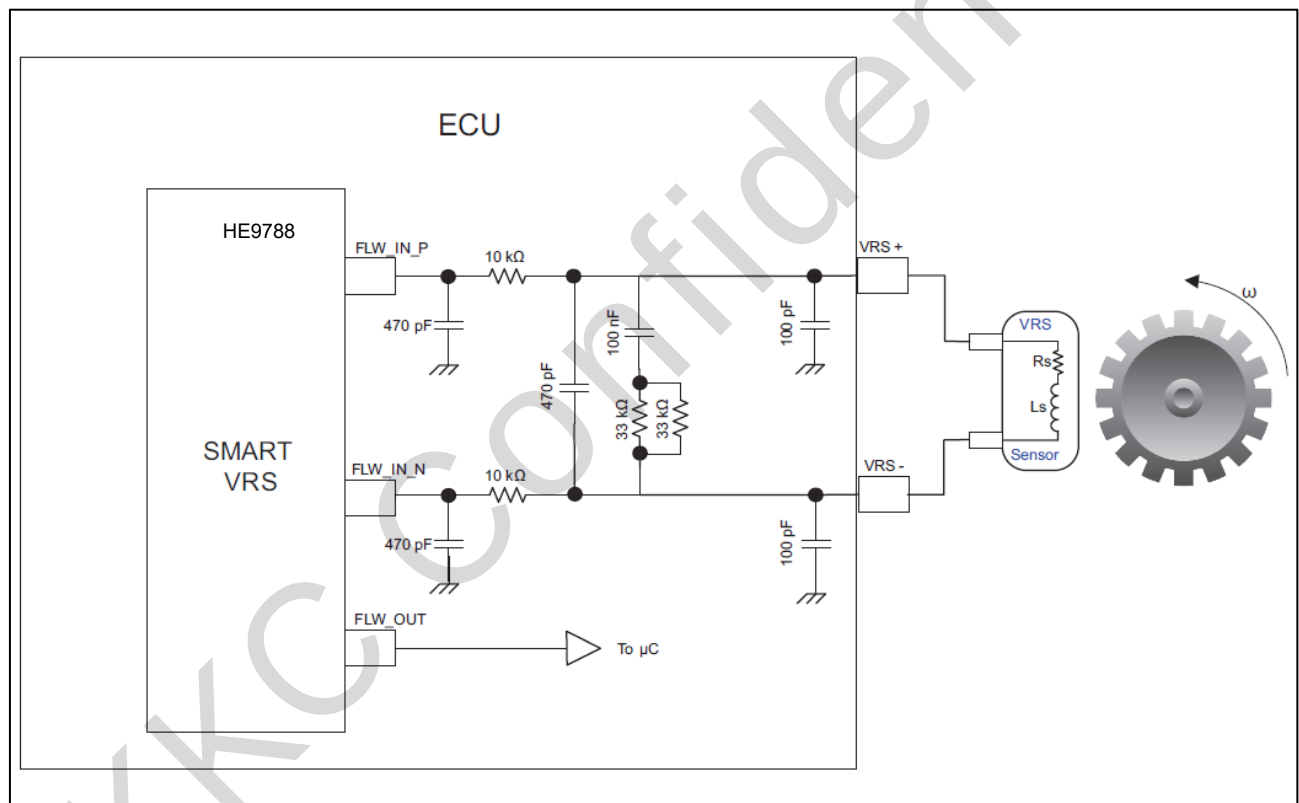
Figure 11-8 Sensor sketch

**Table 11-5 VRS sensor parameters**

| Symbol | Parameter             | Min  | Typ | Max  | Unit     |
|--------|-----------------------|------|-----|------|----------|
| Rs     | Sensor resistance     | 300  | 600 | 1000 | $\Omega$ |
| Ls     | Sensor inductor       | -    | 250 | -    | mH       |
| Vdiff  | Sensor output voltage | -200 | -   | +200 | V        |
| Tout   | Output period         | 100  | -   | 5000 | $\mu$ s  |

The interface handles signals coming from magnetic pick-up sensors, see [Figure 11-9](#) Variable reluctance sensor (VRS), or Hall Effect sensors with two possible configurations, as per [Figure 11-10](#) and [Figure 11-11](#).

The interface feeds the digital signal to microcontroller that extracts flying wheel rotational position, angular speed and acceleration.



**Figure 11-9 Variable reluctance sensor (VRS)**

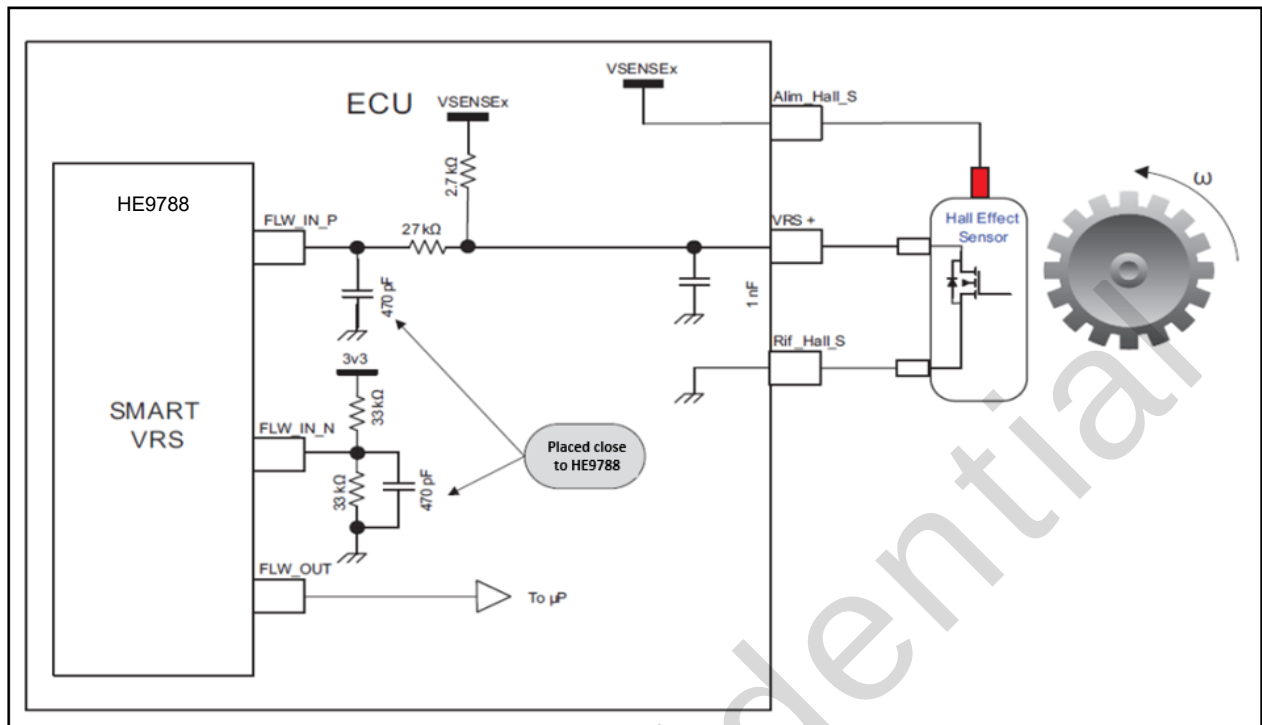


Figure 11-10 Hall effect sensor configuration 1

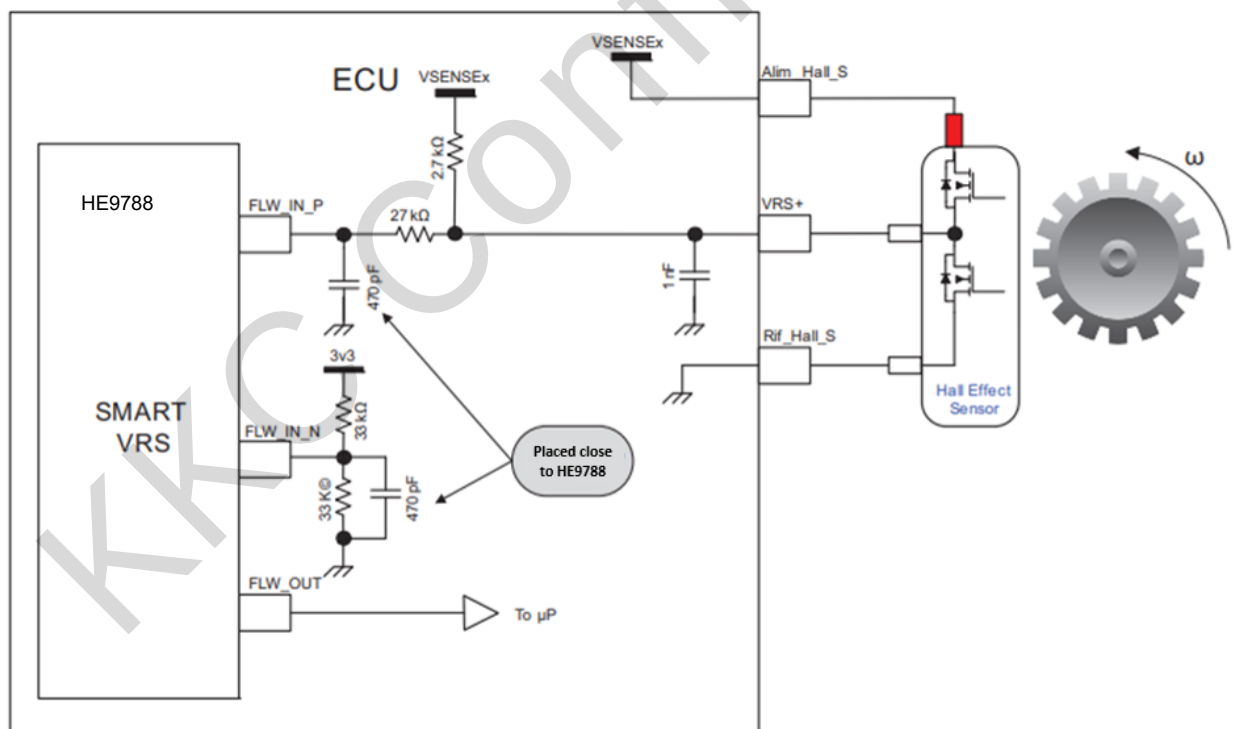


Figure 11-11 Hall effect sensor configuration 2

Conditions:

$5.5\text{ V} \leq V_{B\_IN} \leq 18\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified;

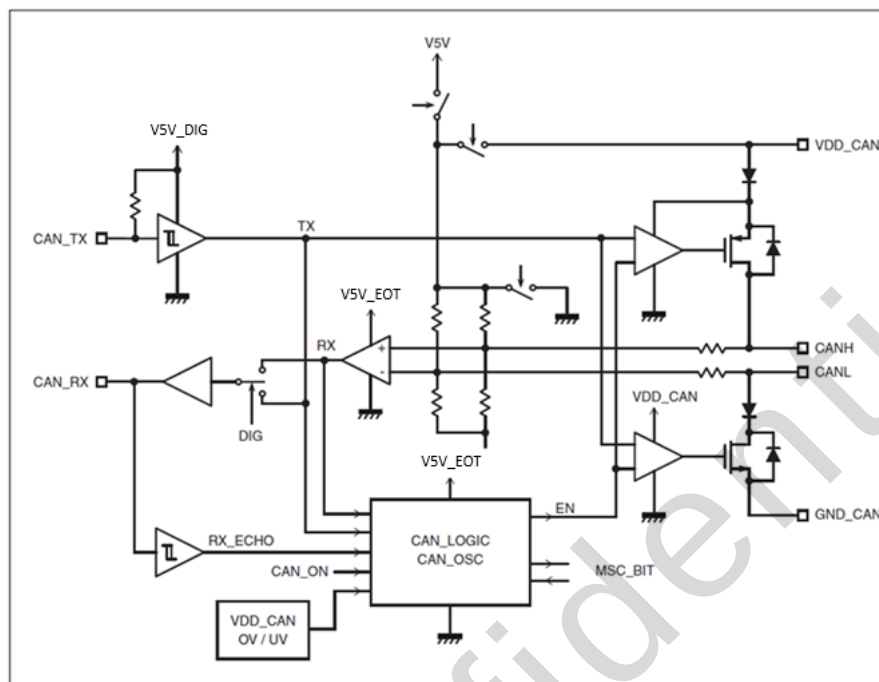
Table 11-6 Flying wheel interface function - Diagnosis test

| Symbol               | Parameter                                        | Test condition                                                | Min          | Typ          | Max     | Unit          | Pin                  |
|----------------------|--------------------------------------------------|---------------------------------------------------------------|--------------|--------------|---------|---------------|----------------------|
| $V_{iThL}$           | Input high-to-low differential threshold voltage | -                                                             | -50          | 0            | 50      | mV            | FLW_IN_P<br>FLW_IN_N |
| $V_{CM}$             | Common mode operating range                      | Not to be tested. It is an application note.                  | 0            | 1.65         | 3       | V             | FLW_IN_P<br>FLW_IN_N |
| $V_{clpH}$           | Input high clamping voltage                      | $ I_{FLW\_IN\_P} = I_{FLW\_IN\_N} =20\text{ mA}$ , device ON  | 5.0-0.5      | -            | 5.0+0.5 | V             | FLW_IN_P<br>FLW_IN_N |
| $V_{clpH}$           | Input high clamping voltage                      | $ I_{FLW\_IN\_P} = I_{FLW\_IN\_N} =20\text{ mA}$ , device OFF | 1            | -            | 2.8     | V             | FLW_IN_P<br>FLW_IN_N |
| $V_{clpL}$           | Input low clamping voltage                       | $ I_{FLW\_IN\_P} = I_{FLW\_IN\_N} =20\text{ mA}$              | -1.5         | -            | -0.3    | V             | FLW_IN_P<br>FLW_IN_N |
| $V_{openload}$       | Output open load voltage                         | FLW_IN_P =<br>FLW_IN_N =<br>$V_{openload}$                    | 1.5          | (3.3)<br>)/2 | 1.8     | V             | FLW_IN_P<br>FLW_IN_N |
| $I_{bvrsp}$          | Input bias current $V_{rsp}$                     | $V_{RS\_INP} \rightarrow$<br>FLW_IN_P                         | -            | -            | 2       | $\mu\text{A}$ | FLW_IN_P<br>FLW_IN_N |
| $I_{bvrsn}$          | Input bias current $V_{rsn}$                     | FLW_IN_N = $V_{openload}$                                     | -            | -            | 2       | $\mu\text{A}$ | FLW_IN_P<br>FLW_IN_N |
| $V_{OL}$             | Output Low Voltage                               | VDD5 = 5 V or 3.3 V<br>$I_{sink}$ current = 2 mA              | -            | -            | 0.5     | V             | FLW_O<br>UT          |
| $V_{OH}$             | Output High Voltage                              | VDD5 = 5 V or 3.3 V<br>$I_{source}$ current = 2 mA            | VDD5<br>-0.5 | -            | -       | V             | FLW_O<br>UT          |
| $I_{lk\_outvrs}$     | Input leakage current to GND                     | -                                                             | -            | -            | 1       | $\mu\text{A}$ | FLW_O<br>UT          |
| $I_{lk\_outvrs}$     | Input leakage current to VDD5                    | -                                                             | -            | -            | 8       | $\mu\text{A}$ | FLW_O<br>UT          |
| $T_{d\_off\_outvrs}$ | Delay on falling edge                            | Test Ext cap = 300 pF                                         | -            | -            | 2       | $\mu\text{s}$ | FLW_O<br>UT          |
| $T_{d\_on\_outvrs}$  | Delay on rising edge                             | Input signal=4ms                                              | -            | -            | 150     | $\mu\text{s}$ | FLW_O<br>UT          |
| $T_{r\_OUT\_vrs}$    | MRX Rise Time                                    | Test Ext cap = 300 pF                                         | -            | -            | 250     | ns            | FLW_O<br>UT          |
| $T_{f\_OUT\_vrs}$    | MRX Fall Time                                    | Test Ext cap = 300 pF                                         | -            | -            | 150     | ns            | FLW_O<br>UT          |
| $V_{outdiag}$        | Output diag voltage                              | FLW_IN_P = open;<br>diag mode                                 | 0.9          | (3.3)<br>)/3 | 1.3     | V             | FLW_O<br>UT          |
| $I_{outdiag}$        | Output diag                                      | FLW_IN_P = open;                                              | 50           | 65           | 80      | $\mu\text{A}$ | FLW_O                |

| Inductive sensor interface   |                                                           |                                                    |     |     |     |   |             |
|------------------------------|-----------------------------------------------------------|----------------------------------------------------|-----|-----|-----|---|-------------|
|                              | Current                                                   | FLW_IN_N= GND; diag mode                           |     |     |     |   | UT          |
| $V_{outsh}$<br>$V_{bdiagth}$ | Output Short-circuit range to VBAT<br>Open Load threshold | FLW_IN_P = open;<br>FLW_IN_N = Vramp;<br>diag mode | 2.8 | 3   | 3.2 | V | FLW_O<br>UT |
| $V_{outshgnddiagth}$         | Output Short-to GND range threshold                       | FLW_IN_P = open;<br>FLW_IN_N = Vramp;<br>diag mode | 1.1 | 1.3 | 1.5 | V | FLW_O<br>UT |

Note: When FLW\_IN\_P and FLW\_IN\_N are both in input high clamping condition, the clamp voltage of FLW\_IN\_N is 30 mV (typical) higher than FLW\_IN\_P.

## 12 CAN FD interface



**Figure 12-1 CAN FD interface diagram**

The pins related to this block are:

- CANH;
- CANL;
- CANTX;
- CANRX;
- GND\_CAN;
- VB\_CAN;
- VDD\_CAN

VDD\_CAN= 5V is the supply for the CAN transceiver block. The voltage on this pin is monitored and if the voltage is below VDD\_CAN\_SUP\_LOW for a `t cansuplow` filter time an undervoltage fault on VDD\_CAN is detected. The fault is latched in a dedicated MSC flag `CAN_SUP_LOW` until an MSC Upstream Read14 is given. While the undervoltage fault is detected on VDD\_CAN the CAN transmitter stays disabled (not MSC read clear access is necessary to restore the transmitter just the fault detection expiration).

If VDD\_CAN is above VDD\_CAN\_SUP\_OV for a *tcansuphigh* filter time an overvoltage fault on VDD\_CAN is detected. The fault is latched on a dedicated MSC flag VDD\_CAN\_OV until an MSC Upstream Read14 is given during the MSC flag VDD\_CAN\_OV stays set the CAN transmitter is disabled

During a VDD\_CAN overvoltage or undervoltage detected fault the biasing voltage is switched from VDD\_CAN to a pre-regulated internal voltage (2.5V) supplied by VB\_STBY.

When the fault ends the biasing voltage switches again on VDD\_CAN. The msc flag *VDD\_CAN\_OV* or *CAN\_SUP\_LOW* remains latched until an MSC Upstream Read14 is given.

The biasing voltage is switched from VDD\_CAN to a pre-regulated internal voltage (2.5V) supplied by VB\_STBY also in CAN\_BIAS\_ON and CAN\_BIAS\_OFF.

VB\_CAN is the signal that enables CAN cell; if this pin is connected to VB\_STBY the CAN is enabled while if it's connected to GND the CAN is disabled

## 12.1 Functional description

General requirements:

- Compliant with C&S CAN FD certification(ISO16845-2:2015);
- Communication Speed up to 2 Mbit/s. (up to 5Mbit/s in "programming" mode);
- Function range from -27 V to +40 V DC at CAN Pins;
- GND disconnection fail safe at module level;
- GND shift operation at system level;
- Microcontroller interface with CMOS compatible I/O Pins;
- Matched output slopes and propagation delay;
- Receive-only mode available.

In order to further reduce the current consumption in low-power mode, the integrated CAN bus interface offers an ultra low current consumption state when the u-chip is off called Sleep mode.

## 12.2 CAN state diagram

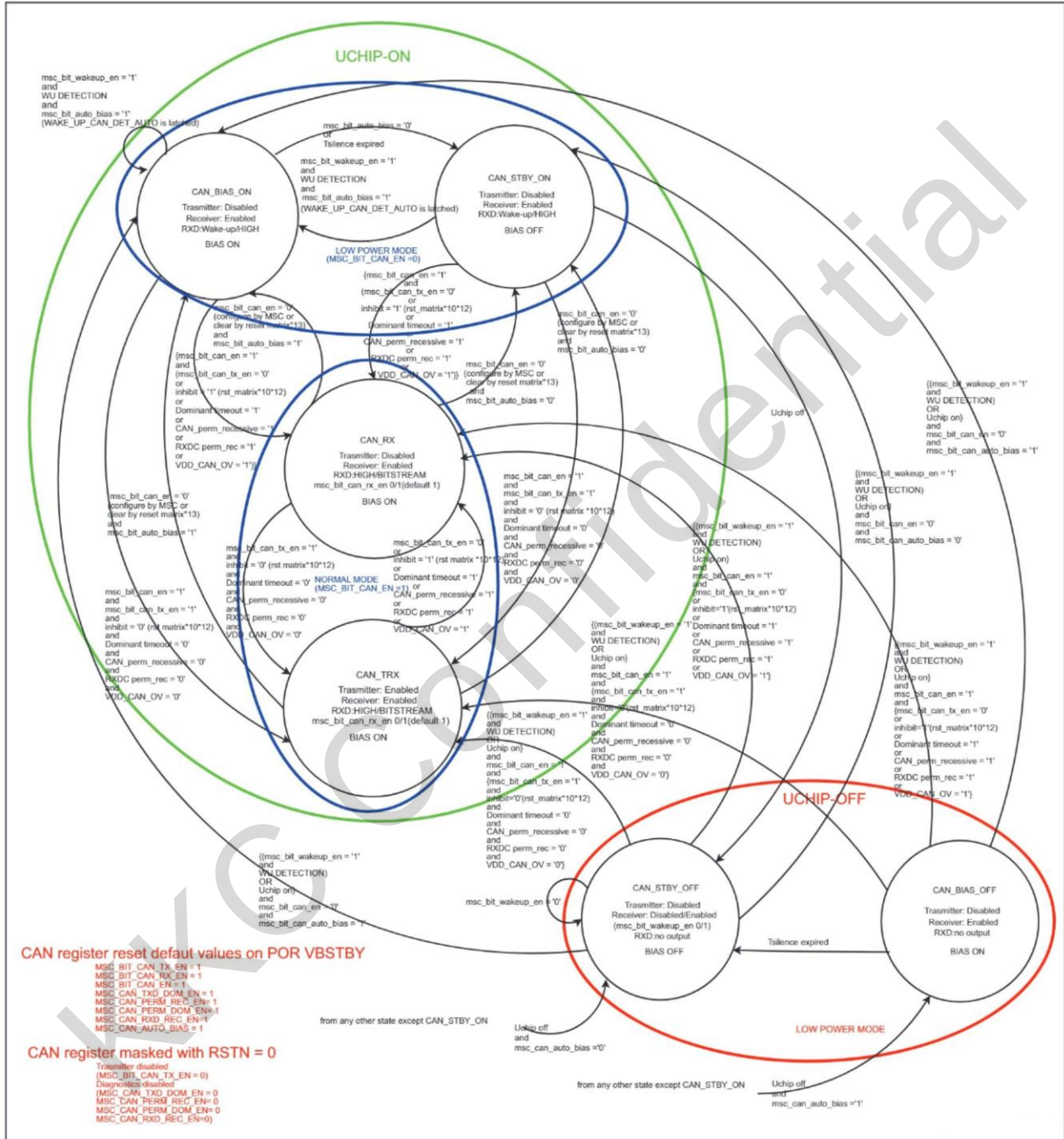


Figure 12-2 CAN state diagram

### 12.3 CAN normal mode

If `CAN_EN` bit `MSC_CONFIG-REG17-1[2]` is set and the U-chip is in ON state the CAN transceiver is in normal mode (referring to state diagram, `CAN_RX` or `CAN_TRX` state). In this state the transmitter and receiver can be configured by MSC (`CAN_TX_EN`, `CAN_RX_EN`) as follows:

TRX Standby: transmitter disabled and `CAN_RX` pin masked to 1 (`CAN_TX_EN` = '0', `CAN_RX_EN` = '0');

TRX Listen: transmitter disabled, receiver enabled (`CAN_TX_EN` = '0', `CAN_RX_EN` = '1');

TRX transmit: transmitter enabled `CAN_RX` pin masked to 1 (`CAN_TX_EN` = '1', `CAN_RX_EN` = '0');

TRX Normal (default): transmitter enabled receiver enabled (`CAN_TX_EN` = '1', `CAN_RX_EN` = '1')

In normal mode the CAN biasing is always active

Wake up by CAN disabled option is allowed only for the application that does not require the transceiver (`CANH/L` not connected to any bus).

### 12.4 CAN low-power mode

When HE9788 is ON it is possible to deactivate the CAN transceiver setting the MSC bit `CAN_EN` (`MSC_CONFIG-REG17_1[2]`) to zero, sending the CAN in Low-power Mode. The CAN transceiver remains deactivated until it will be activated again setting the `CAN_EN` bit. When HE9788 is OFF the CAN transceiver is always in Low-power Mode independently from any other configuration bit except `CAN_WAKEUP_EN`, which if set to zero, sends it to Sleep mode once the transceiver goes to `CAN_STBY_OFF` state. In Low-power mode the biasing can be active or inactive depending on `CAN_AUTO_BIAS` bit value and from the CAN bus activity (see state diagram), furthermore the transmitter is always disabled and the `CAN_RX`

pin is always masked to recessive. When biasing is disabled the receiver input termination will be biased at zero volt.

### 12.5 CAN Sleep Mode

When the CAN transceiver goes in `CAN_STBY_OFF` with `CAN_WAKEUP_EN` = '0' it falls in an ultra low current consumption state called Sleep mode. In this state for the HE9788 is impossible to detect Wake-up activity on CAN bus and consequently no power-up is done for a CAN detection (pattern or no pattern).

### 12.6 CAN error handling

CAN Driver HS and LS are supplied by a dedicated pin (`VDD_CAN`) to avoid disturbances of CAN activity on main supply lines. After internal power on reset of `VB_STBY` the CAN transceiver is configured according to the configuration register bit default values. When the

CAN transceiver is in normal mode, the transmitter is disabled and its state changes to CAN\_RX in case of at least one of the following events:

- Dominant TxDC time out and *CAN\_TXD\_DOM\_EN* MSC bit = 1;
- CAN permanent recessive *CAN\_PERM\_REC\_EN* MSC bit = 1;
- RxDC permanent recessive *CAN\_RXD\_REC\_EN* MSC bit = 1;
- VDD\_CAN overvoltage detection (MSC bit *VDD\_CAN\_OV* = 1);
- VDD\_CAN undervoltage detection active;
- MSC bit *CAN\_TX\_EN* = 0;
- Reset matrix Conditions \*10\*12 (see Reset Matrix section).

The CAN receiver is not disabled in case of any failure condition.

The device provides the following 4 error handling features; the CAN error handling function can be disabled by setting the dedicated corresponding MSC bit. The error handling must be also enabled by setting the *CAN\_EN* bit MSC\_CONFIG-REG17-1[2].

### 12.6.1 Dominant CAN\_TX time out

If CAN\_TX is in dominant state (low) for  $t > t_{TXDC\_DOM\_TO}$  the transmitter will be disabled, *CAN\_TXD\_DOM* status bit will be latched and can be read and cleared by MSC. The transmitter remains disabled until the status register is cleared or until the state returns recessive according to the setting of the bit *CAN\_TX\_DOM\_ERR\_CFG* MSC\_CONFIGREG21 [0]. The detection of this error is enabled by the *CAN\_TXD\_DOM\_EN* bit MSC\_CONFIG\_REG21[3] and by the *CAN\_TX\_EN* bit MSC\_CONFIG\_REG17\_1[3].

### 12.6.2 CAN permanent recessive

If CAN\_TX changes to dominant (low) state but CAN bus does not follow for 4 times, the transmitter will be disabled, *CAN\_PERM\_REC* status bit will be latched and can be read and cleared by MSC. The transmitter remains disabled until the status register is cleared.

The detection of this error is enabled by the *CAN\_PERM\_REC\_EN* bit MSC\_CONFIG\_REG21[4] and by the *CAN\_TX\_EN* bit MSC\_CONFIG\_REG17\_1[3].

### 12.6.3 CAN permanent dominant

If the bus state is dominant (low) for  $t > t_{dom}$  a permanent dominant status will be detected. *CAN\_PERM\_DOM* status bit is latched and can be read and cleared by MSC. The transmitter will not be disabled. The detection of this error is enabled by the *CAN\_PERM\_DOM\_EN* bit MSC\_CONFIG\_REG21[5].

### 12.6.4 CAN\_RX permanent recessive

If CAN\_RX Pin is clamped to recessive (high) state, the controller is not able to recognize a bus dominant state and could start messages at any time, which results in disturbing the overall bus communication. Therefore, if CAN\_RX does not follow CAN\_TX for 4 times the transmitter will be disabled. *CAN\_RXD\_REC* status bit will be latched and can be read and cleared by MSC. The transmitter remains disabled until the status register is cleared. The detection of this error is enabled by the *CAN\_RXD\_REC\_EN* bit MSC\_CONFIG\_REG21[6] and by the *CAN\_TX\_EN* bit MSC\_CONFIG\_REG17\_1[3].

### 12.6.5 Smart reset RSTN effects

In order to avoid spurious interference on CAN Bus and false CAN error detection, during the power-up sequence, as long the Smart Reset RSTN pin signal stays low the following CAN configuration registers are masked to zero:

- `CAN_TX_EN` (transmitter disabled);
- `CAN_TXD_DOM_EN` (Dominant TxDC Time Out disabled);
- `CAN_PERM_REC_EN` (CAN Permanent Recessive disabled);
- `CAN_PERM_DOM_EN` (CAN Permanent Dominant disabled);
- `CAN_RXD_REC_EN` (CAN RXDC Permanent Recessive disabled).

### 12.7 Wake up With U-Chip OFF

When the U-chip is off with CAN wake up option enabled (`CAN_WAKEUP_EN = '1'`), the CAN transceiver can detect wake up activity on bus and then wake up the device. In this situation the MSC bit `WAKE_UP_CAN_DET` is latched (MSC UPSTREAM READ 12, Frame3). If the U-chip is off with CAN wake up option disabled (`CAN_WAKEUP_EN = '0'`) and automatic voltage biasing disabled (`CAN_AUTO_BIAS = '0'`) the transceiver goes in Sleep mode. Otherwise if the U-chip is off with CAN wake up option disabled

(`CAN_WAKEUP_EN = '0'`) and automatic voltage biasing enabled (`CAN_AUTO_BIAS = '1'`) the transceiver goes in `CAN_BIAS_OFF` state and after a period greater than  $T_{\text{silence}}$  without activity on bus goes in Sleep mode.

### 12.8 Wake up with U-Chip ON

When the U-chip is on and the CAN transceiver is in low power mode (`CAN_EN bit = '0'`) and the wake up option is enabled (`CAN_WAKEUP_EN = '1'`), the CAN Transceiver can detect wake up activity on bus. In this situation the MSC bit `WAKE_UP_CAN_DET_AUTO` is latched (MSC UPSTREAM READ 13, Frame3) and a dominant pulse of  $t_{\text{dom\_pulse}}$  is send on pad `CAN_RX`. if the `CAN_AUTO_BIAS` bit MSC\_CONFIG\_REG21[7] is set and the CAN transceiver in the state `CAN_STBY_ON` the detection of a wake up activity will bring the transceiver in the state `CAN_BIAS_ON`

#### 12.8.1 Wake up options

For the wake up feature the device logic differentiates different Wake-up detection conditions.

#### 12.8.2 Normal pattern wake up

Normal pattern wake up can occur when CAN pattern wake up option is enabled (`CAN_PATTERN_EN = '1'`), the wake up enable bit is set (`CAN_WAKEUP_EN = '1'`) and the CAN transceiver is in low-power mode. In order to have a Wake-up detection, the following criteria must be fulfilled:

- The CAN interface wake up receiver must receive a series of two consecutive valid

dominant pulses, each of which must be longer than  $t_{filter}$ ;

- The distance between 2 pulses must be longer than  $t_{filter}$ ;
- The two pulses must occur within a time frame of  $t_{wake}$ .

### 12.8.3 No pattern wake up

If the CAN pattern wake up option is disabled ( $CAN\_PATTERN\_EN = '0'$ ) and the wake up enable bit is set ( $CAN\_WAKEUP\_EN = '1'$ ) and the transceiver is in Low-power mode, a dominant state on CAN BUS for more than  $t_{filter}$  will cause a Wake-up detection.

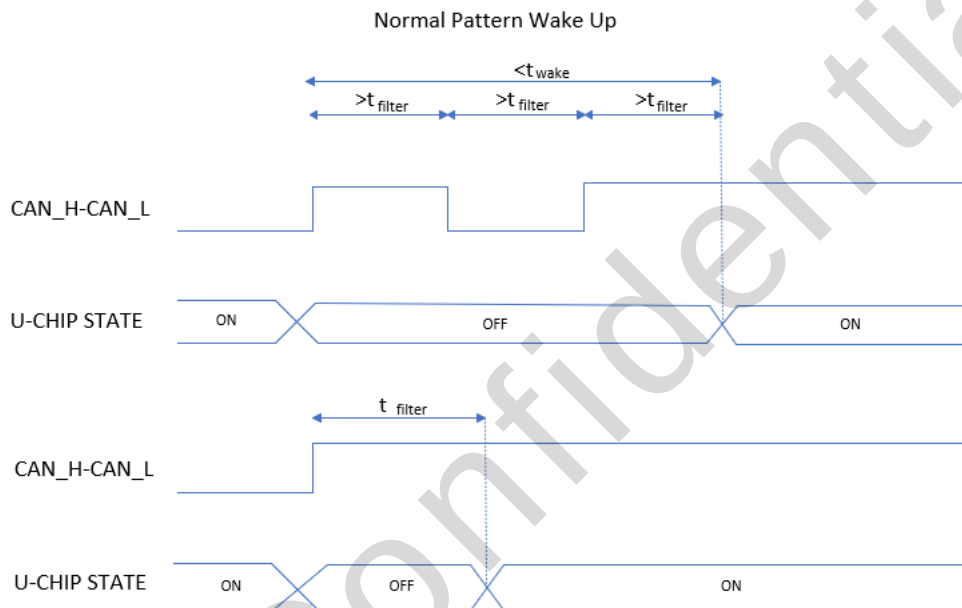


Figure 12-3 CAN wake up options

## 12.9 Automatic voltage biasing

If the CAN transceiver is in Low-power mode ( $CAN\_EN = 0$  or U-CHIP OFF) and the automatic voltage biasing is active ( $CAN\_AUTO\_BIAS = '1'$ , MSC\_CONFIG-REG21[7]) if there has been no activity on the bus for longer than  $t_{SILENCE}$ , the bus lines are biased towards 0 V via the receiver input resistors RCANH/RCANL (biasing disabled) and the MSC flag  $CAN\_SILENT$  is set to 1. If the wake up option is enabled ( $CAN\_WAKEUP\_EN = '1'$ ) and a wake-up activity on the bus lines is detected (wake-up pattern or no pattern), the bus lines are biased to  $V_{CANHrec}$  respectively  $V_{CANLrec}$  via the internal receiver input resistors RCANH/RCANL also the MSC bit  $WAKE\_UP\_CAN\_DET\_AUTO$  is latched (MSC UPSTREAM READ 12, Frame3). The biasing is activated not later than  $t_{Bias}$ .

## 12.10 CAN reset matrix

In the reset matrix table there is a row dedicated to CAN.

Reset matrix \*12 source means that at least one of the following faults is active:

- VDD5\_OV;
- OSC FAULT;
- TNL;
- RST\_N.

When the CAN transceiver is in normal mode and at least one of the above faults occurs the transmitter is disabled.

Reset matrix \*10 source means that at least one of the following fault is active and the *CAN\_TDI* bit is set:

- *ERR\_CNT*>4;
- *ERR\_CNT*>7;
- RST\_PRL;

When the CAN transceiver is in normal mode and at least one of the above faults occurs the transmitter is disabled.

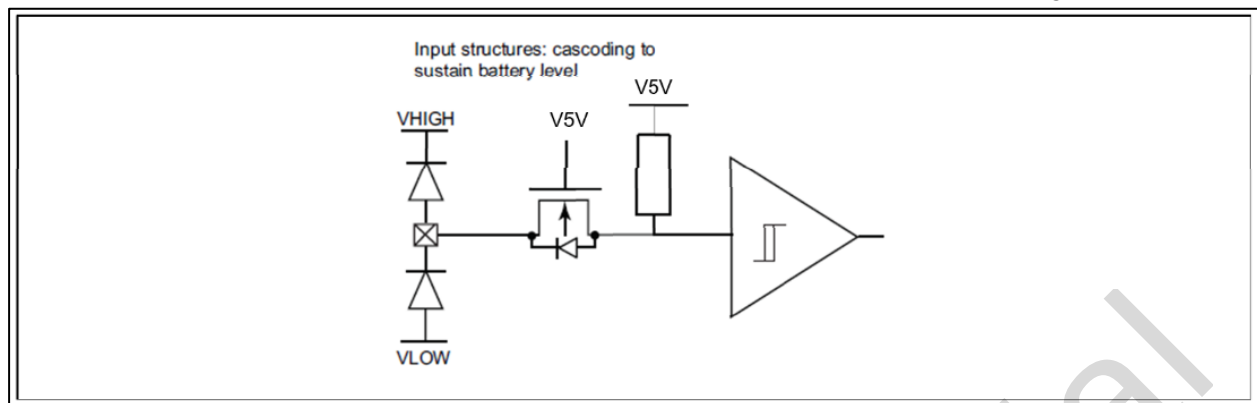
Reset matrix \*13 source means that at least one of the following fault is active:

1. VB\_STBY\_UV
2. VB\_OV  $t > TBOV2$
3. VB\_UV
4. V5VA\_OV, V5VA\_UV, V5VD\_OV and V5VD\_UV
5. VDD5\_UV
6. VDDIO\_UV

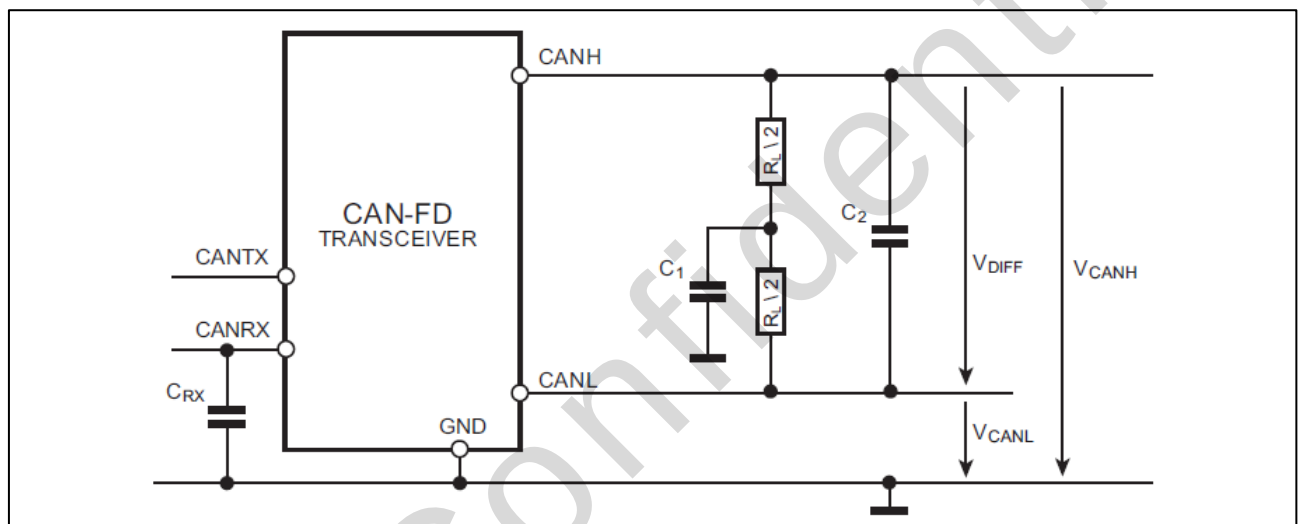
When the CAN transceiver is in normal mode and at least one of the above faults occurs the MSC bit *CAN\_EN* is cleared sending the CAN transceiver in Low-power Mode.

## 12.11 CAN looping mode

If the proper configuration bit (*CAN\_LOOP\_EN*) in control register is set the TxDC input is mapped directly to the RxDC Pin. This mode can be used in combination with the CAN receive-only mode, to run diagnosis for the CAN protocol handler of the microcontroller.



### Figure 12-4 CAN\_TX input structure



### Figure 12-5 CAN transceiver test circuit

**Conditions:**

5.5 V  $\leq$  VB\_IN  $\leq$  18 V , 4.5 V  $\leq$  VDD\_CAN  $\leq$  5.5 V, 5 V  $\leq$  VB\_STBY  $\leq$  18 V, T<sub>j</sub> -40 to 175 °C unless otherwise specified;

### Table 12-1 CAN threshold related Low supply voltage flag

| Symbol          | Parameter                                   | Test condition                  | Min | Typ  | Max | Unit | Pin |
|-----------------|---------------------------------------------|---------------------------------|-----|------|-----|------|-----|
| VDD_CAN_SUP_LOW | Threshold to detect CAN Supply low          | VDD_CAN decreasing              | 4   | 4.65 | 4.8 | V    | -   |
| VB_CAN_TH(1)    | Threshold for CAN cell enable               | Not tested For information only | 0.8 | -    | 2.2 | V    | -   |
| VDD_CAN_SUP_OV  | Threshold to detect CAN Supply over-voltage | VDD_CAN increasing              | 6.1 | 6.85 | 7.5 | V    | -   |

1. Below this threshold all CAN functions are disabled. In case CAN functionality is requested by application VB\_CAN has to be shorted to VB\_stby, otherwise it should be put to GND to disable CAN block.

Table 12-2 CAN communication operating range

| Symbol                   | Parameter                                                                           | Test condition                                                                  | Min | Typ | Max | Unit | Pin |
|--------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----|-----|-----|------|-----|
| VDD_CAN                  | Supply Voltage operating range for CAN transceiver                                  | -                                                                               | 4.5 | -   | 5.5 | V    | -   |
| VCANH,CM                 | Common mode Bus voltage (VCANH + VCANL) / 2                                         | Measured with respect to the ground of the CAN transceiver                      | -12 | -   | 12  | V    | -   |
| ITRCV_REC <sup>(1)</sup> | Transceiver current consumption during normal mode from VDD_CAN, Recessive State    | VTXDC = VTXDCHIGH                                                               | -   | -   | 10  | mA   | -   |
| ITRCV_DOM <sup>(1)</sup> | Transceiver current consumption during normal mode from VDD_CAN Dominate State      | VTXDC = 0 V                                                                     | -   | -   | 60  | mA   | -   |
| ITRCV_short              | Transceiver current consumption during output short from VDD_CAN                    | RL = 50Ω to 65Ω;<br>VCANH = -3 V or<br>VCANL = 40 V                             | -   | -   | 120 | mA   | -   |
| ITRCVLPbias              | Transceiver current consumption during lowpower mode; biasing active from VDD_CAN   | RL = 50Ω to 65Ω;<br>VTXDC = VTXDCHIGH;                                          | -   | -   | 350 | μA   | -   |
| ITRCVLP                  | Transceiver current consumption during lowpower mode; biasing inactive from VDD_CAN | RL = 50 Ω to 65Ω;<br>VTXDC = VTXDCHIGH;                                         | -   | -   | 50  | μA   | -   |
| ITR_VB                   | Transceiver current consumption from VB_CAN                                         | -                                                                               | -   | -   | 5   | μA   | -   |
| BR                       | Supported Bit-rates                                                                 | Supported bit-rates at which all requirements are fulfilled<br>Application info | 5   | -   | -   | Mb/s | -   |

1. To be confirmed after ATE measurements.

**Table 12-3 CAN transmit data input: Pin TxDC**

| Symbol            | Parameter                                     | Test condition                                                                                                                                  | Min | Typ | Max | Unit | Pin    |
|-------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|------|--------|
| Ileak_cantx       | Input leakage current                         | CAN_TX = VDDIO - 1 V                                                                                                                            | -   | -   | 15  | μA   | CAN_TX |
| VTXDCLOW          | Input voltage dominant level                  | -                                                                                                                                               | -   | -   | 0.9 | V    | CAN_TX |
| VTXDCHIGH         | Input voltage recessive level                 | -                                                                                                                                               | 2   | -   | -   | V    | CAN_TX |
| VTXDCCHYS         | VTXDCHIGHVTXDCLOW                             | -                                                                                                                                               | 0.1 | -   | 0.4 | V    | CAN_TX |
| RTXDCEPU          | TxDC pull up resistor                         | -                                                                                                                                               | 20  | 50  | 100 | kΩ   | CAN_TX |
| td,TXDC(dom-rec)  | TxDC - CANH,L Delay Time dominant - recessive | RL = 60 Ω (±1%);<br>C2 = 100 pF (±1%);<br>70%<br>VTXD - VDIFF = 0.5 V;<br>TXDC rise time = 10 ns (10% - 90%)<br>Guaranteed by bench correlation | 0   | -   | 140 | ns   | -      |
| td,TXDC(rec-diff) | TxDC - CANH,L Delay Time recessive - dominant | RL = 60Ω(±1%);<br>C2=100pF (±1%); 30%<br>VTXD - VDIFF = 0.9 V;<br>TXDC fall time = 10ns (90% - 10%)<br>Guaranteed by bench correlation          | 0   | -   | 120 | ns   | -      |
| tTXDC_DOM_TO      | TxDC dominant timeout                         | -                                                                                                                                               | 0.8 | 2   | 5   | ms   | -      |

**Table 12-4 CAN transmit data output: Pin RxDC**

| Symbol               | Parameter                                     | Test condition                                                                 | Min         | Typ        | Max   | Unit | Pin    |
|----------------------|-----------------------------------------------|--------------------------------------------------------------------------------|-------------|------------|-------|------|--------|
| VRXDCLOW             | Output voltage dominant level                 | Active mode,<br>IRXDC = 2 mA                                                   | (0)         | 0.2        | 0.5   | V    | CAN_RX |
| VRXDCHIGH            | Output voltage recessive level                | Active mode,<br>IRXDC = -2 mA                                                  | VDDIO - 0.5 | VDIO - 0.2 | VDDIO | V    | CAN_RX |
| tr, RXDC             | RxDC rise time                                | CRX = 15 pF,<br>30% - 70% VRXDC<br>Guaranteed by bench correlation             | 0           | -          | 25    | ns   | -      |
| tf, RXDC             | RxDC fall time                                | CRX = 15 pF,<br>70% - 30% VRXDC<br>Guaranteed by bench correlation             | 0           | -          | 25    | ns   | -      |
| td, RXDC (dom-rec)   | CANH,L ^ RxDC Delay Time dominant - recessive | CRX = 15 pF,<br>VDIFF = 0.5V - 70%<br>VRXDC<br>Guaranteed by bench correlation | 0           | -          | 120   | ns   | -      |
| td, RXDC (rec - dom) | CANH,L ^ RxDC Delay Time recessive -          | CRX = 15 pF,<br>VDIFF = 0.9 V - 30%<br>VRXDC                                   | 0           | -          | 120   | ns   | -      |

|  |          |  |  |  |  |  |  |
|--|----------|--|--|--|--|--|--|
|  | dominant |  |  |  |  |  |  |
|--|----------|--|--|--|--|--|--|

Table 12-5 CAN transmitter dominant output characteristics

| Symbol             | Parameter                                                                                               | Test condition                                                                                                                                         | Min  | Typ | Max  | Unit | Pin |
|--------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|------|------|-----|
| VCANHdom           | Single Ended CANH voltage level in dominant state                                                       | $V_{TXDC} = V_{TXDCLOW}$ ;<br>$R_L = 50\Omega$ to $65\Omega$ ;                                                                                         | 2.75 | 3.5 | 4.5  | V    | -   |
| VCANLdom           | Single Ended CANL voltage level in dominant state                                                       | $V_{TXDC} = V_{TXDCLOW}$ ;<br>$R_L = 50\Omega$ to $65\Omega$ ;                                                                                         | 0.5  | 1.5 | 2.25 | V    | -   |
| VDIFF, dom         | Differential output voltage in dominant state: VCANHdom-VCANLdom                                        | $V_{TXDC} = V_{TXDCLOW}$ ;<br>$R_L = 50\Omega$ to $65\Omega$ ;                                                                                         | 1.5  | 2.0 | 3    | V    | -   |
| VDIFF, dom_extR    | Differential output voltage in dominant state: VCANHdom-VCANLdom at extended termination resistor range | $V_{TXDC} = V_{TXDCLOW}$ ;<br>$R_L = 45\Omega$ to $70\Omega$ ;                                                                                         | 1.4  | -   | 3.3  | V    | -   |
| VDIFF, dom_Arb     | Differential output voltage in dominant state: VCANHdom-VCANLdom during arbitration                     | $V_{TXDC} = V_{TXDCLOW}$ ;<br>$R_L = 2240\Omega$ ;                                                                                                     | 1.5  | -   | 5    | V    | -   |
| VSym               | Driver Symmetry: $V_{SYM} = (VCANHdom + VCANLdom)/VDD\_CAN$<br>$VDD\_CAN = 5V$                          | Measured over one 1 MHz period (1 $\mu s$ )<br>$R_L = 60\Omega(\pm 1\%)$<br>$f_{TXDC} = 1MHz$ (square wave, 50% duty cycle);<br>$C_1 = 4.7nF(\pm 5\%)$ | 0.9  | 1   | 1.1  | V    | -   |
| IOCANH, dom (-3 V) | CANH output current in dominant state                                                                   | $V_{TXDC} = V_{TXDCLOW}$ ;<br>$VCANH = -3V$ ;                                                                                                          | -115 | -   |      | mA   | -   |
| IOCANL, dom (16 V) | CANL output current in dominant state                                                                   | $V_{TXDC} = V_{TXDCLOW}$ ;<br>$VCANL = 16V$ ;                                                                                                          | -    | -   | 115  | mA   | -   |
| IOCANH, dom (40 V) | CANH output current in dominant state                                                                   | $V_{TXDC} = V_{TXDCLOW}$ ;<br>$VCANH = 40V$ ;<br>$V_s = 40V$                                                                                           | 0    | -   | 5    | mA   | -   |
| IOCANL, dom (40 V) | CANL output current in dominant state                                                                   | $V_{TXDC} = V_{TXDCLOW}$ ;<br>$VCANL = 40V$ ;<br>$V_s = 40V$                                                                                           | (0)  | -   | 115  | mA   | -   |

**Table 12-6 CAN transmitter recessive output characteristics, normal mode**

| Symbol        | Parameter                                                                                 | Test condition                      | Min | Typ | Max | Unit | Pin |
|---------------|-------------------------------------------------------------------------------------------|-------------------------------------|-----|-----|-----|------|-----|
| VCANHrec      | CANH voltage level in recessive state (Normal mode)                                       | $V_{TXDC} = V_{TXDCHIGH}$ ; No Load | 2   | 2.5 | 3   | V    | -   |
| VCANLrec      | CANL voltage level in recessive state (Normal mode)                                       | $V_{TXDC} = V_{TXDCHIGH}$ ; No Load | 2   | 2.5 | 3   | V    | -   |
| VDIFF, recOUT | Differential output voltage in recessive state (Normal mode): $V_{CANHrec} - V_{CANLrec}$ | $V_{TXDC} = V_{TXDCHIGH}$ ; No Load | -50 | -   | 50  | mV   | -   |

**Table 12-7 CAN transmitter recessive output characteristics, low power mode, biasing active**

| Symbol              | Parameter                                                                     | Test condition                                                                          | Min | Typ | Max | Unit | Pin |
|---------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----|-----|-----|------|-----|
| VCANHrecLPbias      | CANH voltage level in recessive state                                         | $V_{TXDC} = V_{TXDCHIGH}$ ; No Load<br>$5.5\text{ V} \leq V_{B\_STBY} \leq 18\text{ V}$ | 2   | 2.5 | 3   | V    | -   |
| VCANLrecLPbias      | CANL voltage level in recessive state                                         |                                                                                         | 2   | 2.5 | 3   | V    | -   |
| VDIFF, recOUTLPbias | Differential output voltage in recessive state<br>$V_{CANHrec} - V_{CANLrec}$ |                                                                                         | -50 | -   | 50  | mV   | -   |

**Table 12-8 CAN transmitter recessive output characteristics, low-power mode, biasing inactive**

| Symbol          | Parameter                                                                     | Test condition                      | Min  | Typ | Max  | Unit | Pin |
|-----------------|-------------------------------------------------------------------------------|-------------------------------------|------|-----|------|------|-----|
| VCANHrecLP      | CANH voltage level in recessive state                                         | $V_{TXDC} = V_{TXDCHIGH}$ ; No Load | -0.1 | 0   | 0.15 | V    | -   |
| VCANLrecLP      | CANL voltage level in recessive state                                         |                                     | -0.1 | 0   | 0.15 | V    | -   |
| VDIFF, recOUTLP | Differential output voltage in recessive state<br>$V_{CANHrec} - V_{CANLrec}$ |                                     | -50  | -   | 50   | mV   | -   |

**Table 12-9 CAN Receiver input characteristics during normal mode**

| Symbol | Parameter                                                           | Test condition                                                                                     | Min | Typ | Max  | Unit | Pin |
|--------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----|-----|------|------|-----|
| VTHdom | Differential receiver threshold voltage recessive to dominant state | $-12\text{ V} \leq V_{CANH} \leq +12\text{ V}$ ,<br>$-12\text{ V} \leq V_{CANL} \leq +12\text{ V}$ | 0.4 | -   | 1.1  | V    | -   |
| VTHrec | Differential                                                        |                                                                                                    | 0.4 | -   | 0.95 | V    | -   |

|  |                                                        |                                                       |  |  |  |  |  |
|--|--------------------------------------------------------|-------------------------------------------------------|--|--|--|--|--|
|  | receiver threshold voltage dominant to recessive state | $-12\text{ V} \leq V_{\text{CANL}} \leq +12\text{ V}$ |  |  |  |  |  |
|--|--------------------------------------------------------|-------------------------------------------------------|--|--|--|--|--|

**Table 12-10 CAN Receiver input characteristics during low power mode, biasing active**

| Symbol       | Parameter                                                           | Test condition                                                                                                   | Min | Typ | Max  | Unit | Pin |
|--------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----|-----|------|------|-----|
| VTHdomLPbias | Differential receiver threshold voltage recessive to dominant state | $-12\text{ V} \leq V_{\text{CANH}} \leq +12\text{ V}$ ,<br>$-12\text{ V} \leq V_{\text{CANL}} \leq +12\text{ V}$ | 0.4 | -   | 0.95 | V    | -   |
| VTHrecLPbias | Differential receiver threshold voltage dominant to recessive state | $-12\text{ V} \leq V_{\text{CANH}} \leq +12\text{ V}$ ,<br>$-12\text{ V} \leq V_{\text{CANL}} \leq +12\text{ V}$ | 0.4 | -   | 0.95 | V    | -   |

**Table 12-11 CAN Receiver input characteristics during low power mode, biasing inactive**

| Symbol   | Parameter                                                           | Test condition                                                                                                 | Min | Typ | Max  | Unit | Pin |
|----------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----|-----|------|------|-----|
| VTHdomLP | Differential receiver threshold voltage recessive to dominant state | $-5\text{ V} \leq V_{\text{CANH}} \leq +12\text{ V}$ ,<br>$-5\text{ V} \leq V_{\text{CANL}} \leq +12\text{ V}$ | 0.4 | -   | 0.95 | V    | -   |
| VTHrecLP | Differential receiver threshold voltage dominant to recessive state | $-5\text{ V} \leq V_{\text{CANH}} \leq +12\text{ V}$ ,<br>$-5\text{ V} \leq V_{\text{CANL}} \leq +12\text{ V}$ | 0.4 | -   | 0.95 | V    | -   |

**Table 12-12 CAN Receiver input resistance**

| Symbol      | Parameter                                | Test condition                                                                                                                                                           | Min   | Typ | Max  | Unit       | Pin |
|-------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|------|------------|-----|
| Rdiff       | Differential internal resistance         | $V_{\text{TXDC}} = V_{\text{TXDCHIGH}}$ ; No Load                                                                                                                        | 12    | -   | 100  | k $\Omega$ | -   |
| RCANH, CANL | Single Ended Internal resistance         | $V_{\text{TXDC}} = V_{\text{TXDCHIGH}}$ ; No Load                                                                                                                        | 6     | -   | 50   | k $\Omega$ | -   |
| mR          | Internal Resistance matching RCANH, CANL | Biasing active;<br>$V_{\text{TXDC}} = V_{\text{TXDCHIGH}}$ ; no load<br>$R = 2 \times (R_{\text{CAN\_H}} - R_{\text{CAN\_L}}) / (R_{\text{CAN\_H}} + R_{\text{CAN\_L}})$ | -0.03 | -   | 0.03 | -          | -   |
| Cin         | Internal capacitance                     | Guaranteed by design                                                                                                                                                     | -     | -   | 66   | pF         | -   |
| Cin, diff   | Differential internal capacitance        | Guaranteed by design                                                                                                                                                     | -     | -   | 33   | pF         | -   |

Table 12-13 CAN transceiver delay

| Symbol           | Parameter                             | Test condition                                                                                                                                                                                                                                                                                                                                                                                                | Min | Typ  | Max  | Unit | Pin |
|------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|------|------|-----|
| tLOOP,hl         | Loop delay TXDC to RXDC (High to Low) | RL = 60Ω(±1%); C2 = 100pF (±1%); 30% VTxDC ^ 30% VRXDC; TXDC fall time = 10ns (90% - 10%); CRX = 15 pF                                                                                                                                                                                                                                                                                                        | -   | -    | 255  | ns   | -   |
| tLOOP,lh         | Loop delay TXDC to RXDC (Low to High) | RL = 60Ω(±1%); C2 = 100pF (±1%); 70% VTxD ^ 70% VRxD; TXDC fall time = 10ns (10% - 90%); CRX = 15 pF                                                                                                                                                                                                                                                                                                          | -   | -    | 255  | ns   | -   |
| TBit(RXD)≤1 Mb/s | Recessive bit symmetry at RXDC        | RL = 60Ω(±1%); VDIFF: 0.5 V(falling) - 0.9 V(rising); C2 = 100pF (±1%); CRX = 15 pF; TXD rise and fall time = 10 ns (10% - 90%, 90% - 10%); Test signal to be applied on the TXD input of the implementation is a square wave signal with a positive duty cycle of 1/6 and a period of six times the nominal recessive bit width. Rectangular pulse signal TTXDC= 6000ns, high pulse 1000ns, low pulse 5000ns | 745 | 1000 | 1255 | ns   | -   |
| TBit(RXD)≤2Mb/s  | Recessive bit symmetry at RXDC        | Rectangular pulse signal TTXDC= 3000ns, high pulse 500ns, low pulse 2500ns                                                                                                                                                                                                                                                                                                                                    | 400 | 500  | 550  | ns   | -   |
| TBit(RXD)≤5Mb/s  | Recessive bit symmetry at RXDC        | Rectangular pulse signal TTXDC = 1200 ns, high pulse 200 ns, low pulse 1000 ns                                                                                                                                                                                                                                                                                                                                | 120 | 200  | 250  | ns   | -   |
| TBit(BUS)≤1Mb/s  | Recessive bit symmetry at CAN-Bus     | RL = 50 Ω to 65Ω; VDIFF: 0.5 V(falling) - 0.9 V(rising); CL = 100 pF; CRXD = 15 pF; TXD rise and fall time = 10 ns (10% - 90%, 90% - 10%); Test signal to be                                                                                                                                                                                                                                                  | 800 | 1000 | 1100 | ns   | -   |

|                                    |                                                                                                                           |                                                                                                                                                                                                                                                          |     |     |      |    |   |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|------|----|---|
|                                    |                                                                                                                           | applied on the TXD input of the implementation is a square wave signal with a positive duty cycle of 1/6 and a period of six times the nominal recessive bit width<br>Rectangular pulse signal<br>TTXDC = 6000 ns, high pulse 1000 ns, low pulse 5000 ns |     |     |      |    |   |
| TBit(BUS) ≤ 2Mb/s                  | Recessive bit symmetry at CAN-Bus                                                                                         | Rectangular pulse signal<br>TTXDC = 3000 ns, high pulse 500 ns, low pulse 2500 ns                                                                                                                                                                        | 435 | 500 | 530  | ns | - |
| TBit(BUS) ≤ 5Mb/s                  | Recessive bit symmetry at CAN-Bus                                                                                         | Rectangular pulse signal<br>TTXDC = 1200 ns, high pulse 200 ns, low pulse 1000 ns;<br>Guaranteed by bench correlation                                                                                                                                    | 155 | 200 | 210  | ns | - |
| $\Delta t_{REC} \leq 2\text{Mb/s}$ | Receiver Timing Symmetry (TBit(RXD) - TBit(BUS))                                                                          | Rectangular pulse signal<br>TTXDC = 3000 ns, high pulse 500 ns, low pulse 2500 ns;<br>Guaranteed by bench correlation                                                                                                                                    | -65 | -   | 40   | ns | - |
| $\Delta t_{REC} \leq 5\text{Mb/s}$ | Receiver Timing Symmetry (TBit(RXD) - TBit(BUS))                                                                          | Rectangular pulse signal<br>TTXDC = 1200 ns, high pulse 200 ns, low pulse 1000 ns;<br>Guaranteed by bench correlation                                                                                                                                    | -45 | -   | 15   | ns | - |
| Tdom                               | CAN permanent dominant time-out                                                                                           | Guaranteed by scan                                                                                                                                                                                                                                       | 500 | 700 | 1000 | μs | - |
| tWUP-RXD                           | time between WUP <sup>(1)</sup> on the CAN bus until RXD is active (i.e. the CAN signal is represented at the RXD output) | Wake-Up according to ISO11898-5<br>RXD output enabled;<br>Guaranteed by scan                                                                                                                                                                             | 0   | -   | 1    | ms | - |

1. Time starts with the end of last dominant phase of the WUP.

**Table 12-14 Maximum leakage currents on CAN\_H and CAN\_L, unpowered**

| Symbol                     | Parameter                  | Test condition                                                    | Min | Typ | Max | Unit | Pin |
|----------------------------|----------------------------|-------------------------------------------------------------------|-----|-----|-----|------|-----|
| I <sub>Leakage, CANH</sub> | Input leakage current CANH | Unpowered device;<br>VCANH = 5V ; VCANL = 5V;<br>VDD_CAN, VB_STBY | -10 | -   | 10  | μA   | -   |

|                            |                            |                                                                                                                                                          |     |   |    |         |   |
|----------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---|----|---------|---|
|                            |                            | connected via 0 $\Omega$ to GND<br>VDD_CAN, VB_STBY connected via 47 k $\Omega$ to GND                                                                   |     |   |    |         |   |
| I <sub>Leakage, CANL</sub> | Input leakage current CANL | Unpowered device;<br>VCANH = 5V ; VCANL = 5V;<br>VDD_CAN, VB_STBY connected via 0 $\Omega$ to GND<br>VDD_CAN, VB_STBY connected via 47 k $\Omega$ to GND | -10 | - | 10 | $\mu$ A | - |

Table 12-15 Biasing control timings

| Symbol                  | Parameter                | Test condition                                                                                                                                                                                               | Min  | Typ | Max  | Unit    | Pin |
|-------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|------|---------|-----|
| t <sub>filter</sub>     | CAN activity filter time | Guaranteed by scan                                                                                                                                                                                           | 0.5  |     | 1.8  | $\mu$ s |     |
| t <sub>wake</sub>       | Wake-up time out         | Guaranteed by scan                                                                                                                                                                                           | 0.35 | 1   | 5    | ms      |     |
| t <sub>Silence</sub>    | CAN timeout              | Guaranteed by scan                                                                                                                                                                                           | 600  | 700 | 1200 | ms      |     |
| t <sub>BIAS</sub>       | Bias Reaction Time       | RL = 60 $\Omega$ ( $\pm 1\%$ );<br>C2 = 100 pF ( $\pm 1\%$ );<br>C1 = 100 pF ( $\pm 1\%$ );<br>VTXDC = VTXDCLOW;<br>50% VDIFF; VSYM = 0.1(1);<br>Transition bias inactive to bias active; Guaranteed by scan | -    | -   | 220  | $\mu$ s | -   |
| t <sub>cansuphigh</sub> | CAN_SUP_LOW filter time  | Guaranteed by scan                                                                                                                                                                                           | 8.5  | 10  | 11.5 | $\mu$ s | -   |
| t <sub>cansuplow</sub>  | VDD_CAN_OV filter time   | Guaranteed by scan                                                                                                                                                                                           | 8.5  | 10  | 11.5 | $\mu$ s | -   |
| t <sub>dom_pulse</sub>  | Dominant interrupt pulse | Guaranteed by scan                                                                                                                                                                                           | 47.6 | 56  | 64.4 | $\mu$ s | -   |

1. Measured from the start of a dominant-recessive-dominant sequence (each phase 6 $\mu$ s) until VSYM  $\geq$  0.1

Table 12-16 Standby current consumption (B attery line @RT)

| Total quiescent current consumption(1)                   | Max value | Unit    |
|----------------------------------------------------------|-----------|---------|
| EOT disabled and CAN block enabled, CAN wake up disabled | 60        | $\mu$ A |
| EOT disabled and CAN block enabled, CAN wake up enabled  | 70        | $\mu$ A |
| EOT enabled and CAN block enabled, CAN wake up disabled  | 70        | $\mu$ A |
| EOT enabled and CAN block enabled, CAN wake up enabled   | 80        | $\mu$ A |

1. HE9788 Total quiescent current consumption at pins VB\_STBY, VB\_IN, VB\_IN\_SW, VB\_SENSE, CP (through 100 nF capacitor), VB\_CAN, MRD, RLY1, LED2, STR2\_DRN, PDR1\_DRN connected to battery. The total current was measured at battery line @RT.

Table 12-17 Standby current consumption (Battery line @HT)

| Total quiescent current consumption(1) | Max value | Unit |
|----------------------------------------|-----------|------|
|----------------------------------------|-----------|------|

|                                                          |     |                        |
|----------------------------------------------------------|-----|------------------------|
| EOT disabled and CAN block enabled, CAN wake up disabled | 140 | CAN FD interface<br>μA |
|----------------------------------------------------------|-----|------------------------|

1. HE9788 Total quiescent current consumption at pins VB\_STBY, VB\_IN, VB\_IN\_SW, VB\_SENSE, CP (through 100 nF capacitor), VB\_CAN, MRD, RLY1, LED2, STR2\_DRN, PDR1\_DRN connected to battery. The total current was measured at battery line @HT.

Expected additional current due to bias ON is:

**Table 12-18** can\_auto\_bias = 1 and can\_wakeup\_en = 1

| -   | 27 °C, I(VB_STBY) | 150 °C, I(VB_STBY) |
|-----|-------------------|--------------------|
| TYP | 548 μA            | 576 μA             |
| MAX | 650 μA            | 675 μA             |

**Table 12-19** can\_auto\_bias = 1 and can\_wakeup\_en = 0

| -   | 27 °C, I(VB_STBY) | 150 °C, I(VB_STBY) |
|-----|-------------------|--------------------|
| TYP | 563 μA            | 591 μA             |
| MAX | 660 μA            | 690 μA             |

## 13 LIN/K-LINE interface

### 13.1 Functional description

This interface is adapted to “LIN Specification Rev.2.1”, and it is also adapted to “K-LINE Specification (ISO9141)”.

The Chip has over-voltage protection, loss of ground protection from -27 V to 40 V. -27 V AMR is guaranteed at room & hot temperature. +40 V AMR is guaranteed in all temp range. Short Circuit to Battery or Ground Protection is provided. If the bus line is short circuit to battery, the over current protection is protected by current limitation function. Upon return of connection, normal operation resumes without any intervention on the LIN bus line.

In case of power loss (VB\_IN pin disconnected or power off) or ground loss (GND pin disconnected) or LIN block in receive only mode (Transmission function disabled), the LIN driver will not disturb the communication of the remaining transceivers connected to LIN or KLINE bus. Upon return of connection, normal operation will resume without any intervention on the LIN bus line.

LIN-TX is internally pulled up to internal 5 V.

All values in this section apply over the entire operating temperature and life of the U-chip.

The LIN block MAX baudrate is 20 Kbit/s.

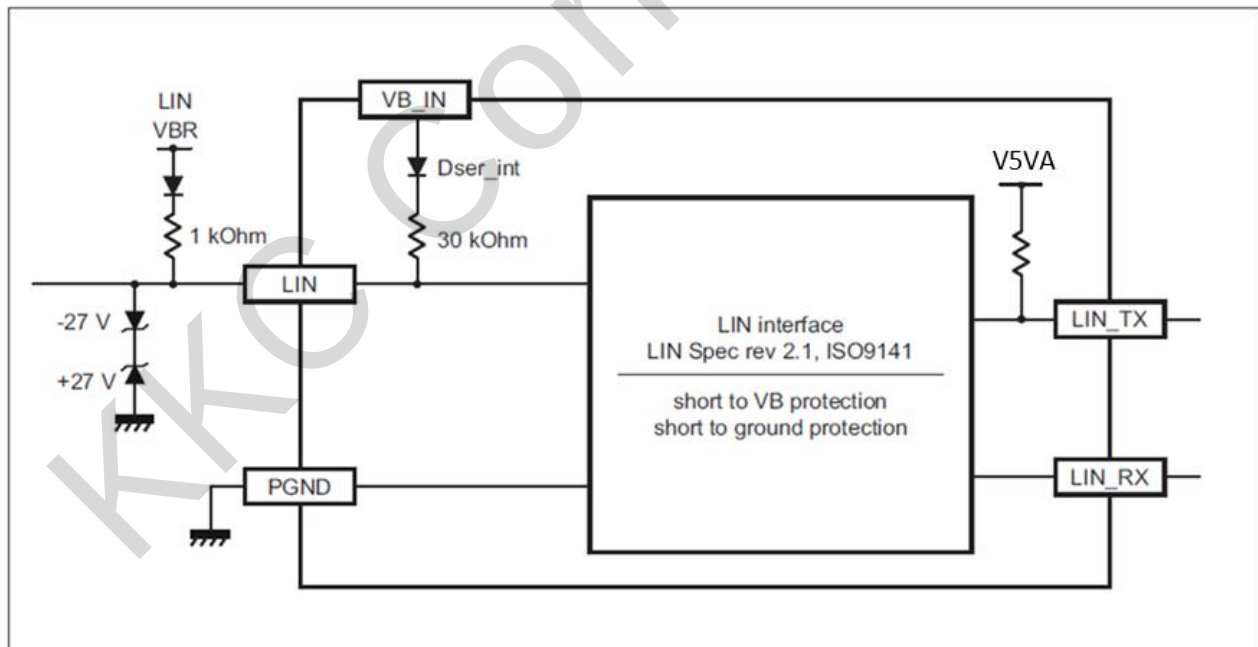


Figure 13-1 Block diagram of LIN interface

### 13.2 LIN receive only mode

There is a MSC bit (*LIN\_TX\_EN* CONFIG-REG 11) to put the LINE block in receive only mode (Transmission function disabled). Refer to reset matrix for the conditions to enter receive only mode.

### 13.3 LIN thermal shut down

The LIN block has a dedicated thermal sensor to protect itself.

To protect LIN block from temperature overheat (high battery range of operation, soft short conditions, etc.) a dedicated thermal sensor placed close to power mos is present in the layout sensing FET temperature; One threshold is implemented (*T\_SD\_H*) with Hysteresis (*T\_SD\_hys*). When the *T\_SD\_H* threshold is detected the Driver switches OFF. Once the power stage has been switched-off for over-temperature detection, it will be able to switch on again when temperature is decreased below thermal shut down threshold plus hysteresis value to avoid high frequency on-off cycling.

### 13.4 LIN error handling

The Chip provides the following 3 error handling features which are not described in the standard LIN Spec Rev2.1, but are realized in different stand alone LIN Interface / microcontrollers to switch the application back to normal operation mode.

The error handling features can be disabled through the *LIN\_ERR\_EN[2:0]* (CONFIG-REG 11 D1~D3)bit.

#### 13.4.1 LIN dominant TXD timeout

If TXDC is in dominant state (low) for  $t > t_{TXDC\_DOM\_TO}=12\text{ms}(\text{typ})$  the transmitter will be disabled, status bit *LIN\_TXD\_DOM* (Upstream Bit Map Read15 frame 1) will be latched and can be read and cleared by MSC. The transmitter remains disabled until the status register is cleared or until the state returns recessive, according to MSC configuration bit *LIN\_TX\_DOM\_ERR\_CFG*.

#### 13.4.2 LIN Permanent Dominant

In case the bus state is dominant (low level) for more than  $t_{LIN2}: 12\text{ ms}(\text{typ})$  a permanent dominant status is detected. The status bit *LIN\_PERM\_DOM* (Upstream Bit Map Read15 frame 1) is latched and can be read and cleared by MSC. The transmitter is not switched off.

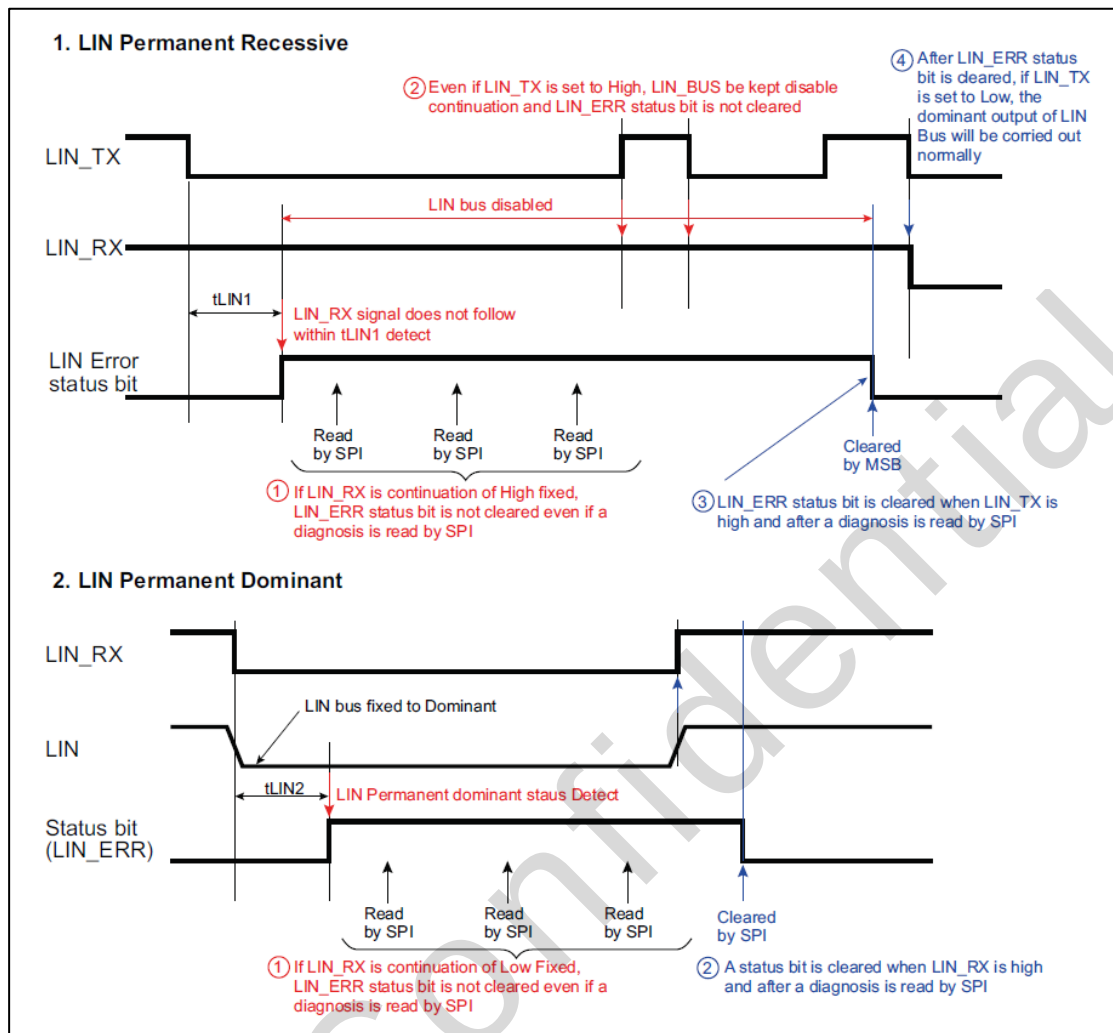


Figure 13-2 LIN TimeOut function

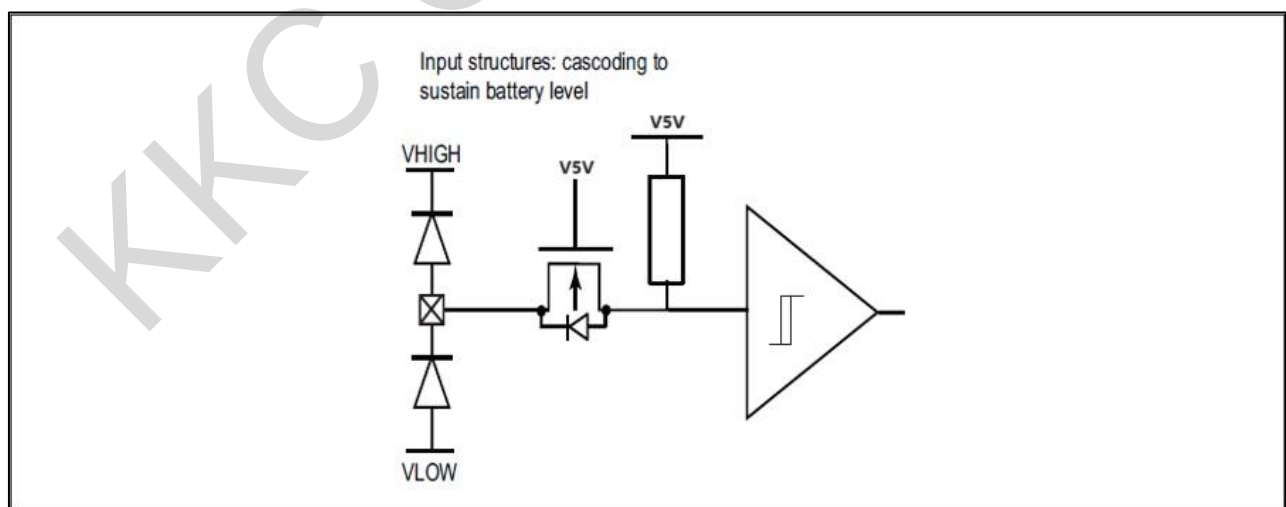


Figure 13-3 LIN\_TX input structure

**Conditions:**

7 V  $\leq$  VB\_IN  $\leq$  18 V, T<sub>j</sub> -40 to 175 °C unless otherwise specified; VB\_IN = 18 to 27 V, T<sub>j</sub>  $\leq$  50°C, all specs in limit. The rest of supply range, parameter are degraded or reset.

If an ECU is not intending to transmit on the LIN bus (e.g. transmit input of a LIN transceiver is recessive), the LIN driver will not drive the LIN bus to dominant state. If the LIN bus is in recessive state, the LIN receiver output will provide a recessive state.

**Table 13-1 Electrical characteristics of LIN interface**

| Symbol                       | Parameter                              | Test condition                                       | Min         | Typ          | Max       | Unit | Pin               |
|------------------------------|----------------------------------------|------------------------------------------------------|-------------|--------------|-----------|------|-------------------|
| I <sub>leak_lintx</sub>      | Input leakage current                  | LIN_TX = VDDIO-1 V                                   | -           | -            | 10        | μA   | LIN_Tx            |
| VIH_TX                       | HIGH_level Input voltage               | -                                                    | 2           | -            | -         | V    | LIN_Tx            |
| VIL_TX                       | LOW_level Input voltage                | -                                                    | -           | -            | 0.9       | V    | LIN_Tx            |
| VHYS_TX                      | Input voltage hysteresis               | -                                                    | 0.1         | -            | 0.4       | V    | LIN_Tx            |
| PULL_TX                      | Pull-up resistance                     | L in_TX = 0 V                                        | 50          | 100          | 200       | kΩ   | LIN_Tx            |
| VOH_RX                       | HIGH-level output voltage              | I <sub>source</sub> = 2 mA                           | VDDIO - 0.6 | VD DIO - 0.2 | -         | V    | LIN_Rx            |
| VOL_RX                       | LOW-level output voltage               | I <sub>sink</sub> = 2 mA                             | -           | 0.2          | 0.5       | V    | LIN_Rx            |
| VOL_LIN                      | LOW-level output voltage               | Pull up = 500 Ω, LIN_Tx = 0 V                        | -           | -            | 0.2 VB_IN | V    | LIN               |
| VOH_LIN                      | HIGH-level output voltage              | Pull up = 500Ω, LIN_Tx = HIGH                        | 0.8 VB_IN   | -            | VB_IN     | V    | LIN               |
| I <sub>bus_dom</sub>         | Dominant Source Current                | LIN_Tx = HIGH, VB_IN = 12 V, LIN = 0V                | -600        | -            | 0         | μA   | LIN               |
| I <sub>bus_res</sub>         | Resessive sink current                 | LIN_Tx=HIGH,VB_IN= 8~18V, LIN=8~18V,LIN $\geq$ VB_IN | 0           | -            | 20        | μA   | LIN               |
| I <sub>bus_stby</sub>        | Standby current                        | VB_IN=8~18V, LIN=8~18V,LIN $\geq$ VB_IN              | 0           | -            | 20        | μA   | LIN               |
| IOVC                         | Current limitation                     | LIN=18V, LIN_Tx=0V                                   | 70          | -            | 150       | mA   | LIN               |
| R <sub>l-g_iso</sub>         | LIN bus to Ground isolation resistance | -                                                    | 500K        | -            | -         | Ω    | LIN               |
| ILK_NG1                      | Ground loss leakage current            | VB_IN=12V, GND = Open,LIN=0~18V                      | -1          | -            | 1         | mA   | LIN               |
| ILK_NB1                      | VB_IN Loss Leakage Current             | VB_IN = Open,LIN = 0~18 V                            | -23         | -            | 23        | μA   | LIN               |
| t <sub>TXDC_DOM_TO</sub>     | TXDC dominant time-out                 | -                                                    | 9           | 12           | 15        | ms   | LIN               |
| t <sub>LIN1</sub> (Time Out) | LIN permanent recessive                | -                                                    | 30          | 40           | 50        | μs   | LIN               |
| t <sub>LIN2</sub>            | LIN permanent dominant                 | -                                                    | 9           | 12           | 15        | ms   | LIN               |
| ViD_LIN                      | Input voltage for dominant state       | Application information                              | -           | -            | 0.4 VB_IN | V    | LIN(Bus Receiver) |

## LIN/K-LINE interface

|                                         |                                         |                                                                                                                       |                |               |                |      |                   |
|-----------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------|---------------|----------------|------|-------------------|
| Vth_DOM                                 | Input voltage for dominant threshold    | Application information                                                                                               | 0.4<br>VB_IN   | 0.45<br>VB_IN | 0.5<br>VB_IN   | V    | LIN(Bus Receiver) |
| ViR_LIN                                 | Input voltage for recessive state       | Application information                                                                                               | 0.6<br>VB_IN   | -             | VB_IN          | V    | LIN(Bus Receiver) |
| Vth_REC                                 | Input voltage for recessive threshold   | -                                                                                                                     | 0.5<br>VB_IN   | 0.55<br>VB_IN | 0.6<br>VB_IN   | V    | LIN(Bus Receiver) |
| Vth_CNT =<br>(Vth_REC +<br>Vth_DOM) / 2 | Input receiver tolerance center voltage | -                                                                                                                     | 0.475<br>VB_IN | 0.5<br>VB_IN  | 0.525<br>VB_IN | V    | LIN(Bus Receiver) |
| Vth_HYS =<br>Vth_REC -<br>Vth_DOM       | Input voltage hysteresis voltage        | -                                                                                                                     | 0.07<br>VB_IN  | 0.1<br>VB_IN  | 0.175<br>VB_IN | V    | LIN(Bus Receiver) |
| tdly(LIN)HL                             | Output delay time HtoL                  | Cbus = 10 nF,<br>Pull up = 500Ω                                                                                       | -              | -             | 50             | μs   | LIN(Bus Receiver) |
| tdly(LIN)LH                             | Output delay time LtoH                  | Cbus = 10 nF,<br>Pull up = 500Ω                                                                                       | -              | -             | 50             | μs   | LIN(Bus Receiver) |
| tDLY_HL                                 | Input delay time HtoL                   | CRXD=20pF                                                                                                             | -              | -             | 6              | μs   | LIN(Bus Receiver) |
| tDLY_LH                                 | Input delay time LtoH                   | CRXD=20pF                                                                                                             | -              | -             | 6              | μs   | LIN(Bus Receiver) |
| tDLY                                    | Input delay time                        | tDLY_HL- tDLY_LH                                                                                                      | -2             | -             | 2              | μs   | LIN(Bus Receiver) |
| D1                                      | Duty Cycle                              | Thresmax = 0.744VB_IN,<br>Thdommax =<br>0.581VB_IN,<br>VB_IN = 7-18 V,<br>tbit = 50 μs,<br>D1 = tbus_resmin/(2xtbit)  | 0.396          | -             | -              | -    | LIN(Bus Receiver) |
| D2                                      | Duty Cycle                              | Thresmin = 0.422VB_IN,<br>Thdommin = 0.284VB_IN,<br>VB_IN = 7.6-18V,<br>tbit = 50 μs,<br>D2 = tbus_resmax/(2xtbit)    | -              | -             | 0.581          | -    | LIN(Bus Receiver) |
| D3                                      | Duty Cycle                              | Thresmax = 0.778VB_IN,<br>Thdommax =<br>0.616VB_IN,<br>VB_IN = 7-18 V,<br>tbit = 96 μs,<br>d3 = tbus_resmin/(2xtbit)  | 0.417          | -             | -              | -    | LIN(Bus Receiver) |
| D4                                      | Duty Cycle                              | Thresmin = 0.389 VB_IN,<br>Thdommin = 0.251<br>VB_IN,<br>VB_IN = 7.6-18 V,<br>tbit=96μs,<br>D4 = tbus_resmax/(2xtbit) | -              | -             | 0.59           | -    | LIN(Bus Receiver) |
| trise_LIN                               | Slew Rate(LIN)                          | Cbus = 2.2nF, Rbus =<br>1kΩ,<br>40-60%,VB_IN=13.5V                                                                    | 0.8            | 2             | 3              | V/μs | LIN(Bus Receiver) |
| Tfall_LIN                               | Slew Rate(LIN)                          | Cbus = 2.2nF, Rbus =<br>1kΩ,<br>40-60%,VB_IN=13.5V                                                                    | 0.8            | 2             | 3              | V/μs | LIN(Bus Receiver) |
| trise_K_LINE                            | Slew Rate(K_LINE)                       | Cbus = 7.2 nF + 2 nF,<br>Rbus = 510Ω,<br>20-80%,VB_IN = 18 V                                                          | 0.8            | -             | 5              | V/μs | LIN(Bus Receiver) |
| Tfall_K_LINE                            | Slew Rate(K_LINE)                       | Cbus = 7.2 nF + 2 nF,<br>Rbus = 510Ω,<br>20-80%,VB_IN = 18 V                                                          | 1.2            | -             | 11             | V/μs | LIN(Bus Receiver) |
| T_SD_HIGH                               | Temperature shut                        | -                                                                                                                     | 185            | -             | 200            | °C   | LIN               |

|                          |                                                                 |                                       |     |    |     |    |     |
|--------------------------|-----------------------------------------------------------------|---------------------------------------|-----|----|-----|----|-----|
|                          | down                                                            |                                       |     |    |     |    |     |
| T_SD_LOW                 | Temperature shut down recover                                   | -                                     | 175 | -  | 190 | °C | LIN |
| T_SD_hys                 | Temperature shut down hysteresis                                | -                                     | 5   | -  | 10  | °C | LIN |
| -                        | Thermal shutdown analog filter time                             | Guaranteed by design                  | 1.5 | -  | 4   | μs | LIN |
| -                        | Digital deglitch filter time on Temperature shut down detection | Guaranteed by scan                    | -   | 10 | -   | μs | LIN |
| Rslave <sup>(1)</sup>    | -                                                               | -                                     | 20  | 30 | 60  | kΩ | LIN |
| VserDiode <sup>(1)</sup> | Diode drop + pullup resistor drop                               | Ibias = 10 μA                         | 0.7 | 1  | 1.3 | V  | LIN |
| VserDiode_drop           | Diode drop                                                      | Ibias= 550 μA<br>Guaranteed by design | 0.7 | 1  | 1.3 | V  | LIN |
| C <sub>LINIO</sub>       | LINIO input capacitance                                         | Guaranteed by design                  | -   | -  | 25  | pF | LIN |

1. LIN Physical Layer Spec Revision 2.1 November 24, 2006; Page 119

### 13.4.3 Timing diagram

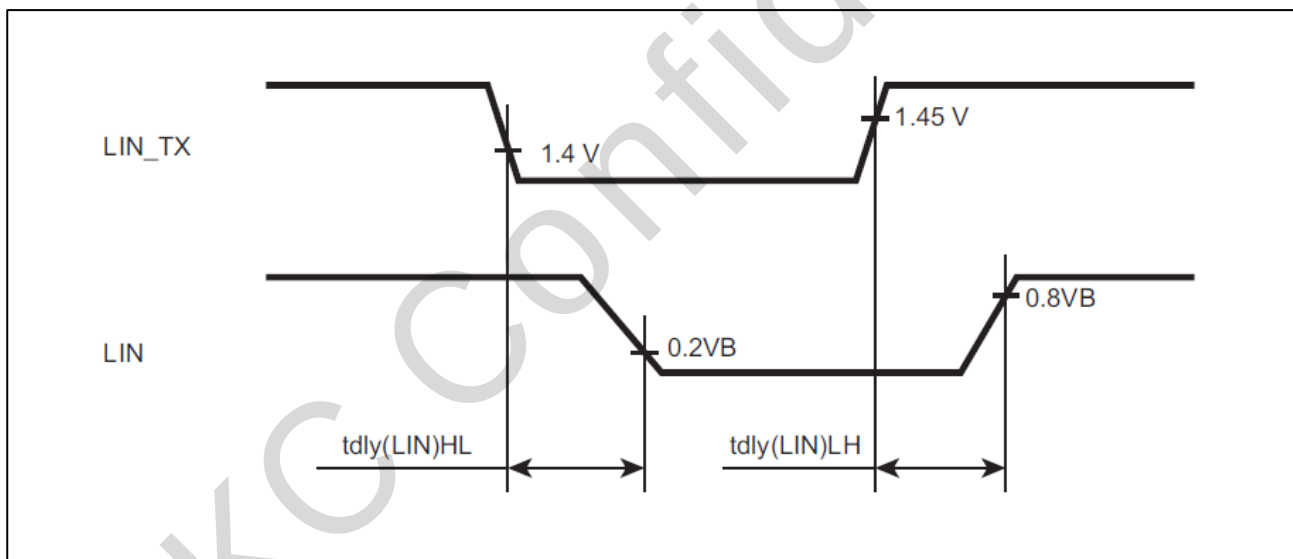


Figure 13-4 LIN transmission timing chart

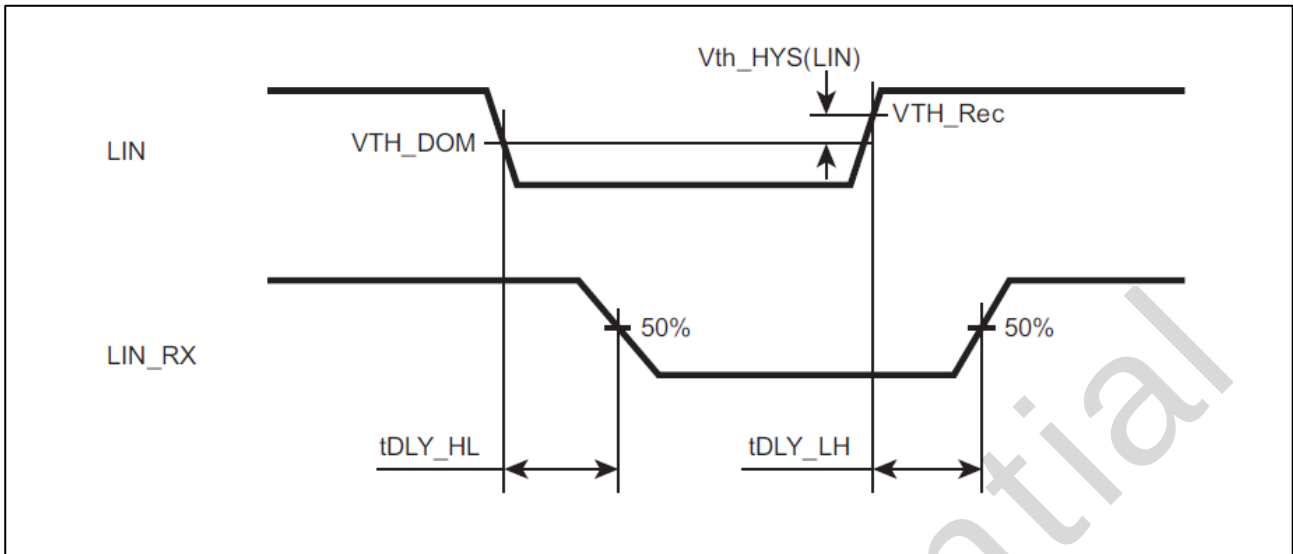


Figure 13-5 LIN reception timing chart

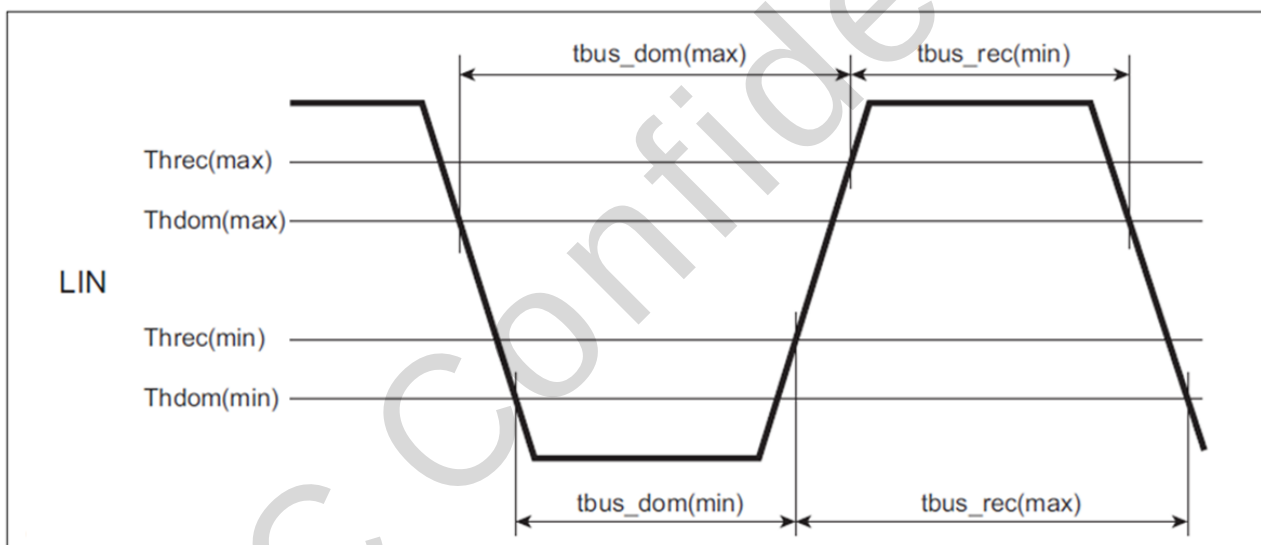


Figure 13-6 LIN duty cycle timing chart

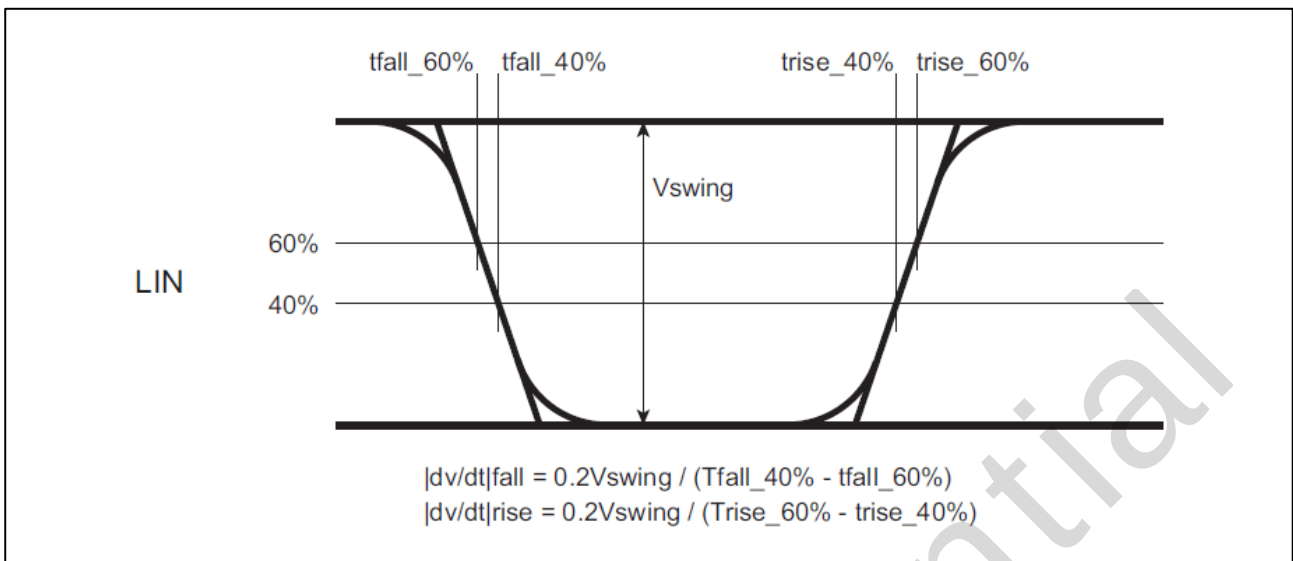


Figure 13-7 LIN slew rate timing chart

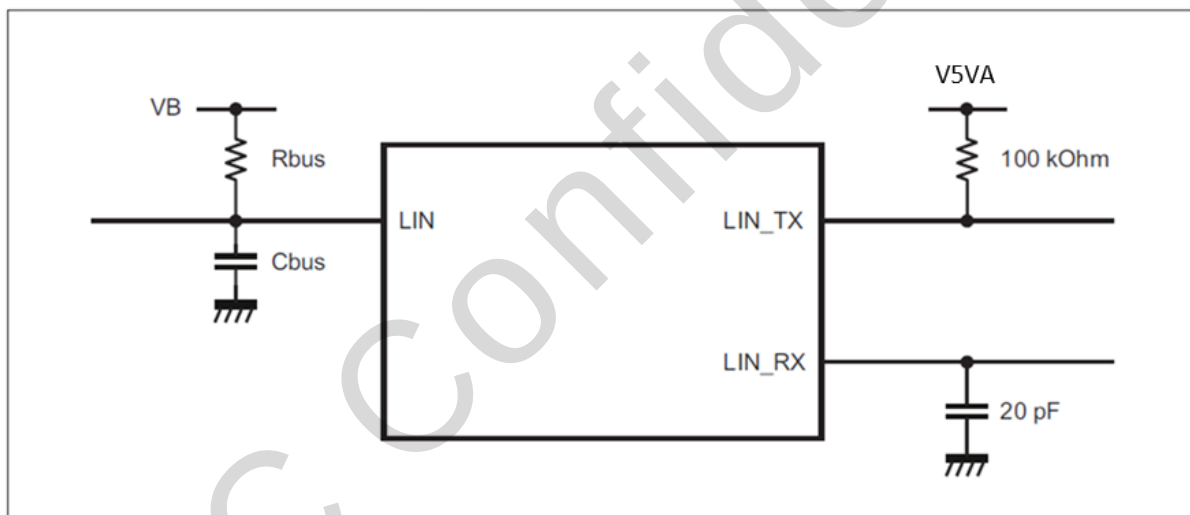


Figure 13-8 Test circuit for measurement of slew rate

#### 13.4.4 LIN permanent recessive

In case TXD changes to dominant (low level) state but RXD signal does not follow within  $t_{LIN1}$ : 40  $\mu$ s (typ), the transmitter is disabled. The status bit *LIN\_PERM\_REC* (Upstream Bit Map Read15 frame 1) is latched and can be read and cleared by MSC. The transmitter remains disabled until the status register is cleared.

### 14 Built in self test

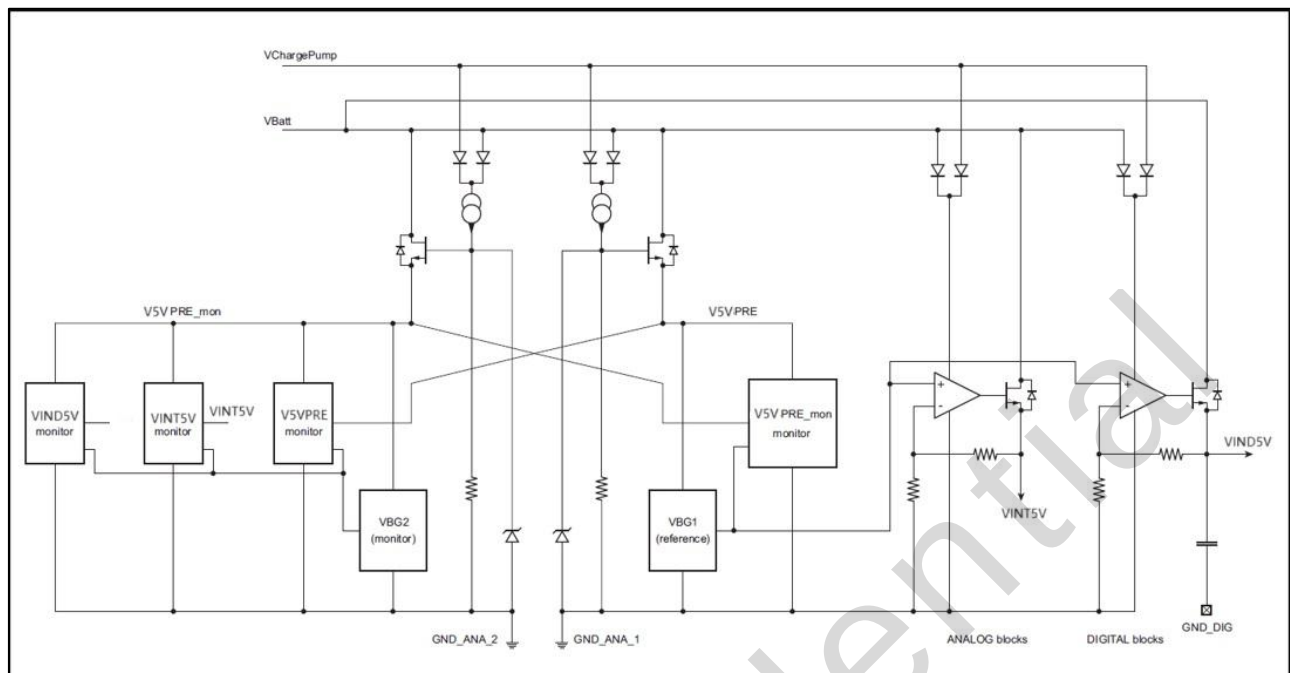
#### 14.1 Power supply independency and voltage monitors

In order to guarantee independence between monitor functions of the HE9788 and monitored functions of the same HE9788 two independent band-gap references are used; a simple field of application of this concept is the under/over voltage detection of MCU power supply generated by the same HE9788.

The basic schematic of the band-gap used to generate the main function and of the bandgap used to monitor it, including their own supply architecture, is shown in [Figure 14-1](#).

Two band-gaps VBG1 and VBG2 are generated starting from protected battery line: VBG1 is used as reference voltage to generate internal analog and digital supply lines and to generate external power supply voltages; VBG2 is used as reference for the uv/ov monitors of these supplies.

The concept to guarantee independence between the two references is that both of them have independent supply and ground architectures to ensure that a single point failure in one of the two structures will not affect the other; independence is achieved by means of physical isolation between the two band-gaps and their supply circuits (A and B areas), isolation is done layouting the circuits in different floor plan areas and by usage of deep trench isolation in between.



**Figure 14-1 Band-gap supply architecture**

A cross-referenced monitor is used to detect eventual failures in any of the band-gap supply and enter a safe state: VBG1 is used as reference for the monitor of the supply of VBG2 and VBG2 is used as monitor of the supply of VBG1.

The concept is needed to solve dependencies of band-gap voltage from the supply line: let us suppose that V5VPRE\_mon in the above picture has a fault impacting VBG2 voltage reference, since V5VPRE\_mon is also the supply for monitor circuits a failure in this region will cause the loss of monitor functionality with an undetected latent fault; the insertion of a monitor supplied from V5VPRE and referenced to VBG1 allows the detection of this latent.

The same concept applies on the vice-versa.

The remaining common points of the internal architecture can be summarized as:

- protected battery line,
- charge pump supply,
- V5VPRE\_mon monitor input (netA),
- V5VPRE monitor input (netB)

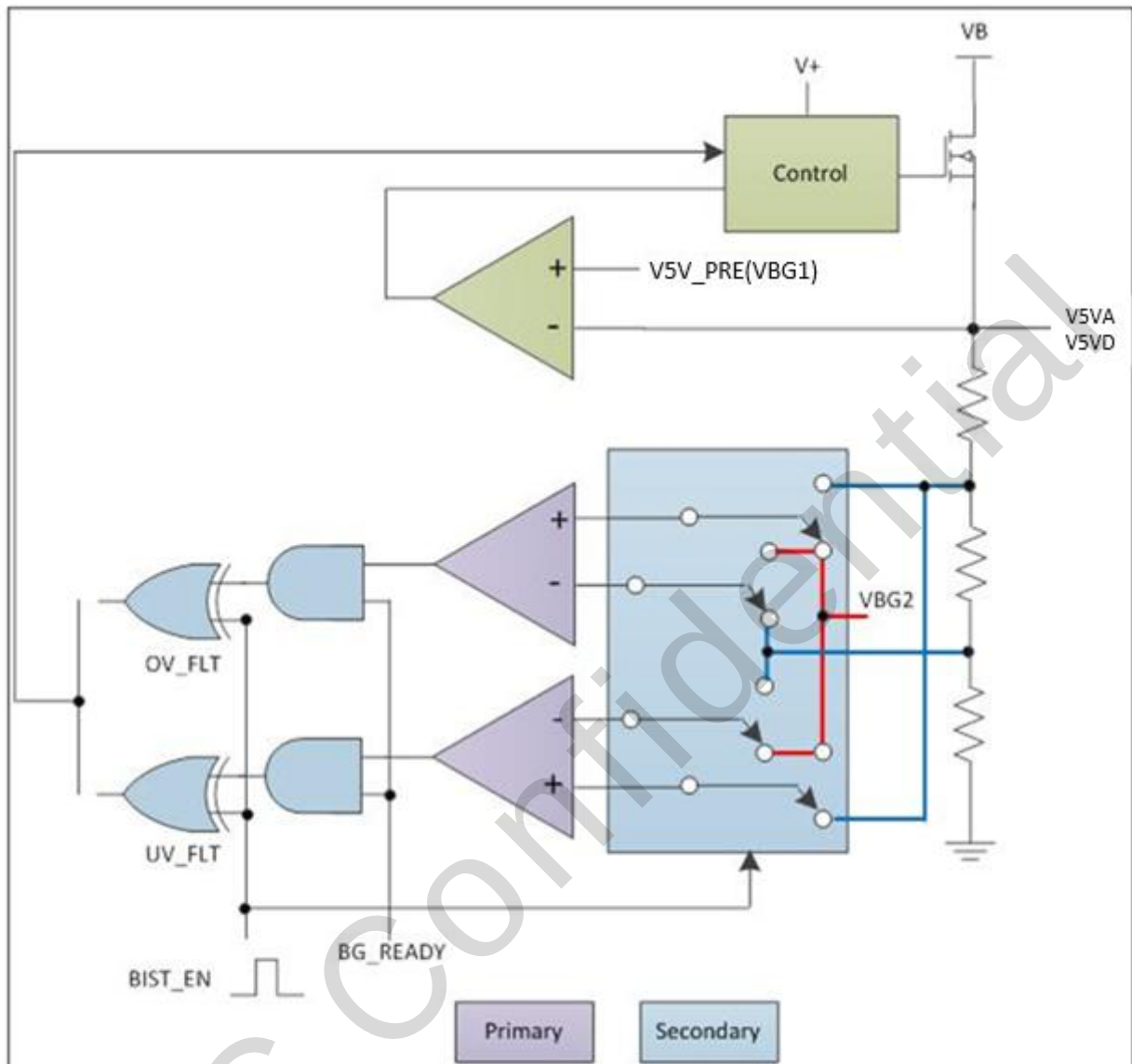
The above points can be tolerated because:

- Protected battery line is the main supply of the HE9788, used to generate supply for analog and digital core. All the circuits connected to this rail are capable of high voltage operation, in case the connection is lost no logic supply can be generated and the device will be stuck in reset condition.
- Charge pump is used to guarantee proper functionality during battery cranking only and is useless in normal mode. All the circuits connected to this line are capable of high voltage, in case the line is open it can be detected by dedicated monitor (not shown here) during normal operation or at power-up of the device.

- In case netA is shorted to ground there is fault detection while short to supply line is normal condition, short to VBG1 through V5VPRE\_mon monitor inputs is not possible since the inputs are cascaded; in case a double fault is present (shorted cascade) it will cause an increase in analog and digital supply lines (VINT5V and VIND5V) detected by their monitors using VBG2 reference and triggering safety switch-off path for safety relevant outputs.
- In case netB is shorted to ground there is fault detection while short to supply line is normal condition, short to VBG2 through V5VPRE monitor inputs is not possible since the inputs are cascaded; in case a double fault is present (shorted cascade) it will cause a failure on the monitor circuit only, without affecting normal functionality until a third fault occurs.

## 14.2 Analog comparators BIST

For all the safety relevant monitors a self test is implemented to verify that the comparator can detect a variation of the monitored signal applied at its input. The basic implementation scheme is shown in Figure 14-2: comparator inputs are chopped by a diagnostic clock and the output of the comparator is xored with diagnostic clock itself: if the comparator is not able to toggle due to fault condition (internal or external) its output will start toggling following diagnostic clock and causing a self test failure.



**Figure 14-2 Analog bist implementation**

The analog bist is applied to the following analog comparators:

V5VA\_OV, V5VA\_UV, V5VD\_OV, V5VD\_UV (internal supply monitors) VDD5\_OV, VDD5\_UV (VDD5 monitors)

The bist is run on MSC command *BIST\_EN*. The result is *read and clear by MSC*.

- On driver section BIST is implemented for diagnostic of main outputs:
- OVC, OPL, STG comparators (injectors)
- STG comparator (igniter)
- The bist is run on MSC command. The result is read and clear by MSC.

## 15 Stand-by memory

This memory is a general purpose memory registers array for microcontroller-data which is supposed to be saved during standby when ECU is switched off. Intention is to reduce the number of write cycles in microcontroller-flash.

In case of standby-power-supply failure the data is no longer valid. This condition can be detected reading flag *VB\_STBY\_UV* MSC\_READ13.FRAME2[0].

As long as the standby-power-supply is valid, no data is lost.

There are 15 memory-registers which can be used for application data. The 16th register is an address-register which cannot be used for application data.

Before writing application data into one of the memory-registers, the address of the memory-register must be written into the address-register with the *STBY\_NVM\_ADD\_REG* command.

After setting the address a *STBY\_NVM\_ADD\_REG* command can write the corresponding array byte.

Using the *STBY\_NVM\_ADD\_REG* command and thus updating the address-register is not necessary in case the previously written address is still correct, e.g. when application writes consecutively to same register.

The memory-register-addresses can have values from 0 to 14. Value 15 is not used for write commands. This means *STBY\_NVM\_ADD\_REG* commands will have no effect when writing to address 15.

The application data is written with mentioned *STBY\_NVM\_ADD\_REG* command which writes one byte (8bit) of application data into the memory-register. To write more than one register, *STBY\_NVM\_ADD\_REG* and *STBY\_NVM\_ADD\_REG* sequences are necessary.

As the other registers above, the upstream reading of the registers is not done register-wise but in blocks of four registers. *MEM\_REG1 ... 4*, *MEM\_REG5 ... 8*, *MEM\_REG9 ... 12* and *MEM\_REG13 ... 15* are grouped in the four upstream blocks of Read16.

The address-register *MEM\_ADR\_REG16* is located at the end of the memory and is the 16th register. As mentioned, it cannot be used for application data. It is the fourth byte of the upstream Read16 and can be read back with a read command for this block.

Before upstream-reading, the register-address must be set with the *STBY\_NVM\_ADD\_REG* command. If the previously written address is already the correct one, the *STBY\_NVM\_ADD\_REG* command is not necessary.

For upstream reading of the *MEM\_REGx* and *MEM\_ADR\_REG16* registers, the command *RD\_COMMAND16* is used. The *RD\_COMMAND16* command is the same for all four upstream blocks as the block to be read is defined via the address-register. The read command will not only read the addressed register, but will read the complete four-byte upstream-block which contains the addressed register.

The *STBY\_NVM\_ADD\_REG* command allows to write a *MEM\_VALID* bit (D6) that can be set to 1 by the microcontroller to signal that the memory has been written and validated reading back data. If standby-power-supply is removed *MEM\_VALID* bit is reset.

## 16 DAC

Device integrates a 2 bit DAC, with 4 possible output voltages between 0V and VDD5\_IN. Selected level is determined by 2 bits DAC[0:1] into *CONFIG\_REG\_7*.

**Conditions:**

$5.5\text{ V} \leq \text{VB\_IN} \leq 18\text{ V}$ ,  $4.75\text{ V} \leq \text{VDD\_IO} \leq 5.25\text{ V}$ ,  $T_j -40$  to  $175\text{ }^{\circ}\text{C}$  unless otherwise specified.

**Table 16-1 DAC electrical parameters**

| Symbol | Parameter         | Test condition                        | Min | Typ | Max | Unit          | Pin         |
|--------|-------------------|---------------------------------------|-----|-----|-----|---------------|-------------|
| V00    | Output voltage 00 | -                                     | 0.7 |     | 1   | V             | AD_TES<br>T |
| V01    | Output voltage 01 | -                                     | 1.5 |     | 1.8 | V             | AD_TES<br>T |
| V10    | Output voltage 10 | -                                     | 2.2 |     | 2.8 | V             | AD_TES<br>T |
| V11    | Output voltage 11 | -                                     | 3.6 |     | 4.7 | V             | AD_TES<br>T |
| trise  | Rising time       | Transition from 00 to 11 10%-90% open | -   | -   | 10  | $\mu\text{s}$ | AD_TES<br>T |
| tfall  | Falling time      | Transition from 11 to 00 90%-10% open | -   | -   | 10  | $\mu\text{s}$ | AD_TES<br>T |

## 17 ADC

A 10 bit ADC converter is integrated into device in order to give information about silicon temperature.

10 bit conversion can be read with *READ\_COMMAND12* MSC frame. Conversion formula is  $T = \text{MSCcode} * 290 / 1024 - 63$  (°C).

**Conditions:**

$5.5 \text{ V} \leq \text{VB\_IN} \leq 18 \text{ V}$ ,  $4.75 \text{ V} \leq \text{VDD\_IO} \leq 5.25 \text{ V}$ ,  $T_j - 40$  to  $175$  °C unless otherwise specified.

**Table 17-1 ADC electrical parameters**

| Symbol   | Parameter                           | Test condition       | Min | Typ | Max | Unit | Pin |
|----------|-------------------------------------|----------------------|-----|-----|-----|------|-----|
| ADC_bits | ADC effective number of bits        | Guaranteed by design | -   | 10  | -   | -    | -   |
| Trange   | Internal junction temperature range | Guaranteed by design | -50 | -   | 200 | °C   | -   |
| Tacc     | Temperature accuracy                | Guaranteed by design | -10 | -   | 10  | °C   | -   |

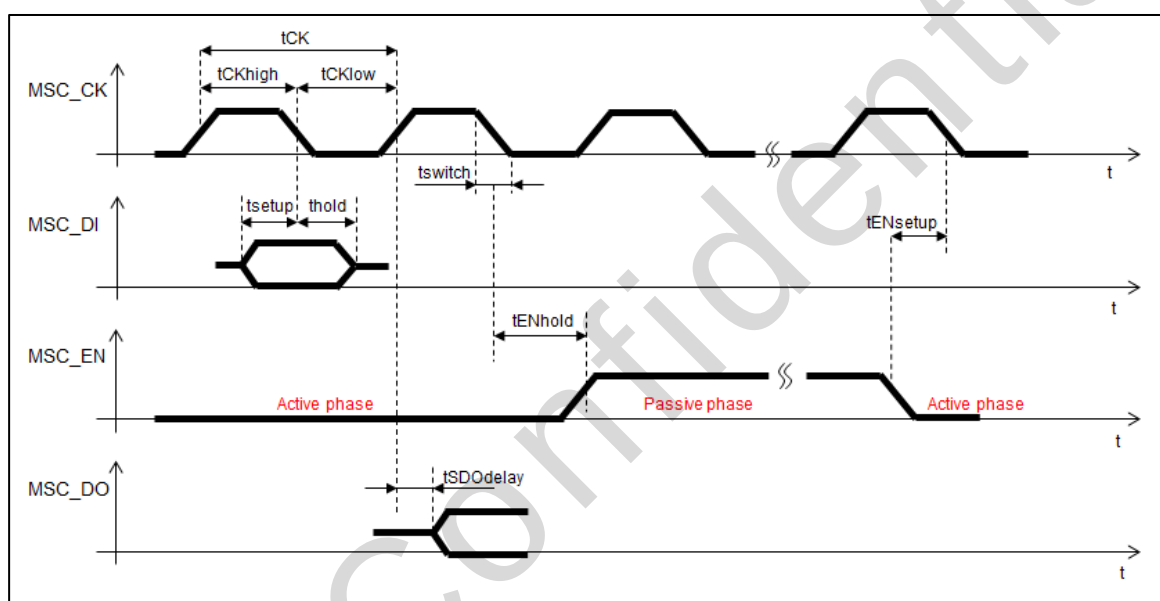
## 18 Micro second channel (MSC) interface

### 18.1 Function description

The Micro Second Channel interface is an adapted high performance serial bus for power switch device.

HE9788 has one Micro Second Channel slave controller capable to receive up to 35 MHz downstream and to transmit up to 2.18 Mbaud upstream.

MSC Clock is supported with synchronous continuous mode. Master clock and data are implemented by LVDS. Slave data out signal is implemented by LVTTTL.



**Figure 18-1** MSC communication timing diagram

#### Conditions:

$5.5\text{ V} \leq V_{B\_IN} \leq 18\text{ V}$ ,  $4.75\text{ V} \leq V_{DD\_IO} \leq 5.25\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified.

**Table 18-1** MSC communication timing parameters

| Symbol  | Parameter                                                  | Test condition  | Min | Typ | Max | Unit | Pin |
|---------|------------------------------------------------------------|-----------------|-----|-----|-----|------|-----|
| tCK     | Cycle time                                                 | Tested by SCAN/ | 25  | -   | -   | ns   | -   |
| tsetup  | Data setup time                                            | Tested by SCAN/ | 5   | -   | -   | ns   | -   |
| thold   | Data hold time                                             | Tested by SCAN/ | 5   | -   | -   | ns   | -   |
| Tswitch | Switching time for CL and SI measured between 0.1*VDD3 and | Tested by SCAN/ | -   | -   | 3   | ns   | -   |

|           |                                                                                                            |                 |     |                       |     |     |   |
|-----------|------------------------------------------------------------------------------------------------------------|-----------------|-----|-----------------------|-----|-----|---|
|           | 0.9*VDD3                                                                                                   |                 |     |                       |     |     |   |
| tCKlow    | CK low time                                                                                                | Tested by SCAN/ | 7.5 | -                     | -   | ns  | - |
| tCKhigh   | CK high time                                                                                               | Tested by SCAN/ | 7.5 | -                     | -   | ns  | - |
| tENsetup  | EN setup time (i.e. time between falling edge of EN and next falling edge of CK)                           | Tested by SCAN/ | 5   | -                     | -   | ns  | - |
| tENhold   | EN hold time (i.e. time between falling edge of CK and next rising edge of EN)                             | Tested by SCAN/ | 5   | -                     | -   | ns  | - |
| tSDO /tCK | Data out cycle time F_DO_SEL (CONFIG_REG2 [1:2]) = "11"<br>F_DO_SEL="10"<br>F_DO_SEL="01"<br>F_DO_SEL="00" | Tested by SCAN/ | -   | 128<br>64<br>32<br>16 | -   | -   | - |
| fCK       | Clock range at CK as long as there is a clock at pins MSC_CK_P, MSC_CK_N. tCPP $\geq 2 * t_{CL}$           | Tested by SCAN/ | -   | -                     | 35  | MHz | - |
|           | Clock range at CK as long as there is a clock at pins MSC_CK_P, MSC_CK_N. tCPP $\geq 4 * t_{CL}$           | Tested by SCAN/ | -   | -                     | 40  | MHz | - |
| -         | tSDOdelay                                                                                                  | Tested by SCAN/ | -   | -                     | 160 | ns  | - |

The MSC is used to receive the input command and data from CPU and to transmit an output data to CPU. Four signals are used according to the timing chart of Figure 0-2:

#### EN: Bus Enable

There is one input for chip select at pin [EN] This signal is LVTTTL Interface from Master to Slave. MSC uses inverted polarity for EN: a logic '1' is a 'passive level' and a logic '0' is a 'active level'. It is possible to drive multiple power devices with shared CL and DI lines and individual EN signal.

#### CL: Synchronous Serial Clock

The clock pins are [CLP] and [CLN], the differential clock. [CLP]-[CLN] is referred to as CL. The maximum downstream clock rate is CL= 35 MHz. There is an internal resistor between pins [CLP] and [CLN]. This signals are LVDS Interface from Master to Slave.

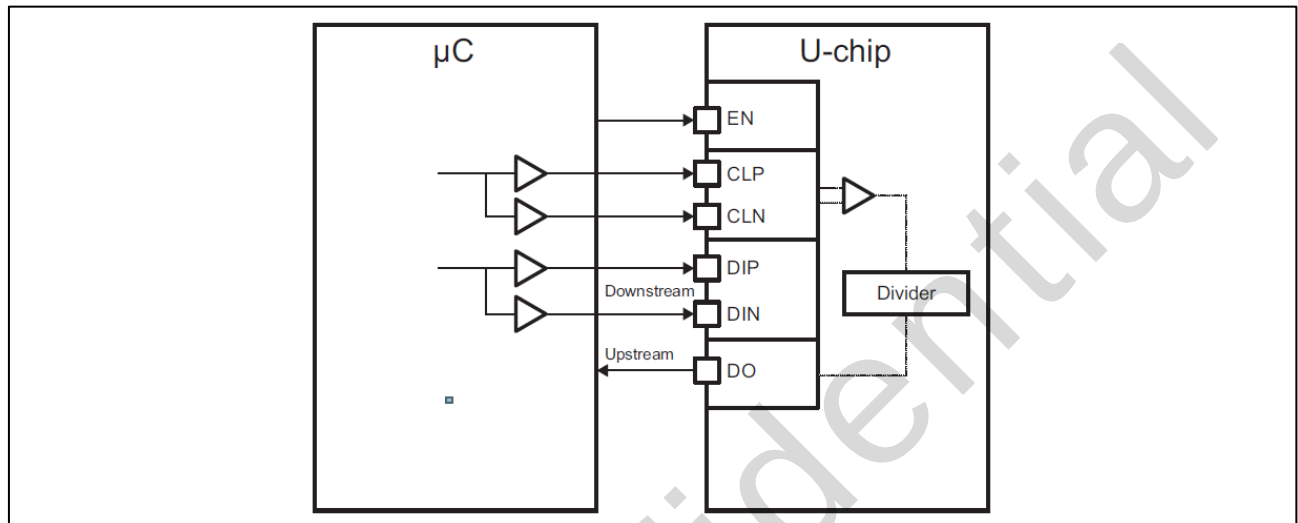
#### DI: Serial Input Data

Differential inputs for downstream data are pins [DIP] and [DIN]; the differential input signal [DIP]-[DIN] is referred to as DI. There is an internal resistor between pins [DIPP] and [DIN]. These signals are LVDS Interface from Master to Slave

#### DO: Serial Output Data

There is one push-pull output for upstream data at pin [DO]. Upstream is done with a lower

clock rate  $f_{SDO}$ , selectable by the microcontroller; after a reset the upstream clock rate is  $f_{SDO} = CL/64$ . The upstream clock is synchronous with CL since it is derived from a clock divider. Therefore the CL signal must be always running independently whether a downstream transmission is running or not. This signal is LVTTTL Interface from Slave to Master.



**Figure 18-2** Communication diagram between  $\mu C$  and U-CHIP

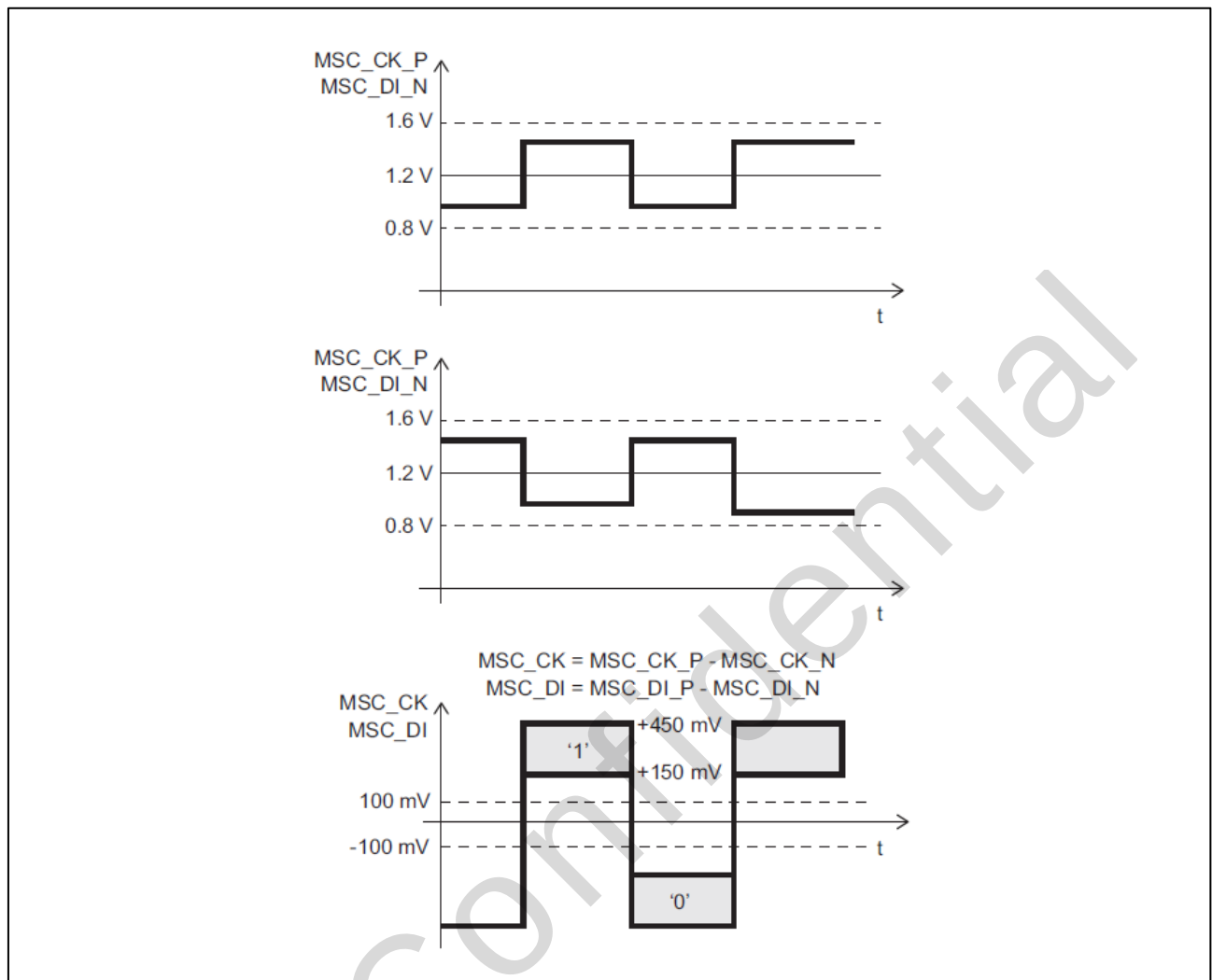


Figure 18-3 MSC voltage levels diagram

**Conditions:**

$5.5\text{ V} \leq V_{B\_IN} \leq 18\text{ V}$ ,  $4.75\text{ V} \leq V_{DD\_IO} \leq 5.25\text{ V}$ ,  $T_j$  -40 to 175 °C unless otherwise specified.

Table 18-2 MSC electrical characteristics

| Symbol      | Parameter                                                    | Test condition                              | Min | Typ | Max | Unit | Pin                  |
|-------------|--------------------------------------------------------------|---------------------------------------------|-----|-----|-----|------|----------------------|
| VCK_P,VCK_N | Input voltage range                                          | Not to be tested. It is an application note | 0.8 | -   | 1.6 | V    | MSC_CK_P<br>MSC_CK_N |
| VCKdiff     | Differential input voltage<br>$VCK_{diff} =  VCK_P - VCK_N $ | Not to be tested. It is an application note | 150 | -   | 450 | mV   | MSC_CK_P<br>MSC_CK_N |
| VCKoff      | Input voltage offset                                         | Not to be tested. It is an application note | 1   | -   | 1.4 | V    | MSC_CK_P             |

Micro second channel (MSC) interface

|           |                                                                                           |                                             |      |     |     |            |                      |
|-----------|-------------------------------------------------------------------------------------------|---------------------------------------------|------|-----|-----|------------|----------------------|
|           | $VCK_{off} = 0.5 \cdot (VCK\_P + VCK\_N)$                                                 |                                             |      |     |     |            | MSC_CK_N             |
| Rck       | EXTERNAL Resistor between CK_P and CK_N                                                   | Not to be tested. It is an application note |      | 100 | -   | $\Omega$   | MSC_CK_P<br>MSC_CK_N |
| Rpu_N     | Internal pull-up resistor                                                                 | -                                           | 100  | 200 | 400 | k $\Omega$ | MSC_CK_P<br>MSC_CK_N |
| Rpd_P     | Internal pulldown resistor                                                                | -                                           | 100  | 200 | 400 | k $\Omega$ | MSC_CK_P<br>MSC_CK_N |
| VCK_high  | Differential input high detection level<br>$VCK\_high = VCK\_P\_high \wedge VCK\_N\_high$ | -                                           | -    | -   | 100 | mV         | MSC_CK_P<br>MSC_CK_N |
| VCK_low   | Differential input low detection level<br>$VCK\_low = VCK\_P\_low \wedge VCK\_N\_low$     | -                                           | -100 | -   | -   | mV         | MSC_CK_P<br>MSC_CK_N |
| VDI+,VDI- | Input voltage range                                                                       | Not to be tested. It is an application note | 0.8  | -   | 1.6 | V          | MSC_DI_P<br>MSC_DI_N |
| VDIdiff   | Differential input voltage<br>$VDIdiff =  VDI\_P - VDI\_N $                               | Not to be tested. It is an application note | 150  | -   | 450 | mV         | MSC_DI_P<br>MSC_DI_N |
| VDIoff    | Input voltage offset<br>$VDIoff = 0.5 \cdot (VDI\_P + VDI\_N)$                            | Not to be tested. It is an application note | 1    | -   | 1.4 | V          | MSC_DI_P<br>MSC_DI_N |
| Rcl       | Resistor between DI_P and DI_N                                                            | Not to be tested. It is an application note | -    | 100 | -   | $\Omega$   | MSC_DI_P<br>MSC_DI_N |
| Rpu_N     | Internal pull-up resistor                                                                 | -                                           | 100  | 200 | 400 | k $\Omega$ | MSC_DI_P<br>MSC_DI_N |
| Rpd_P     | Internal pulldown resistor                                                                | -                                           | 100  | 200 | 400 | k $\Omega$ | MSC_DI_P<br>MSC_DI_N |
| VDI_high  | Differential input high detection level<br>$VDI\_high = VDI\_P\_high - VDI\_N\_high$      | -                                           | -    | -   | 100 | mV         | MSC_DI_P<br>MSC_DI_N |
| VDI_low   | Differential input low detection level                                                    | -                                           | -100 | -   | -   | mV         | MSC_DI_P             |

Micro second channel (MSC) interface

|       | VDI_low=<br>VDI_P_low-<br>VDI_N_low |                                                     |                   |            |              |     | MSC_DI<br>_N |
|-------|-------------------------------------|-----------------------------------------------------|-------------------|------------|--------------|-----|--------------|
| VDO_L | DO output low level                 | VDDIO=5V or 3.3V<br>I <sub>source</sub> current=2mA | -                 | -          | 0.5          | V   | MSC_DO       |
| VDO_H | DO output high level                | VDDIO =5V or 3.3V<br>I <sub>sink</sub> current=2mA  | VDDI<br>O<br>-0.5 | -          | -            | V   | MSC_DO       |
| fDO   | Maximum frequency                   | Tested by SCAN                                      | fCK/1<br>28       | fCK/<br>64 | fCK/1<br>6   | MHz | MSC_DO       |
| ENL   | Low input level                     | -                                                   | -0.3              | -          | 1.1          | V   | MSC_EN       |
| ENH   | High input level                    | -                                                   | 2.3               | -          | VDD5<br>+0.3 | V   | MSC_EN       |
| VHYS  | Hysteresis                          | -                                                   | 0.1               | -          |              | V   | MSC_EN       |
| IIN   | Input current                       | -                                                   |                   | -          | 32           | μA  | MSC_EN       |
| RPU   | Pull up resistor                    | -                                                   | 50                | -          | 250          | kΩ  | MSC_EN       |

## 18.2 Downstream communication

The enable input is active with inverted polarity - i.e. 'low level' during the active phases of command or data frames. An active enable signal validates the DI input signal. Outside the active phase (enable line is at high level) invalid data may occur at DI. The active phase of a downstream frame starts with the falling edge of the enable signal and ends with the rising edge of the enable signal. The enable signal changes its state with the rising edge of the clock CL. DI changes its state on rising edge and it is latched by HE9788 on the falling edge of CL. Downstream frames are synchronous serial frames. They support enable signal and command/data selection bit as part of the frame. Command/data selection bit allows distinguishing frames as command and data frames in the receiver circuit. Command frames and data frames may be sent in any sequence with a passive phase of at least 2 CL cycles after each frame.

### 18.2.1 Command frame

A command frame always starts with a high level bit (command selection bit). The number of the command bit of the active phase of a command frame NCB is fixed to 16. If the number of the command bit is not equal to NCB = 16 the frame will be ignored, the command will not be executed and the error flag (*TRANS\_L*) will be set. If the MSC command frame has an invalid command the flag "*CMD\_ERROR*" is set but no action on outputs is taken. The command frame is ignored.

The length of the command frame's passive phase tCPP must be a minimum of 2 \* tCL.

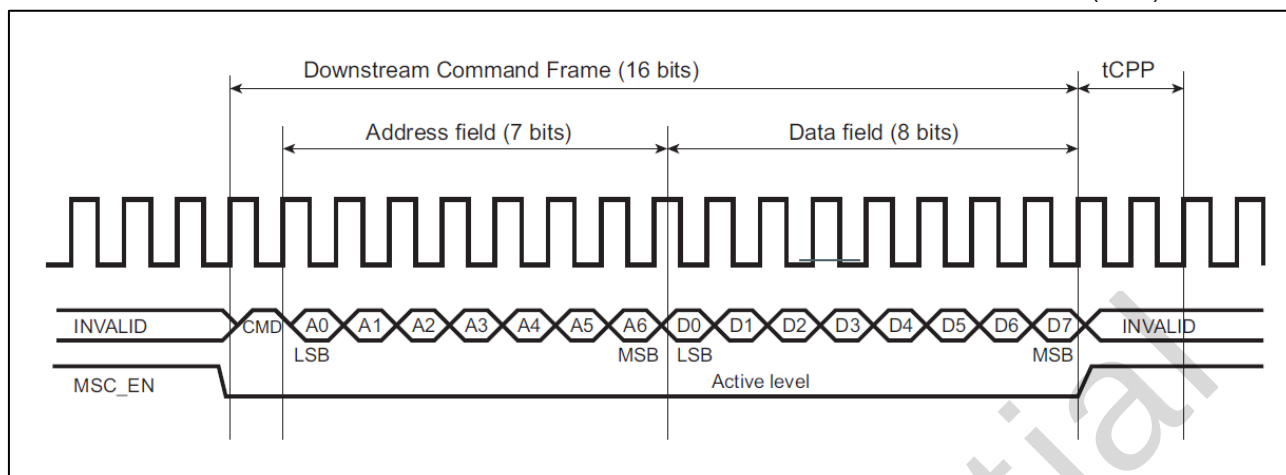


Figure 18-4 MSC Command Frame bit stream

Table 18-3 MSC Interface command frame

| bit  | Description                      |
|------|----------------------------------|
| 0    | "1": command selection bit       |
| 1-7  | Command. LSB first!              |
| 8-15 | Data for the command. LSB first! |

### 18.2.2 Data frame

A data frame always starts with a low level bit (data selection bit). The number of the data bit of the active phase of a data frame NDB is fixed to 31 bits:

If the number of the data bit is not equal to NDB = 31 the frame will be ignored and the error flag (*TRANS\_L*) will be set, but no action on outputs is taken.

The length of the data frame's passive phase tCPP must be a minimum of  $2 * tCL$ .

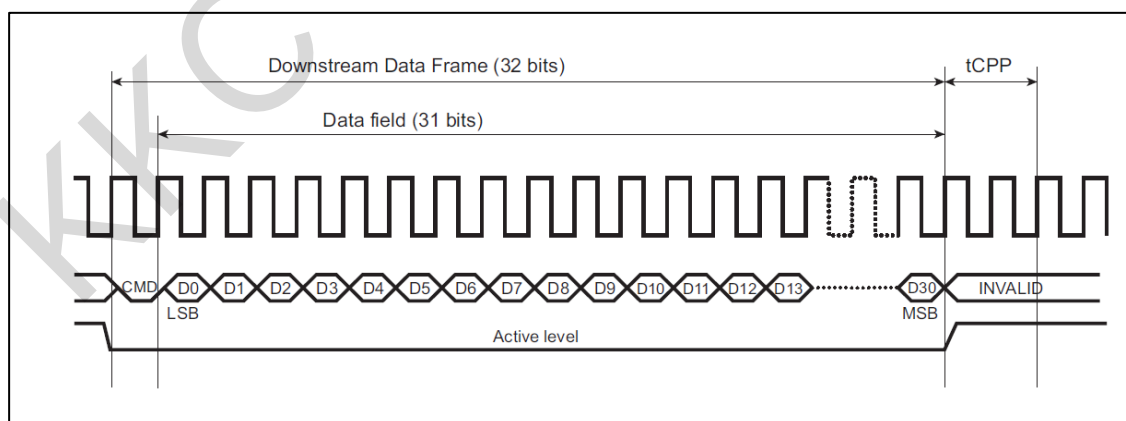


Figure 18-5 MSC Data Frame bit stream

Content of DATA frame (transmitted LSB first)

**Table 18-4 MSC Interface Data frame**

| bit  | Description                        |
|------|------------------------------------|
| 0    | = "0": DATA selection bit          |
| 1-31 | D0-D30 CONTROL REGISTER LSB first! |

**Table 18-5 Control register**

| Control Register |              |              |              |          |          |          |          |
|------------------|--------------|--------------|--------------|----------|----------|----------|----------|
| D0               | D1           | D2           | D3           | D4       | D5       | D6       | D7       |
| LSD-INJ1         | LSD-INJ2     | LSD-INJ3     | LSD-INJ4     | LSD-O2H1 | -        | LSD-O2H2 | -        |
| D8               | D9           | D10          | D11          | D12      | D13      | D14      | D15      |
| LSD-SOL1         | LSD-SOL2     | LSD-LED1     | LSD-LED2     | LSD-RLY1 | LSD-RLY2 | LSD-RLY3 | LSD-RLY4 |
| D16              | D17          | D18          | D19          | D20      | D21      | D22      | D23      |
| LSD-RLY5         | LSD-HSD-STR1 | LSD-HSD-STR2 | LSD-HSD-STR3 | FET-PRD1 | FET-PRD2 | FET-PRD3 | FET-PRD4 |
| D24              | D25          | D26          | D27          | D28      | D29      | D30      | -        |
| FET-PRD5         | IGN1         | IGN2         | IGN3         | IGN4     | IGN5     | IGN6     | -        |

*Note: If RLY4, STR2, STR3 are not configured as starter their status is controlled by the data in control register.*

*If they are configured as starter their status is controlled by MSC dedicated command.*

### 18.3 Upstream communication

The serial data output [DO] is the synchronous serial data signal of the upstream channel.

The polarity for [DO] is ,normal polarity'- i.e. a low level bit at [DO] is stored in the  $\mu$ C as a logic ,0', and a high level bit at [DO] is stored in the  $\mu$ C as a logic ,1'. The serial data output is single-ended.

The frequency is derived from CL by an internal divider to typ.  $f_{SDO} = CL/64$ . It can be adjusted via *MSC* to  $f_{SDO} = CL/16 \dots CL/128$ . The time for a bit is  $T_{SDO} = 1/f_{SDO}$ .

Each upstream frame consists of 16 bit:

- 1 start bit, always '0'
- 4-bit-upstream address field (A[0..3] with LSB first)
- 8 bit data upstream data field (D[0..7] with LSB first)
- 1 upstream parity bit (with even parity for the complete data frame)
- 2  $f_{SDO}$  stop bit, always '1'.

There are 16 commands to perform read accesses.

All the read command READ-N Upstream Block are composed of 4 continued frames sent to  $\mu$ C.

Within the execution of these read commands an upstream data frame is sent after the 2 stop bit of the prior upstream data frame and one additional interframe bit waiting time. If a new read command is received while the up-stream communication is active, the 16 bit upstream on-going is completed and the new read command is canceled. At the end of the upstream frame the latched flags contained in the register are cleared automatically, if the frame is interrupted by new read command, flags will not be cleared. The time from the read command to the first upstream frame of the answer is less than 100  $\mu$ s.

Outside the upstream frame the DO output is high impedance

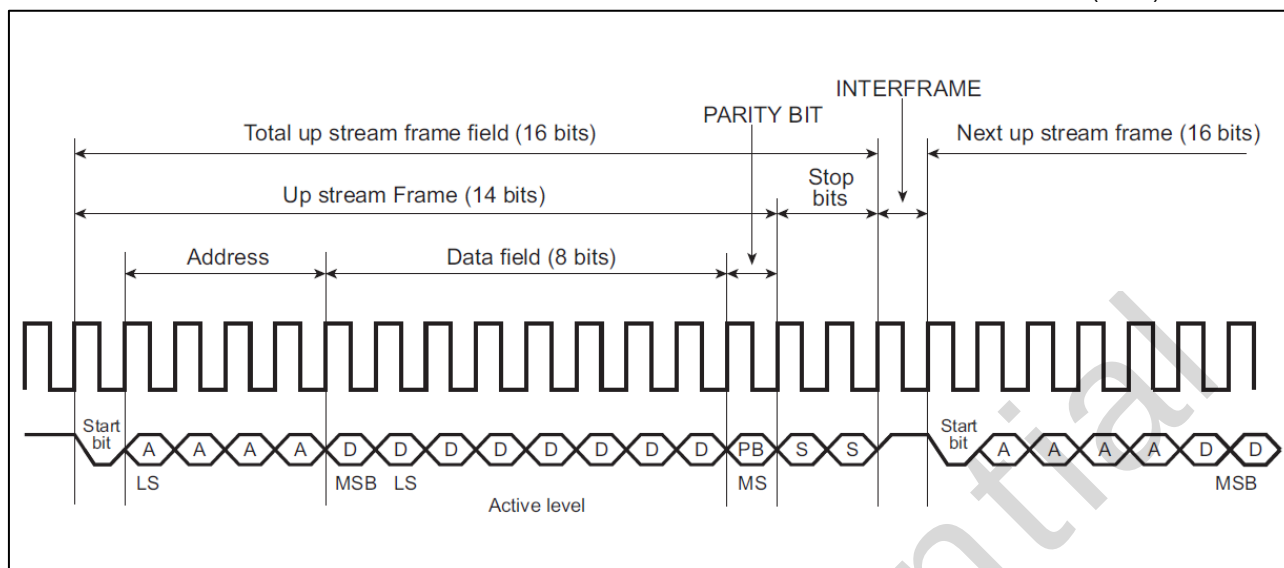


Figure 18-6 MSC upstream

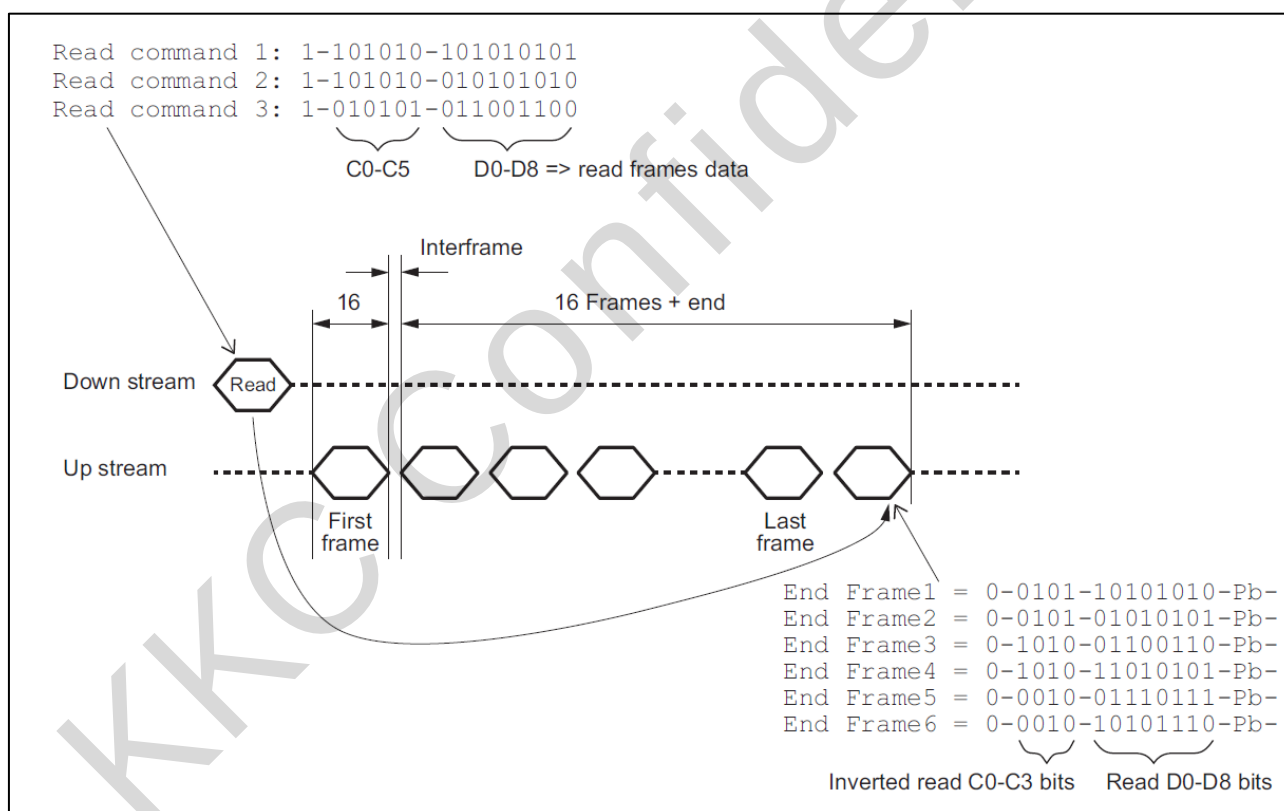


Figure 18-7 MSC upstream commands example

## 18.4 Micro Second Channel activity watchdog

MSC data frames are monitored to be sent in intervals shorter than tMSC\_mon. If HE9788

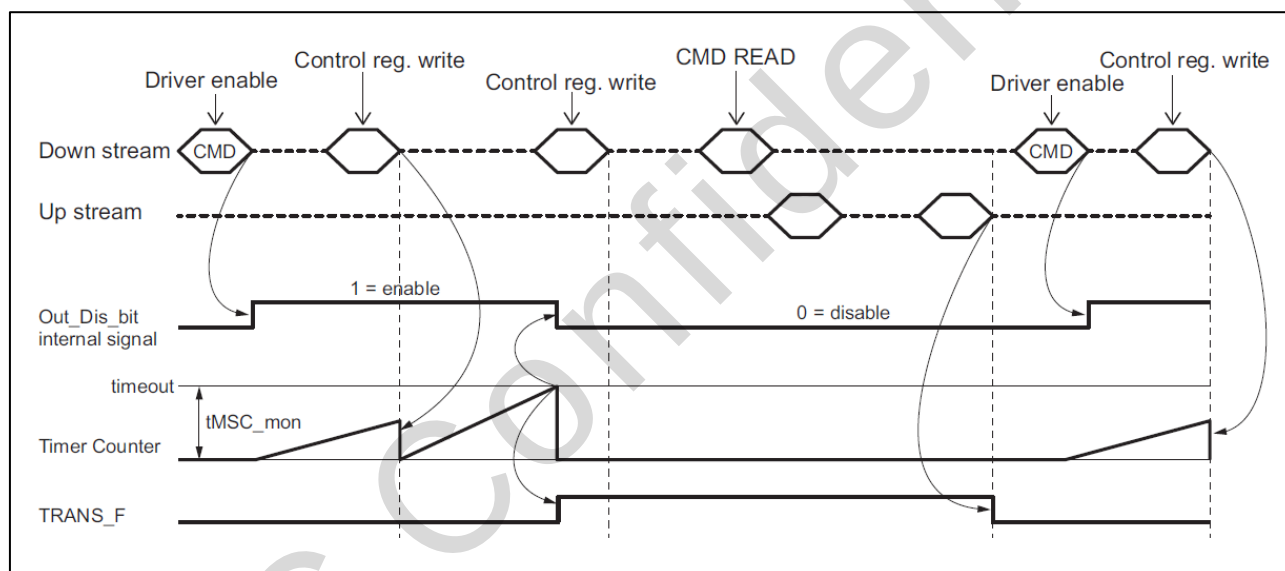
## Micro second channel (MSC) interface

receives no valid data frame for longer than  $t_{MSC\_mon}$ , it will switch off all the drivers, removing driver enable command, and the error flag ( $TRANS\_F$ ) is set to 1 and internal signal "OUT\_DIS" will be set to "0". Note:  $OUT\_DIS = driver\_en$  (Read11 frame 1);  $t_{MSC\_mon}$  is the time needed to send OUT\_DIS, it is different from time needed to turn off drivers (which can be different depending on driver type, the maximum time to turn off all drivers is about 300  $\mu s$ ).

To enable again the outputs, the  $\mu C$  has to read the  $TRANS\_F$  and then send the command Enable\_Driver, and then outputs are reactivated with the first correct data frame. If the fault flag is not cleared the Enable\_Driver command is ignored.

By default the Micro Second Channel activity watchdog is enabled and the monitoring time will start after writing of the internal signal OUT\_DIS bit by Enable\_Driver command. Each time HE9788 receives a valid data frame the  $t_{MSC\_on}$  timer is reset. This means that micro controller can drive the outputs only when the monitoring module is active.

It is also possible to enable /disable the Micro Second Channel activity watchdog through  $MSC\_ACT\_EN$  bit (CONFIG\_REG2 D0), 0= disable and 1 = enable (reset value).



**Figure 18-8** MSC activity watchdog time diagram

**Table 18-6** MSC Interface Micro Second Channel activity watchdog

| Parameter      | min | typ | max | unit    |
|----------------|-----|-----|-----|---------|
| $t_{MSC\_mon}$ | 100 | 142 | 185 | $\mu s$ |

## 18.5 Downstream frame Bit Map

**Table 18-7 MSC Interface DOWN STREAM FRAME Bit Map**

| ADDRESS      |    |    |    |    |    |    | DATA |             |                |               |                  |          |               |              |    |
|--------------|----|----|----|----|----|----|------|-------------|----------------|---------------|------------------|----------|---------------|--------------|----|
|              | C0 | C1 | C2 | C3 | C4 | C5 | C6   | D0          | D1             | D2            | D3               | D4       | D5            | D6           | D7 |
| RD_COMMAND1  | 1  | 0  | 1  | 0  | 1  | 0  | 1    | 0           | 1              | 0             | 1                | 0        | 1             | 0            | 1  |
| RD_COMMAND2  | 1  | 0  | 1  | 0  | 1  | 0  | 1    | 1           | 0              | 1             | 0                | 1        | 0             | 1            | 0  |
| RD_COMMAND3  | 0  | 1  | 0  | 1  | 0  | 1  | 0    | 0           | 1              | 0             | 1                | 0        | 1             | 0            | 1  |
| RD_COMMAND4  | 0  | 1  | 0  | 1  | 0  | 1  | 0    | 1           | 0              | 1             | 0                | 1        | 0             | 1            | 0  |
| RD_COMMAND5  | 1  | 0  | 1  | 0  | 1  | 0  | 1    | 1           | 0              | 0             | 1                | 1        | 0             | 0            | 1  |
| RD_COMMAND6  | 1  | 0  | 1  | 0  | 1  | 0  | 1    | 0           | 1              | 1             | 0                | 0        | 1             | 1            | 0  |
| RD_COMMAND7  | 0  | 1  | 0  | 1  | 0  | 1  | 0    | 1           | 0              | 0             | 1                | 1        | 0             | 0            | 1  |
| RD_COMMAND8  | 0  | 1  | 0  | 1  | 0  | 1  | 0    | 0           | 1              | 1             | 0                | 0        | 1             | 1            | 0  |
| RD_COMMAND9  | 1  | 0  | 1  | 0  | 1  | 0  | 1    | 1           | 1              | 1             | 0                | 1        | 1             | 1            | 0  |
| RD_COMMAND10 | 1  | 0  | 1  | 0  | 1  | 0  | 1    | 0           | 0              | 0             | 1                | 0        | 0             | 0            | 1  |
| RD_COMMAND11 | 0  | 1  | 0  | 1  | 0  | 1  | 0    | 1           | 1              | 1             | 0                | 1        | 1             | 1            | 0  |
| RD_COMMAND12 | 0  | 1  | 0  | 1  | 0  | 1  | 0    | 0           | 0              | 0             | 1                | 0        | 0             | 0            | 1  |
| RD_COMMAND13 | 1  | 0  | 1  | 0  | 1  | 0  | 1    | 0           | 1              | 1             | 1                | 0        | 1             | 1            | 1  |
| RD_COMMAND14 | 1  | 0  | 1  | 0  | 1  | 0  | 1    | 1           | 0              | 0             | 0                | 1        | 0             | 0            | 0  |
| RD_COMMAND15 | 0  | 1  | 0  | 1  | 0  | 1  | 0    | 0           | 1              | 1             | 1                | 0        | 1             | 1            | 1  |
| RD_COMMAND16 | 0  | 1  | 0  | 1  | 0  | 1  | 0    | 1           | 0              | 0             | 0                | 1        | 0             | 0            | 0  |
| CONFIG-REG 1 | 0  | 0  | 1  | 1  | 1  | 1  | 1    | PHOLD_EN    | PHOLD_TIM[0:1] | LED1_PD_EN    | LED2_PD_EN       | BOOST_EN | WAKE_UP_TIMER |              |    |
| CONFIG-REG 2 | 0  | 1  | 1  | 1  | 1  | 0  | 1    | MSC_AC_T_EN | F_DO_SEL[0:1]  | VRS_MO_DE_SEL | IGN_CUR_RENT_CFG | IGN_DIAG | IPUP_PD_EN    | IPUP_DM_CODE |    |

| ADDRESS                 | DATA      |           |           |           |         |         |                  |                  |               |               |
|-------------------------|-----------|-----------|-----------|-----------|---------|---------|------------------|------------------|---------------|---------------|
|                         | RLY4_DLY_ | LSD/HSD_D | LSD/HSD_D | LSD/HSD_D | SRT2_EN | STR3_EN | T_SEO_DELAY(0:1) |                  |               |               |
| CONFIG-REG 3            | 1         | 1         | 1         | 1         | 0       | 0       | 1                | 0                | 0             | 1             |
| CONFIG-REG 4            | 1         | 1         | 0         | 0         | 0       | 0       | 1                | PRD1_IDRV(0:1)   |               |               |
| CONFIG-REG 5            | 0         | 0         | 0         | 0         | 1       | 1       | 1                | PRD5_IDRV(0:1)   |               |               |
| CONFIG-REG 6            | 1         | 1         | 0         | 0         | 1       | 1       | 1                | PRD3_VDS(0:2)    |               |               |
| CONFIG-REG 7            | 0         | 0         | 1         | 1         | 0       | 0       | 1                | PRD5_VDS(0:2)    | TRK_EN[1]     | TRK_EN[2]     |
| CONFIG-REG 8            | 1         | 0         | 1         | 1         | 0       | 1       | 1                | VRS_DIAG         | VRS_MODE(0:1) | VRS_HYS(20:2) |
| CONFIG-REG 9-0          | 0         | 1         | 0         | 0         | 1       | 0       | 1                | 0                | 1             |               |
| CONFIG-REG 9-1          | 0         | 1         | 0         | 0         | 1       | 0       | 1                | 1                | 0             |               |
| CONFIG-REG 10           | 0         | 0         | 0         | 1         | 1       | 0       | 1                | WDA_RESPTIM(0:7) |               |               |
| CONFIG-REG 11           | 0         | 1         | 1         | 0         | 1       | 1       | 1                | LIN/KLINE SEL    | LIN_E_RR_E    | LIN_E_RR_E    |
| CONFIG-REG 12           | 1         | 0         | 0         | 1         | 0       | 0       | 1                | TM_L_RST_EN      | VB_IN         | VDD5_         |
| CONFIG-REG 13           | 1         | 1         | 0         | 1         | 1       | 0       | 1                |                  | WDA_P         | IPUPD_EN_H    |
| CONFIG-REG 14           | 1         | 1         | 1         | 0         | 0       | 1       | 1                | RLY1_UC          | RLY2_UC       | RLY3_UC       |
| CONFIG-REG15-0 (WUPT 0) | 1         | 0         | 0         | 0         | 1       | 1       | 1                | 0                | 0             |               |

|                         | C0 | C1 | C2 | C3 | C4 | C5 | C6 | D0                   | D1         | D2                         | D3              | D4               | D5               | D6                   | D7             |  |  |
|-------------------------|----|----|----|----|----|----|----|----------------------|------------|----------------------------|-----------------|------------------|------------------|----------------------|----------------|--|--|
| CONFIG-REG15-1 (WUPT_1) | 1  | 0  | 0  | 0  | 0  | 1  | 0  | 0                    | 1          | Wake up timer_SET_1[6:11]  |                 |                  |                  |                      |                |  |  |
| CONFIG-REG15-2 (WUPT_2) | 1  | 0  | 0  | 0  | 0  | 1  | 0  | 1                    | 0          | Wake up timer_SET_2[12:17] |                 |                  |                  |                      |                |  |  |
| CONFIG-REG15-2 (WUPT_3) | 1  | 0  | 0  | 0  | 0  | 1  | 1  | 1                    | 1          | Wake up timer_SET_3[18:23] |                 |                  |                  |                      |                |  |  |
| CONFIG-REG16-0          | 0  | 0  | 1  | 0  | 0  | 1  | 1  | 0                    | 1          | PRD1_BLK[0:1]              | PRD2_BLK[0:1]   |                  | PRD3_BLK[0:1]    |                      |                |  |  |
| CONFIG-REG16-1          | 0  | 0  | 1  | 0  | 0  | 1  | 1  | 1                    | 0          | PRD4_BLK[0:1]              | PRD5_BLK[0:1]   |                  | O2H2_S R         |                      | O2H2_O C_TH    |  |  |
| CONFIG-REG17-0          | 0  | 0  | 1  | 0  | 0  | 1  | 0  | 0                    | 0          |                            | O2H1_SR         | O2H1_O C_TH      | O2H2_S R         |                      | O2H2_O C_TH    |  |  |
| CONFIG-REG17-1          | 0  | 0  | 1  | 0  | 0  | 1  | 0  | 1                    | 1          | CAN_EN                     | CAN_TX_EN       | CAN_WA KEUP_EN   | CAN_PAT TERN_EN  | CAN_RX_EN            | CAN_2_5_MB     |  |  |
| STBY_NVM_ADD_REG        | 0  | 1  | 0  | 0  | 1  | 1  | 0  | 0                    | 1          | STBY_NVM_ADD[0:3]          |                 |                  |                  | MEM_VA LID           |                |  |  |
| STBY_NVM_DATA_REG       | 1  | 0  | 1  | 1  | 0  | 0  | 0  | STBY_NVM_DATA[0:7]   |            |                            |                 |                  |                  |                      |                |  |  |
| CONFIG-REG20            | 1  | 1  | 1  | 1  | 1  | 1  | 0  | BIST_EN              | FIN _WA KE | WAKE_U P_TIMER             | KEY_OC RT       | CAN_LO OP_EN     | CAN_TDI          | LIN_TX _DOM_ER R_CFG | PDRV_O 2H_DLY  |  |  |
| CONFIG-REG21            | 0  | 0  | 0  | 0  | 0  | 1  | 1  | CAN_TX _DOM_ER R_CFG | OL _RE D   | BUCK_SL OW_SR              | CAN_TX D_DOM_EN | CAN_PE RM_REC_EN | CAN_PE RM_DOM_EN | CAN_RX D_REC_EN      | CAN_AU TO_BIAS |  |  |
| LOCK                    | 1  | 0  | 0  | 1  | 1  | 1  | 1  | 0                    | 1          | 0                          | 1               | 0                | 1                | 0                    | 1              |  |  |
| UNLOCK                  | 1  | 0  | 0  | 1  | 1  | 1  | 1  | 1                    | 0          | 1                          | 0               | 1                | 0                | 1                    | 0              |  |  |
| WDA_EN                  | 1  | 0  | 0  | 1  | 1  | 1  | 1  | 0                    | 1          | 1                          | 0               | 0                | 1                | 1                    | 0              |  |  |
| WDA_DIS                 | 1  | 0  | 0  | 1  | 1  | 1  | 1  | 1                    | 0          | 0                          | 1               | 1                | 0                | 0                    | 1              |  |  |
| SW-RESET                | 1  | 0  | 1  | 1  | 1  | 0  | 1  | 0                    | 1          | 1                          | 0               | 1                | 1                | 0                    | 1              |  |  |

|                 | ADDRESS |    |    |    |    |    | DATA |    |    |    |    |    |    |    |    |
|-----------------|---------|----|----|----|----|----|------|----|----|----|----|----|----|----|----|
|                 | C0      | C1 | C2 | C3 | C4 | C5 | C6   | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
| EN-DRIVERS      | 0       | 1  | 1  | 0  | 0  | 0  | 1    | 1  | 0  | 0  | 1  | 1  | 0  | 0  | 1  |
| DISABLE-DRIVERS | 0       | 1  | 1  | 0  | 0  | 0  | 1    | 0  | 1  | 1  | 0  | 0  | 1  | 1  | 0  |
| MRD_ON          | 0       | 1  | 1  | 0  | 0  | 0  | 1    | 1  | 1  | 0  | 0  | 0  | 1  | 1  | 1  |
| MRD_OFF         | 0       | 1  | 1  | 0  | 0  | 0  | 1    | 0  | 0  | 1  | 1  | 1  | 0  | 0  | 0  |
| MRD_UC          | 0       | 1  | 1  | 0  | 0  | 0  | 1    | 1  | 1  | 0  | 1  | 1  | 0  | 1  | 1  |
| STR2_ON         | 0       | 1  | 0  | 0  | 1  | 0  | 0    | 0  | 0  | 1  | 0  | 1  | 0  | 1  | 0  |
| STR2_OFF        | 0       | 1  | 0  | 0  | 1  | 0  | 0    | 1  | 1  | 0  | 1  | 0  | 1  | 0  | 1  |
| STR3_ON         | 0       | 1  | 0  | 0  | 1  | 0  | 0    | 0  | 0  | 0  | 0  | 1  | 1  | 0  | 0  |
| STR3_OFF        | 0       | 1  | 0  | 0  | 1  | 0  | 0    | 1  | 1  | 1  | 1  | 0  | 0  | 1  | 1  |
| RLY4_ON         | 0       | 1  | 0  | 0  | 1  | 0  | 0    | 0  | 0  | 1  | 1  | 0  | 0  | 1  | 1  |
| RLY4_OFF        | 0       | 1  | 0  | 0  | 1  | 0  | 0    | 1  | 1  | 0  | 0  | 1  | 1  | 0  | 0  |

**Legenda:**

|            |                        |         |               |                                                                       |
|------------|------------------------|---------|---------------|-----------------------------------------------------------------------|
| Locked bit | Bit generated as pulse | Address | Default value | Bit in satellite logic, not affected by reset matrix clear conditions |
|------------|------------------------|---------|---------------|-----------------------------------------------------------------------|

Locked bit: Bit generated as pulse Bit in satellite logic, not affected by reset matrix clear conditions.

## 18.5.1 CONFIG-REG 1

|           | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                                                                                                                                    | D1              | D2 | D3         | D4         | D5       | D6                            | D7 |
|-----------|----------------|----|----|----|----|----|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----|------------|------------|----------|-------------------------------|----|
| CONFREG 1 | 0              | 0  | 1  | 1  | 1  | 1  | 1  | PHOLD_EN                                                                                                                                                                                              | PHOLD_TIME[0:1] |    | LED1_PD_EN | LED2_PD_EN | BOOST_EN | WAKE_UP_TIMER_START_STOP[0:1] |    |
| RW        | DEFAULT        |    |    |    |    |    |    | 0                                                                                                                                                                                                     | 0               | 0  | 0          | 0          | 1        | 1                             | 0  |
|           | RESET SOURCE   |    |    |    |    |    |    | D0:D5<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN<br>D6:D7<br>the read bits are from satellite logic on VB_STBY and are not affected by reset matrix |                 |    |            |            |          |                               |    |
|           | ACTIVATION     |    |    |    |    |    |    | D0:D7 bits active as level                                                                                                                                                                            |                 |    |            |            |          |                               |    |
|           | CONTROL ACCESS |    |    |    |    |    |    | not locked bits                                                                                                                                                                                       |                 |    |            |            |          |                               |    |

[6:7] WAKE\_UP\_TIMER\_START\_STOP [0:1]

Start/stop Wake\_up\_timer counter

10: (stop) wake up timer stops (default)

01: (start) → 10 (stop) wake up timer counter value hold

10: (stop) → 01 (start) wake up timer starts from h'000001

01: (start) wake up timer counter is running

Application note: after a start/stop command 500us must be waited to have the command be effective in the EOT clock domain.

[5] BOOST\_EN

Boost enable.

1: enabled (default)

0: disabled

[4] LED2\_PD\_EN

Enable the pull down current (open load diagnosis not available) for LED2 driver

0: pull down current disabled (LED mode enabled) (default)

1: enable

[3] LED1\_PD\_EN

Enable the pull down current (open load diagnosis not available) for LED1 driver

0: pull down current disabled (LED mode enabled) (default)

1: enable

[2:1] PHOLD\_TIME[1:0]

Set the power hold timeout.

00: timeout disabled (default)

01: 500 ms

10: 5 minutes

11: 20 minutes

[0] PHOLD\_EN

Enable the power hold function.

0: disabled (default)

1: enabled

## 18.5.2 CONFIG-REG 2

|              | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                                 | D1            | D2 | D3           | D4                | D5       | D6       | D7         |
|--------------|----------------|----|----|----|----|----|----|----------------------------------------------------------------------------------------------------|---------------|----|--------------|-------------------|----------|----------|------------|
| CONFIG-REG 2 | 0              | 1  | 1  | 1  | 1  | 0  | 1  | MSC_ACT_EN                                                                                         | F_DO_SEL[0:1] |    | VRS_MODE_SEL | IGN_CURR_RENT_CFG | IGN_DIAG | IPUPD_EN | IPUPD_MODE |
| RW           | DEFAULT        |    |    |    |    |    |    | 1                                                                                                  | 1             | 0  | 0            | 0                 | 0        | 0        | 1          |
|              | RESET SOURCE   |    |    |    |    |    |    | D0:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN |               |    |              |                   |          |          |            |
|              | ACTIVATION     |    |    |    |    |    |    | D0:D7 bits active as level                                                                         |               |    |              |                   |          |          |            |
|              | CONTROL ACCESS |    |    |    |    |    |    | not locked bits                                                                                    |               |    |              |                   |          |          |            |

[7] IPUPD\_MODE

Enable the fast charge current I\_LS\_PU1 required by driver fast off diagnosis.

0: fast charge current disabled

1: fast charge current enabled (default)

[6] IPUPD\_EN

Enable the driver off diagnosis and the currents I\_LS\_PU2, I\_LS\_PD1 required by it.

0: off diagnosis disabled (default)

1: off diagnosis enabled

[5] IGN\_DIAG.

0: IGN diagnosis off (default)

1: IGN diagnosis on

[4] IGN\_CURRENT\_CFG.

0: I1\_HS\_IGN current selected 15-30 mA (default)

1: I2\_HS\_IGN current selected 5-15 mA

[3] VRS\_MODE\_SEL: VRS mode

0: limited adaptive (default)

1: full adaptive

[2:1] F\_DO\_SEL [1:0] Upstream clock ratio selection.

00: fDO= fCL/ 16

01: fDO= fCL/ 32(default)

10: fDO= fCL/ 64

11: fDO= fCL/ 128

[0] MSC\_ACT\_EN: MSC activity monitoring enable

0: MSC activity monitoring function is disabled

1: MSC activity monitoring function is enabled (default)

## 18.5.3 CONFIG-REG 3

|              | C0      | C1 | C2 | C3 | C4 | C5 | C6 | D0              | D1                 | D2                 | D3                 | D4      | D5      | D6               | D7 |
|--------------|---------|----|----|----|----|----|----|-----------------|--------------------|--------------------|--------------------|---------|---------|------------------|----|
| CONFIG-REG 3 | 1       | 1  | 1  | 1  | 0  | 0  | 1  | RLY4_DLY_OFF_EN | LSD/H SD_D RV1-CFG | LSD/H SD_D RV2-CFG | LSD/H SD_D RV3-CFG | SRT2_EN | STR3_EN | T_SEO_DELAY[0:1] |    |
| RW           | DEFAULT |    |    |    |    |    |    | 1               | 1                  | 1                  | 1                  | 1       | 1       | 0                | 0  |

|  |                |                                                                                                    |
|--|----------------|----------------------------------------------------------------------------------------------------|
|  | RESET SOURCE   | D0:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN |
|  | ACTIVATION     | D0:D7 bits active as level                                                                         |
|  | CONTROL ACCESS | not locked bits                                                                                    |

[7:6] SEO\_DELAY\_DELAY[1:0]

Delay from KEY\_OFF for SEO function activation.

00: 100 ms (default)

01: 200 ms

10: 400 ms

11: 800 ms

[5] STR3\_EN

0: disable

1: starter functions (DELAY OFF) enable for driver STR[3] (default)

[4] STR2\_EN

0: disable

1: starter functions (DELAY OFF) enable for driver STR[2] (default)

[3] LSD/HSD\_DRV3-CFG

1: LOW-SIDE configuration for STR3 driver (default)

0: HIGH-SIDE configuration for STR3 driver

[2] LSD/HSD\_DRV2-CFG

1: LOW-SIDE configuration for STR2 driver (default)

0: HIGH-SIDE configuration for STR2 driver

[1] LSD/HSD\_DRV1-CFG

1: LOW-SIDE configuration for STR1 driver (default)

0: HIGH-SIDE configuration for STR1 driver

[0] RLY4\_DLY\_OFF\_EN

Enable delay off for RLY4 driver.

0: disabled

1: enabled (default)

#### 18.5.4 CONFIG-REG 4

|                            | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                                 | D1 | D2             | D3 | D4             | D5 | D6             | D7 |
|----------------------------|----------------|----|----|----|----|----|----|----------------------------------------------------------------------------------------------------|----|----------------|----|----------------|----|----------------|----|
| CO<br>NFI<br>GR<br>EG<br>4 | 1              | 1  | 0  | 0  | 0  | 0  | 1  | PRD1_IDRV[0:1]                                                                                     |    | PRD2_IDRV[0:1] |    | PRD3_IDRV[0:1] |    | PRD4_IDRV[0:1] |    |
| RW                         | DEFAULT        |    |    |    |    |    |    | 0                                                                                                  | 0  | 0              | 0  | 0              | 0  | 0              | 0  |
|                            | RESET SOURCE   |    |    |    |    |    |    | D0:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN |    |                |    |                |    |                |    |
|                            | ACTIVATION     |    |    |    |    |    |    | D0:D7 bits active as level                                                                         |    |                |    |                |    |                |    |
|                            | CONTROL ACCESS |    |    |    |    |    |    | not locked bits                                                                                    |    |                |    |                |    |                |    |

[7:6] PRD4\_IDRV[1:0]

MOSFET pre-driver current driving strength

00: IDR\_V\_00 20mA (default)

01: IDR\_V\_01 10mA

10: IDR\_V\_10 4mA

11: IDR\_V\_11 2mA

[5:4] PRD3\_IDRV[1:0]

MOSFET pre-driver current driving strength

00: IDR\_V\_00 20mA (default)

01: IDR\_V\_01 10mA

10: IDR\_V\_10 4mA

11: IDR\_V\_11 2mA

[3:2] PRD2\_IDRV[1:0]

MOSFET pre-driver current driving strength

00: IDR\_V\_00 20mA (default)

01: IDR\_V\_01 10mA

10: IDR\_V\_10 4mA

11: IDR\_V\_11 2mA

[1:0] PRD1\_IDRV[1:0]

MOSFET pre-driver current driving strength  
 00: IDR\_V\_00 20mA (default)  
 01: IDR\_V\_01 10mA  
 10: IDR\_V\_10 4mA  
 11: IDR\_V\_11 2mA

### 18.5.5 CONFIG-REG 5

|          | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                                 | D1 | D2            | D3 | D4 | D5            | D6 | D7 |
|----------|----------------|----|----|----|----|----|----|----------------------------------------------------------------------------------------------------|----|---------------|----|----|---------------|----|----|
| CONFREG5 | 0              | 0  | 0  | 0  | 1  | 1  | 1  | PRD5_IDRV[0:1]                                                                                     |    | PRD1_VDS[0:2] |    |    | PRD2_VDS[0:2] |    |    |
| RW       | DEFAULT        |    |    |    |    |    |    | 0                                                                                                  | 0  | 0             | 0  | 0  | 0             | 0  | 0  |
|          | RESET SOURCE   |    |    |    |    |    |    | D0:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN |    |               |    |    |               |    |    |
|          | ACTIVATION     |    |    |    |    |    |    | D0:D7 bits active as level                                                                         |    |               |    |    |               |    |    |
|          | CONTROL ACCESS |    |    |    |    |    |    | not locked bits                                                                                    |    |               |    |    |               |    |    |

[7:5] PRD2\_VDS[2:0]

MOSFET pre-driver VDS threshold

000: VTH\_DS\_000 150mV (default)

001: VTH\_DS\_001 245mV

010: VTH\_DS\_010 325mV

011: VTH\_DS\_011 405mV

100: VTH\_DS\_100 525mV

101: VTH\_DS\_101 660mV

110: VTH\_DS\_110 950mV

[4:2] PRD1\_VDS[2:0]

MOSFET pre-driver VDS threshold

000: VTH\_DS\_000 150mV (default)

001: VTH\_DS\_001 245mV

010: VTH\_DS\_010 325mV

011: VTH\_DS\_011 405mV

100: VTH\_DS\_100 525mV

101: VTH\_DS\_101 660mV

110: VTH\_DS\_110 950mV

[1:0] PRD5\_IDRV[1:0]

MOSFET pre-driver current driving strength

00: IDR\_V\_00 20mA (default)

01: IDR\_V\_01 10mA

10: IDR\_V\_10 4mA

11: IDR\_V\_11 2mA

### 18.5.6 CONFIG-REG 6

|          | C0           | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                                                                                                                                  | D1 | D2 | D3            | D4 | D5 | D6       | D7              |
|----------|--------------|----|----|----|----|----|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---------------|----|----|----------|-----------------|
| CONFREG6 | 1            | 1  | 0  | 0  | 1  | 1  | 1  | PRD3_VDS[0:2]                                                                                                                                                                                       |    |    | PRD4_VDS[0:2] |    |    | EOT_MODE | IGN_L<br>SD_DIS |
| RW       | DEFAULT      |    |    |    |    |    |    | 0                                                                                                                                                                                                   | 0  | 0  | 0             | 0  | 0  | 1        | 1               |
|          | RESET SOURCE |    |    |    |    |    |    | D0:D5, D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN<br>D6<br>the read bit is from satellite logic on VB_STBY and is not affected by reset matrix |    |    |               |    |    |          |                 |

|  |                |                            |
|--|----------------|----------------------------|
|  | ACTIVATION     | D0:D7 bits active as level |
|  | CONTROL ACCESS | not locked bits            |

## [7] IGN\_LSD\_DIS

Disable for LSD stage in igniter driver to be used with IGBT load

0: LSD enable

1: LSD disabled (default)

## [6] EOT\_MODE

Bit going to Wake Up Timer Logic

Selection for EOT function

0: counter without wake up function EOT function 1

1: counter with wake up function EOT function 2 (default)

Application note: EOT\_MODE bit can be written only when WAKE\_UP\_TIMER\_EN\_SEL=0

(CONFIG\_REG20 D2, MSC Start/Stop) and WAKE\_UP\_TIMER\_START\_STOP[0:1]=10

(CONFIG\_REG1 D7-D6, Stop condition) when WAKE\_UP\_TIMER\_START\_STOP[0:1] is

effective in EOT clock domain

## [5:3] PRD4\_VDS[2:0]

MOSFET pre-driver VDS threshold

000: VTH\_DS\_000 150mV (default)

001: VTH\_DS\_001 245mV

010: VTH\_DS\_010 325mV

011: VTH\_DS\_011 405mV

100: VTH\_DS\_100 525mV

101: VTH\_DS\_101 660mV

110: VTH\_DS\_110 950mV

## [2:0] PRD3\_VDS[2:0]

MOSFET pre-driver VDS threshold

000: VTH\_DS\_000 150mV (default)

001: VTH\_DS\_001 245mV

010: VTH\_DS\_010 325mV

011: VTH\_DS\_011 405mV

100: VTH\_DS\_100 525mV

101: VTH\_DS\_101 660mV

110: VTH\_DS\_110 950mV

## 18.5.7 CONFIG-REG 7

|                            | C              | C | C | C | C | C | C | D0                                                                                                 | D1 | D2 | D3        | D4        | D5        | D6       | D7 |
|----------------------------|----------------|---|---|---|---|---|---|----------------------------------------------------------------------------------------------------|----|----|-----------|-----------|-----------|----------|----|
|                            | 0              | 1 | 2 | 3 | 4 | 5 | 6 |                                                                                                    |    |    |           |           |           |          |    |
| CO<br>NFI<br>GR<br>EG<br>7 | 0              | 0 | 1 | 1 | 0 | 0 | 1 | PRD5_VDS[0:2]                                                                                      |    |    | TRK_EN[1] | TRK_EN[2] | TRK_EN[3] | DAC[0:1] |    |
| RW                         | DEFAULT        |   |   |   |   |   |   | 0                                                                                                  | 0  | 0  | 1         | 1         | 1         | 1        | 1  |
|                            | RESET SOURCE   |   |   |   |   |   |   | D0:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN |    |    |           |           |           |          |    |
|                            | ACTIVATION     |   |   |   |   |   |   | D0:D7 bits active as level                                                                         |    |    |           |           |           |          |    |
|                            | CONTROL ACCESS |   |   |   |   |   |   | not locked bits                                                                                    |    |    |           |           |           |          |    |

## [7:6] DAC

A/D test output voltage selection

00: 0.25\*Vref

01: 0.50\*Vref

10: 0.75\*Vref

11: 1.0\*Vref (default)

## [5] TRK\_EN[3]

Enable for tracking regulator 3

0: disabled

1: enabled (default)

## [4] TRK\_EN[2]

Enable for tracking regulator 2

0: disabled

1: enabled (default)

## [3] TRK\_EN[1]

Enable for tracking regulator 1

0: disabled

1: enabled (default)  
 [2:0] PRD5\_VDS[2:0]  
 MOSFET pre-driver VDS threshold  
 000: VTH\_DS\_000 150mV (default)  
 001: VTH\_DS\_001 245mV  
 010: VTH\_DS\_010 325mV  
 011: VTH\_DS\_011 405mV  
 100: VTH\_DS\_100 525mV  
 101: VTH\_DS\_101 660mV  
 110: VTH\_DS\_110 950mV

### 18.5.8 CONFIG-REG 8

|             | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                                 | D1            | D2 | D3            | D4 | D5 | D6                | D7         |
|-------------|----------------|----|----|----|----|----|----|----------------------------------------------------------------------------------------------------|---------------|----|---------------|----|----|-------------------|------------|
| CONFGR EG 8 | 1              | 0  | 1  | 1  | 0  | 1  | 1  | VRS_D IAG                                                                                          | VRS_MODE[0:1] |    | VRS_HYS2[0:2] |    |    | EN_FALLING_FILTER | HYS_FB_SEL |
| RW          | DEFAULT        |    |    |    |    |    |    | 0                                                                                                  | 0             | 1  | 0             | 0  | 0  | 0                 | 0          |
|             | RESET SOURCE   |    |    |    |    |    |    | D0:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN |               |    |               |    |    |                   |            |
|             | ACTIVATION     |    |    |    |    |    |    | D0:D7 bits active as level                                                                         |               |    |               |    |    |                   |            |
|             | CONTROL ACCESS |    |    |    |    |    |    | not locked bits                                                                                    |               |    |               |    |    |                   |            |

[7] HYS\_FB\_SEL:  
 0: VRS hyst. Feedback connected before adaptive filter (default)  
 1: VRS hyst. Feedback connected after adaptive filter  
 [6] EN\_FALLING\_FILTER:  
 0: Falling edge filter disabled (default)  
 1: Falling edge filter enabled  
 [5:3] VRS\_HYST[2:0]  
 000: Hys current = 17  $\mu$ A (Hys VRS = 347 mV with 10 k $\Omega$  ext resistors) (default)  
 001: Hys current = 5  $\mu$ A (Hys VRS=100mV with 10 k $\Omega$  ext resistors)  
 010: Hys current = 10  $\mu$ A (Hys VRS=200mV with 10 k $\Omega$  ext resistors)  
 011: Hys current = 17  $\mu$ A (Hys VRS=347mV with 10 k $\Omega$  ext resistors)  
 100: Hys current = 32  $\mu$ A (Hys VRS=644mV with 10 k $\Omega$  ext resistors)  
 101: Hys current = 51  $\mu$ A (Hys VRS=967mV with 10 k $\Omega$  ext resistors)  
 110: Hys current = 17  $\mu$ A (Hys VRS=347mV with 10 k $\Omega$  ext resistors)  
 111: Hys current = 0  $\mu$ A (used only for test purpose)  
 [2:1] VRS\_MODE[1:0]  
 Internal auto-adaptive hysteresis OFF allows to configure hysteresis by MSC. Internal autoadaptive hysteresis ON selects higher hysteresis between the one configured by MSC and the one internally computed by peak voltage.  
 Internal auto-adaptive filter time ON works properly only in full adaptive mode, in limited adaptive mode filter time ON is fixed to 4 us. Internal auto-adaptive filter time OFF allows to bypass the filter in both modalities.  
 00: Internal auto-adaptive hysteresis OFF, internal auto-adaptive filter time OFF  
 01: Internal auto-adaptive hysteresis ON, internal auto-adaptive filter time OFF  
 10: Internal auto-adaptive hysteresis OFF, internal auto-adaptive filter time ON (default)  
 11: Internal auto-adaptive hysteresis ON, internal auto-adaptive filter time ON  
 [0] VRS\_DIAG: VRS diagnosis enable, only for full adaptive mode  
 0: diagnosis function is disabled (default)  
 1: diagnosis function is enabled

### 18.5.9 CONFIG-REG 9

|  | C0 | C1 | C2 | C3 | C4 | C5 | C6 | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
|--|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
|--|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|

Micro second channel (MSC) interface

|                              |                |   |   |   |   |   |   |                                                                                                                           |   |                   |   |   |   |   |   |  |
|------------------------------|----------------|---|---|---|---|---|---|---------------------------------------------------------------------------------------------------------------------------|---|-------------------|---|---|---|---|---|--|
| CO<br>NFI<br>GR<br>EG<br>9_0 | 0              | 1 | 0 | 0 | 1 | 0 | 1 | 0                                                                                                                         | 1 | WDA_RESPTIME[0:5] |   |   |   |   |   |  |
| RW                           | DEFAULT        |   |   |   |   |   |   |                                                                                                                           |   | 1                 | 1 | 1 | 1 | 1 | 1 |  |
|                              | RESET SOURCE   |   |   |   |   |   |   | D2:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_ov, vdd5_uv, vddio_uv,<br>err_cnt>7, sw_reset, after_run_reset, RSTN_IN |   |                   |   |   |   |   |   |  |
|                              | ACTIVATION     |   |   |   |   |   |   | D2:D7 bits active as level                                                                                                |   |                   |   |   |   |   |   |  |
|                              | CONTROL ACCESS |   |   |   |   |   |   | D2:D7 not locked bits                                                                                                     |   |                   |   |   |   |   |   |  |

[7:2] WDA\_RESPTIME[5:0]

Response-time = WDA\_RESPTIME[5:0] \* 1.6 ms

The error counter is incremented by one on a controller write access to this register

|                              |                |   |   |   |   |   |   |                                                                                                                           |    |                     |              |    |    |    |    |
|------------------------------|----------------|---|---|---|---|---|---|---------------------------------------------------------------------------------------------------------------------------|----|---------------------|--------------|----|----|----|----|
|                              | C              | C | C | C | C | C | C | D0                                                                                                                        | D1 | D2                  | D3           | D4 | D5 | D6 | D7 |
|                              | 0              | 1 | 0 | 0 | 1 | 0 | 1 |                                                                                                                           |    |                     |              |    |    |    |    |
| CO<br>NFI<br>GR<br>EG<br>9_1 | 0              | 1 | 0 | 0 | 1 | 0 | 1 | 1                                                                                                                         | 0  | WDA_<br>WIN<br>_SEL | WDA_I<br>NIT |    |    |    |    |
| RW                           | DEFAULT        |   |   |   |   |   |   |                                                                                                                           |    | 1                   | 0            |    |    |    |    |
|                              | RESET SOURCE   |   |   |   |   |   |   | D2:D3<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_ov, vdd5_uv, vddio_uv,<br>err_cnt>7, sw_reset, after_run_reset, RSTN_IN |    |                     |              |    |    |    |    |
|                              | ACTIVATION     |   |   |   |   |   |   | D2:D3 bits active as level                                                                                                |    |                     |              |    |    |    |    |
|                              | CONTROL ACCESS |   |   |   |   |   |   | D2:D3 locked bits                                                                                                         |    |                     |              |    |    |    |    |

[3] WDA\_INIT

Monitoring module reset

0: disabled (default)

1: enabled

[2] WDA\_WIN\_SEL

Select the time base for window watchdog generation

0: 39 kHz

1: 64 kHz (default)

### 18.5.10 CONFIG-REG 10

|                             |                |   |   |   |   |   |   |                                                                                                                           |    |    |    |    |    |    |    |
|-----------------------------|----------------|---|---|---|---|---|---|---------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|
|                             | C              | C | C | C | C | C | C | D0                                                                                                                        | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
|                             | 0              | 1 | 2 | 3 | 4 | 5 | 6 |                                                                                                                           |    |    |    |    |    |    |    |
| CO<br>NFI<br>GR<br>EG<br>10 | 0              | 0 | 0 | 1 | 1 | 0 | 1 | WDA_RESP[0:7]                                                                                                             |    |    |    |    |    |    |    |
| RW                          | DEFAULT        |   |   |   |   |   |   | 0                                                                                                                         | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
|                             | RESET SOURCE   |   |   |   |   |   |   | D0:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_ov, vdd5_uv, vddio_uv,<br>err_cnt>7, sw_reset, after_run_reset, RSTN_IN |    |    |    |    |    |    |    |
|                             | ACTIVATION     |   |   |   |   |   |   | D0:D7 bits active as level                                                                                                |    |    |    |    |    |    |    |
|                             | CONTROL ACCESS |   |   |   |   |   |   | D0:D7 not locked bits                                                                                                     |    |    |    |    |    |    |    |

WDA\_RESP[0:7]

Q&amp;A WD response according to Table 4-3.

## 18.5.11 CONFIG-REG 11

|               | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                                 | D1           | D2           | D3           | D4        | D5                     | D6           | D7          |
|---------------|----------------|----|----|----|----|----|----|----------------------------------------------------------------------------------------------------|--------------|--------------|--------------|-----------|------------------------|--------------|-------------|
| CONFIG-REG 11 | 0              | 1  | 1  | 0  | 1  | 1  | 1  | LIN/KLIN_SEL                                                                                       | LIN_ERR_N[0] | LIN_ERR_N[1] | LIN_ERR_N[2] | LIN_TX_EN | LIN_TX_DIS_FOR_WDA_ERR | VDD5_OFF_SEL | O2H2_OC_FLT |
| RW            | DEFAULT        |    |    |    |    |    |    | 0                                                                                                  | 0            | 0            | 0            | 0         | 1                      | 0            | 0           |
|               | RESET SOURCE   |    |    |    |    |    |    | D0:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN |              |              |              |           |                        |              |             |
|               | ACTIVATION     |    |    |    |    |    |    | D0:D7 bits active as level                                                                         |              |              |              |           |                        |              |             |
|               | CONTROL ACCESS |    |    |    |    |    |    | D0:D7 not locked bits                                                                              |              |              |              |           |                        |              |             |

[7] O2H2\_OC\_FLT

Select the OVC filter time for O2H2.

0: select  $I_{LS\_ocv\_flt}$  (default)1: select  $I_{LS\_ocv\_flt} \times 2$ 

[6] VDD5\_OFF\_SEL

VDD5 disabled by VB\_IN UV:

0 VDD5 depends on VB\_IN (default)

1: VDD5 does not depend on VB\_IN

[5] LIN\_TX\_DIS\_FOR\_WDA\_ERR

Disable the LIN transmitting when Watchdog ERR\_CNT&gt;4

0: LIN transmitting stays enabled when Watchdog ERR\_CNT&gt;4

1: LIN transmitting disabled when Watchdog ERR\_CNT&gt;4 (default)

[4] LIN\_TX\_EN

Enable the LIN transmission function

0: disabled (default)

1: enabled

[3] LIN\_ERR\_EN[2]

LIN error handling permanent dominant

0: LIN error handling disabled (default)

1: LIN error handling enabled

[2] LIN\_ERR\_EN[1]

LIN error handling permanent recessive

0: LIN error handling disabled (default)

1: LIN error handling enabled

[1] LIN\_ERR\_EN[0]

LIN error handling permanent TX timeout

0: LIN error handling disabled (default)

1: LIN error handling enabled

[0] LIN/KLIN\_SEL

Select between LIN/K-LINE sel

0: LIN Mode (default)

1: K-LINE Mode

## 18.5.12 CONFIG-REG 12

|               | C0           | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                                 | D1              | D2             | D3          | D4          | D5              | D6              | D7        |
|---------------|--------------|----|----|----|----|----|----|----------------------------------------------------------------------------------------------------|-----------------|----------------|-------------|-------------|-----------------|-----------------|-----------|
| CONFIG-REG 12 | 1            | 0  | 0  | 1  | 0  | 0  | 1  | TNL_RST_EN                                                                                         | VB_IN_OVR_ST_EN | VDD5_OV_RST_EN | O2H1_OC_FLT | FILTER_MODE | WAKE_UP_CAN_RST | WAKE_UP_EOT_RST | WK_IN_RST |
| RW            | DEFAULT      |    |    |    |    |    |    | 1                                                                                                  | 1               | 1              | 0           | 0           |                 | 0               | 0         |
|               | RESET SOURCE |    |    |    |    |    |    | D0:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN |                 |                |             |             |                 |                 |           |
|               | ACTIVATION   |    |    |    |    |    |    | D0:D7 bits active as level                                                                         |                 |                |             |             |                 |                 |           |

|  |                |                       |
|--|----------------|-----------------------|
|  | CONTROL ACCESS | D0:D7 not locked bits |
|--|----------------|-----------------------|

## [7] WK\_IN\_RST

Activation of WK\_IN\_RST, active on level

1: clearing of WK\_IN\_DET

0: no functionality (default)

## [6] WAKE\_UP\_EOT\_RST

Activation of WAKE\_UP\_EOT\_RST, active on level

1: clearing of WAKE\_UP\_EOT\_DET

0: no functionality (default)

## [5] WAKE\_UP\_CAN\_RST

Activation of WAKE\_UP\_CAN\_RST, active on level

1: clearing of WAKE\_UP\_CAN\_DET

0: no functionality (default)

## [4] FILTER\_MODE

Select the filter time (Tflt\_diagoff) of all the drivers.

0: selected Tflt\_diagoff1 (100us) (default)

1: selected Tflt\_diagoff2 (600us)

## [3] O2H1\_OC\_FLT

Select the OVC filter time for O2H1.

0: select I\_LS\_ovcflt (default)

1: select I\_LS\_ovcflt x 2

## [2] VDD5\_OV\_RST\_EN

Enable the contribution of VDD5\_OV in the reset matrix

0: disabled

1: enabled (default)

## [1] VB\_IN\_OV\_RST\_EN

Enable the contribution of VB\_IN\_OV on BUCK /VDD5/ VTRK regulator in the reset matrix

0: disabled

1: enabled (default)

## [0] TNL\_RST\_EN

enable the RSTN activation for TNL time on the positive edge of KEY\_IN

1: enable (default)

0: disable

## 18.5.13 CONFIG-REG 13

|                             | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                                    | D1                                  | D2                     | D3                     | D4                     | D5                     | D6                     | D7                      |
|-----------------------------|----------------|----|----|----|----|----|----|-------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|-------------------------|
| CO<br>NFI<br>GR<br>EG<br>13 | 1              | 1  | 0  | 1  | 1  | 0  | 1  |                                                                                                       | WDA_<br>PWR_<br>CNT_O<br>FF_DI<br>S | IPUPD_<br>EN_S<br>TR 1 | IPUPD_<br>EN_S<br>TR 2 | IPUPD_<br>EN_S<br>TR 3 | IDIAG_<br>HIGH_<br>INJ | IDIAG_<br>HIGH_<br>SOL | IDIAG_<br>HIGH_<br>PDRV |
| RW                          | DEFAULT        |    |    |    |    |    |    | 0                                                                                                     | 0                                   | 0                      | 0                      | 0                      | 1                      | 1                      | 1                       |
|                             | RESET SOURCE   |    |    |    |    |    |    | D1:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset,<br>after_run_reset, RSTN_IN |                                     |                        |                        |                        |                        |                        |                         |
|                             | ACTIVATION     |    |    |    |    |    |    | D1:D7 bits active as level                                                                            |                                     |                        |                        |                        |                        |                        |                         |
|                             | CONTROL ACCESS |    |    |    |    |    |    | D1 locked bit<br>D2:D7 not locked bits                                                                |                                     |                        |                        |                        |                        |                        |                         |

## [7] IDIAG\_HIGH\_PDRV

Increase the currents required for pre-drivers off diagnosis.

0: I\_LS\_PU2 (40 µA (min)) I\_LS\_PD1(60 µA (min))

1: I\_LS\_PU3 (100 µA (min)) I\_LS\_PD2(350 µA (min)) (default)

## [6] IDIAG\_HIGH\_SOL

Increase the currents required for Solenoids off diagnosis.

0: I\_LS\_PU2 (40 µA (min)) I\_LS\_PD1(60 µA (min))

1: I\_LS\_PU3 (100 µA (min)) I\_LS\_PD2(350 µA (min)) (default)

## [5] IDIAG\_HIGH\_INJ

Increase the currents required for Solenoids off diagnosis.

0: I\_LS\_PU2 (40 µA (min)) I\_LS\_PD1(60 µA (min))

1: I\_LS\_PU3 (100 µA (min)) I\_LS\_PD2(350 µA (min)) (default)

## [4] IPUPD\_EN\_STR3

Enable the driver off diagnosis and the currents required by it.

0: off diagnosis disabled (default)

1: off diagnosis enabled

[3] IPUPD\_EN\_STR2

Enable the driver off diagnosis and the currents required by it.

0: off diagnosis disabled (default)

1: off diagnosis enabled

[2] IPUPD\_EN\_STR1

Enable the driver off diagnosis and the currents required by it.

0: off diagnosis disabled (default)

1: off diagnosis enabled

[1] WDA\_PWR\_CNT\_OFF\_DIS

Disable the power down due to watchdog pwr\_cnt overflow.

0: enabled (default)

1: disabled

[0] Not used

### 18.5.14 CONFIG-REG 14

|                             | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                                    | D1          | D2          | D3          | D4          | D5          | D6          | D7          |
|-----------------------------|----------------|----|----|----|----|----|----|-------------------------------------------------------------------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| CO<br>NFI<br>GR<br>EG<br>14 | 1              | 1  | 1  | 0  | 0  | 1  | 1  | RLY1_<br>UC                                                                                           | RLY2_<br>UC | RLY3_<br>UC | RLY4_<br>UC | RLY5_<br>UC | STR1_<br>UC | STR2_<br>UC | STR3_<br>UC |
| RW                          | DEFAULT        |    |    |    |    |    |    | 0                                                                                                     | 0           | 0           | 0           | 0           | 0           | 0           | 0           |
|                             | RESET SOURCE   |    |    |    |    |    |    | D0:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset,<br>after_run_reset, RSTN_IN |             |             |             |             |             |             |             |
|                             | ACTIVATION     |    |    |    |    |    |    | D0:D7 bits active as level                                                                            |             |             |             |             |             |             |             |
|                             | CONTROL ACCESS |    |    |    |    |    |    | D0:D7 not locked bits                                                                                 |             |             |             |             |             |             |             |

[7] STR3\_UC

0→1: micro command for fast diagnostic activation for relay driver STR3

0: no functionality (default)

1: functionality

[6] STR2\_UC

0→1: micro command for fast diagnostic activation for relay driver STR2

0: no functionality (default)

1: functionality

[5] STR1\_UC

0→1: micro command for fast diagnostic activation for relay driver STR1

0: no functionality (default)

1: functionality

[4] RLY5\_UC

0→1: micro command for fast diagnostic activation for relay driver RLY5

0: no functionality (default)

1: functionality

[3] RLY4\_UC

0→1: micro command for fast diagnostic activation for relay driver RLY4

0: no functionality (default)

1: functionality

[2] RLY3\_UC

0→1: micro command for fast diagnostic activation for relay driver RLY3

0: no functionality (default)

1: functionality

[1] RLY2\_UC

0→1: micro command for fast diagnostic activation for relay driver RLY2

0: no functionality (default)

1: functionality

[0] RLY1\_UC

0→1: micro command for fast diagnostic activation for relay driver RLY1

0: no functionality (default)

1: functionality

Note: If you want to enable this function again, reconfigure it to 0 and then to 1.

## 18.5.15 CONFIG-REG 15

|                       | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                              | D1 | D2                         | D3 | D4 | D5 | D6 | D7 |
|-----------------------|----------------|----|----|----|----|----|----|-------------------------------------------------------------------------------------------------|----|----------------------------|----|----|----|----|----|
| CONFIGREG15-0(WUPT_0) | 1              | 0  | 0  | 0  | 0  | 1  | 1  | 0                                                                                               | 0  | Wake up timer_SET_0[0:5]   |    |    |    |    |    |
| RW                    | DEFAULT        |    |    |    |    |    |    |                                                                                                 |    | 0                          | 0  | 0  | 0  | 0  | 0  |
| CONFIGREG15-1(WUPT_1) | 1              | 0  | 0  | 0  | 0  | 1  | 0  | 0                                                                                               | 1  | Wake up timer_SET_1[6:11]  |    |    |    |    |    |
| RW                    | DEFAULT        |    |    |    |    |    |    |                                                                                                 |    | 0                          | 0  | 0  | 0  | 0  | 0  |
| CONFIGREG15-2(WUPT_2) | 1              | 0  | 0  | 0  | 0  | 0  | 1  | 1                                                                                               | 0  | Wake up timer_SET_2[12:17] |    |    |    |    |    |
| RW                    | DEFAULT        |    |    |    |    |    |    |                                                                                                 |    | 0                          | 0  | 0  | 0  | 0  | 0  |
| CONFIGREG15-3(WUPT_3) | 1              | 0  | 0  | 0  | 0  | 1  | 1  | 1                                                                                               | 1  | Wake up timer_SET_3[23:28] |    |    |    |    |    |
| RW                    | DEFAULT        |    |    |    |    |    |    |                                                                                                 |    | 0                          | 0  | 0  | 0  | 0  | 0  |
|                       | RESET SOURCE   |    |    |    |    |    |    | D2:D7<br>The read bits are from satellite logic on VB_STBY and are not affected by reset matrix |    |                            |    |    |    |    |    |
|                       | ACTIVATION     |    |    |    |    |    |    | D2:D7 bits active as level                                                                      |    |                            |    |    |    |    |    |
|                       | CONTROL ACCESS |    |    |    |    |    |    | D2:D7 not locked bits                                                                           |    |                            |    |    |    |    |    |

Application note: the minimum time distance between two consecutive access to the Wake up timer SET registers is 2.5 us.

Wake up timer\_SET\_x Bits go to and are stored in Wake Up Timer Logic.

## 18.5.16 CONFIG-REG 16

|               | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                                 | D1 | D2            | D3 | D4            | D5 | D6            | D7          |
|---------------|----------------|----|----|----|----|----|----|----------------------------------------------------------------------------------------------------|----|---------------|----|---------------|----|---------------|-------------|
| CONFIGREG16-0 | 0              | 0  | 1  | 0  | 0  | 1  | 1  | 0                                                                                                  | 1  | PRD1_BLK[0:1] |    | PRD2_BLK[0:1] |    | PRD3_BLK[0:1] |             |
| RW            | DEFAULT        |    |    |    |    |    |    |                                                                                                    |    | 1             | 0  | 1             | 0  | 1             | 0           |
| CONFIGREG16-1 | 0              | 0  | 1  | 0  | 0  | 1  | 1  | 1                                                                                                  | 0  | PRD4_BLK[0:1] |    | PRD5_BLK[0:1] |    | O2H_P DRV_1   | O2H_P DRV_3 |
| RW            | DEFAULT        |    |    |    |    |    |    |                                                                                                    |    | 1             | 0  | 1             | 0  | 0             | 0           |
|               | RESET SOURCE   |    |    |    |    |    |    | D2:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN |    |               |    |               |    |               |             |
|               | ACTIVATION     |    |    |    |    |    |    | D2:D7 bits active as level                                                                         |    |               |    |               |    |               |             |
|               | CONTROL ACCESS |    |    |    |    |    |    | D2:D7 not locked bits                                                                              |    |               |    |               |    |               |             |

[7] O2H\_PDRV\_3

VDS comparator selection for ON diagnosis

0: PDRV3 takes VDS3 comparator and PDRV4 works as independent driver

1: PDRV3 takes VDS4 comparator and PDRV4 driver cannot be used

[6] O2H\_PDRV\_1

VDS comparator selection for ON diagnosis

0: PDRV1 takes VDS1 comparator and PDRV2 works as independent driver

1: PDRV1 takes VDS2 comparator and PDRV2 driver cannot be used

PRDx\_BLK[1:0]

Blanking time on VDS detection in ON state for Predriver

00: 6  $\mu$ s01: 12  $\mu$ s (default)10: 18  $\mu$ s11: 24  $\mu$ s**18.5.17 CONFIG-REG 17**

|                 | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                                 | D1 | D2 | D3      | D4         | D5 | D6      | D7         |
|-----------------|----------------|----|----|----|----|----|----|----------------------------------------------------------------------------------------------------|----|----|---------|------------|----|---------|------------|
| CONFIG-REG 17_0 | 0              | 0  | 1  | 0  | 0  | 1  | 0  | 0                                                                                                  | 0  |    | O2H1_SR | O2H1_OC_TH |    | O2H2_SR | O2H2_OC_TH |
| RW              | DEFAULT        |    |    |    |    |    |    |                                                                                                    |    | 0  | 0       | 1          | 0  | 0       | 1          |
|                 | RESET SOURCE   |    |    |    |    |    |    | D2:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN |    |    |         |            |    |         |            |
|                 | ACTIVATION     |    |    |    |    |    |    | D2:D7 bits active as level                                                                         |    |    |         |            |    |         |            |
|                 | CONTROL ACCESS |    |    |    |    |    |    | D2:D7 not locked bits                                                                              |    |    |         |            |    |         |            |

[7] O2H2\_OC\_TH

Select OVC threshold

0: threshold1 I\_ovc1\_o2h 4.5A typ

1: threshold2 I\_ovc2\_o2h 10A typ (default)

[6] O2H2\_SR

Select SR threshold

0: typ 0,5 V/ $\mu$ s (default)1: typ 4 V/ $\mu$ s

[4] O2H1\_OC\_TH

Select OVC threshold

0: threshold1 I\_ovc1\_o2h 4.5A typ

1: threshold2 I\_ovc2\_o2h 10A typ (default)

[3] O2H1\_SR

Select SR threshold

0: typ 0,5 V/ $\mu$ s (default)1: typ 4 V/ $\mu$ s

|                 | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                              | D1 | D2     | D3        | D4             | D5             | D6        | D7         |
|-----------------|----------------|----|----|----|----|----|----|-------------------------------------------------------------------------------------------------|----|--------|-----------|----------------|----------------|-----------|------------|
| CONFIG-REG 17_1 | 0              | 0  | 1  | 0  | 0  | 1  | 0  | 1                                                                                               | 1  | CAN_EN | CAN_TX_EN | CAN_WAK_EUP_EN | CAN_PATTERN_EN | CAN_RX_EN | CAN_2_5_MB |
| RW              | DEFAULT        |    |    |    |    |    |    |                                                                                                 |    | 1      | 1         | 1              | 1              | 1         | 0          |
|                 | RESET SOURCE   |    |    |    |    |    |    | D2:D7<br>The read bits are from satellite logic on VB_STBY and are not affected by reset matrix |    |        |           |                |                |           |            |
|                 | ACTIVATION     |    |    |    |    |    |    | D2:D7 bits active as level                                                                      |    |        |           |                |                |           |            |
|                 | CONTROL ACCESS |    |    |    |    |    |    | D2:D6 not locked bits<br>D7 locked bit                                                          |    |        |           |                |                |           |            |

[7] CAN\_2\_5\_MB

CAN mode selection:

0: CAN with 2Mbaud (default)

1: CAN FD with 5Mbaud with reduced EMC performance.

[6] CAN\_RX\_EN:

CAN Receiver enable:

0: disable

1: enable (default)

[5] CAN\_PATTERN\_EN

Normal pattern wake up enable:

0: disable

1: enable (default)  
 [4] CAN\_WAKEUP\_EN  
 Enable wake up by CAN:  
 0: disable  
 1: enable (default)  
 [3] CAN\_TX\_EN  
 CAN Transmitter enable:  
 0: disable  
 1: enable (default)  
 [2] CAN\_EN  
 CAN enable  
 0: disable  
 1: enable (default)  
 All Bits go to and are stored in CAN Logic

### 18.5.18 CONFIG-REG STBY\_NVM

|               | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                              | D1 | D2                | D3 | D4 | D5 | D6        | D7 |
|---------------|----------------|----|----|----|----|----|----|-------------------------------------------------------------------------------------------------|----|-------------------|----|----|----|-----------|----|
| STBY_NVM_ADDR | 0              | 1  | 0  | 0  | 1  | 1  | 0  | 0                                                                                               | 1  | STBY_NVM_ADD[0:3] |    |    |    | MEM_VALID |    |
| RW            | DEFAULT        |    |    |    |    |    |    |                                                                                                 |    | 0                 | 0  | 0  | 0  | 0         |    |
|               | RESET SOURCE   |    |    |    |    |    |    | D2:D6<br>The read bits are from satellite logic on VB_STBY and are not affected by reset matrix |    |                   |    |    |    |           |    |
|               | ACTIVATION     |    |    |    |    |    |    | D2:D6 bits active as level                                                                      |    |                   |    |    |    |           |    |
|               | CONTROL ACCESS |    |    |    |    |    |    | D2:D6 not locked bits                                                                           |    |                   |    |    |    |           |    |

[6] MEM\_VALID  
 STBY NVM memory valid bit:  
 0: not valid (default)  
 1: valid  
 [5:2] STBY\_NVM\_ADD  
 STBY NVM write address  
 0000: address of the byte to write (default)  
 All Bits go to and are stored in Wake Up Timer Logic

|               | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                              | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
|---------------|----------------|----|----|----|----|----|----|-------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|
| STBY_NVM_DATA | 1              | 0  | 1  | 1  | 0  | 0  | 0  | STBY_NVM_DATA[0:7]                                                                              |    |    |    |    |    |    |    |
| RW            | DEFAULT        |    |    |    |    |    |    |                                                                                                 |    | 0  | 0  | 0  | 0  | 0  |    |
|               | RESET SOURCE   |    |    |    |    |    |    | D0:D7<br>The read bits are from satellite logic on VB_STBY and are not affected by reset matrix |    |    |    |    |    |    |    |
|               | ACTIVATION     |    |    |    |    |    |    | D0:D7 bits active as level                                                                      |    |    |    |    |    |    |    |
|               | CONTROL ACCESS |    |    |    |    |    |    | D0:D7 not locked bits                                                                           |    |    |    |    |    |    |    |

[7:0] STBY\_NVM\_ADD  
 Byte to write in STBY NVM  
 All Bits go to and are stored in Wake Up Timer Logic

## 18.5.19 CONFIG-REG 20

|                         | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                                                                                                                                        | D1       | D2                   | D3                  | D4          | D5      | D6                 | D7           |
|-------------------------|----------------|----|----|----|----|----|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------|---------------------|-------------|---------|--------------------|--------------|
| CON<br>FIGR<br>EG<br>20 | 1              | 1  | 1  | 1  | 1  | 1  | 0  | BIST_EN                                                                                                                                                                                                   | FIN_WAKE | WAKE_UP_TIMER_EN_SEL | KEY_OC_RETRY_MAX_EN | CAN_LOOP_EN | CAN_TDI | LIN_TX_DOM_ERR_CFG | PDRV_O2H_DLY |
| RW                      | DEFAULT        |    |    |    |    |    |    | 0                                                                                                                                                                                                         | 0        | 0                    | 1                   | 0           | 0       | 0                  | 0            |
|                         | RESET SOURCE   |    |    |    |    |    |    | D0:D1; D4:D7<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN<br>D2<br>the read bits are from satellite logic on VB_STBY and are not affected by reset matrix |          |                      |                     |             |         |                    |              |
|                         | ACTIVATION     |    |    |    |    |    |    | D0 bit active as pulse<br>D1:D7 bits active as level                                                                                                                                                      |          |                      |                     |             |         |                    |              |
|                         | CONTROL ACCESS |    |    |    |    |    |    | D0:D3 not locked bits<br>D4:D7 locked bits                                                                                                                                                                |          |                      |                     |             |         |                    |              |

[7] PDRV\_O2H\_DLY

Selection of delay time for switching from low to high current in predriver on/off transitions in o2h configuration.

0: 150 us selected (default)

1: 300 us selected

[6] LIN\_TX\_DOM\_ERR\_CFG

LIN Tx enable after dominant error timeout:

0: re-enable as soon as Tx become recessive (default)

1: wait for error flag read

[5] CAN\_TDI

CAN transmission depends on WDA:

0: WDA does not affect CAN (default)

1: disable transmission if WDA\_INT active

[4] CAN\_LOOP\_EN

CAN looping mode enable:

0: disable (default)

1: enable

[3] KEY\_OC\_RETRY\_MAX\_EN

Enable a maximum number of retry after OVC, when wake up by key:

0: disable

1: enable(default)

Bits going to Wake Up Timer Logic

[2] WAKE\_UP\_TIMER\_EN\_SEL

Select the source of activation for the wake\_up\_timer

0: from MSC start/stop commands (default)

1: from KEY\_IN 1-&gt;0

Bits going to Wake Up Timer Logic

[1] FIN\_WAKE

0: satellite logics execute auto clear 1 second after power up (default)

1: micro confirmed power up, no auto clear after 1 second

[0] BIST\_EN

Bist enable for regulators

0: disable (default)

1: enable

## 18.5.20 CONFIG-REG 21

|               | C0             | C1 | C2 | C3 | C4 | C5 | C6 | D0                                                                                                                                                                                                    | D1     | D2           | D3             | D4              | D5              | D6             | D7            |
|---------------|----------------|----|----|----|----|----|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------|----------------|-----------------|-----------------|----------------|---------------|
| CONFIG-REG 21 | 0              | 0  | 0  | 0  | 0  | 0  | 1  | CAN_TX_DOM_ERR_CFG                                                                                                                                                                                    | OL_RED | BUCK_SLOW_SR | CAN_TXD_DOM_EN | CAN_PERM_REC_EN | CAN_PERM_DOM_EN | CAN_RXD_REC_EN | CAN_AUTO_BIAS |
| RW            | DEFAULT        |    |    |    |    |    |    | 0                                                                                                                                                                                                     | 0      | 0            | 1              | 1               | 1               | 1              | 1             |
|               | RESET SOURCE   |    |    |    |    |    |    | D0:D2<br>V5Va_ov, V5Va_uv, V5Vd_ov, V5Vd_uv, vdd5_uv, vddio_uv, sw_reset, after_run_reset, RSTN_IN<br>D3:D7<br>the read bits are from satellite logic on VB_STBY and are not affected by reset matrix |        |              |                |                 |                 |                |               |
|               | ACTIVATION     |    |    |    |    |    |    | D0:D7 bits active as level                                                                                                                                                                            |        |              |                |                 |                 |                |               |
|               | CONTROL ACCESS |    |    |    |    |    |    | D0 locked bit<br>D1:D7 not locked bits                                                                                                                                                                |        |              |                |                 |                 |                |               |

[7] CAN\_AUTO\_BIAS

0: auto biasing disabled

1: auto biasing enabled (default)

[6] CAN\_RXD\_REC\_EN

0: error handling disable

1: error handling enable (default)

[5] CAN\_PERM\_DOM\_EN

0: error handling disable

1: error handling enable (default)

[4] CAN\_PERM\_REC\_EN

0: error handling disable

1: error handling enable (default)

[3] CAN\_TXD\_DOM\_EN

0: error handling disable

1: error handling enable (default)

All Bits going to CAN Logic

[2] BUCK\_SLOW\_SR

0: fast slew rate selected (default)

1: slow slew rate selected

[1] OL\_RED

Select open load current:

0: 0.8 mA max (default)

1: 0.2 mA max

[0] CAN\_TX\_DOM\_ERR\_CFG

CAN Tx enable after dominant error timeout:

0: re-enable as soon as Tx become recessive (default)

1: wait for error flag read

### 18.5.21 Commands

#### RD\_COMMAND1.. RD\_COMMAND16

These commands allow the reading of information that will be output on Upstream.

**Table 18-8 Commands**

| Name         | ADDRESS |    |    |    |    |    |    | DATA |    |    |    |    |    |    |    |
|--------------|---------|----|----|----|----|----|----|------|----|----|----|----|----|----|----|
|              | C0      | C1 | C2 | C3 | C4 | C5 | C6 | D0   | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
| RD_COMMAND1  | 1       | 0  | 1  | 0  | 1  | 0  | 1  | 0    | 1  | 0  | 1  | 0  | 1  | 0  | 1  |
| RD_COMMAND2  | 1       | 0  | 1  | 0  | 1  | 0  | 1  | 1    | 0  | 1  | 0  | 1  | 0  | 1  | 0  |
| RD_COMMAND3  | 0       | 1  | 0  | 1  | 0  | 1  | 0  | 0    | 1  | 0  | 1  | 0  | 1  | 0  | 1  |
| RD_COMMAND4  | 0       | 1  | 0  | 1  | 0  | 1  | 0  | 1    | 0  | 1  | 0  | 1  | 0  | 1  | 0  |
| RD_COMMAND5  | 1       | 0  | 1  | 0  | 1  | 0  | 1  | 1    | 0  | 0  | 1  | 1  | 0  | 0  | 1  |
| RD_COMMAND6  | 1       | 0  | 1  | 0  | 1  | 0  | 1  | 0    | 1  | 1  | 0  | 0  | 1  | 1  | 0  |
| RD_COMMAND7  | 0       | 1  | 0  | 1  | 0  | 1  | 0  | 1    | 0  | 0  | 1  | 1  | 0  | 0  | 1  |
| RD_COMMAND8  | 0       | 1  | 0  | 1  | 0  | 1  | 0  | 0    | 1  | 1  | 0  | 0  | 1  | 1  | 0  |
| RD_COMMAND9  | 1       | 0  | 1  | 0  | 1  | 0  | 1  | 1    | 1  | 1  | 0  | 1  | 1  | 1  | 0  |
| RD_COMMAND10 | 1       | 0  | 1  | 0  | 1  | 0  | 1  | 0    | 0  | 0  | 1  | 0  | 0  | 0  | 1  |
| RD_COMMAND11 | 0       | 1  | 0  | 1  | 0  | 1  | 0  | 1    | 1  | 1  | 1  | 0  | 1  | 1  | 0  |
| RD_COMMAND12 | 0       | 1  | 0  | 1  | 0  | 1  | 0  | 0    | 0  | 0  | 1  | 0  | 0  | 0  | 1  |
| RD_COMMAND13 | 1       | 0  | 1  | 0  | 1  | 0  | 1  | 0    | 1  | 1  | 1  | 0  | 1  | 1  | 1  |
| RD_COMMAND14 | 1       | 0  | 1  | 0  | 1  | 0  | 1  | 1    | 0  | 0  | 0  | 1  | 0  | 0  | 0  |
| RD_COMMAND15 | 0       | 1  | 0  | 1  | 0  | 1  | 0  | 0    | 1  | 1  | 1  | 0  | 1  | 1  | 1  |
| RD_COMMAND16 | 0       | 1  | 0  | 1  | 0  | 1  | 0  | 1    | 0  | 0  | 0  | 1  | 0  | 0  | 0  |

#### Lock Command

Registers that have this function are locked as default.

Registers are unlocked by the UNLOCK command and locked again by the LOCK command. Before the LOCK command the registers stay unlocked.

| W    | C0 | C1 | C2 | C3 | C4 | C5 | C6 | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
|------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| LOCK | 1  | 0  | 0  | 1  | 1  | 1  | 1  | 0  | 1  | 0  | 1  | 0  | 1  | 0  | 1  |

#### Unlock Command

Enables the write right to registers under lock.

| W      | C0 | C1 | C2 | C3 | C4 | C5 | C6 | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| UNLOCK | 1  | 0  | 0  | 1  | 1  | 1  | 1  | 1  | 0  | 1  | 0  | 1  | 0  | 1  | 0  |

#### Question and Answer Watch Dog Enable command

WD is enabled as default.

| W      | C0 | C1 | C2 | C3 | C4 | C5 | C6 | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| WDA_EN | 1  | 0  | 0  | 1  | 1  | 1  | 1  | 0  | 1  | 1  | 0  | 0  | 1  | 1  | 0  |

#### Question and Answer Watch Dog Disable command

| W       | C0 | C1 | C2 | C3 | C4 | C5 | C6 | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
|---------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| WDA_DIS | 1  | 0  | 0  | 1  | 1  | 1  | 1  | 1  | 0  | 0  | 1  | 1  | 0  | 0  | 1  |

### SW-Reset command

This command generates internal reset initiated by the CPU's software that clears all the configuration and diagnostic registers and switches off all the drivers.

| W        | C0 | C1 | C2 | C3 | C4 | C5 | C6 | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| SW-RESET | 1  | 0  | 1  | 1  | 1  | 0  | 1  | 0  | 1  | 1  | 0  | 1  | 1  | 0  | 1  |

### EN\_Drivers command

The command Enable Drivers sets the bit Drivers\_EN to “1” (UPS1 Frame 10 bit D1).

With Drivers\_EN =1 all drivers controlled through control register can be activated using the Data Frame.

After a reset or command Disable Drivers\_EN = “0” and all drivers are disabled.

| W          | C0 | C1 | C2 | C3 | C4 | C5 | C6 | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
|------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| EN-DRIVERS | 0  | 1  | 1  | 0  | 0  | 0  | 1  | 1  | 0  | 0  | 1  | 1  | 0  | 0  | 1  |

### Disable Driver

| W               | C0 | C1 | C2 | C3 | C4 | C5 | C6 | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
|-----------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| DISABLE-DRIVERS | 0  | 1  | 1  | 0  | 0  | 0  | 1  | 0  | 1  | 1  | 0  | 0  | 1  | 1  | 0  |

### Main relay Driver ON

| W      | C0 | C1 | C2 | C3 | C4 | C5 | C6 | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| MRD_ON | 0  | 1  | 1  | 0  | 0  | 0  | 1  | 1  | 1  | 0  | 0  | 0  | 1  | 1  | 1  |

### Main relay Driver OFF

| W       | C0 | C1 | C2 | C3 | C4 | C5 | C6 | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
|---------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| MRD_OFF | 0  | 1  | 1  | 0  | 0  | 0  | 1  | 0  | 0  | 1  | 1  | 1  | 0  | 0  | 0  |

### MRD\_OFF\_UC command

The Main Relay Driver is switched off for the time required to perform the off diagnosis and then switched on again. The command can be activated if firstly is unlocked.

| W      | C0 | C1 | C2 | C3 | C4 | C5 | C6 | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
|--------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| MRD_UC | 0  | 1  | 1  | 0  | 0  | 0  | 1  | 1  | 1  | 0  | 1  | 1  | 0  | 1  | 1  |

### STR2 driver ON/OFF

It has effect only if delay-off function for this driver is active (Config3[4] = 1).

|          | C0 | C1 | C2 | C3 | C4 | C5 | C6 | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| STR2_ON  | 0  | 1  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 1  | 0  | 1  | 0  | 1  | 0  |
| STR2_OFF | 0  | 1  | 0  | 0  | 1  | 0  | 0  | 1  | 1  | 0  | 1  | 0  | 1  | 0  | 1  |

### STR3 driver ON/OFF

It has effect only if delay-off function for this driver is active (Config3[5] = 1).

|          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|----------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| STR3_ON  | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| STR3_OFF | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 |

**RLY4 driver ON/OFF**

It has effect only if delay-off function for this driver is active (Config3[0] = 1).

|          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|----------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RLY4_ON  | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| RLY4_OFF | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 |

**18.6 Upstream frame**

See downstream paragraph for configuration register bits description. Bits highlighted in orange are clear on read. Bits highlighted in green come from satellite logics, they need VB\_STBY in order to be correctly read.

**18.6.1 Upstream Bit Map Read1 (CONFIG-REG 1-4)****Table 18-9 MSC Interface Upstream Bit Map Read1**

|        | ADDRESS |    |    |    | DATA                        |    |    |    |    |    |    |    |
|--------|---------|----|----|----|-----------------------------|----|----|----|----|----|----|----|
|        | A0      | A1 | A2 | A3 | D0                          | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
| FRAME1 | 0       | 0  | 0  | 0  | CONFIG-REG 1                |    |    |    |    |    |    |    |
| FRAME2 | 0       | 0  | 1  | 0  | CONFIG-REG 2                |    |    |    |    |    |    |    |
| FRAME3 | 0       | 0  | 0  | 1  | CONFIG-REG 3                |    |    |    |    |    |    |    |
| FRAME4 | 0       | 0  | 1  | 1  | CONFIG-REG 4                |    |    |    |    |    |    |    |
| source |         |    |    |    | logic on VB_IN (main logic) |    |    |    |    |    |    |    |
| access |         |    |    |    | read                        |    |    |    |    |    |    |    |

**Frame[1]**

[7:6] WAKE\_UP\_TIMER\_START\_STOP:

Read start "10" or stop "01" status, (never read "00" or "11").

These bits are valid only if WUT is driven by MSC.

*application note: Bits [7:6](D7D6) is "01" after Power on, For EOT counting status, refer to the flag bit EOT STATUS, While the counter is counting the EOT STATUS will be 1. If it's not counting EOT STATUS will be 0.*

**18.6.2 Upstream Bit Map Read2 (CONFIG-REG 5-8)****Table 18-10 MSC Interface Upstream Bit Map Read2**

|        | ADDRESS |    |    |    | DATA         |    |    |    |    |    |                     |    |
|--------|---------|----|----|----|--------------|----|----|----|----|----|---------------------|----|
|        | A0      | A1 | A2 | A3 | D0           | D1 | D2 | D3 | D4 | D5 | D6                  | D7 |
| FRAME1 | 0       | 0  | 0  | 0  | CONFIG-REG 5 |    |    |    |    |    |                     |    |
| FRAME2 | 0       | 0  | 1  | 0  | CONFIG-REG 6 |    |    |    |    |    |                     |    |
|        |         |    |    |    |              |    |    |    |    |    | EOT<br>_MO<br>DE(1) |    |
| FRAME3 | 0       | 0  | 0  | 1  | CONFIG-REG 7 |    |    |    |    |    |                     |    |
|        |         |    |    |    |              |    |    |    |    |    |                     |    |

Micro second channel (MSC) interface

|        |   |   |   |   |                                                                                                                                    |  |  |  |  |  |  |  |
|--------|---|---|---|---|------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| FRAME4 | 0 | 0 | 1 | 1 | CONFIG-REG 8                                                                                                                       |  |  |  |  |  |  |  |
| source |   |   |   |   | FRAME[1][3][4] FRAME[2]D[0:5] FRAME[2]D[7]<br>logic on VB_IN (main logic)<br>FRAME[2]D[6]<br>logic on VB_STBY(Wake Up Timer logic) |  |  |  |  |  |  |  |
| access |   |   |   |   | read                                                                                                                               |  |  |  |  |  |  |  |

1. Bit in satellite logic.

### 18.6.3 Upstream Bit Map Read3 (CONFIG-REG 11-13, 16-0)

Table 18-11 MSC Interface Upstream Bit Map Read3

|        | ADDRESS |    |    |    | DATA                         |          |                 |    |    |    |    |    |
|--------|---------|----|----|----|------------------------------|----------|-----------------|----|----|----|----|----|
|        | A0      | A1 | A2 | A3 | D0                           | D1       | D2              | D3 | D4 | D5 | D6 | D7 |
| FRAME1 | 0       | 0  | 0  | 0  | CONFIG-REG 11                |          |                 |    |    |    |    |    |
|        |         |    |    |    |                              |          |                 |    |    |    |    |    |
| FRAME2 | 0       | 0  | 1  | 0  | CONFIG-REG 12                |          |                 |    |    |    |    |    |
|        |         |    |    |    |                              |          |                 |    |    |    |    |    |
| FRAME3 | 0       | 0  | 0  | 1  | CONFIG-REG 13                |          |                 |    |    |    |    |    |
|        |         |    |    |    |                              |          |                 |    |    |    |    |    |
| FRAME4 | 0       | 0  | 1  | 1  | STR_DELAY<br>_OFF_STAT<br>US |          | CONFIG-REG 16-0 |    |    |    |    |    |
|        |         |    |    |    | STR<br>2                     | STR<br>3 |                 |    |    |    |    |    |
| source |         |    |    |    | logic on VB_IN (main logic)  |          |                 |    |    |    |    |    |
| access |         |    |    |    | read                         |          |                 |    |    |    |    |    |

**FRAME[4]:**

[1] 1:STR3 delay-off function active

0: STR3 delay-off function inactive

[0] 1: STR2 delay-off function active

0: STR2 delay-off function inactive

### 18.6.4 Upstream Bit Map Read4 (CONFIG-REG 16-1 + Wake up timer\_SET)

Table 18-12 MSC Interface Upstream Bit Map Read4

|        | ADDRESS |    |    |    | DATA                                          |                         |                 |    |    |    |    |    |
|--------|---------|----|----|----|-----------------------------------------------|-------------------------|-----------------|----|----|----|----|----|
|        | A0      | A1 | A2 | A3 | D0                                            | D1                      | D2              | D3 | D4 | D5 | D6 | D7 |
| FRAME1 | 0       | 0  | 0  | 0  | RLY<br>4_D<br>ELA<br>Y_O<br>FF_<br>STA<br>TUS | MRD<br>_ST<br>_ATU<br>S | CONFIG-REG 16_1 |    |    |    |    |    |
|        |         |    |    |    | RLY<br>4                                      | MRD                     |                 |    |    |    |    |    |
|        |         |    |    |    | Wake up timer_SET [23:0]                      |                         |                 |    |    |    |    |    |
| FRAME2 | 0       | 0  | 1  | 0  | Wake up timer_SET_0[0:7](1)                   |                         |                 |    |    |    |    |    |
| FRAME3 | 0       | 0  | 0  | 1  | Wake up timer_SET_1[8:15](1)                  |                         |                 |    |    |    |    |    |

Micro second channel (MSC) interface

|        |   |   |   |   |                                                                                                            |  |  |  |  |  |  |  |
|--------|---|---|---|---|------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| FRAME4 | 0 | 0 | 1 | 1 | Wake up timer_SET_2[16:23](1)                                                                              |  |  |  |  |  |  |  |
| source |   |   |   |   | FRAME[1][0:7]<br>logic on VB_IN (main logic)<br>FRAME[2:4]D[0:7]<br>logic on VB_STBY (Wake Up Timer logic) |  |  |  |  |  |  |  |
| access |   |   |   |   | read                                                                                                       |  |  |  |  |  |  |  |

1. Bit in satellite logic.

**FRAME[1]:**

[1] 1: Main Relay Driver On

0: Main Relay Driver Off

[0] 1: RLY4 delay-off function active

0: RLY4 delay-off function inactive

FRAME[2:4]: Wake Up Timer setting

Bits coming from WUT Logic

### 18.6.5 Upstream Bit Map Read5 (CONFIG-REG 17-0 + WAKE\_UP\_TIMER VALUE)

Table 18-13 MSC Interface Upstream Bit Map Read5

|        | ADDRESS |    |    |    | DATA                                                                                                        |                           |                 |    |    |    |    |    |
|--------|---------|----|----|----|-------------------------------------------------------------------------------------------------------------|---------------------------|-----------------|----|----|----|----|----|
|        | A0      | A1 | A2 | A3 | D0                                                                                                          | D1                        | D2              | D3 | D4 | D5 | D6 | D7 |
| FRAME1 | 0       | 0  | 0  | 0  |                                                                                                             | TNL                       | CONFIG-REG 17_0 |    |    |    |    |    |
|        |         |    |    |    |                                                                                                             | TNL<br>_RE<br>_SET<br>(1) |                 |    |    |    |    |    |
|        |         |    |    |    | TIMER CONT VALUE[23:0]                                                                                      |                           |                 |    |    |    |    |    |
| FRAME2 | 0       | 0  | 1  | 0  | Wake up timer _CNT_0[0:7](2)                                                                                |                           |                 |    |    |    |    |    |
| FRAME3 | 0       | 0  | 0  | 1  | Wake up timer _CNT_1[8:15](2)                                                                               |                           |                 |    |    |    |    |    |
| FRAME4 | 0       | 0  | 1  | 1  | Wake up timer _CNT_2[16:23](2)                                                                              |                           |                 |    |    |    |    |    |
| source |         |    |    |    | FRAME[1]D[0:7]<br>logic on VB_IN (main logic)<br>FRAME[2:3]D[0:7]<br>logic on VB_STBY (Wake Up Timer logic) |                           |                 |    |    |    |    |    |
| access |         |    |    |    | FRAME[1]D[1]<br>clear-on-read<br>FRAME[1]D[0] FRAME[1]D[2:7] FRAME[2:4]D[0:7]<br>read                       |                           |                 |    |    |    |    |    |

1. Clear on Read.

2. Bit in satellite logic.

**FRAME[1]:**

[1] 1: After run reset (TNL) happened

0: After run reset (TNL) not happened, clear on read

FRAME[2:4]: Wake Up Timer counting

### 18.6.6 Upstream Bit Map Read6 (DRIVER DIAGNOSIS)

Table 18-14 MSC Interface Upstream Bit Map Read6

|        | ADDRESS |    |    |    | DATA      |    |    |    |           |    |    |    |
|--------|---------|----|----|----|-----------|----|----|----|-----------|----|----|----|
|        | A0      | A1 | A2 | A3 | D0        | D1 | D2 | D3 | D4        | D5 | D6 | D7 |
| FRAME1 | 0       | 0  | 0  | 0  | DIAG INJ1 |    |    |    | DIAG INJ2 |    |    |    |

Micro second channel (MSC) interface

|        |   |   |   |   | STA                         | STG | OPL | OVC | STA       | STG | OPL | OVC |
|--------|---|---|---|---|-----------------------------|-----|-----|-----|-----------|-----|-----|-----|
| FRAME2 | 0 | 0 | 1 | 0 | DIAG INJ3                   |     |     |     | DIAG INJ4 |     |     |     |
|        |   |   |   |   | STA                         | STG | OPL | OVC | STA       | STG | OPL | OVC |
| FRAME3 | 0 | 0 | 0 | 1 | DIAG O2H1                   |     |     |     |           |     |     |     |
|        |   |   |   |   | STA                         | STG | OPL | OVC |           |     |     |     |
| FRAME4 | 0 | 0 | 1 | 1 | DIAG O2H2                   |     |     |     |           |     |     |     |
|        |   |   |   |   | STA                         | STG | OPL | OVC |           |     |     |     |
| source |   |   |   |   | logic on VB_IN (main logic) |     |     |     |           |     |     |     |
| access |   |   |   |   | clear-on-read               |     |     |     |           |     |     |     |

Clear on Read

FRAME[1:4]: Driver diagnosis for driver

STA: driver diagnosis: error in driver status

The driver status is detected by reading the STG comparator in on and off, it should be aligned with command (msc command and enabling).

0: aligned, no error

1: not aligned, error

STG/STB: driver diagnosis: short to ground/battery

0: no fault (default)

1: for low-side and igniter drivers: short to ground fault

for high-side driver: short to battery fault

OPL: driver diagnosis: open load

0: no fault (default)

1: open load fault

OVC: driver diagnosis: over current

0: no fault (default)

1: over current fault

## 18.6.7 Upstream Bit Map Read7 (DRIVER DIAGNOSIS)

Table 18-15 MSC Interface Upstream Bit Map Read7

|        | ADDRESS |    |    |    | READ3 DATA                  |     |     |     |           |     |     |     |
|--------|---------|----|----|----|-----------------------------|-----|-----|-----|-----------|-----|-----|-----|
|        | A0      | A1 | A2 | A3 | D0                          | D1  | D2  | D3  | D4        | D5  | D6  | D7  |
| FRAME1 | 0       | 0  | 0  | 0  | DIAG LED1                   |     |     |     | DIAG LED2 |     |     |     |
|        |         |    |    |    | STA                         | STG | OPL | OVC | STA       | STG | OPL | OVC |
| FRAME2 | 0       | 0  | 1  | 0  | MRD DIAG                    |     |     |     | DIAG RLY1 |     |     |     |
|        |         |    |    |    | STA                         | STG | OPL | OVC | STA       | STG | OPL | OVC |
| FRAME3 | 0       | 0  | 0  | 1  | DIAG RLY2                   |     |     |     | DIAG RLY3 |     |     |     |
|        |         |    |    |    | STA                         | STG | OPL | OVC | STA       | STG | OPL | OVC |
| FRAME4 | 0       | 0  | 1  | 1  | DIAG RLY4                   |     |     |     | DIAG RLY5 |     |     |     |
|        |         |    |    |    | STA                         | STG | OPL | OVC | STA       | STG | OPL | OVC |
| source |         |    |    |    | logic on VB_IN (main logic) |     |     |     |           |     |     |     |
| access |         |    |    |    | clear-on-read               |     |     |     |           |     |     |     |

Clear on Read

## 18.6.8 Upstream Bit Map Read8 (DRIVER DIAGNOSIS)

Table 18-16 MSC Interface Upstream Bit Map Read8

| ADDRESS |    |    |    |    | READ4 DATA |         |     |     |           |         |     |     |
|---------|----|----|----|----|------------|---------|-----|-----|-----------|---------|-----|-----|
|         | A0 | A1 | A2 | A3 | D0         | D1      | D2  | D3  | D4        | D5      | D6  | D7  |
| FRAME1  | 0  | 0  | 0  | 0  | DIAG STR1  |         |     |     | DIAG STR2 |         |     |     |
|         |    |    |    |    | STA        | STG/STB | OPL | OVC | STA       | STG/STB | OPL | OVC |
| FRAME2  | 0  | 0  | 1  | 0  | DIAG STR3  |         |     |     | DIAG PRD1 |         |     |     |

Micro second channel (MSC) interface

|        |   |   |   |   |                             |         |     |     |           |     |     |     |
|--------|---|---|---|---|-----------------------------|---------|-----|-----|-----------|-----|-----|-----|
|        |   |   |   |   | STA                         | STG/STB | OPL | OVC | STA       | STG | OPL | OVC |
| FRAME3 | 0 | 0 | 0 | 1 | DIAG PRD2                   |         |     |     | DIAG PRD3 |     |     |     |
|        |   |   |   |   | STA                         | STG     | OPL | OVC | STA       | STG | OPL | OVC |
| FRAME4 | 0 | 0 | 1 | 1 | DIAG PRD4                   |         |     |     | DIAG PRD5 |     |     |     |
|        |   |   |   |   | STA                         | STG     | OPL | OVC | STA       | STG | OPL | OVC |
| source |   |   |   |   | logic on VB_IN (main logic) |         |     |     |           |     |     |     |
| access |   |   |   |   | clear-on-read               |         |     |     |           |     |     |     |

Clear on Read

### 18.6.9 Upstream Bit Map Read9 (DRIVER DIAGNOSIS)

Table 18-17 MSC Interface Upstream Bit Map Read9

|        | ADDRESS |    |    |    | READ4 DATA                  |     |     |     |           |     |     |     |
|--------|---------|----|----|----|-----------------------------|-----|-----|-----|-----------|-----|-----|-----|
|        | A0      | A1 | A2 | A3 | D0                          | D1  | D2  | D3  | D4        | D5  | D6  | D7  |
| FRAME1 | 0       | 0  | 0  | 0  | DIAG IGN1                   |     |     |     | DIAG IGN2 |     |     |     |
|        |         |    |    |    | STA                         | STG | OPL | OVC | STA       | STG | OPL | OVC |
| FRAME2 | 0       | 0  | 1  | 0  | DIAG IGN3                   |     |     |     | DIAG IGN4 |     |     |     |
|        |         |    |    |    | STA                         | STG | OPL | OVC | STA       | STG | OPL | OVC |
| FRAME3 | 0       | 0  | 0  | 1  | DIAG IGN5                   |     |     |     | DIAG IGN6 |     |     |     |
|        |         |    |    |    | STA                         | STG | OPL | OVC | STA       | STG | OPL | OVC |
| FRAME4 | 0       | 0  | 1  | 1  | DIAG SOL1                   |     |     |     | DIAG SOL2 |     |     |     |
|        |         |    |    |    | STA                         | STG | OPL | OVC | STA       | STG | OPL | OVC |
| source |         |    |    |    | logic on VB_IN (main logic) |     |     |     |           |     |     |     |
| access |         |    |    |    | clear-on-read               |     |     |     |           |     |     |     |

Clear on Read

### 18.6.10 Upstream Bit Map Fast Read10 (SAFETY WDA)

Table 18-18 MSC Interface Upstream Bit Map Read10

|        | ADDRESS |    |    |    | DATA                                                    |         |      |              |              |            |               |          |
|--------|---------|----|----|----|---------------------------------------------------------|---------|------|--------------|--------------|------------|---------------|----------|
|        | A0      | A1 | A2 | A3 | D0                                                      | D1      | D2   | D3           | D4           | D5         | D6            | D7       |
| FRAME1 | 0       | 0  | 0  | 0  | WDA REQULO                                              |         |      |              |              |            |               |          |
|        |         |    |    |    | REQU[0:3]                                               |         |      |              | ERR_CNT[0:2] |            |               | WDA_INT  |
| FRAME2 | 0       | 0  | 1  | 0  | WDA REQUHI                                              |         |      |              |              |            |               |          |
|        |         |    |    |    | RESP_TOO_EA_RLY                                         | NO_RESP | WRES | CHRT         | RESP_Z0      | RESP_ERR   | RESP_CNT[0:1] |          |
| FRAME3 | 0       | 0  | 0  | 1  | WDA PWR_RST_CNT                                         |         |      |              |              |            |               |          |
|        |         |    |    |    | RST_CNT[0:2]                                            |         |      | PWR_CNT[0:2] |              |            |               | WDA_RST  |
| FRAME4 | 0       | 0  | 1  | 1  | WDA RESPTIME                                            |         |      |              |              |            |               |          |
|        |         |    |    |    | RESPTIME[0:5]                                           |         |      |              |              | WDA_WINSEL |               | WDA_INIT |
| source |         |    |    |    | logic on VB_IN (main logic)                             |         |      |              |              |            |               |          |
| access |         |    |    |    | FRAME[1]D[0:7] FRAME[2]D[0] FRAME[2]D[2] FRAME[2]D[4:7] |         |      |              |              |            |               |          |

|  |                                                                                     |
|--|-------------------------------------------------------------------------------------|
|  | FRAME[3]D[0:7] FRAME[4]D[0:7]<br>read<br>FRAME[2]D[1] FRAME[2]D[3]<br>clear-on-read |
|--|-------------------------------------------------------------------------------------|

## Clear on Read

FRAME[1]: WATCHDOG REQULO

[7] WDA\_INT

1: ERR\_CNT[2:0]&gt;4

Default: 1

[6:4] ERR\_CNT[2:0]

Value of ERR\_CNT[2:0]

Default: 0x6

[3:0] REQU[3:0]

4-bit question

FRAME[2]: WATCHDOG REQUHI

[7:6] RESP\_CNT[1:0]

Counter for receiving the 4 response bytes

Default: 0xC

[5] RESP\_ERR

1: 1 byte of 32-bit response is incorrect: one of the response bytes in the current sequencer run is wrong, reading WDA status register after the 4th byte write implies that this flag will always be read as 0.

reset to zero at each sequencer-run

[4] RESP\_0

1: Controller set response time to 0ms

a correct response within the time window nevertheless increments the error counter by one

0: Response-time is greater than 0ms

[3] CHRT

1: Controller has changed response time

reset to zero after a read access and after the next sequencer run

[2] W\_RESP

1: if one of the RESP\_BYTE<sub>x</sub> was incorrect during the previous sequencer run;

0: otherwise

[1] NO\_RESP

1: in case of no response at all timer is restarted automatically

reset to zero after a read access

[0] RESP\_TOO\_EARLY

1: in case the 4 response bytes arrive before time window starts during the previous

sequencer run;

Reset to zero at each sequencer run

FRAME[3]: WATCHDOG PWR\_RST\_CNT

[7] WDA\_RST

See reset matrix

[5:3] PWR\_CNT[2:0]

Current value of PWR\_CNT register

[2:0] RST\_CNT[2:0]

Current value of RST\_CNT register

FRAME[4]: WATCHDOG RESPTIME

[7:6] CONFIG-REG9\_1

[5:0] RESPTIME[5:0]

CONFIG-REG9\_0

## 18.6.11 Upstream Bit Map Read11 (SAFETY+BIST+VRS+LIN+ASIC\_REV)

Table 18-19 MSC Interface Upstream Bit Map Read11

|        | ADDRESS |    |    |    | DATA       |         |         |          |               |       |        |          |
|--------|---------|----|----|----|------------|---------|---------|----------|---------------|-------|--------|----------|
|        | A0      | A1 | A2 | A3 | D0         | D1      | D2      | D3       | D4            | D5    | D6     | D7       |
| FRAME1 | 0       | 0  | 0  | 0  | SAFETY/VRS |         |         |          |               |       |        |          |
|        |         |    |    |    | TRANS_L    | TRANS_F | CMD_ERR | FREQ_ERR | CLK_MON_FAULT | OSC_S | OL_RED | VRS_DIAG |

|         |   |   |   |   |                                                                                                                                |      |      |      |      |           |                  |      |  |
|---------|---|---|---|---|--------------------------------------------------------------------------------------------------------------------------------|------|------|------|------|-----------|------------------|------|--|
|         |   |   |   |   |                                                                                                                                |      |      |      |      |           | TUCK             |      |  |
| FRAM E2 | 0 | 0 | 1 | 0 | BIST                                                                                                                           |      |      |      |      |           |                  |      |  |
|         |   |   |   |   | V5V                                                                                                                            | VDD5 | IGN1 | IGN2 | IGN3 | IGN4      | IGN5             | IGN6 |  |
| FRAM E3 | 0 | 0 | 0 | 1 | BIST/LIN                                                                                                                       |      |      |      |      |           |                  |      |  |
|         |   |   |   |   | BIST_END                                                                                                                       | INJ1 | INJ2 | INJ3 | INJ4 | DRIVER_EN | LIN/KLINES TATUS | LOCK |  |
| FRAM E4 | 0 | 0 | 1 | 1 | ASIC REVISION[0:7]                                                                                                             |      |      |      |      |           |                  |      |  |
|         |   |   |   |   | 1                                                                                                                              | 0    | 0    | 0    | 0    | 0         | 1                | 0    |  |
| source  |   |   |   |   | logic on VB_IN (main logic)                                                                                                    |      |      |      |      |           |                  |      |  |
| access  |   |   |   |   | FRAME[1]D[6] FRAME[3][5:7] FRAME[4]D0:7]<br>read<br>FRAME[1]D[0:5] FRAME[1]D[7] FRAME[2]D[0:7] FRAME[3]D[0:4]<br>clear-on-read |      |      |      |      |           |                  |      |  |

## Clear on Read

FRAME[1]: SAFETY/VRS

[0] TRANS\_L

Wrong command frame or data frame down stream length (longer then 16 bits)

0: no fault (default)

1: down stream frame length incorrect

[1] TRANS\_F

No valid the data frame for longer than tMSC\_mon

0: no fault (default)

1: no data stream within tMSC\_mon time out

[2] CMD\_ERR

This bit is address error(C0-C5) of previous command Frame

0: no fault (default)

1: command error

[3] FREQ\_ERR

Main OSC and checker OSC running with more than +/- 20% freq difference

0: no fault (default)

1: clocks out of frequency or stucked

[4] CLK\_MON\_FAULT

Main OSC and checker OSC running with more than +/- 30% freq difference

0: no fault (default)

1: clocks out of frequency or stucked (drivers disabled)

[5] OSC\_STUCK

Main OSC or checker OSC stuck

0: no fault (default)

1: clocks stucked (drivers disabled)

[6] OL\_RED

CONFIG\_REG21 BIT 1

[7] VRS\_DIAG

VRS diagnosis result

0: no fault detected (default)

1: generic fault detected

FRAME[2]: BIST

[0] BIST\_V5V

Bist result for V5V regulator

0: bist pass (default)

1: bist fail

[1] BIST\_VDD5

Bist result for VDD5 regulator

0: bist pass (default)

1: bist fail

[2:7] BIST\_IGN

Bist result for Igniter drivers

0: bist pass (default)

1: bist fail

FRAME[3]: BIST/LIN

[0] BIST\_END

End of Bist operation

0: bist not end

1: bist end

[1:4] BIST\_INJ

Bist result for INJECTOR drivers

0: bist pass (default)

1: bist fail

[5] DRIVER\_EN

Drivers enabled or disabled by MSC command

0: drivers disabled (default)

1: drivers enabled

[6] LIN\_KLINE\_STATUS

LIN / K-LINE Mode (CONFIG-REG11 [0])

0: LIN mode (default)

1: KLINE mode

[7] MSC LOCK

MSC command and bit lock

0: unlocked

1: locked (default)

FRAME [4]: ASIC REVISION FOR DA

[0:7] ASIC\_REVISION

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## 18.6.12 Upstream Bit Map Read12 (PHOLD+POWER\_UP+ADC)

Table 18-20 MSC Interface Upstream Bit Map Read12

1. Bit in satellite logic.

|        | ADDRESS |    |    |    | DATA                                                                                                                                                                   |                 |                 |                  |                  |                    |                  |                  |  |
|--------|---------|----|----|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|------------------|------------------|--------------------|------------------|------------------|--|
|        | A0      | A1 | A2 | A3 | D0                                                                                                                                                                     | D1              | D2              | D3               | D4               | D5                 | D6               | D7               |  |
| FRAME1 | 0       | 0  | 0  | 0  | PHOLD                                                                                                                                                                  |                 |                 |                  |                  |                    |                  |                  |  |
|        |         |    |    |    | PHOLD_TIMER_STATUS                                                                                                                                                     | PHOLD_TIMER [0] | PHOLD_TIMER [1] | PHOLD_TIMER [2]  | PHOLD_TIMER [3]  | PHOLD_TIMER [4]    | PHOLD_TIMER [5]  | PHOLD_TIMER [6]  |  |
| FRAME2 | 0       | 0  | 1  | 0  | PHOLD                                                                                                                                                                  |                 |                 |                  |                  |                    |                  |                  |  |
|        |         |    |    |    | PHOLD_TIMER [7]                                                                                                                                                        | PHOLD_TIMER [8] | PHOLD_TIMER [9] | PHOLD_TIMER [10] | PHOLD_TIMER [11] | PHOLD_TIMER [12]   | PHOLD_TIMER [13] | PHOLD_TIMER [14] |  |
| FRAME3 | 0       | 0  | 0  | 1  | POWER UP                                                                                                                                                               |                 |                 |                  |                  | EOT_STATUS         |                  | ADC              |  |
|        |         |    |    |    | KEY_INTERRUPT                                                                                                                                                          | WK_INTERRUPT    | WAKE_UP_EVENT   | WAKE_UP_CANCEL   | FAULT_WARN       | EOT <sup>(1)</sup> | ADC[0]           | ADC[1]           |  |
| FRAME4 | 0       | 0  | 1  | 1  | ADC                                                                                                                                                                    |                 |                 |                  |                  |                    |                  |                  |  |
|        |         |    |    |    | ADC[2]                                                                                                                                                                 | ADC[3]          | ADC[4]          | ADC[5]           | ADC[6]           | ADC[7]             | ADC[8]           | ADC[9]           |  |
| source |         |    |    |    | FRAME[1]D[0:7] FRAME[2]D[0:7] FRAME[3]D[0:4] FRAME[3]D[6:7]<br>FRAME[4]D[0:7]<br>logic on VB_IN (main logic)<br>FRAME[3]D[5]<br>logic on VB_STBY (Wake Up Timer logic) |                 |                 |                  |                  |                    |                  |                  |  |
| access |         |    |    |    | read                                                                                                                                                                   |                 |                 |                  |                  |                    |                  |                  |  |

FRAME[1:2]: POWER\_HOLD TIMER

It is the counter used to reach the timeout selected by PHOLD\_TIME[0:1] configuration bits.

The run time is derived from the PHOLD\_TIMER[14:0] according to the formula:

run time = PHOLD\_TIMER[14:0] \* 279 ms

FRAME[3]: POWER\_UP

[0] KEY\_IN\_DET

Latch status of activation of KEY\_IN signal

0: low level detected on KEY\_IN signal

1: high level on KEY\_IN signal

[1] WK\_IN\_DET

Latch status of activation of WK\_IN signal.

0: no positive edge detected on WK\_IN signal

1: positive edge on WK\_IN signal

[2] WAKE\_UP\_EOT\_DET

Latch status of activation of Wake\_Up\_Timer counter

0: Wake\_Up\_Timer end counting not detected

1: Wake\_Up\_Timer end counting detected

[3] WAKE\_UP\_CAN\_DET

Latch status of activation of CAN signal

0: no wake up activity detected on CAN

1: wake up activity detected on CAN

[4] FAULT\_WARN

0: no fault warning active

1: at least one of fault warning contributors (oscillator fault, V5Vpre\_mon\_ov,

V5Vpre\_mon\_uv, V5Vpre\_ov, V5Vpre\_uv, V5Vana\_ov, V5Vana\_uv, V5Vdig\_ov, V5Vdig\_uv, gnd\_dig\_loss) is active.

[5] EOT\_STATUS:

1: EOT active (WAKE\_UP\_TIMER\_START\_STOP[0:1] = 01, start)

0: EOT inactive (WAKE\_UP\_TIMER\_START\_STOP[0:1] = 10, stop)

[6:7] ADC[0:1]

FRAME[4]

[0:7] ADC[2:9]

Temperature= (ADC[9:0] \* 290 / 1024) - 63 [°C]

### 18.6.13 Upstream Bit Map Read13 (SUPPLY UV/OV)

Table 18-21 MSC Interface Upstream Bit Map Read13

| ADDRESS |    |    |    |    | DATA                                                                                                                                                 |               |               |               |            |               |               |               |
|---------|----|----|----|----|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------------|------------|---------------|---------------|---------------|
|         | A0 | A1 | A2 | A3 | D0                                                                                                                                                   | D1            | D2            | D3            | D4         | D5            | D6            | D7            |
| FRAME1  | 0  | 0  | 0  | 0  | SUPPLY OVERVOLTAGE/UNDERVOLTAGE                                                                                                                      |               |               |               |            |               |               |               |
|         |    |    |    |    |                                                                                                                                                      | V5VA_UV       | V5VA_OV       | V5VD_UV       | V5VD_OV    | VB_IN_UV      | VB_IN_OV1     | VB_IN_OV2     |
| FRAME2  | 0  | 0  | 1  | 0  | SUPPLY OVERVOLTAGE/UNDERVOLTAGE                                                                                                                      |               |               |               |            |               |               |               |
|         |    |    |    |    | VB_STBY_UV <sup>(1)</sup>                                                                                                                            | GND_DIG_LOSS  | VDDIO_UV      | VDD5_UV       | VDD5_OV    | VPRE_UV       | VPRE_OV       |               |
| FRAME3  | 0  | 0  | 0  | 1  | SUPPLY OVERVOLTAGE/UNDERVOLTAGE                                                                                                                      |               |               |               |            |               |               |               |
|         |    |    |    |    | WAKE_UP_CAN_DET_AUTO                                                                                                                                 | TRIM_VALID    | TRK_STB[1]    | TRK_STB[2]    | TRK_STB[3] | TRK_OVC[1]    | TRK_OVC[2]    | TRK_OVC[3]    |
| FRAME4  | 0  | 0  | 1  | 1  | SUPPLY OVERVOLTAGE/UNDERVOLTAGE                                                                                                                      |               |               |               |            |               |               |               |
|         |    |    |    |    | EXT_TRK_UV                                                                                                                                           | INT_TRK_UV[1] | INT_TRK_UV[2] | INT_TRK_UV[3] | EXT_TRK_OV | INT_TRK_OV[1] | INT_TRK_OV[2] | INT_TRK_OV[3] |
| source  |    |    |    |    | FRAME[1]D[0:7] FRAME[2]D[1:7] FRAME[3]D[0:7] FRAME[4]D[0:7]<br>logic on VB_IN (main logic)<br>FRAME[2]D[0]<br>logic on VB_STBY (Wake Up Timer logic) |               |               |               |            |               |               |               |
| access  |    |    |    |    | FRAME[3]D[1]<br>read<br>FRAME[1]D[0:7] FRAME[2]D[0:7] FRAME[3]D[0] FRAME[3]D[2:7] FRAME[4]D[0:7]<br>clear-on-read                                    |               |               |               |            |               |               |               |

1. Bit in satellite logic.

Clear on Read

FRAME[1:4]: SUPPLY OVERVOLTAGE/UNDERVOLTAGE

0: no fault (default)

1: overvoltage/undervoltage detected on the corresponding pin

VB\_STBY\_UV is functional only if VB\_STBY is present, if VB\_STBY is totally disconnected

no diagnosis related to VB\_STBY will be available (bit latched in wut logic).

GND\_DIG\_LOSS comparator for digital ground level

0: the digital ground supply is connected to ground.

1: the digital ground supply is disconnected from ground.

TRIM\_VALID

0: trimming content not written or not valid (parity check fail) (default)

(in this case the trimming content is anyway used)

1: trimming content written and valid

WAKE\_UP\_CAN\_DET\_AUTO

0: No Wakeup pattern detected in sleep mode with UCHIP-ON and CAN\_AUTO\_BIAS =

1(CONFIG-REG 21 D7) (default)

1: Wakeup pattern detected

### 18.6.14 Upstream Bit Map Read14 (THERMAL WARNING AND CAN)

Table 18-22 MSC Interface Upstream Bit Map Read14

| ADDRESS |    |    |    |    | DATA                                                                                                                                         |               |                 |                   |                   |                 |                |                   |
|---------|----|----|----|----|----------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------|-------------------|-------------------|-----------------|----------------|-------------------|
|         | A0 | A1 | A2 | A3 | D0                                                                                                                                           | D1            | D2              | D3                | D4                | D5              | D6             | D7                |
| FRAME1  | 0  | 0  | 0  | 0  | OVERTEMPERATURE                                                                                                                              |               |                 |                   |                   |                 |                |                   |
|         |    |    |    |    | OVT [1]                                                                                                                                      | OVT [2]       | OVT [3]         | OVT [4]           | OVT [5]           | OVT [6]         | OVT [7]        | OVT [8]           |
| FRAME2  | 0  | 0  | 1  | 0  | OVERTEMPERATURE                                                                                                                              |               |                 |                   |                   |                 |                |                   |
|         |    |    |    |    | OVT [9]                                                                                                                                      | OVT [10]      | OVT [11]        | OVT [12]          | OVT [13]          | OVT [14]        | OVT [15]       | OVT [16]          |
| FRAME3  | 0  | 0  | 0  | 1  | OVERTEMPERATURE                                                                                                                              |               |                 |                   | CONFIG-REG 17_1   |                 |                |                   |
|         |    |    |    |    | OVT [17]                                                                                                                                     | OVT [18]      | OVT [19]        | VDD_CAN_OV (1)    | CAN_EN (1)        | CAN_TX_EN (1)   | CAN_RX_EN (1)  | CAN_WAKEUP_EN (1) |
| FRAME4  | 0  | 0  | 1  | 1  | CONFIG-REG 17_1                                                                                                                              |               |                 | CAN DIAG          |                   |                 |                |                   |
|         |    |    |    |    | CAN_PATTERN_EN (1)                                                                                                                           | CAN_25_MB (1) | CAN_TXD_DOM (1) | CAN_PERRM_REC (1) | CAN_PERRM_DOM (1) | CAN_RXD_REC (1) | CAN_SILENT (1) | CAN_SUP_LOW (1)   |
| source  |    |    |    |    | FRAME[1]D[0:7] FRAME[2]D[0:7] FRAME[3]D[0:2]<br>logic on VB_IN (main logic)<br>FRAME[3]D[4:7] FRAME[4]D[0:7]<br>logic on VB_STBY (Can logic) |               |                 |                   |                   |                 |                |                   |
| access  |    |    |    |    | FRAME[3]D[4:7] FRAME[4]D[0:1]<br>read<br>FRAME[1]D[0:7] FRAME[2]D[0:7] FRAME[3]D[0:3] FRAME[4]D[2:7]<br>clear on read                        |               |                 |                   |                   |                 |                |                   |

1. Bit in satellite logic.

Clear on Read

FRAME[1:3]: OVERTEMPERATURE

0: no fault (default)

1: overtemperature detected on the corresponding block

FRAME[3:4]:

[3]: VDD\_CAN\_OV

0: no fault

1: VDD\_CAN\_OV overvoltage detected

CONFIG\_REG 17-1

FRAME[4]: CAN DIAG, see CAN section for detailed description.

### 18.6.15 Upstream Bit Map Read15 (MEM\_STBY\_CONFIG+CONFIG\_REG20+UCs)

Table 18-23 MSC Interface Upstream Bit Map Read15

|        | ADDRESS |    |    |    | DATA      |          |         |                       |                  |    |    |           |
|--------|---------|----|----|----|-----------|----------|---------|-----------------------|------------------|----|----|-----------|
|        | A0      | A1 | A2 | A3 | D0        | D1       | D2      | D3                    | D4               | D5 | D6 | D7        |
| FRAME1 | 0       | 0  | 0  | 0  | LIN_ERROR |          |         |                       | STBY_NVM_ADD_REG |    |    |           |
|        |         |    |    |    | LIN_PERM  | LIN_PERM | LIN_TXD | STBY_NVM_ADD[0:3] (1) |                  |    |    | MEM_VALID |

|                |   |   |   |   | REC                                                                                                                    | DOM                               | DOM                                     |                                 |                               |                              |                                |                         | (1) |  |
|----------------|---|---|---|---|------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------------------------|---------------------------------|-------------------------------|------------------------------|--------------------------------|-------------------------|-----|--|
| FR<br>AM<br>E2 | 0 | 0 | 1 | 0 | CONFIG_REG_21                                                                                                          |                                   |                                         |                                 |                               |                              |                                |                         |     |  |
|                |   |   |   |   | BUCK_SL<br>OW_SR                                                                                                       | CAN_T<br>XD_DO<br>M<br>_EN<br>(1) | CAN_P<br>ERM_<br>REC_E<br>N<br>(1)      | CAN_PER<br>M_<br>DOM_EN<br>(1)  | CAN_RX<br>D_REC_<br>EN<br>(1) | CAN_<br>AUTO<br>_BIAS(<br>1) | KEY_IN_P<br>IN<br>_FLT         | WAKE_I<br>N_PIN_F<br>LT |     |  |
| FR<br>AM<br>E3 | 0 | 0 | 0 | 1 |                                                                                                                        | CONFIG-REG 20                     |                                         |                                 |                               |                              |                                |                         |     |  |
|                |   |   |   |   | CAN_TX_<br>DOM_ER<br>R_<br>CFG(1)                                                                                      | FIN_W<br>AKE                      | WAKE_<br>UP_TI<br>MER_E<br>N_SEL(<br>1) | KEY_OC_<br>RERTY_M<br>A X_EN(1) | CAN_LO<br>OP_EN               | CAN_<br>TDI                  | LIN_TX_D<br>OM_ERR_<br>C<br>FG | PDRV_O<br>2H_DLY        |     |  |
| FR<br>AM<br>E4 | 0 | 0 | 1 | 1 | CONFIG-REG 14 (UCOMMANDS STATUS)                                                                                       |                                   |                                         |                                 |                               |                              |                                |                         |     |  |
|                |   |   |   |   | RLY1_UC                                                                                                                | RLY2_<br>UC                       | RLY3_<br>UC                             | RLY4_U C                        | RLY5_U<br>C                   | STR1_<br>UC                  | STR2_UC                        | STR3_U<br>C             |     |  |
| source         |   |   |   |   | FRAME[1]D[0:2] FRAME[2]D[0] FRAME[2]D[6:7] FRAME[3]D[1] FRAME[3][4:7]<br>FRAME[4]D[0:7]<br>logic on VB_IN (main logic) |                                   |                                         |                                 |                               |                              |                                |                         |     |  |
| access         |   |   |   |   | FRAME[1]D[3:7] FRAME[2]D[0:7] FRAME[3]D[0:7] FRAME[4]D[0:7]<br>read<br>FRAME[1]D[0:2]<br>clear-on-read                 |                                   |                                         |                                 |                               |                              |                                |                         |     |  |

1. Bit in satellite logic.

Clear on Read

FRAME [1]

[7] MEM\_VALID (see 4.19)

1: memory has been written and validated by micro by reading back data.

0: memory was not validated or standby-power-supply is removed

[6:3] STBY\_NVM\_ADD

[2] 1: LIN TxD dominant timeout error occurs

0: no LIN TxD dominant timeout error occurs

Clear on read

[1] 1: LIN permanent dominant error occurs

0: no LIN permanent dominant error occurs

Clear on read

[0] 1: LIN permanent recessive error occurs

0: no LIN permanent recessive error occurs

Clear on read

FRAME[2]

[7] 1: The WK\_IN pin (filtering on low to high transition) is at high state.

0: The WK\_IN pin (filtering on high to low transition) is at low state.

[6] 1: The KEY\_IN pin (filtering on low to high transition) is at high state.

0: The KEY\_IN pin (filtering on high to low transition) is at low state.

FRAME[3]

FRAME[4]

[7:0] µcommands:

1 if driver µcommand is still active and driver is off, 0 otherwise

0 if driver µcommand is still active and driver is on, 1 otherwise

See driver section for detailed description of µcommands.

### 18.6.16 Upstream Bit Map Read16 (MEM\_STBY\_DATA)

**Table 18-24** MSC Interface Upstream Bit Map Read16

|        | ADDRESS |    |    |    | DATA                                                  |    |    |    |    |    |    |    |
|--------|---------|----|----|----|-------------------------------------------------------|----|----|----|----|----|----|----|
|        | A0      | A1 | A2 | A3 | D0                                                    | D1 | D2 | D3 | D4 | D5 | D6 | D7 |
| FRAME1 | 0       | 0  | 0  | 0  | MEM_STBY_DATA(Byte0) <sup>(1)</sup>                   |    |    |    |    |    |    |    |
| FRAME2 | 0       | 0  | 1  | 0  | MEM_STBY_DATA(Byte1) <sup>(1)</sup>                   |    |    |    |    |    |    |    |
| FRAME3 | 0       | 0  | 0  | 1  | MEM_STBY_DATA(Byte2) <sup>(1)</sup>                   |    |    |    |    |    |    |    |
| FRAME4 | 0       | 0  | 1  | 1  | MEM_STBY_DATA(Byte3) <sup>(1)</sup>                   |    |    |    |    |    |    |    |
| source |         |    |    |    | logic on VB_STBY (Wake Up Timer logic) <sup>(1)</sup> |    |    |    |    |    |    |    |
| access |         |    |    |    | read                                                  |    |    |    |    |    |    |    |

1. Bit in satellite logic.

FRAME[1:4]: BYTES READ FROM MEM+STBY

## 19 Package information.

### ETQFP package information

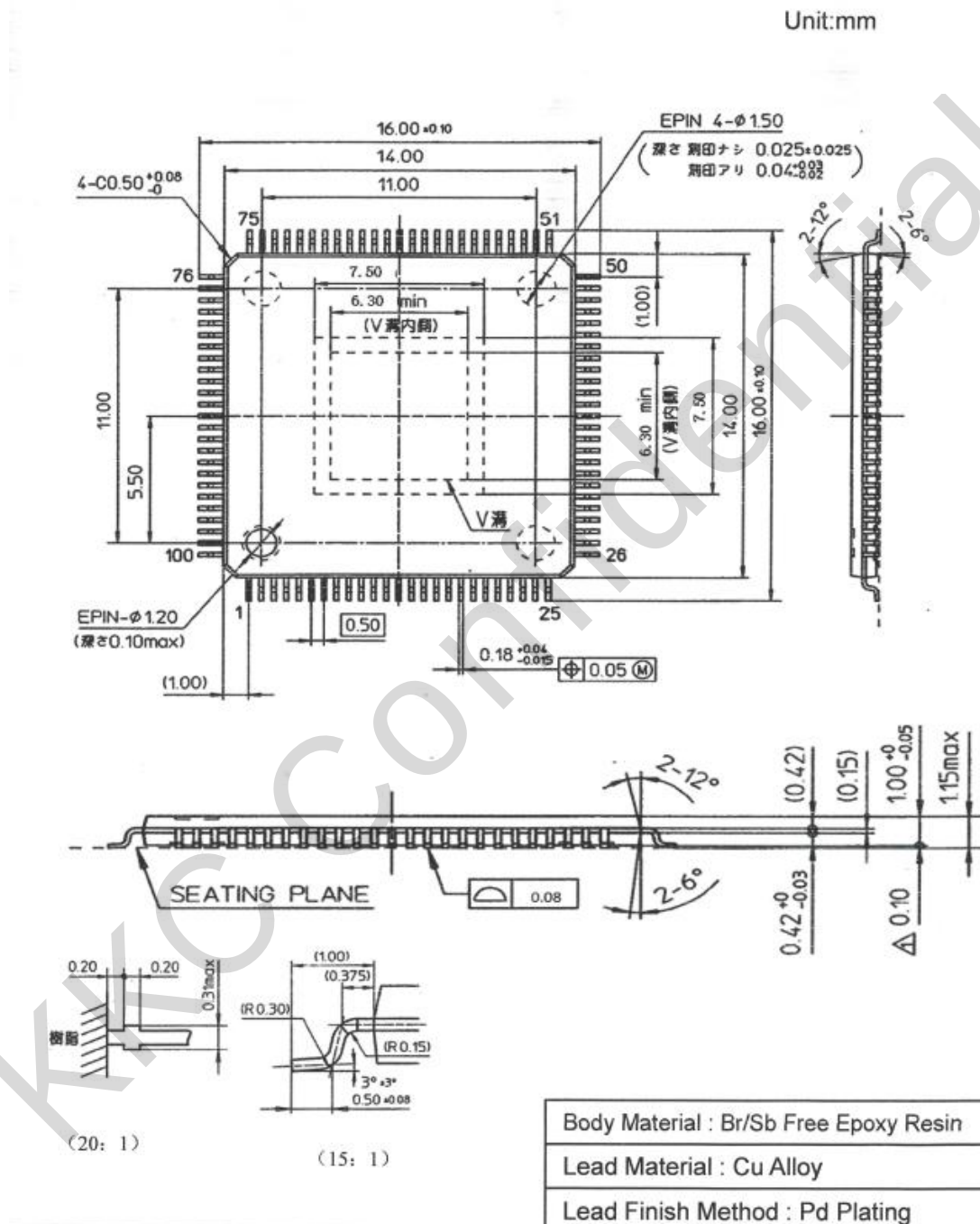
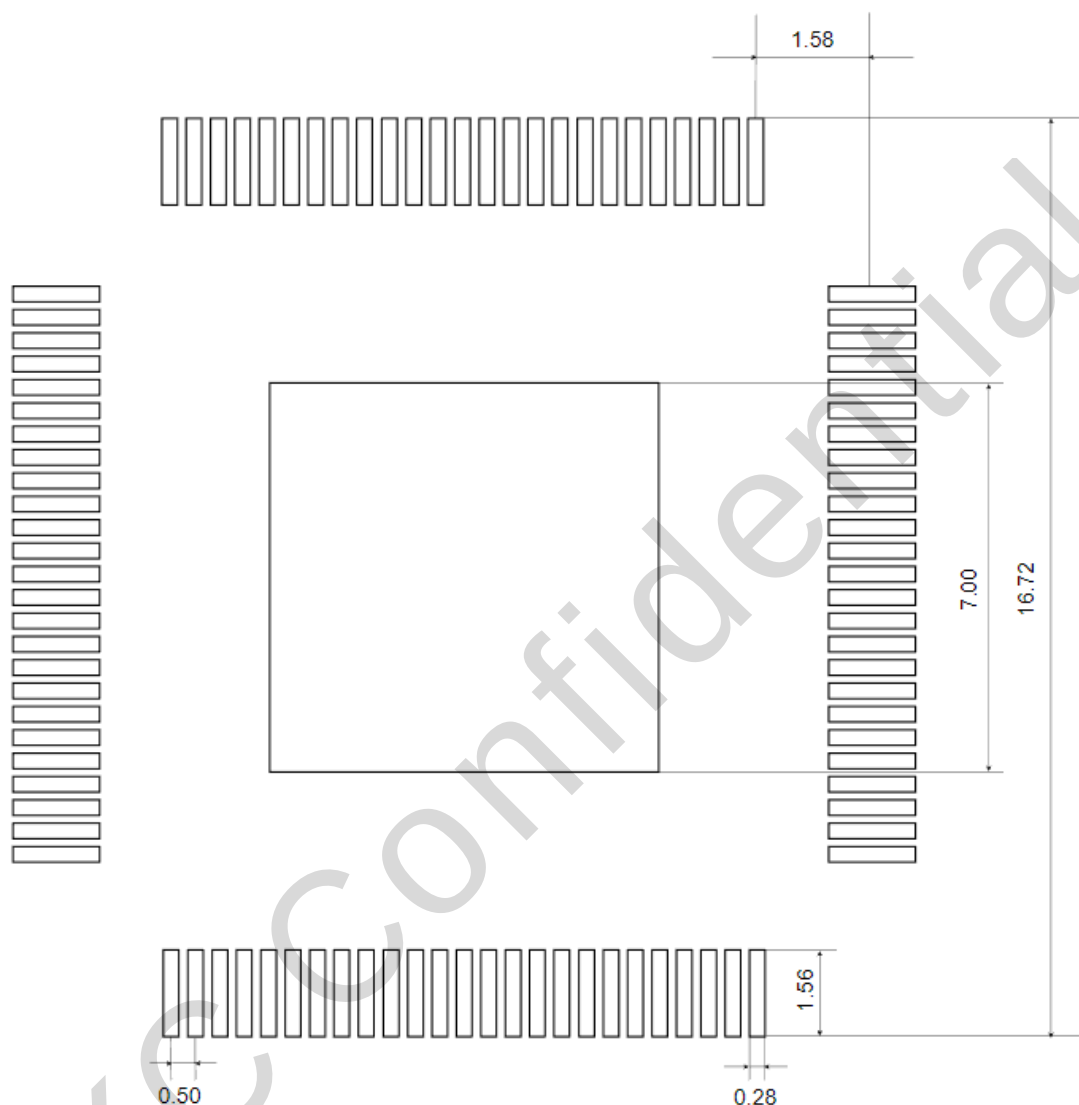


Figure 19-1 ETQFP package outline

## 20 Recommended pad size



Unit: mm

**Figure 20-1 Recommended pad size**

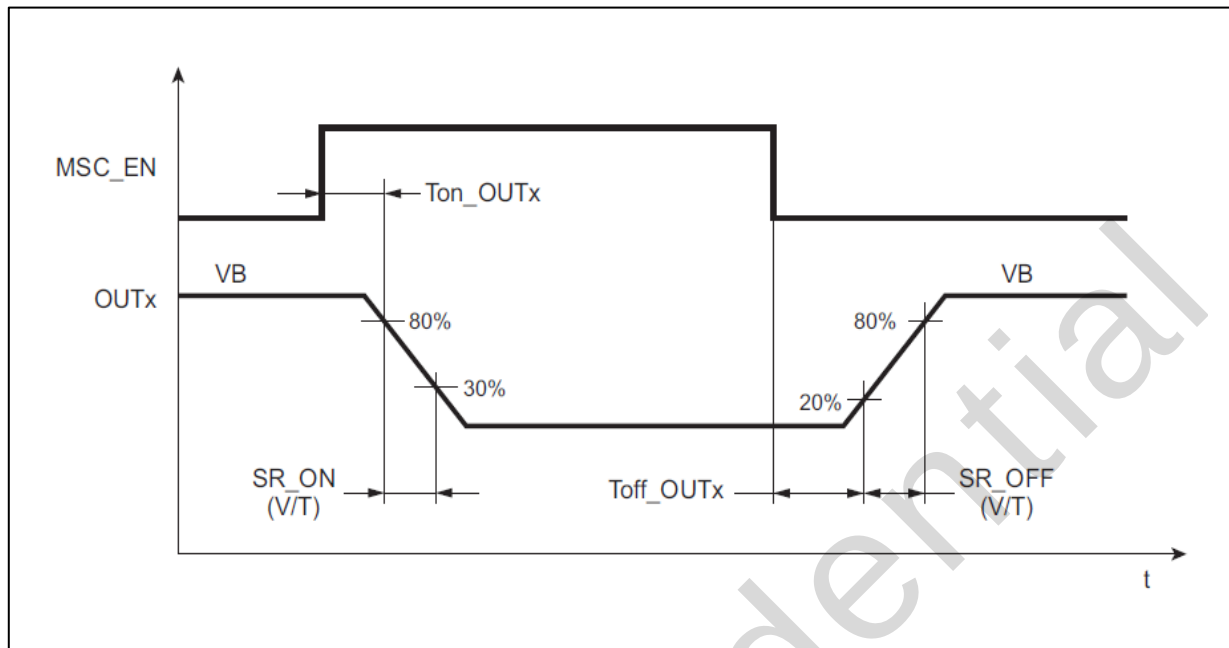
## 21 Appendix A Device behavior

### 21.1 Summary of all drivers block

**Table 21-1**Summary of all drivers block

| Drivers |    | Min OVC value(A) | Rdson ( $\Omega$ ) | Vclamp Typ(V) | STB | STG | OPL | OVC |
|---------|----|------------------|--------------------|---------------|-----|-----|-----|-----|
| INJ     |    | 3                | 0.6                | 55            | √   | √   | √   | √   |
| SOL     |    | 3                | 0.47               | 55            | √   | √   | √   | √   |
| O2H     |    | 3/7.8            | 0.2                | 50            | √   | √   | √   | √   |
| RLY     |    | 1                | 1.5                | 50            | √   | √   | √   | √   |
| HSLs    | HS | 1                | 1.5                | -3.5          | √   | √   | √   | √   |
|         | LS | 1                | 1.5                | 45            | √   | √   | √   | √   |
| LED     |    | 0.07             | 20                 | 45            | √   | √   | √   | √   |
| MRD     |    | 1                | -                  | 50            | √   | √   | √   | √   |
| IGN     |    | 0.1              | 7                  | -             | √   | √   | √   |     |
| PreMOS  |    | -                | -                  | -             | √   | √   | √   | √   |

## 21.2 Slew rate and on/off delay time



**Figure 0-1 Slew rate and on/off delay time**

$Ton\_OUTx$  is the on delay time of  $OUTx$ ;

$Toff\_OUTx$  is the off delay time of  $OUTx$ ;

$SR\_ON$  is the on slew rate of  $OUTx$ ;

$SR\_OFF$  is the off slew rate of  $OUTx$ .

### 21.3 Power up/down scenarios

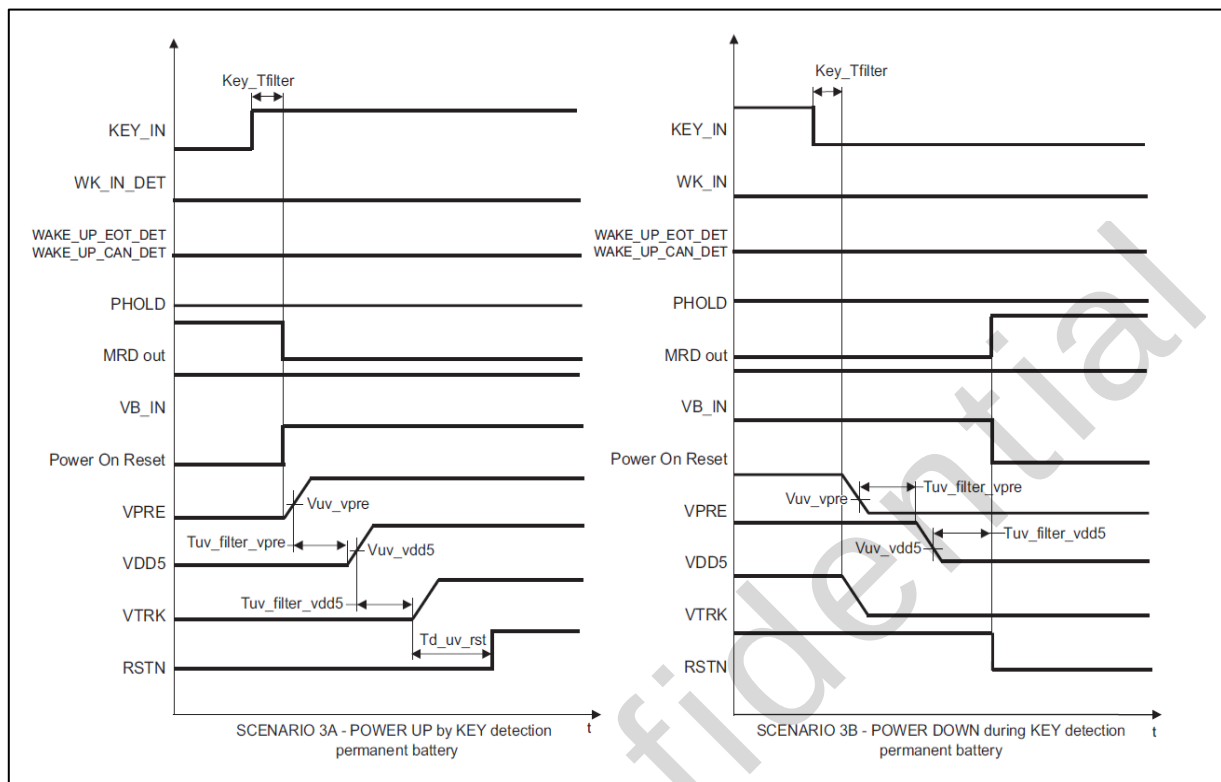
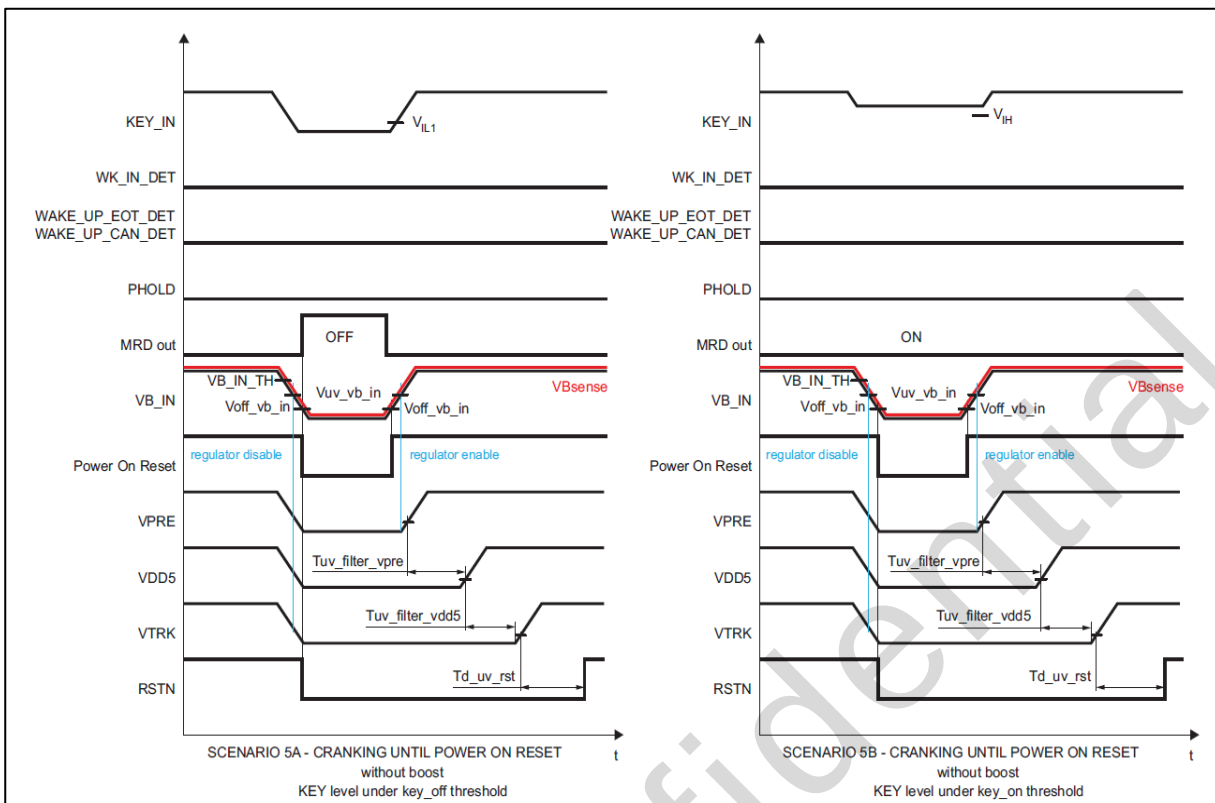
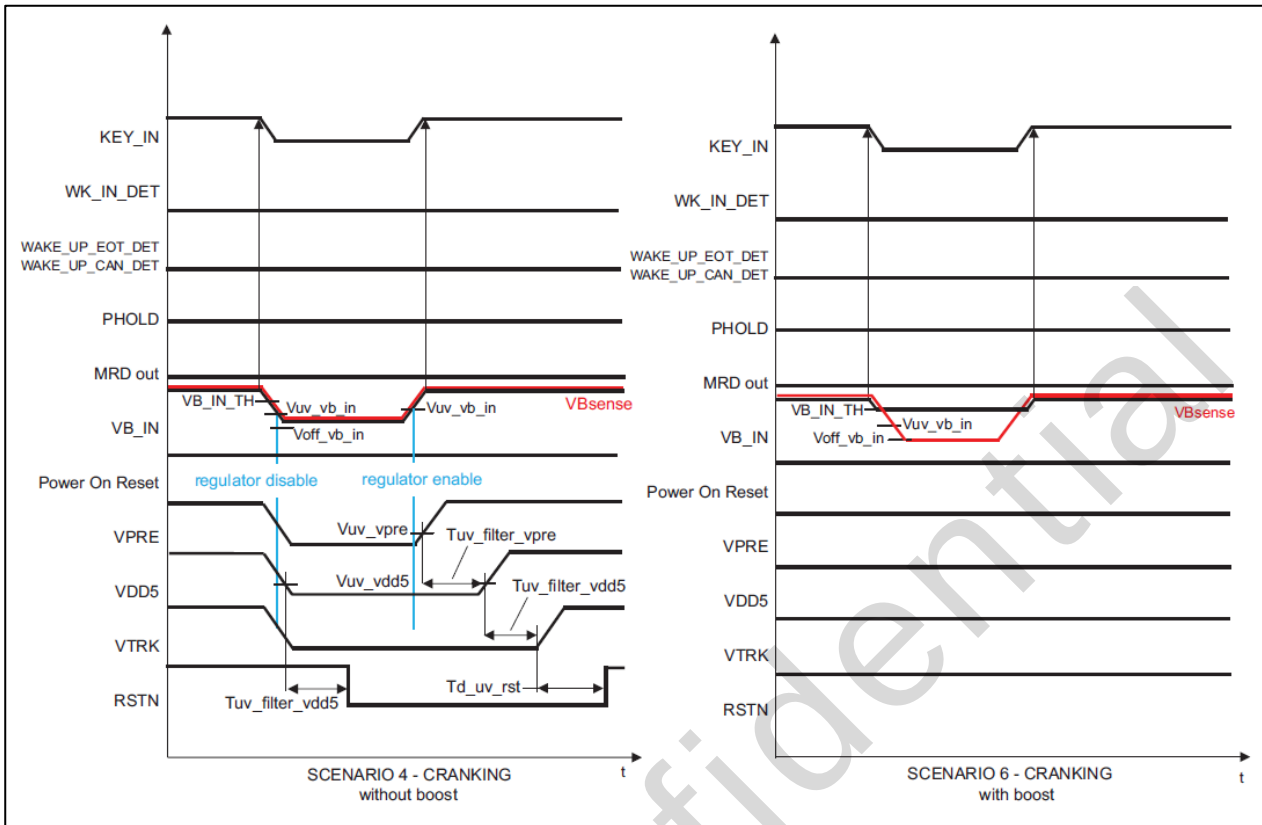


Figure 0-2 Slew rate and on/off delay time



**Figure 0-3 Power up permanent battery without boost and cranking**



**Figure 0-4 Power up permanent battery with boost and cranking**

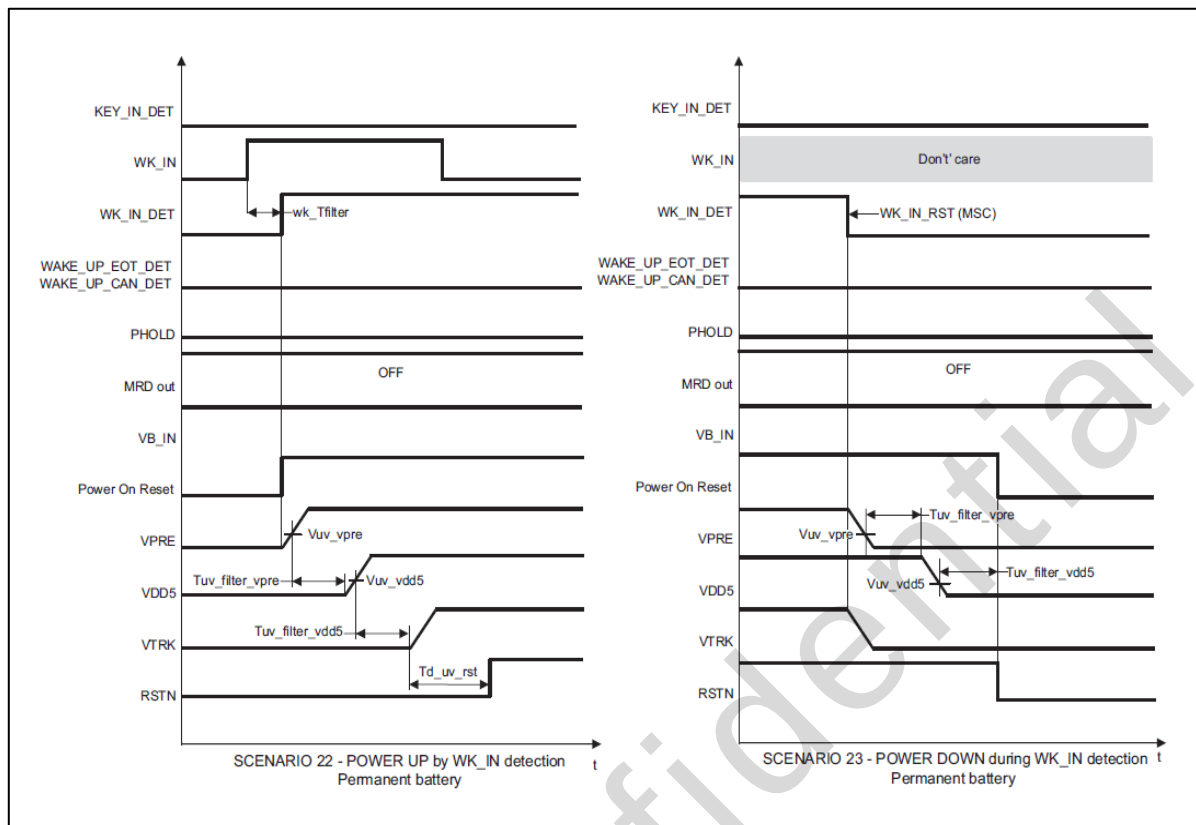


Figure 0-5 Power up with WK\_IN and power down with WK\_IN in permanent battery condition

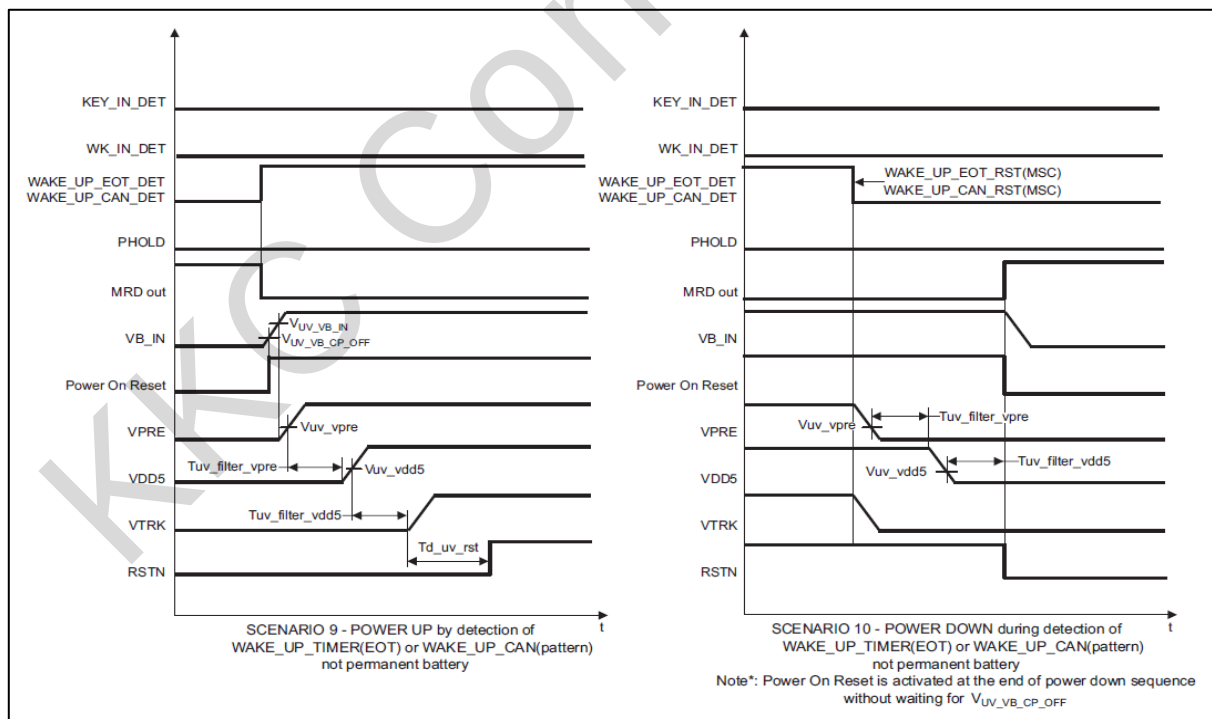
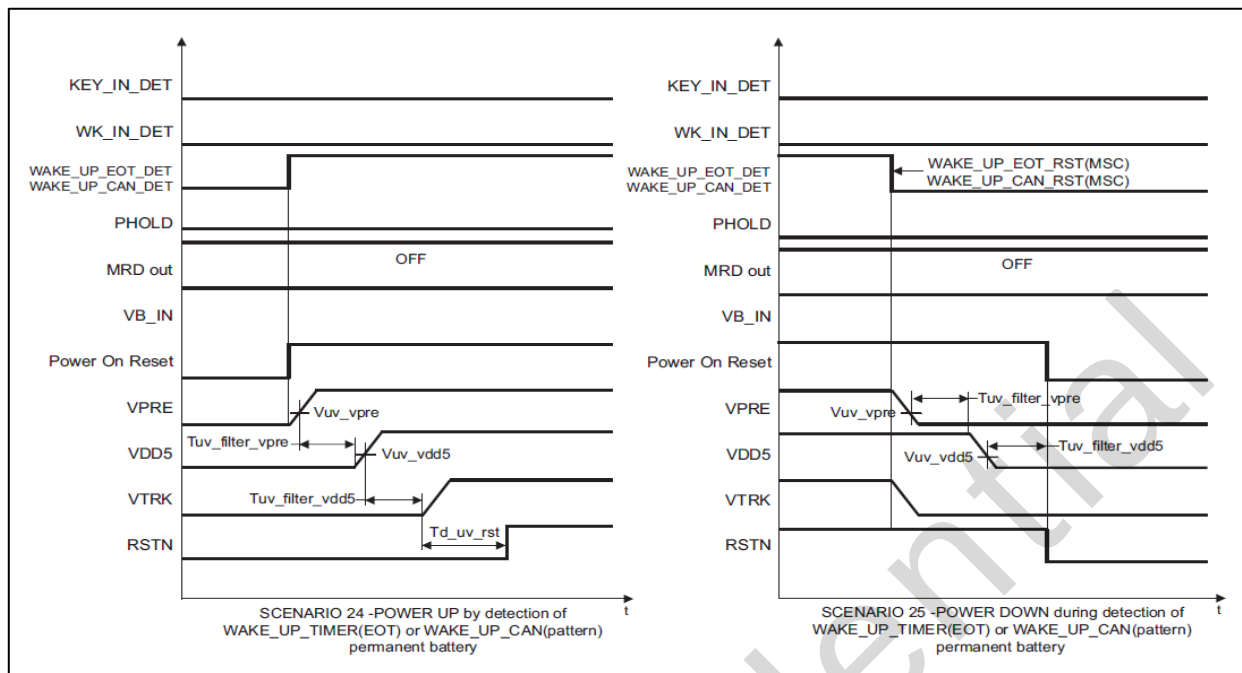
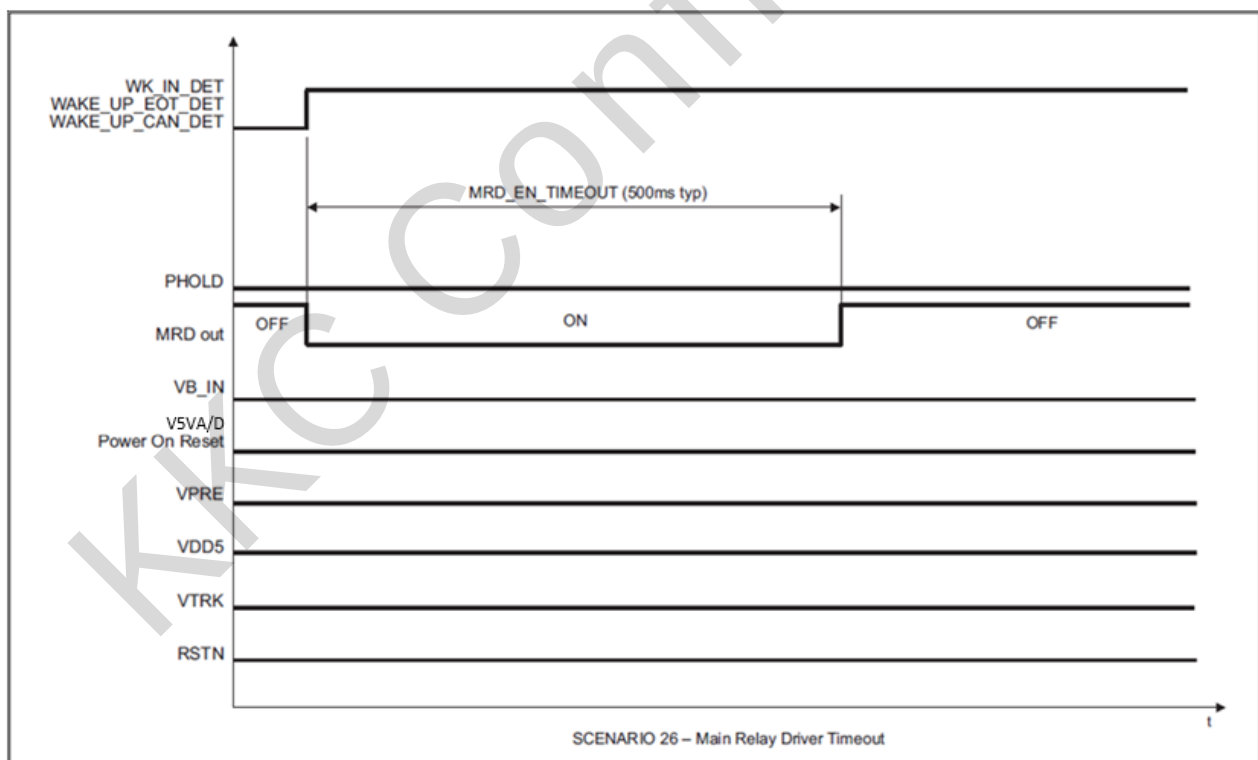


Figure 0-6 Power up with WAKE\_UP\_EOT/CAN and power down with WAKE\_UP\_EOT/CAN



**Figure 0-7** Power up with WAKE\_UP\_EOT/CAN and power down with WAKE\_UP\_EOT/CAN



**Figure 0-8** Main relay driver timeout

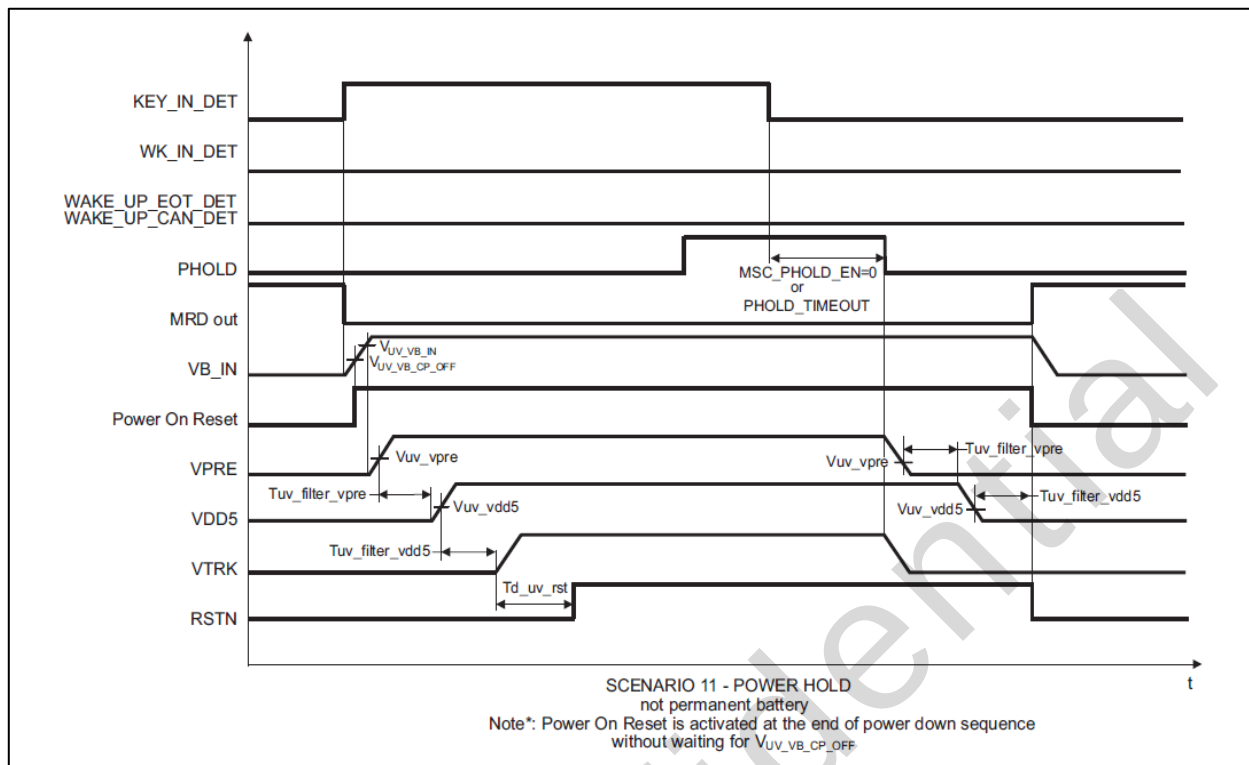


Figure 0-9 Power HOLD

According to reset matrix power down is also produced by Watchdog PWR\_CNT counter overflow during KEY\_IN low. In these cases the power down sequence is actuated without considering VDD5 undervoltage

## A.4 Main relay scenario

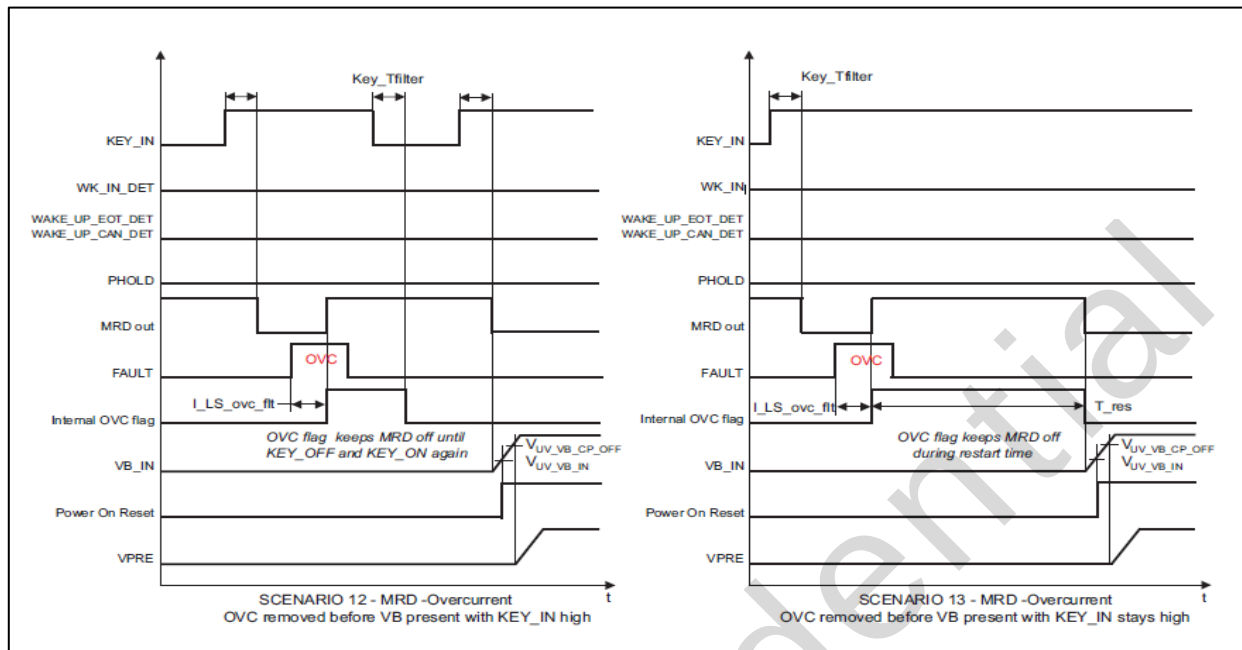


Figure 0-10 Overcurrent OVC removed before VB present with KEY\_IN high and Overcurrent OVC

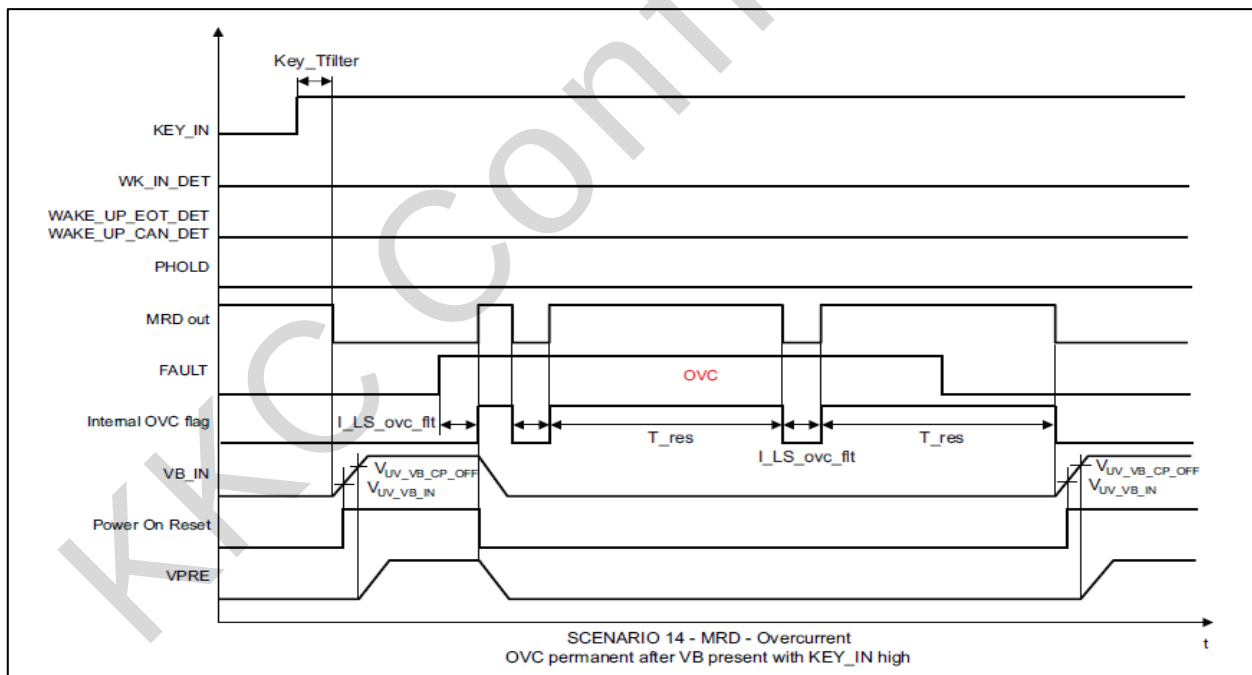
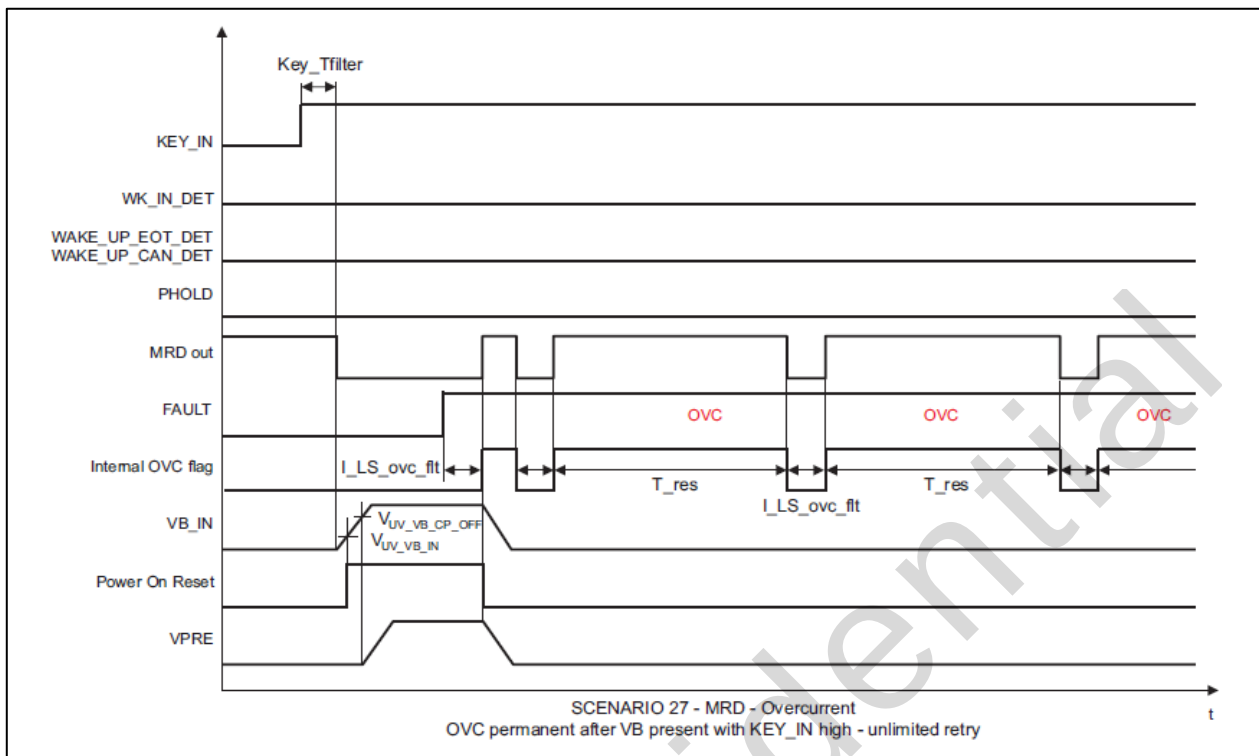
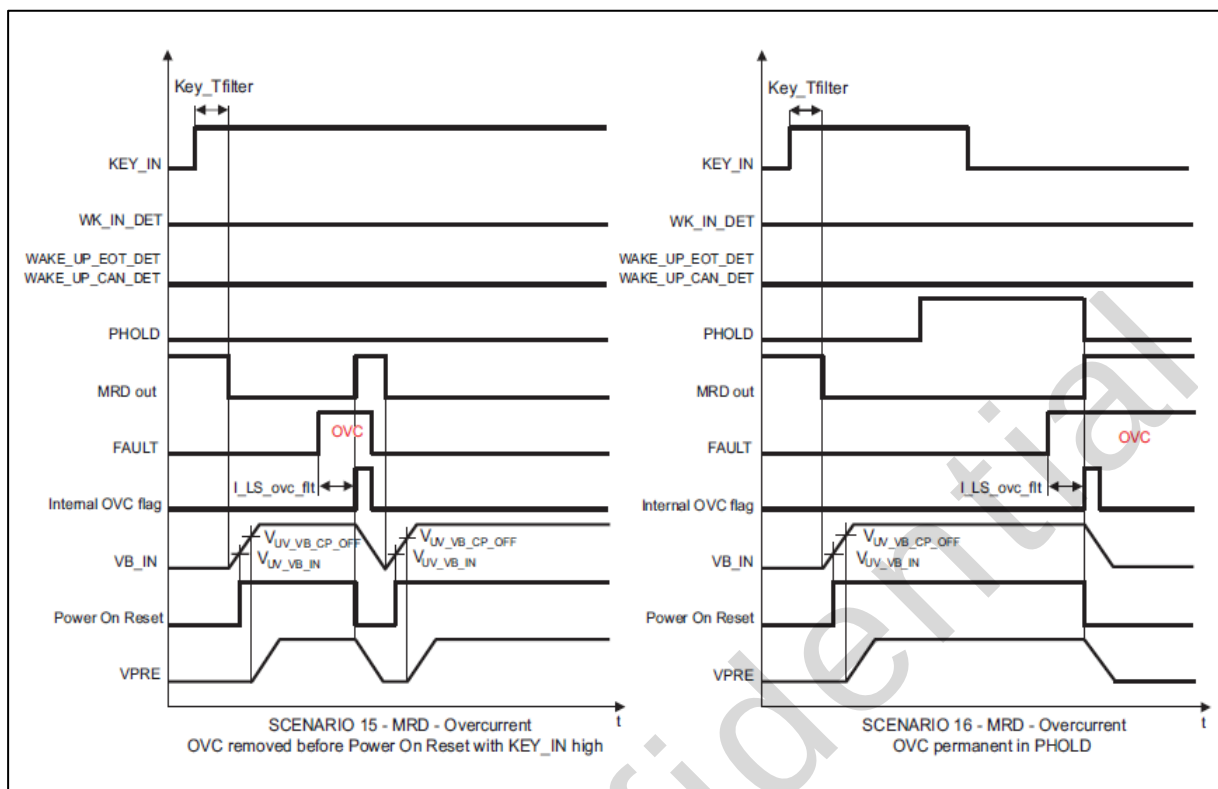


Figure 0-11 Overcurrent OVC permanent after VB present with KEY\_IN high

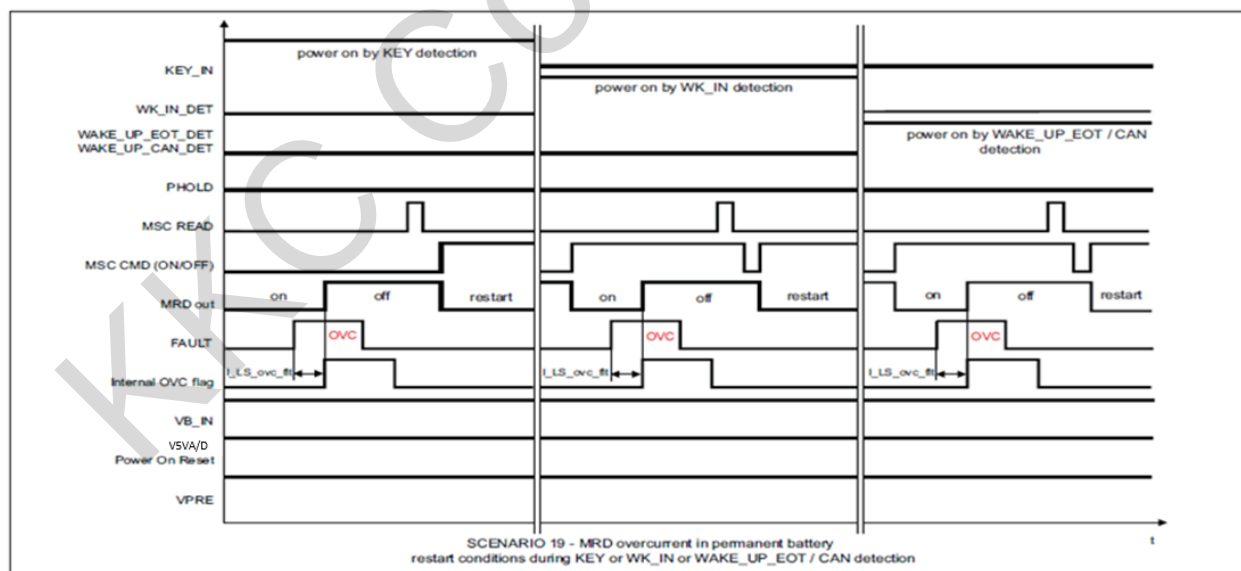


**Figure 0-12** Overcurrent OVC permanent after VB present with KEY\_IN high - unlimited retry



**Figure 0-13** Overcurrent OVC removed before Tres activation with KEY\_IN high and Overcurrent

## Appendix B Application diagrams



**Figure 0-14** Overcurrent in permanent battery restart conditions during KEY or WK\_IN or

In permanent battery when power on by KEY the MSC CMD can only restart driver after OVC. The MSC CMD OFF has no effect

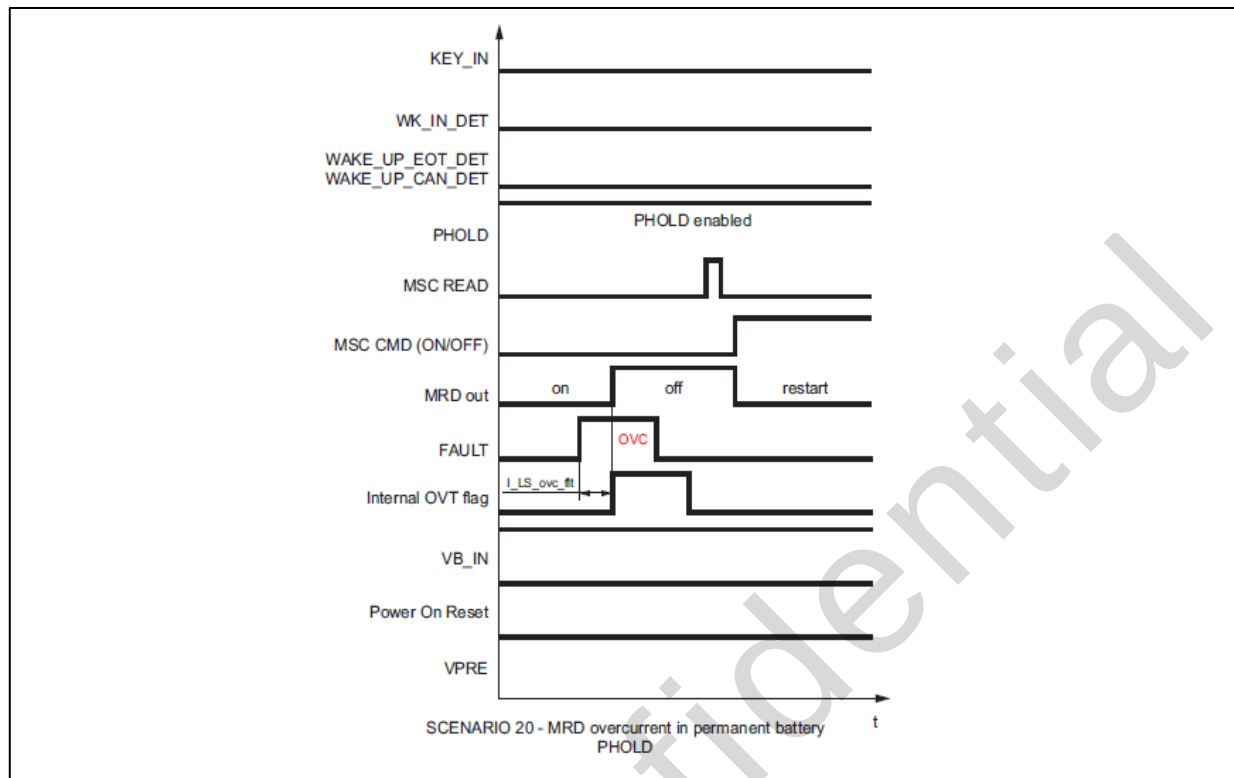


Figure 0-15 Overcurrent in permanent battery PHOLD

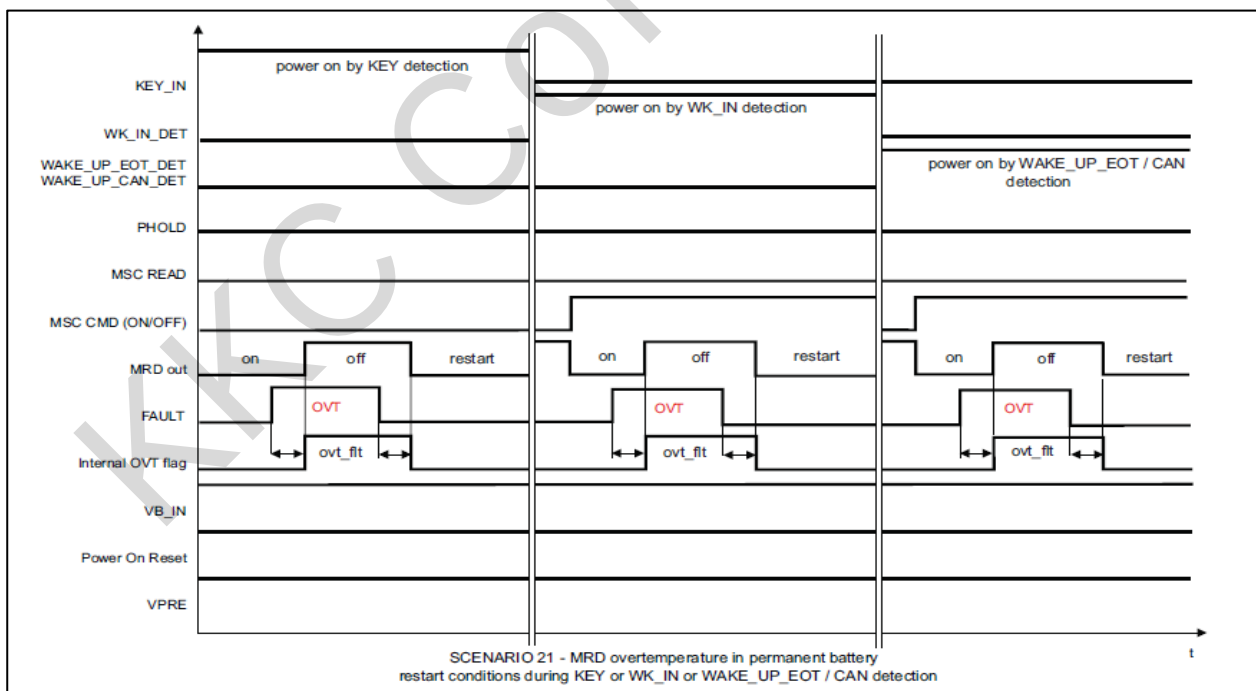
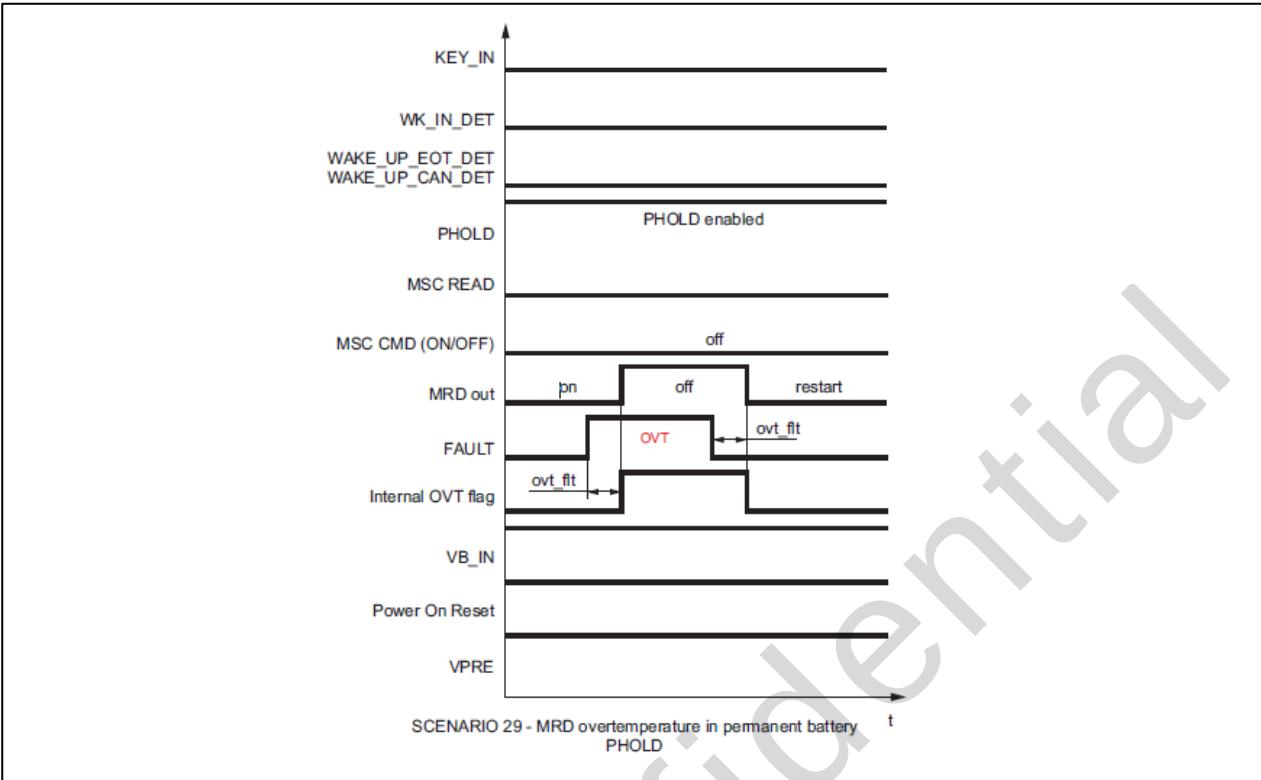


Figure 0-16 Overtemperature in permanent battery restart conditions during KEY or WK\_IN or



**Figure 0-17** Overtemperature in permanent battery PHOLD

# HE9788

Multifunction IC for automotive engine management system



## Appendix A Device behavior

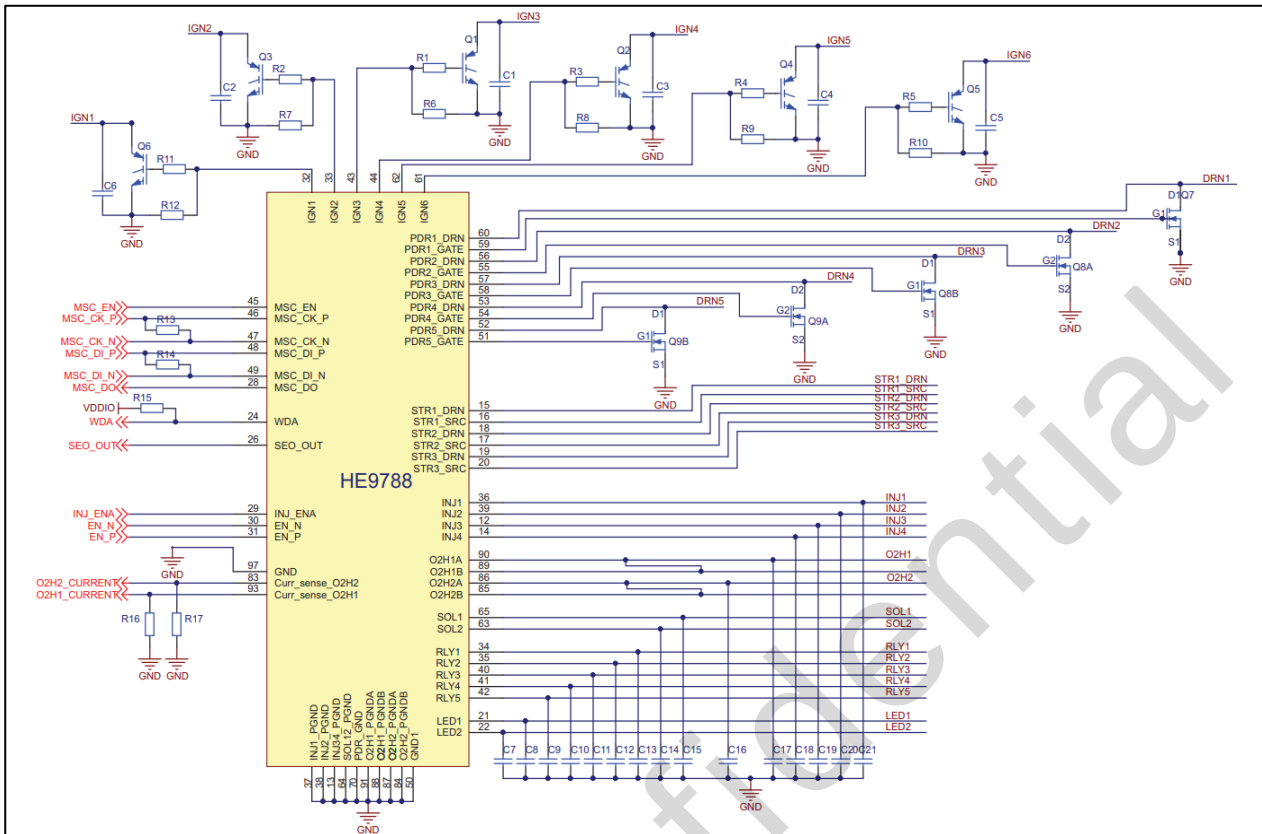
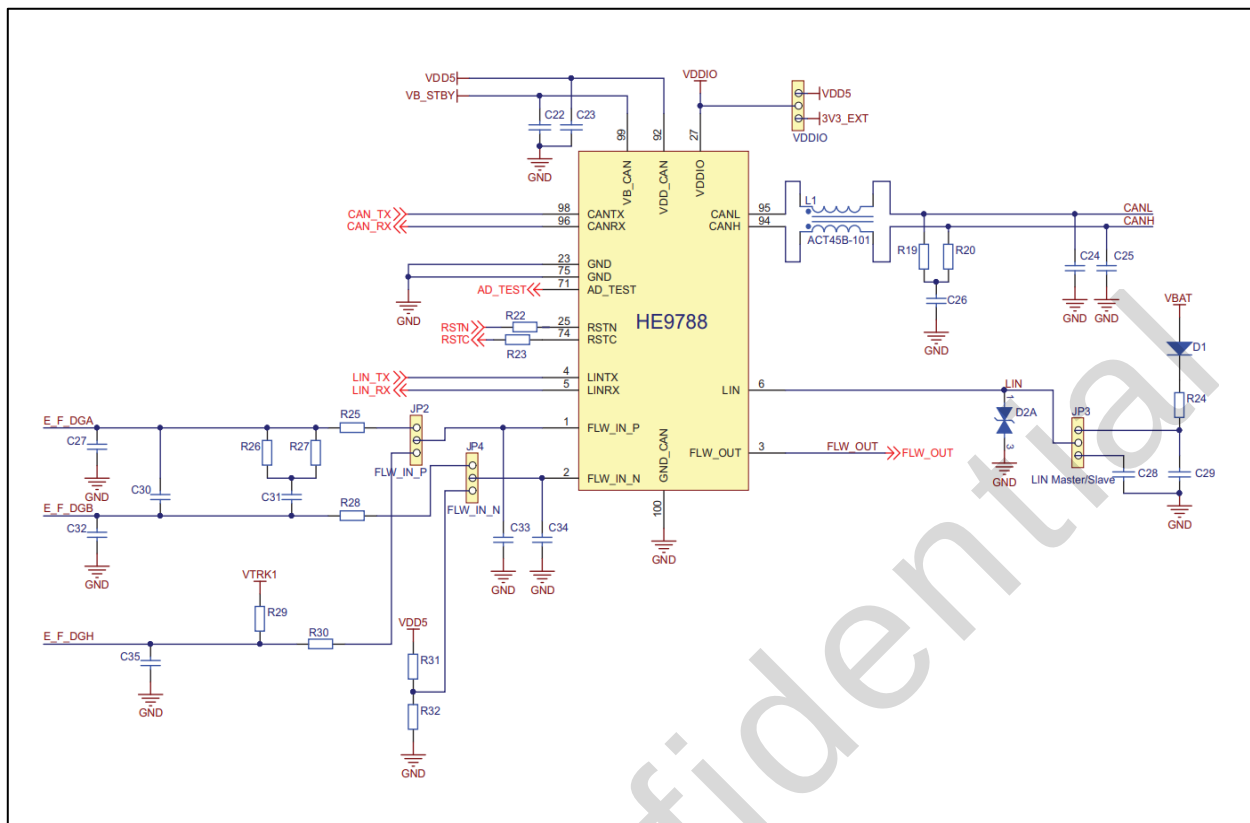
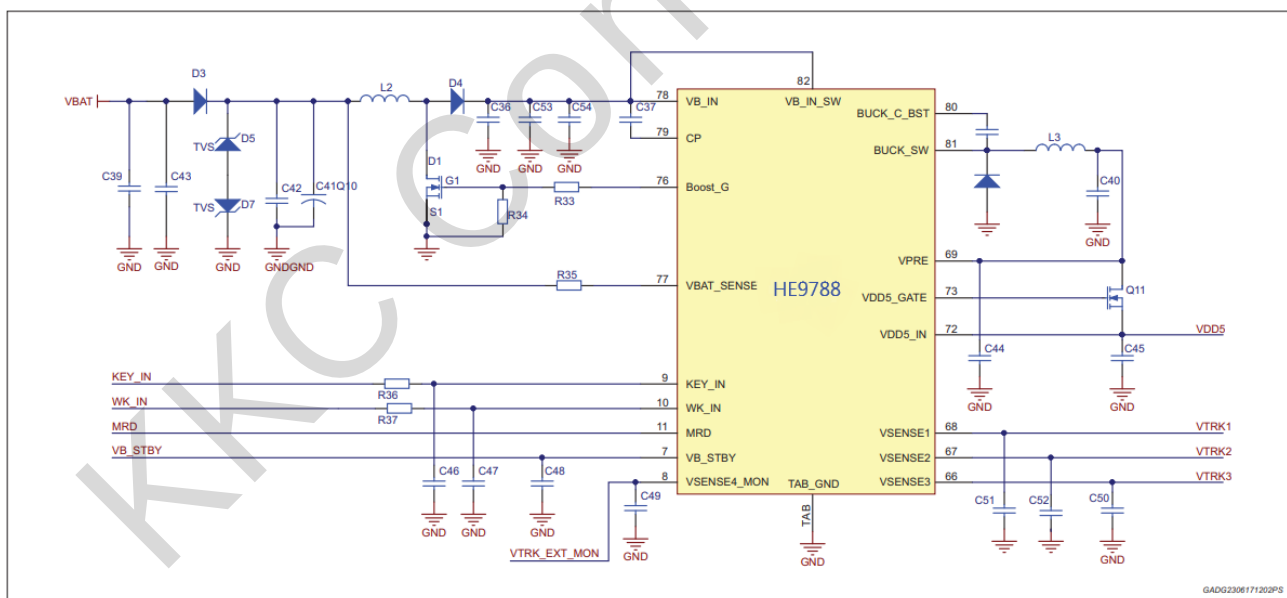


Figure 0-18 Driver/predriver section application diagram



**Figure 0-19** Interface section application diagram



**Figure 0-20 Multi regulator supply section application diagram**

## 21.4 Bill of material

Table 21-2 Bill of material

| Symbol                                                      | Parameter                                   | Min  | Typ  | Max | Unit | Note                                              |
|-------------------------------------------------------------|---------------------------------------------|------|------|-----|------|---------------------------------------------------|
| C39                                                         | Battery filter capacitor                    | -    | 4.7  | -   | nF   | -                                                 |
| D3                                                          | Reverse battery protection diode            | -    | -    | -   | -    | Choice: STTH4R02                                  |
| D5,D7                                                       | TVS diode                                   | -    | -    | -   | -    | Choice: TPSMB36A                                  |
| C42                                                         | battery decoupling capacitor                | -    | 100  | -   | nF   | -                                                 |
| C41                                                         | battery decoupling capacitor                | -    | 100  | -   | μF   | -                                                 |
| L2                                                          | Boost inductance                            | -    | -    | -   | -    | see Table 6-2                                     |
| Q10                                                         | Boost MOSFET                                | -    | -    | -   | -    | choice: STD20NF06L<br>see Table 6-1 and Table 6-2 |
| D4                                                          | Boost diode                                 | -    | -    | -   | -    | choice: STPS10H100C see Table 6-2                 |
| C36                                                         | Cout_boost                                  | 36   | 40   | 150 | μF   | see Table 6-2                                     |
| C53                                                         | Cout_boost_ALU                              | 0    | 330  | 470 | μF   | see Table 6-2                                     |
| R33                                                         | Rg                                          | -    | 2.2  | -   | Ω    | see Table 6-2                                     |
| R34                                                         | Rg_pd                                       | -    | 150  | -   | kΩ   | see Table 6-2                                     |
| C37                                                         | External charge pump capacitor              | 80   | 100  | 120 | nF   | see Table 6-13                                    |
| R35                                                         | Rsense                                      | 10   | 22   | 30  | kΩ   | see Table 6-2                                     |
| C43                                                         | filter capacitor                            | -    | 100  | -   | nF   | Optional                                          |
| R36                                                         | KEY_IN protection resistor                  | -    | 1    | -   | kΩ   | -                                                 |
| R37                                                         | WK_IN protection resistor                   | -    | 1    | -   | kΩ   | -                                                 |
| C46,C47,C48,C49,C44,C54                                     | filter capacitor                            | -    | 100  | -   | nF   | -                                                 |
| C38                                                         | CBST                                        | -    | 47   | -   | nF   | see Table 6-4                                     |
| L3                                                          | Lvpre                                       | 15   | 22   | 29  | μH   | see Table 6-4                                     |
| D6                                                          | Dvbuck_CTRL                                 | -    | -    | -   | -    | Choice: STPS5L60-Y<br>see Table 6-4               |
| C40                                                         | Cvpre                                       | 15   | 40   | 150 | μF   | see Table 6-4                                     |
| Q11                                                         | -                                           | -    | -    | -   | -    | choice: STD20NF06L<br>see Table 6-8 and Table 6-9 |
| C45                                                         | CVDD5                                       | 5    | 20   | 60  | μF   | see Table 6-9                                     |
| C50,C51,C52                                                 | CVS1/2/3                                    | 0.38 | 0.47 | 20  | μF   | see Table 6-9                                     |
| R1,R2,R3,R4,R5, R11                                         | IGBT gate resistor                          | -    | 1    | -   | kΩ   | -                                                 |
| R6,R7,R8,R9,R10, R12                                        | IGBT Gate discharge resistor                | -    | 4.7  | 0   | kΩ   | -                                                 |
| C1,C2,C3,C4,C5,C6                                           | filter capacitor                            | -    | 1.5  | -   | nF   | -                                                 |
| R13,R14                                                     | MSC differential line adapting resistor     | -    | 100  | -   | Ω    | -                                                 |
| R15                                                         | WDA pull-up external resistor               | 50   | -    | 250 | kΩ   | See Table 4-6                                     |
| Q7, Q8A, Q8B, Q9A, Q9B                                      | -                                           | -    | -    | -   | -    | choice: STL15DN4F                                 |
| C7,C8,C9, C10,C11, C12,C13,C14,C15,C16, C17,C18,C19,C20,C21 | filter capacitor                            | -    | 4.7  | -   | nF   | -                                                 |
| R16, R17                                                    | -                                           | -    | 5.1  | -   | kΩ   | See Table 7-6                                     |
| C22,C23                                                     | decoupling capacitor                        | -    | 100  | -   | nF   | -                                                 |
| R22,R23                                                     | protection resistor                         | -    | 1    | -   | kΩ   | -                                                 |
| C27, C32                                                    | filter capacitor (VRS sensor configuration) | -    | 100  | -   | pF   | -                                                 |
| C30                                                         | filter capacitor (VRS sensor configuration) | -    | 470  | -   | pF   | -                                                 |
| R26,R27                                                     | filter resistor (VRS sensor                 | -    | 33   | -   | kΩ   | -                                                 |

|          | configuration)                                                |   |          |   |    |                                    |
|----------|---------------------------------------------------------------|---|----------|---|----|------------------------------------|
| C31      | filter capacitor (VRS sensor configuration)                   | - | 100      | - | nF | -                                  |
| R25,R28  | Protection and reference resistors (VRS sensor configuration) | - | 10<br>1% | - | kΩ | see Section 11.2                   |
| C35      | filter capacitor (Hall sensor configuration)                  | - | 1        | - | nF | -                                  |
| R29      | pull-up resistor (Hall sensor configuration)                  | - | 27       | - | kΩ | -                                  |
| R30      | Protection and reference resistor (Hall sensor configuration) | - | 27       | - | kΩ | -                                  |
| R31,R32  | Protection and reference resistor (Hall sensor configuration) | - | 33       | - | kΩ | -                                  |
| C33,C34  | filter capacitor                                              | - | 470      | - | pF | place close to pin                 |
| L1       | CAN Choke                                                     | - | -        | - | -  | choice: ACT45B-101                 |
| R19, R20 | CAN termination resistor                                      | - | 60       | - | Ω  | -                                  |
| C26      | CAN termination capacitor                                     | - | 4.7      | - | nF | -                                  |
| C24,C25  | -                                                             | - | 47       | - | pF | -                                  |
| D2A      | -                                                             | - | -        | - | -  | 27V TVS                            |
| D1       | Vser Diode                                                    | - | -        | - | -  | Choice: BAS21                      |
| R24      | Lin pull-up resistor                                          | - | 1        | - | kΩ | -                                  |
| C28      | -                                                             | - | -        | - | -  | Choice: 220pF                      |
| C29      | -                                                             | - | -        | - | -  | Choice: 1 nF (K-Line),680 pF (LIN) |

## 21 Customer Service

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## 22 Revision history

**Table 22-1 Document revision history**

| Date       | Revision | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023/08/01 | 2.0      | 1. Format updated to 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2023/11/04 | 2.1      | 2. Minor changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2024/02/04 | 0.3.1    | <b>Format updated to 0.3.1</b><br>1、Figure 33&44: Change FAULT in the bottom right corner of the figure to NO FAUL.<br>2、Table 99: Add STB.<br>3、Table 1: 'outpour' is corrected to 'output'.<br>4、Figure 91、95、105-107、112-115: clear content.<br>5、Table 2: The format is centered.<br>6、VB_IN: All parameters below 4.1V were modified.<br>7、Digital and analog supply: All 3V3 is replaced by 5V.<br>8、VUV_VB_CP_ON: min Typ max three-term value +1.<br>9、3.1.5 Section: Delete the part about not permanent.<br>10、Table 104: V3V3 is changed to V5V.<br>11、Vline_tran: Unit error.<br>12、Figure 21&24: Delete.<br>13、VGS(th): Unit error.<br>14、Figure 78: Modify V3V3 to V5V, with VIND, VINT.<br>15、Table 25: Fast ON diagnostic OFF→ON→OFF is changed to ON→OFF→ON<br>16、Change the Package diagram.<br><b>VPRE:</b><br>17、1.6-<2.2 A maximum output current.<br>18、The upper and lower limits of <i>ILIMVPRE</i> are changed to 3.0A and 4.4A.<br><b>O2H1/2</b><br>19、The upper limit of <i>LS_RdsON</i> Tj = 25 °C are set to 0.04Ω.<br>12、The lower limits of <i>LS_RdsON</i> Tj = -40 °C are set to 0.03Ω. |
| 2024/6/17  | 0.3.2    | 1、Add recommended pad size                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2024/7/22  | 1.0.0    | 1. Format updated to 1.0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2024/8/2   | 1.0.1    | <b>The following parameters have been modified:</b><br><b>VB_IN:</b><br>1. The upper limit of the <i>IVB_IN_STBY</i> is changed to 13μA<br>2. The upper limit of <i>IVB_IN</i> is changed to 80mA<br><b>VPRE:</b><br>3. The upper limit of the <i>VPRE_slope</i> is changed to 40V/ms<br>4. The upper limit of the <i>RVPRE</i> is changed to 250 mΩ<br><b>VSENSE1/2/3</b><br>5. The upper and lower limits of ΔVTRK are changed to VDD5-15mV and VDD5+20mV<br>6. The upper and lower limits of <i>Vline_VS</i> are changed to -15mV and 30mV<br>7. The upper and lower limits of <i>Vload_VS</i> are changed to -15mV and 30mV<br><b>VDD5_IN</b><br>8. The lower limit of <i>VOV_VDD5</i> is changed to 5.25V<br><b>INJ1/2/3/4</b><br>9. The upper limit and lower limit of <i>LS_RdsON</i> Tj = 150 °C are changed to 0.6Ω and 0.3Ω.<br>10. The upper limit of <i>LS_RdsON</i> Tj = 25°C is changed to 0.45Ω<br>11. The upper limit of <i>LS_RdsON</i> Tj = -40 °C is changed to 0.3Ω<br><b>O2H1/2</b><br>12. The upper limit and lower limit of <i>LS_RdsON</i> Tj = 150 °C are set to 0.23Ω and 0.04Ω.               |

|            |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |       | <p>13. The upper limit of <math>LS\_RdsON</math> <math>T_j = 25\text{ }^{\circ}\text{C}</math> is set to <math>0.2\Omega</math></p> <p>14. The upper limit of <math>LS\_RdsON</math> <math>T_j = -40\text{ }^{\circ}\text{C}</math> is set to <math>0.13\Omega</math></p> <p><b>SOL1/2</b></p> <p>15. The upper limit of <math>LS\_RdsON</math> <math>T_j = 25\text{ }^{\circ}\text{C}</math> is changed to <math>0.35\Omega</math></p> <p>16. The upper limit of <math>LS\_RdsON</math> <math>T_j = -40\text{ }^{\circ}\text{C}</math> is set to <math>0.23\Omega</math></p> <p><b>STR1/2/3</b></p> <p>17. The upper limit of <math>I_{rev\_OUTx}</math> is changed to <math>1.5\text{V}</math></p> <p><b>PRD[1:5]</b></p> <p>18. The upper and lower limits of <math>I_{source\_02H}</math> are changed to <math>-0.355\text{mA}</math> and <math>-0.155\text{mA}</math></p> <p>19. The upper and lower limits of <math>I_{sink\_02H}</math> are changed to <math>0.155\text{mA}</math> and <math>0.355\text{mA}</math></p> <p><b>IGN1/2/3/4/5/6</b></p> <p>20. The lower limit of <math>I_{leak\_STB\_IGN}</math> is changed to <math>-1\text{mA}</math></p> <p><b>FLW_OUT</b></p> <p>21. The upper limit of <math>Td\_off\_outvrs</math> is set to <math>2\mu\text{s}</math></p> <p>22. The upper limit of <math>T\_r\_OUT\_vrs</math> is set to <math>250\text{ns}</math></p> <p><b>CAN</b></p> <p>23. The lower limit of <math>VDD\_CAN\_SUP\_LOW</math> is set to <math>4\text{V}</math></p> <p>24. The upper and lower limits of <math>VTHdom</math> are changed to <math>0.4\text{V}</math> and <math>1.1\text{V}</math></p> <p>25. The upper and lower limits of <math>VTHrec</math> are changed to <math>0.4\text{V}</math> and <math>0.95\text{V}</math></p> <p>26. The upper and lower limits of <math>VTHdomLPbias</math> are changed to <math>0.4\text{V}</math> and <math>0.95\text{V}</math></p> <p>27. The upper and lower limits of <math>VTHrecLPbias</math> are changed to <math>0.4\text{V}</math> and <math>0.95\text{V}</math></p> <p>28. The upper and lower limits of <math>VTHdomLP</math> are changed to <math>0.4\text{V}</math> and <math>0.95\text{V}</math></p> <p>29. The upper and lower limits of <math>VTHrecLP</math> are changed to <math>0.4\text{V}</math> and <math>0.95\text{V}</math></p> <p><b>LIN</b></p> <p>30. The lower limit of <math>trise\_LIN</math> is changed to <math>0.8\text{V}/\mu\text{s}</math></p> <p>31. The upper and lower limits of <math>VserDiode</math> are changed to <math>0.7\text{V}</math> and <math>1.3\text{V}</math>, and the typical value is changed to <math>1\text{V}</math>.</p> <p>32. The upper and lower limits of <math>VserDiode\_drop</math> are changed to <math>0.7\text{V}</math> and <math>1.3\text{V}</math>, and the typical value is changed to <math>1\text{V}</math>.</p> <p><b>ESD</b></p> <p>33. Table 2-2: Revised ESD pin and voltage range.</p> <p><b>Main oscillator</b></p> <p>34. Example Delete information about freq err.</p> <p><b>STR</b></p> <p>35. Delete the STA specification for STR diagnostics and the STA bit register.</p> <p><b>Delay-off function</b></p> <p>36. Added the description in the delay-off function section.</p> <p><b>CAN</b></p> <p>37. Changed CAN_RX pin masked to 1 to 0.</p> <p><b>MSC</b></p> <p>38. Add Tswitch2 parameters and add comments.</p> <p><b>Tracking</b></p> <p>39. Added a note that only track1_OV diagnostics are valid and removed register INT_TRK_OV[2]&amp;[3] bit.</p> <p><b>KEY_OC_RETRY_MAX_EN</b></p> <p>40. Example Change default to 1.</p> |
| 2024/11/29 | 1.0.2 | <p><b>Modification details are as follows:</b></p> <p><b>ESD</b></p> <p>1. Table 2-2: Revised ESD pin and voltage range.</p> <p><b>Main oscillator</b></p> <p>2. Add information about Freq err.</p> <p><b>STR</b></p> <p>3. Added STA specifications for STR diagnostics and STA bit registers.</p> <p><b>Delay-off function</b></p> <p>4. Delete the description of the delay function section.</p> <p><b>CAN</b></p> <p>5. Changed CAN_RX pin masked to 0 to 1.</p> <p>6. The upper limit of <math>TBit(RXD) \leq 5\text{Mb/s}</math> is set to <math>250\text{ns}</math>.</p> <p><b>MSC</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|            |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |       | <p>7. Delete Tswitch2 parameters and comments.</p> <p><b>Tracking</b></p> <p>8. Remove the comment that only track1_OV diagnostics are valid, and add registers INT_TRK_OV[2] and [3] bits.</p> <p><b>Part Number</b></p> <p>9. The number is changed from HE9788YIAG to HE9788YIAJ.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2024/12/20 | 1.0.3 | <p><b>CAN</b></p> <p>1. The upper limit of <i>VCANHrecLP</i>&amp;<i>VCANLrecLP</i> is changed to 0.15V</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2025/3/3   | 2.0   | <p><b>Update the format and the following:</b></p> <ol style="list-style-type: none"> <li>The upper and lower limits of <i>ESD HBM Global CAN</i> are changed to <math>\pm 4KV</math>.</li> <li>Delete the description about enabling or detecting MRD on WK_IN. For details, see section 3.1.5.</li> <li>Delete the MRD EN TIMEOUT parameter.</li> <li>Delete the description about enabling or detecting MRD on WK_IN. For details, see section 3.1.7.</li> <li>Delete the description about enabling or detecting MRD on WK_IN. For details, see section 3.1.11.</li> <li>Delete the SEO_OUT_L parameter.</li> <li>Add description: "The MRD is automatically restarted every T res if KEY IN is till active in permanent battery supply scheme.", see section 5.1.1.</li> <li>Delete the description about enabling or detecting MRD on WK_IN. For details, see section 5.1.1.</li> <li>Change -12V of the Test condition in Table 12-11 to -6V.</li> <li>The lower limit of <i>TRVPRE_slow</i> is changed to 5 ns.</li> <li>The lower limit of <i>V_OUT_0p5A</i> is changed to 0.3V.</li> <li>The upper limit of <i>Vth_DS_001</i> is set to 273 mV.</li> <li>The lower limit of <i>VUV_VB_H</i> is changed to 4.75V.</li> <li>The upper limit of <i>Ith_OL_IGN</i> is set to 1.25 mA.</li> <li>The lower limit of <i>trise_K_LINE</i> is changed to 0.8 V/<math>\mu</math>s.</li> <li>The upper limit of <i>ILIMVPRE</i> is set to 5 A.</li> <li>Change fram 4 from 0X31 to 0X41 in Upstream Bit Map Read11.</li> <li>The upper limit of <i>Vth_DS_000</i> is set to 190 mV.</li> <li>The upper limit of <i>Vth_DS_001</i> is set to 285 mV.</li> <li>The upper limit of <i>Vth_DS_010</i> is set to 360 mV.</li> <li>The upper limit of <i>Vth_DS_011</i> is set to 445 mV.</li> <li>The upper limit of <i>Vth_DS_100</i> is set to 560 mV.</li> <li>The upper limit of <i>Vth_DS_101</i> is set to 710 mV.</li> <li>Update the Package information</li> <li>Delete the duplicate parameter items of <i>Vline/Vload_VPRE</i> and <i>Rgate_source</i>.</li> <li>Delete part of the description of CONFIG-REG 15.</li> <li>The functional descriptions regarding count up and Wake Up Timer start/stop commands have been corrected.</li> <li>Add application note to the read 1 register.</li> <li>Correct the illustration in figure 3-2.</li> <li>Delete part of the description in config 20.</li> <li>The upper and lower limits for the <i>VUV_VB_CP_ON</i> and <i>VUV_VB_CP_OFF</i> parameters have been adjusted.</li> <li>IOCANH, dom (40 V) Parameter correction of current direction.</li> <li>Description of ViR_LIN Parameter Increase.</li> <li>VDO_L Parameter Item Optimization Test Conditions.</li> <li>The <i>VB_IN_OFF</i> parameter has been changed from 4.1V to 4.25V, and this change applies to all parameters related to <i>VB_IN_OFF</i>.</li> <li>The upper limit of the <b>IMAX_VS</b> is changed to 300 mA.</li> </ol> |
| 2025/6/30  | 2.1   | <p><b>Update the format and the following:</b></p> <ol style="list-style-type: none"> <li>Change -6V of the Test condition in Table 12-11 to -5V.</li> <li>The upper limit of <b>IMAX_VS</b> is set to 300 mA.</li> <li>The BUCK output current of the "Features" module has been changed to 2.3A, and the VDD5 output current has been adjusted to 1.6A.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2025/7/4   | 2.2   | <ol style="list-style-type: none"> <li>Delete the <i>VIL2</i> parameter.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 2025/8/25  | 2.3   | <ol style="list-style-type: none"> <li><b>IOH2_HS_IGN</b> parameter item adds test condition description.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|          |     |                                                                                                                                                                                                                                                                                                                                                                        |
|----------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |     | <ol style="list-style-type: none"><li>Revise the description of the <b>MRD</b> section.</li><li>Optimization of electrical parameters.</li><li>The internal pull-up voltage of <b>LIN_TX</b> is changed from 3.3V to 5V.</li><li>The upper and lower limits of <b>ILIMVPRE</b> are changed to 3.5A and 5.1A.</li></ol>                                                 |
| 2026/2/2 | 2.4 | <ol style="list-style-type: none"><li>Optimize some parameter descriptions.</li><li>Optimize the block diagram.</li></ol>                                                                                                                                                                                                                                              |
| 2026/3/5 | 2.5 | <ol style="list-style-type: none"><li>The naming and descriptions of the parameters of the same type are unified: <b>Vclamp_OUTX</b>, <b>VOUTOPEN</b>, <b>I_OVC_XXX</b>.</li><li>Optimize the parameters for sink and source.</li><li>Modify the function description and error description.</li><li>Add a description of SPI communication on the home page</li></ol> |

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